Dynamic Solutions - International, LLC

THE ENVIRONMENTAL FLUID DYNAMICS CODE:

THEORETICAL & COMPUTATIONAL ASPECTS OF EFDC+

2017

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1 INTRODUCTION

1.1 Development History

The Environmental Fluid Dynamics Code (EFDC) is a public domain, open source, surface water modeling system, which includes one dimensional, two dimensional, and/or three dimensional hydrodynamics, sediment transport, water column constituent transport, water quality kinetics, submerged aquatic vegetation and sediment diagenesis integrated in a single source code implementation. EFDC has been applied to many hundreds of water bodies including rivers, lakes, reservoirs, wetlands, estuaries, and coastal ocean regions in support of environmental assessment and management and regulatory requirements.

EFDC was originally developed at the Virginia Institute of Marine Science (VIMS) and School of Marine Science of The College of William and Mary, by Dr. John M. Hamrick beginning in 1988. The U.S. Environmental Protection Agency including the Office of Science and Technology, the Office of Research and Development and Regions 1 and 4 have provided continued support for EFDC. The ongoing evolution of the EFDC model has to a great extent been application driven by a diverse group of EFDC users in the academic, governmental, and private sectors.

This EFDC+ Theory Guide relies to a significant extent on a number of documents from John Hamrick, especially "A Three-Dimensional Environmental Fluid Dynamics Computer Code: Theoretical and Computational Aspects" (Hamrick, 1992a). Other important documents referenced in this guide include:

- A User's Manual for the Environmental Fluid Dynamics Computer Code (EFDC), (Hamrick, J. M., 1996)
- A Three-dimensional Hydrodynamic-Eutrophication model (HEM3D): Description of Water Quality and Sediment Processes Submodels. (Park, K., *et al* 1995)
- Sandia National Laboratories Environmental Fluid Dynamics Code: Sediment Transport User Manual (Thanh, P.H.X., M.D. Grace, and S.C. James. 2008).

Since 1998, Dynamic Solutions-International, LLC (DSI) has been providing ongoing enhancement and development of EFDC for various surface water, sediment transport, water and quality projects. This includes adding many new features to EFDC, the theory for which is described in this guide. In 2016 DSI released a major update to the EFDC code, which is now named EFDCPlus (or EFDC+).

1.2 Model Overview

The Environmental Fluid Dynamics Code (EFDC) is a general-purpose modeling package for simulating three dimensional (3-D) flow, transport, and biogeochemical processes in surface water systems including: rivers, lakes, estuaries, reservoirs, wetlands, and near-shore to shelf-scale coastal regions.

In addition to hydrodynamic and salinity, temperature and dye transport simulation capabilities, EFDC is capable of simulating cohesive and non-cohesive sediment transport, the transport and fate of toxic contaminants in the water and sediment phases, and the dissolved oxygen/nutrient process (i.e. eutrophication). Special enhancements to the hydrodynamics of the code, including vegetation resistance, drying and wetting, hydraulic structure representation, wave current boundary layer interaction, and wave-induced currents, refined modeling of wetland and marsh systems, controlled-flow systems, and near-field and far-field discharge dilution from multiple sources.

The structure of the EFDC model includes six major modules (Figure 1-1):

(1) Hydrodynamics

(2) Water Column Constituent Transport

(3) Water Quality Parameter Kinetics

(4) Sediment Erosion/Deposition

(5) Toxics (Fate and transport)

(6) Lagrangian Particle Tracking

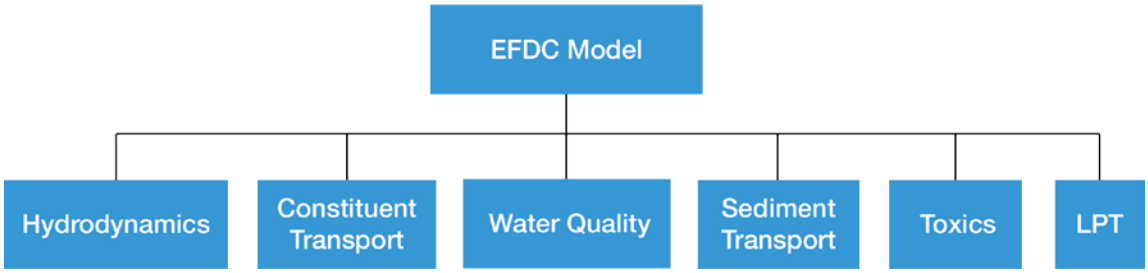


Figure 1-1: Primary modules of the EFDC model

1.2.1. Hydrodynamics

The EFDC hydrodynamic model simulates near field plume, wind generated and externally linked wave models. In the hydrodynamics, temperature and salinity are optionally incorporated to address density effects. The hydrodynamic model is linked to sub models such as dye / age of water, sediments, toxics, water quality and Lagrangian particle tracking. (see Figure 1-2). EFDC+ is a coupled model which solves both the hydrodynamics, transport and kinetics in an integrated code, which eliminates the need for external coupling between hydrodynamics and transport modules.

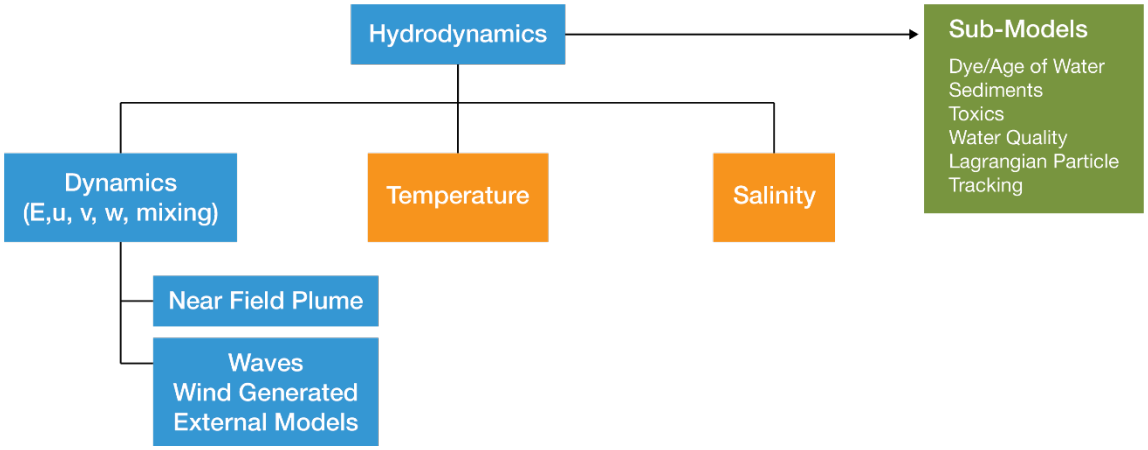


Figure 1-2: Structure of the EFDC hydrodynamic model.

1.2.2. Sediment Transport

EFDC+ supports two separate erosion/deposition approaches:

1. Original EFDC Sediment Transport
2. SEDZLJ Sediment Transport

Both approaches treat the water column transport the same way. The differences in these two approaches are in the way the water column sediment interacts with the sediment bed, i.e. erosion and/or deposition and in the treatment of the sediment bed itself. The first approach is referred to as the "original" sediment transport option, which was added by Hamrick (2002). This approach simulates transports and fate of multiple size classes of cohesive and non-cohesive suspended sediment including bed deposition and resuspension (see Figure 1-3). The SEDZLJ approach (Thanh, et.al., 2008) uses bed shear and erosion rate data developed from testing sediment cores with the SEDFlume apparatus as shown in Figure 1-4.

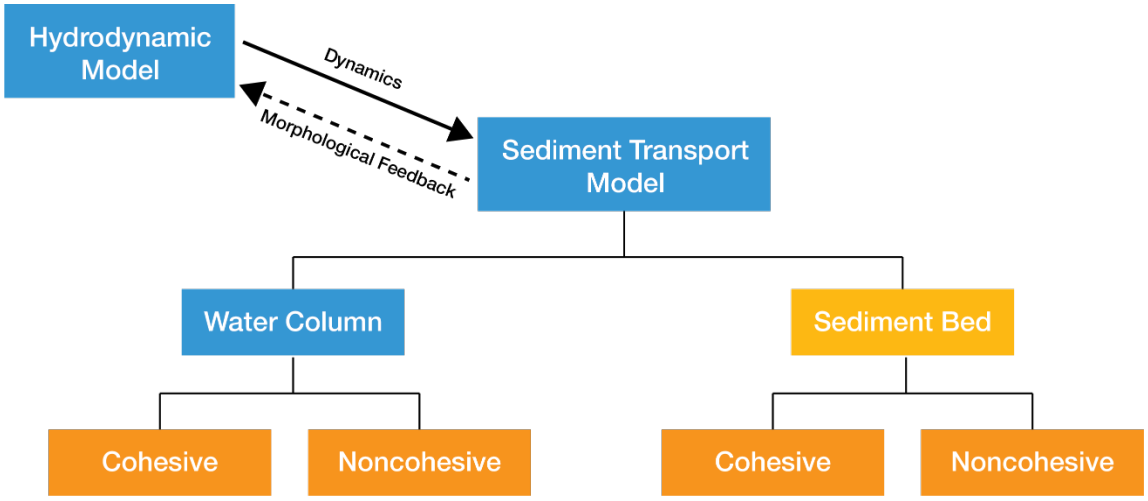


Figure 1-3: Structure of the EFDC sediment transport model.

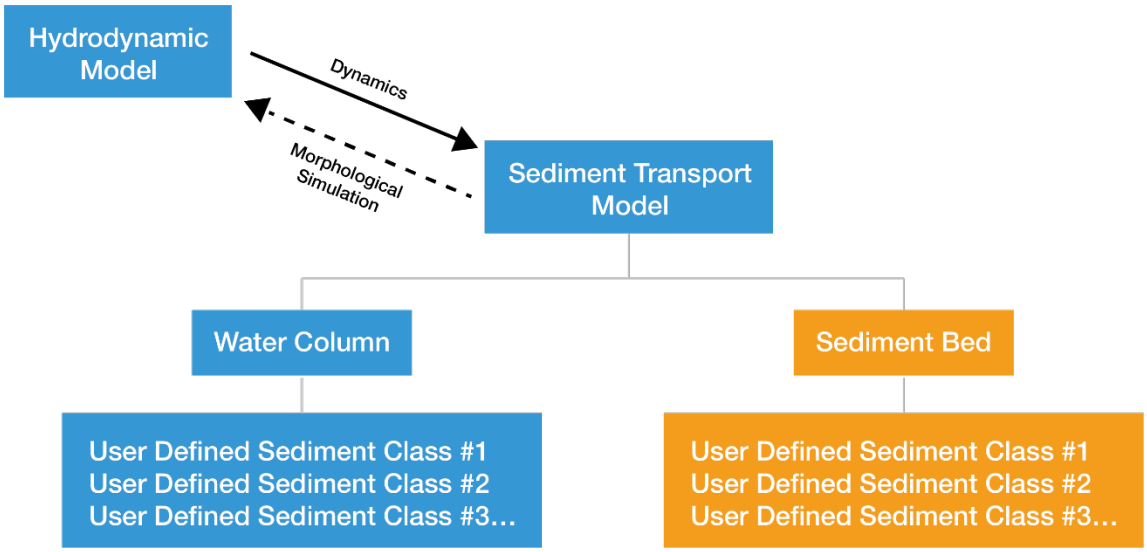


Figure 1-4: Structure of the EFDC+ SEDZLJ sediment transport model.

1.2.3. Water Quality

In 1995, water quality with full sediment diagenesis (Park, et.al 1995) was added to EFDC. The original version of this coupled water quality model was called Hydrodynamic Eutrophication Model Three Dimensional (HEM3D). This original water quality model allowed the simulation of 21 state variables. In 2000 the model was modified to include "macroalgae" (Tetra Tech, 2000). The model simulates spatial and temporal distributions of water quality parameters including dissolved oxygen, suspended algae (3 groups), various components of carbon, nitrogen, phosphorus and silica cycles, and fecal coliform bacteria. A sediment diagenesis process model with 27 state variables was also developed for EFDC. The coupling of the sediment diagenesis model with the water quality model not only enhances the model's predictive capability of water quality parameters but also enables it to simulate the long-term changes in water quality conditions in response to changes in nutrient loadings.

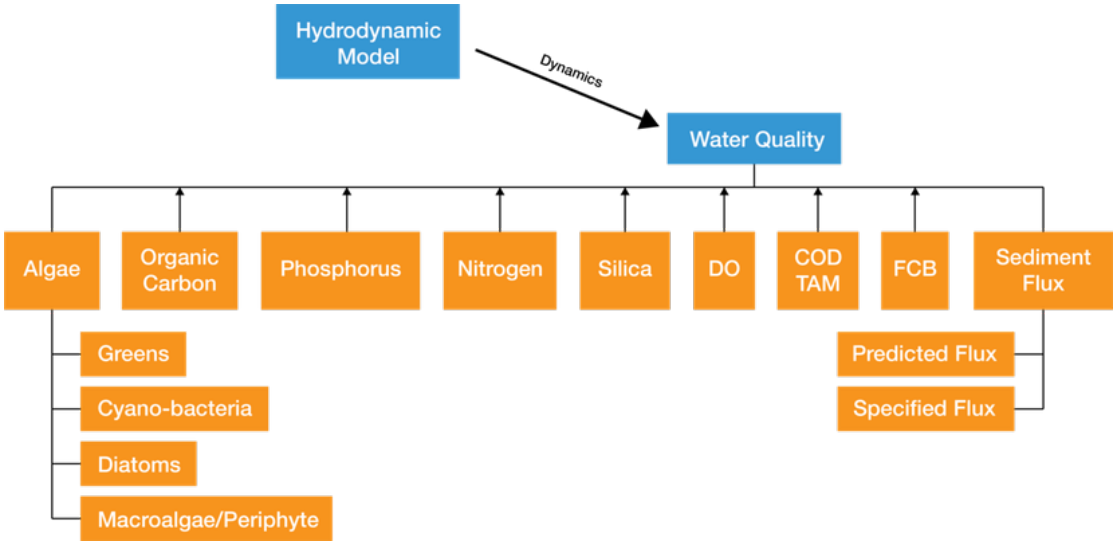


Figure 1-5: Structure of the EFDC water quality model

Toxics (Fate and Transport)

When the "Original" sediment transport capability was added (Tetra Tech, 2002), toxic contaminant transport was also added. This module allows for optional toxic contaminant partitioning onto water column and sediment bed solids. Prior to 2015, even though multiple partitioning options were available, only one approach could be used for all the toxics included in a simulation. DSI updated the toxic partitioning module to allow each toxic constituent to use its own unique partitioning approach. Figure 1-6 provides a schematic of the basic model approach.

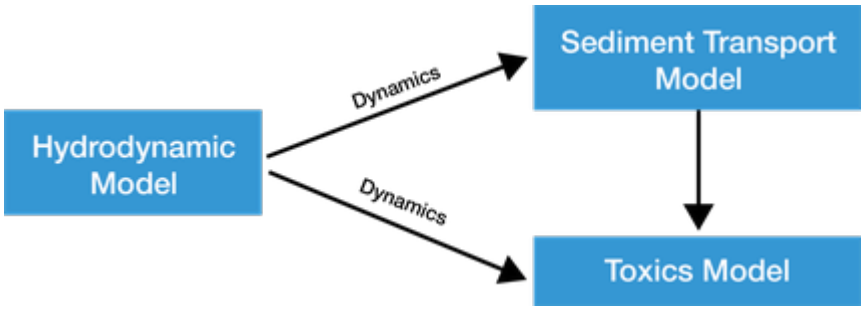


Figure 1-6: Linkage between hydrodynamic, sediment transport and toxics model.

1.2.4. Lagrangian particle tracking

Lagrangian particle tracking (LPT) is developed as a tool to solve many problems on simulation and prediction of drifters in fluid dynamics arising from practical necessity, such as

- Trajectory of floating objects in rivers, lakes and the sea in general.
- Oil spill

The advantage of this approach is that it is possible to track the process of movement for each specific particle in more detail and more accurately in comparison with the method of determining average concentration for grid cells. The unique difficulty for this approach is that when the number of particles is too large due to LPT depending on the speed of computational processing as well as the large amount of memory to be distributed for the variables. Fortunately, nowadays the development of information technology both on hardware and software is considerable and such a problem is completely possible to solve quickly, especially applying parallel processors. The movement of solid particles is decided by a field of fluid velocity, therefore it is necessary to couple it to a fluid flow model.

1.3 EFDC+ Advancements

Since 2002, DSI has been steadily improving and enhancing the original EPA version of EFDC code (https://www.epa.gov/exposure-assessment-models/efdc). The improvements to the EFDC code had become so extensive that in 2016 the DSI version of EFDC was renamed as the EFDCPlus or EFDC+ model. Key enhancements to the EFDC code include the following:

- **Dynamic memory allocation:** Dynamic memory allocation allows the user to use the same executable code for applications to different water bodies. Dynamic allocation eliminates the need to re-compile the EFDC code for different applications because of different maximum array sizes required to specify the computational grid domain and time series input data sets. Dynamic allocation also helps prevent inadvertent errors and provides better traceability for source code development.
- **OpenMP - Multithreading:** OpenMP provides vastly improved model run times. The Intel® OpenMP* Runtime Library binds OpenMP threads to physical processing units. Depending on the machine topology, application, and operating system, thread affinity can have a substantial impact on the application speed. DSI's EFDC+ typically produces run times up to 4 times faster on a six core processor than the conventional single-threaded EFDC model.

- **Sigma-Zed Layering:** An improved version of the EFDC code helps deal with pressure gradient errors that occur in simulations that have steep changes in bed elevation. The Sigma Zed code contrasts with the conventional EFDC code which uses a sigma coordinate transformation in the vertical direction and uses the same number of layers for all cells in the domain. In the EFDC_SGZ model, the vertical layering scheme has been modified to allow for the number of layers to vary over the model domain. This approach is computationally efficient and provides significantly improved simulation of thermal stratification.

- **Hydraulic Structures:** Equations governing hydraulic structures such as culverts, weirs, sluice gates, and orifices are implemented in EFDC+. This additional feature is different to the previous head lookup table used to describe the relationship between head and flow for a hydraulic structure.

- **Enhanced heat exchange**: Heat exchange options that use equilibrium temperatures for the water and atmospheric interface and spatially variable sediment bed temperatures.
- **Ice Formation and Melt**
- **Lagrangian Particle Tracking** (LPT) sub-model has been added. Among useful applications are for oil spill modeling and emergency response simulations.
- **Improved / Simplified external wave model linkage**

- **SEDZLJ Toxics Implementation:** Sandia National Lab's SEDZLJ sedflume model has undergone significant improvement for mass balance, and a whole new toxic modeling capability has been implemented.
- **RPEM module:** A Rooted Plant and Epiphyte Model (RPEM) has been incorporated into EFDC to better simulate water quality interactions with submerged aquatic vegetation (epiphytic algae and macrophytes).
- **Internal wind wave generation :** A wind generated wave sub-model has been added to enable the computation of wind wave generated bed shear stress on sediment resuspension and wave induced currents.
- **High Frequency Output :** New output snapshot controls for targeting specific periods for high frequency output within the standard regular output frequency.
- **Restart/Continuation run options**: Gives more options to run a model at the beginning or continue to run from the previous stop.
- **Streamlining the code** for quicker execution times.
- **Model Linkages :** Customizing linkage of model results for the Windows-based EFDC_Explorer graphical pre- and post-processor for EFDC.
- **MHK Linkage:** Incorporated Marine and Hydro-kinetic (MHK) friendly module for simulation of placement and potential effects of installing and operating turbines and wave energy converters in rivers, tidal channels, ocean currents, and other waterbodies. This code comes from Sandia National Laboratories modified Environmental Fluid Dynamics Code (SNL-EFDC) (James, 2008).
- **NetCDF:** Output EFDC in NetCDF file formats

2 HYDRODYNAMICS

The section is primarily based on Hamrick (1992) and Ji (2008) with updates from DSI and others. The basic governing equations for the EFDC hydrodynamics are presented and discussed.

2.1 Governing Equations

The fundamental principles of the hydrodynamic model in EFDC are the conservation laws of mass, momentum and energy for the flows. With the basic assumption that ambient environmental flows are characterized by horizontal length scales which are orders of magnitude greater than their vertical length scales, the formulation of the governing equations begins with the vertically hydrostatic, boundary layer form of the turbulent equations of motion for an incompressible, variable density fluid. The governing equations of EFDC include Navier-Stokes for fluid flow, the advection-diffusion equations for salinity, temperature, dye, toxicants, constituents in water quality and suspended sediment transport (Hamrick, 1992, 1996; Hamrick and Wu, 1997). In the horizontal direction, the equations are presented in the curvilinear coordinate system and Sigma transformation (at the bed and at the water surface) for the vertical direction. They are descretized with the finite difference method based on an explicit scheme.

2.1.1. Horizontal and Vertical Coordinate Systems

To accommodate realistic horizontal boundaries, it is convenient to formulate the equations such that the horizontal coordinates, x and y, are curvilinear and orthogonal.

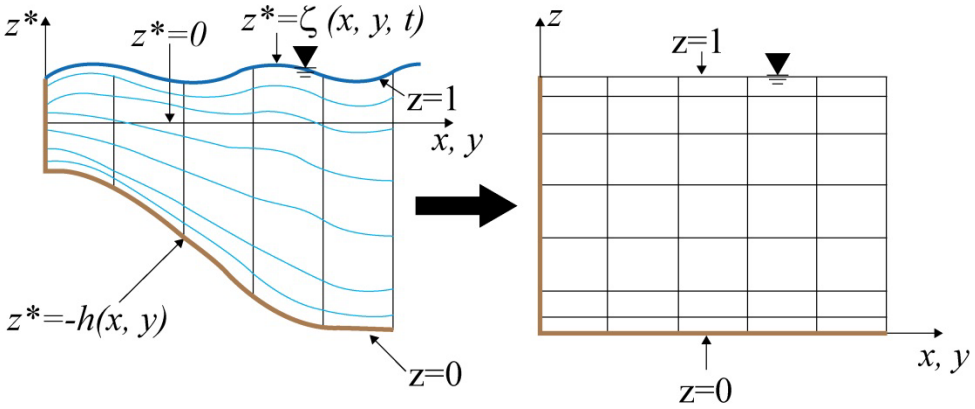


Figure 2-1: The stretched vertical coordinate system

To provide uniform resolution in the vertical direction, aligned with the gravitational vector and bounded by bottom topography and a free surface permitting long wave motion, a time variable mapping or stretching transformation is desirable. The mapping or stretching is given by:

$$z = \frac{z^* + h}{\zeta + h} = \frac{z^* + h}{H} \tag{2.1}$$

where:

z is the sigma coordinate (dimensionless),

z^* is the vertical coordinate with respect to the vertical reference level (datum) (m),

h is the water depth bellow the vertical reference level (m),

ζ i is the water surface elevation above the vertical reference level (m), and

Figure 2-1 provides a schematic of the vertical coordinate system in the physical space in the left panel and the sigma space in the right panel.

EFDC supports sigma stretched and sigma-zed (SGZ) grids for the vertical discretization of water column. Details on the sigma-zed vertical layering options are described in Section 2.6. Details of the transformation may be found in Vinokur (1974), Blumberg and Mellor (1987) or Hamrick (1986).

2.1.2. Basic Hydrodynamic Equations

Transforming the vertically hydrostatic boundary layer form of the turbulent equations of motion and utilizing the Boussinesq approximation for variable density results in the momentum and continuity equations and the transport equations for salinity and temperature in the following form:

The momentum equation in the *x* direction:

$$\begin{aligned}
&\frac{\partial}{\partial t}\left(m_x m_y H u\right)+\frac{\partial}{\partial x}\left(m_y H u u\right)+\frac{\partial}{\partial y}\left(m_x H v u\right)+\frac{\partial}{\partial z}\left(m_x m_y w u\right)-m_x m_y f H v \\
&-\left(v\frac{\partial m_y}{\partial x}-u\frac{\partial m_x}{\partial y}\right)H v=-m_y H\frac{\partial}{\partial x}\left(g\zeta+p+P_{atm}\right)-m_y\left(\frac{\partial h}{\partial x}-z\frac{\partial H}{\partial x}\right)\frac{\partial p}{\partial z} \\
&+\frac{\partial}{\partial x}\left(\frac{m_y}{m_x}HA_H\frac{\partial u}{\partial x}\right)+\frac{\partial}{\partial y}\left(\frac{m_x}{m_y}HA_H\frac{\partial u}{\partial y}\right)+\frac{\partial}{\partial z}\left(\frac{m_x m_y}{H}A_v\frac{\partial u}{\partial z}\right) \\
&-m_x m_y c_p D_p u\sqrt{u^2+v^2}+S_u
\end{aligned} \tag{2.2}$$

The momentum equation in the *y* direction:

$$\begin{aligned}
&\frac{\partial}{\partial t}\left(m_x m_y H v\right)+\frac{\partial}{\partial x}\left(m_y H u v\right)+\frac{\partial}{\partial y}\left(m_x H v v\right)+\frac{\partial}{\partial z}\left(m_x m_y w v\right)+m_x m_y f H u \\
&+\left(v\frac{\partial m_y}{\partial x}-u\frac{\partial m_x}{\partial y}\right)H u=-m_x H\frac{\partial}{\partial y}\left(g\zeta+p+P_{atm}\right)-m_x\left(\frac{\partial h}{\partial y}-z\frac{\partial H}{\partial y}\right)\frac{\partial p}{\partial z} \\
&+\frac{\partial}{\partial x}\left(\frac{m_y}{m_x}HA_H\frac{\partial v}{\partial x}\right)+\frac{\partial}{\partial y}\left(\frac{m_x}{m_y}HA_H\frac{\partial v}{\partial y}\right)+\frac{\partial}{\partial z}\left(\frac{m_x m_y}{H}A_v\frac{\partial v}{\partial z}\right) \\
&-m_x m_y c_p D_p v\sqrt{u^2+v^2}+S_v
\end{aligned} \tag{2.3}$$

The momentum equation in the *z* direction:

$$\frac{\partial p}{\partial z} = -gH\frac{\rho - \rho_0}{\rho_0} = -gHb \tag{2.4}$$

The continuity equations (internal and external modes)

$$\frac{\partial}{\partial t}\left(m_x m_y \zeta\right) + \frac{\partial}{\partial x}\left(m_y H u\right) + \frac{\partial}{\partial y}\left(m_x H v\right) + \frac{\partial}{\partial z}\left(m_x m_y w\right) = S_h \tag{2.5}$$

$$\frac{\partial}{\partial t}\left(m_x m_y \zeta\right) + \frac{\partial}{\partial x}\left(m_y H U\right) + \frac{\partial}{\partial y}\left(m_x H V\right) = S_h \tag{2.6}$$

The equation of state

$$\rho = \rho\left(p, S, T, C\right) \tag{2.7}$$

where:

$$U = \int_0^1 u dz, \quad V = \int_0^1 v dz \tag{2.8}$$

$$P = m_y H u, \quad Q = m_x H v \tag{2.9}$$

u, v are the horizontal velocity components in the curvilinear coordinates (m/s),

x, y are the orthogonal curvilinear coordinates in the horizontal direction (m),

z is the sigma coordinate (dimensionless),

t is time (s),

m_x, m_y are the square roots of the diagonal components of the metric tensor (m),

m is the Jacobian, $m = m_x m_y$ (m^2),

p is the physical pressure in excess of the reference density hydrostatic pressure (m^2/s^2),

P_{atm} is the barotropic pressure (Pa),

ρ_o is the reference water density (kg/m^3),

b is the buoyancy

f is the Coriolis parameter (1/s),

A_H is the horizontal momentum and mass diffusivity (m^2/s),

A_v is the vertical turbulent eddy viscosity (m^2/s),

c_p is the vegetation resistance coefficient (dimensionless),

D_p is the dimensionless projected vegetation area normal to the flow per unit horizontal area (dimensionless),

S_u and S_v are the source/sink terms for the horizontal momentum in the x and y directions, respectively (m²/s²),

S_h is the source/sink terms for the mass conservation equation (m³/s),

S is salinity (ppt),

T is temperature (°C),

C is total inorganic suspended solids (TSS) (g/m³), and

U and V are the depth averaged velocity components in the x and y directions, respectively (m/s).

P and Q are the mass flux components in the x and y directions, respectively (m²/s).

The vertical velocity, with physical units, in the stretched, dimensionless vertical coordinate z is:

$$w = w^* - z\left(\frac{\partial \zeta}{\partial t} + \frac{u}{m_x}\frac{\partial \zeta}{\partial x} + \frac{v}{m_y}\frac{\partial \zeta}{\partial y}\right) + (1-z)\left(\frac{u}{m_x}\frac{\partial h}{\partial x} + \frac{v}{m_y}\frac{\partial h}{\partial y}\right) \tag{2.10}$$

where:

w is the vertical velocity component in sigma coordinate (m/s), and

w^* is the physical vertical velocity (m/s).

The pressure p is the physical pressure in excess of the reference density hydrostatic pressure, $\rho_o g H(1 - z)$, divided by the reference density, ρ_o. In the momentum Eqs. (2.2) and (2.3) the momentum source/sink terms S_u and S_v will be later modeled as subgrid scale horizontal diffusion. The density ρ is in general a function of temperature T and salinity S for hydrospheric flows and water vapor for atmospheric flows. Density can be a weak function of pressure, but will be treated as an incompressible fluid in the continuity equation under the anelastic approximation (Mellor, 1991; Clark and Hall, 1991). The buoyancy b as defined in Eq. (2.4), is the normalized deviation of density from the reference value. The continuity Eq. (2.5) has been integrated with respect to z over the interval (0,1) to produce the depth integrated continuity Eq. (2.6) using the vertical boundary conditions, $w = 0$, at $z = (0,1)$, which follows from the kinematic conditions and Eq. (2.8). It is noted that constraining the free surface displacement to be time independent and spatially constant yields the equivalent of the rigid lid ocean circulation equations employed by Smetner (1974) and equations similar to the terrain following equations used by Clark (1977) to model mesoscale atmospheric flow.

2.1.3. Equation of state

In case the water density is dependant on temperature and salinity, the UNESCO's equation of state reads

$$\begin{aligned}\rho &= 999.842594+6.793952\times10^{-2}T-9.095290\times10^{-3}T^2 \\ &+1.001685\times10^{-4}T^3-1.120083\times10^{-6}T^4+6.536332\times10^{-9}T^5 \\ &+\left(0.824493-4.0899\times10^{-3}T+7.6438\times10^{-5}T^2-8.2467\times10^{-7}T^3+5.3875\times10^{-9}T^4\right)S \\ &+\left(-5.72466\times10^{-3}+1.0227\times10^{-4}T-1.6546\times10^{-6}T^2\right)S^{1.5}+4.8314\times10^{-4}S^2\end{aligned} \tag{2.11}$$

where:

ρ is the water density (kg/m³),

T is the water temperature (°C), and

S is the water salinity (ppt).

With the present of sediment in the water column, the water density and the buoyancy are corrected using a correction factor. The correction factor for the water density is

$$C_{TSS} = 1-\sum_{j}^{N}\rho_{s,j}C_j+\sum_{j}^{N}\left(s_j-1\right)\rho_{s,j}C_j \tag{2.12}$$

where:

C_{TSS} is the correction factor taking into account the influence of sediment on water density (dimensionless),

$\rho_{s,j}$ is the sediment density of the sediment class j (kg/m³),

C_j is the concentration of the sediment class j (g/m³),

s_j is the specific gravity of the sediment class j (dimensionless), and

N is the number of sediment classes.

2.1.4. Vertical Turbulent Closure

The system of eight equations from Eqs. (2.2) to (2.10) provides a closed system for the variables *u*, *v*, *w*, *p*, *ζ*, *ρ*, and *C*, provided that the vertical turbulent viscosity and diffusivity and the source and sink terms are specified. To provide the vertical turbulent viscosity and diffusivity, the second moment turbulence closure model developed by Mellor - Yamada (1982) and modified by Galperin *et al*. (1988) can be used. The model relates the vertical turbulent viscosity and diffusivity to the turbulent intensity (q) and a turbulent length scale (l), and a Richardson number R_q by:

The vertical turbulent momentum diffusion coefficient

$$A_v = \phi_A A_0 q l \tag{2.13}$$

where Φ_A is the stability viscosity coefficient

$$\phi_A = \frac{\left(1 + R_q / R_1\right)}{\left(1 + R_q / R_2\right)\left(1 + R_q / R_3\right)} \tag{2.14}$$

and

$$A_0 = A_1\left(1 - 3C_1 - \frac{6A_1}{B_1}\right) = \frac{1}{B_1^{1/3}} \tag{2.15}$$

$$\frac{1}{R_1} = 3A_2 \frac{\left(B_2 - 3A_2\right)\left(1 - \frac{6A_1}{B_1}\right) - 3C_1\left(B_2 + 6A_1\right)}{1 - 3C_1 - \frac{6A_1}{B_1}} \tag{2.16}$$

$$\frac{1}{R_2} = 9A_1A_2 \tag{2.17}$$

$$\frac{1}{R_3} = 3A_2\left[6A_1 + B_2\left(1 - C_3\right)\right] \tag{2.18}$$

The vertical mass diffusion coefficient

$$A_b = \phi_K K_0 q l \tag{2.19}$$

Where ϕ_K is the stability diffusivity coefficient

$$\phi_K = \frac{1}{\left(1 + R_q / R_3\right)} \tag{2.20}$$

and K_0 is the dimensionless coefficient

$$K_0 = A_2\left(1-\frac{6A_1}{B_1}\right) \tag{2.21}$$

The Richardson number

$$R_q = \frac{gH}{q^2}\frac{l^2}{H^2}\frac{\partial b}{\partial z} \tag{2.22}$$

where:

q^2 is the turbulent intensity (m^2/s^2),

l is the turbulent length scale (m),

R_q is the Richardson number.

Mellor and Yamada (1982) specify the constants A_1 = 0.92, B_1 = 16.6, C_1 = 0.08, A_2 = 0.74, and B_2 = 10.1.

However, the values of R_1, R_2, R_3 calculated by Galperin, Kantha and Clayson are different from Mellor-Yamada as in Table 2.1.

Table 2.1: Parameters for different turbulent models

Formulation	K_0	R_1^{-1}	R_2^{-1}	R_3^{-1}
Mellor and Yamada (1982)	0.493928	7.846436	34.676400	6.127200
Galperin et al. (1988)	0.493928	7.760050	34.676440	6.127200
Kantha and Clayson (2000)	0.493928	8.679790	30.192000	6.127200
Kantha	0.490025	14.509100	24.388300	3.236400

The so-called stability functions Φ_A and Φ_K account for reduced and enhanced vertical mixing or transport in stable and unstable vertically density stratified environments, respectively. The turbulence intensity and the turbulence length scale are determined by a pair of Mellor-Yamada equations:

$$\frac{\partial}{\partial t}\left(mHq^2\right)+\frac{\partial}{\partial x}\left(Pq^2\right)+\frac{\partial}{\partial y}\left(Qq^2\right)+\frac{\partial}{\partial z}\left(mwq^2\right)=$$
$$\frac{\partial}{\partial z}\left(m\frac{A_q}{H}\frac{\partial q^2}{\partial z}\right)2m\frac{A_v}{H}\left[\left(\frac{\partial u}{\partial z}\right)^2+\left(\frac{\partial v}{\partial z}\right)^2\right]++2mgA_b\frac{\partial b}{\partial z}-2m\frac{Hq^3}{B_1 l}+S_b \tag{2.23}$$

$$\frac{\partial}{\partial t}\left(mHq^2l\right)+\frac{\partial}{\partial x}\left(Pq^2l\right)+\frac{\partial}{\partial y}\left(Qq^2l\right)+\frac{\partial}{\partial z}\left(mwq^2l\right)=$$
$$\frac{\partial}{\partial z}\left[m\frac{A_{ql}}{H}\frac{\partial}{\partial z}\left(q^2l\right)\right]+mlE_1\left\{\frac{A_v}{H}\left[\left(\frac{\partial u}{\partial z}\right)^2+\left(\frac{\partial v}{\partial z}\right)^2\right]+E_3gA_b\frac{\partial b}{\partial z}\right\} \tag{2.24}$$
$$-mE_2\frac{Hq^3}{B_1}\left[1+E_4\left(\frac{l}{\kappa Hz}\right)^2+E_5\left(\frac{l}{\kappa H(1-z)}\right)^2\right]+S_l$$

$$\frac{1}{L}=\frac{1}{H}\left(\frac{1}{z}+\frac{1}{1-z}\right) \tag{2.25}$$

where

E_1 = 1.8, E_2 = 1.0, E_3 = 1.8, E_4 = 1.33, and E_5 = 0.25 are empirical constants,

S_q the source-sink term for turbulent intensity equation,

S_l the source-sink term for turbulent length scale equation,

A_q the vertical turbulent diffusivity for turbulent intensity equation,

A_{ql} the vertical turbulent diffusivity for turbulent length scale equation.

The vertical diffusivity for turbulence intensity, A_q, is set to $0.2ql$ following Mellor and Yamada (1982). For stable stratification, Galperin et al. (1988) suggest limiting the length scale such that the square root of R_q is less than 0.52. When horizontal turbulent viscosity, A_H and diffusivity are included in the momentum and transport equations, they are determined independently using Smagorinsky's (1963) subgrid scale closure formulation:

$$A_H=\Delta x\Delta y\sqrt{\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial v}{\partial y}\right)^2+\frac{1}{2}\left(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\right)^2} \tag{2.26}$$

where Δx, Δy are the grid sizes in x and y directions, respectively.

The terms S_q and S_l may represent additional source-sink terms such as subgrid scale horizontal turbulent diffusion, wave and vegetation effects.

2.2 Boundary Conditions and External Forcings

The vertical boundary conditions for the solution of the momentum equations (2.2)-(2.3) are based on the specification of the kinematic shear stresses at the free water surface and at the bed.

Vertical boundary conditions for the turbulent kinetic energy and length scale equations are:

$$q^2 = B_1^{2/3}\sqrt{\tau_{sx}^2 + \tau_{sy}^2}, \quad l = 0, \quad \text{at } z = 1 \tag{2.27}$$

$$q^2 = B_1^{2/3}\sqrt{\tau_{bx}^2 + \tau_{by}^2}, \quad l = 0, \quad \text{at } z = 0 \tag{2.28}$$

Equation (2.28) can become inappropriate under a number of conditions associated with either or both high near bottom sediment concentrations and high frequency surface wave activity.

2.2.1. Bottom Friction

At the bed, the stress components are related to the near bed or bottom layer velocity components by the quadratic resistance formulation:

$$\frac{1}{\rho}\begin{bmatrix}\tau_{xz}\\ \tau_{yz}\end{bmatrix} = \frac{1}{\rho}\begin{bmatrix}\tau_{bx}\\ \tau_{by}\end{bmatrix} = C_b\sqrt{u_1^2 + v_1^2}\begin{bmatrix}u_1\\ v_1\end{bmatrix} \tag{2.29}$$

where the subscript 1 denotes bottom layer values. Under the assumption that the near bottom velocity profile is logarithmic at any instant of time, the bottom stress coefficient is given by

$$C_b = \left[\frac{\kappa}{\ln\left(\Delta_1 / 2z_0\right)}\right]^2 \tag{2.30}$$

where κ, is the von Karman constant, Δ_1 is the dimensionless thickness of the bottom layer, and $z_o = z_o^*/H$ is the dimensionless roughness height.

2.2.2. Vegetation

Content to updated at a later date.

2.2.3. Wind Forcings

The influences of winds on hydrodynamics are due to the wind shear stresses exerted on the water surface.

At the free surface, the x and y components of the stress are specified by the wind stress:

$$\frac{1}{\rho}\begin{bmatrix}\tau_{xz}\\ \tau_{yz}\end{bmatrix} = \frac{1}{\rho}\begin{bmatrix}\tau_{sx}\\ \tau_{sy}\end{bmatrix} = C_D\frac{\rho_a}{\rho}W_s\begin{bmatrix}U_w\\ V_w\end{bmatrix} \tag{2.31}$$

$$W_s = \sqrt{U_w^2 + V_w^2} \tag{2.32}$$

where W_s, U_w and V_w are the wind velocity and x- and y- components of the wind velocity (m/s) at 10 meters above the water surface, respectively; C_D is the wind drag coefficient; ρ_a and ρ_w are air and water densities, respectively.

The wind drag coefficient is determined based on the wind speed as follows

$$C_D = 0.065\times10^{-3}W_s + 0.8\times10^{-3} \tag{2.33}$$

In case of magnitude sheltering and no directional sheltering, the wind drag coefficient can be calculated as

$$C_D = \begin{cases} 3.83111\times10^{-5}W_s^{-3}\text{-}0.000308715W_s^{-2}+0.00116012W_s^{-1}+0.000899602, & W_s < 5\text{m/s} \\ \text{-}5.37642\times10^{-6}W_s^{3}+0.000112556W_s^{2}\text{-}0.000721203W_s+0.00259657, & 5\text{m/s} \le W_s < 7\text{m/s} \\ \text{-}3.99677\times10^{-7}W_s^{2}+7.32937\times10^{-5}W_s+0.000726716, & W_s \ge 7\text{m/s} \end{cases} \tag{2.34}$$

2.2.4. Wave Action

The action of short waves on the field velocity of flow in a large water body could be an important aspect that may not be ignored, especially in estuaries and coastal areas. As is commonly known, both longshore currents and undertow are generated by waves. The asymmetry of wave velocity in its orbital plane is one of the causes of mass transport, such as sediment. Waves may be generated either by local wind or by distant storms with longer time periods. In this document wind-induced wave theory is be presented as applied in EFDC. In EFDC there are two options to include wave effects: 1) by an internal wind-generated waves sub-model inside EFDC, and 2) by providing wave parameters are computed by an external wave model such as SWAN, REF/DIF, STWAVE, etc.

In the case of waves, apart from the forces from currents, it is also necessary to add the forces from waves for the whole water column, such as radiation stresses or stresses due to the roller in breaking waves (Mengguo and Chongren, 2003; Robert J. Weaver, 2004). However, EFDC takes into account for only radiation stresses. The additional wave-induced momentum exerted on the flow field can be accounted for through wave radiation streses (Longuet-Higgins and Stewart, 1964):

$$S_{xx} = \left(n\cos^2\theta + n - \frac{1}{2} \right) E \tag{2.35}$$

$$S_{xy} = S_{yx} = \left(n\cos\theta\sin\theta \right) E \tag{2.36}$$

$$S_{yy} = \left(n\sin^2\theta + n - \frac{1}{2} \right) E \tag{2.37}$$

where

S_{xx}, S_{xy}, S_{yx}, S_{yy} are the components of wave radiation streses;

E is the wave energy (kg/s²),

$$E = \frac{1}{8}\rho g H_S^2 \tag{2.38}$$

H_s is the wave height (m)

θ is wave direction angle with respect to the x axis (counter clockwise) (radians),

k is the wave number,

$$k = \frac{2\pi}{L} \tag{2.39}$$

L is the wave length (m)

C is the wave celerity (m/s), and

$$n = \frac{1}{2}\left[1 + \frac{2kH}{\sinh(2kH)}\right] \tag{2.40}$$

In general, the wave length *L* (m) can be computed by solving the non-linear equation for the dispersion relation.

$$L = \frac{gT^2}{2\pi}\tanh\left(\frac{2\pi}{L}H\right) \tag{2.41}$$

This dispersion relation can be solved for the wave length using approximations or iterative methods, for example, EFDC computes wave length by using an approximate formula (Hunt, 1979):

$$L \approx T\sqrt{\frac{1}{\delta}gH} \tag{2.42}$$

where

$$\delta = \gamma + \frac{1}{\left(1 + 0.6522\gamma + 0.4622\gamma^2 + 0.0864\gamma^4 + 0.0675\gamma^5\right)} \tag{2.43}$$

and

$$\gamma = \omega^2\frac{H}{g} \tag{2.44}$$

where ω is the wave angular frequency (1/s).

$$\omega = \frac{2\pi}{T} \tag{2.45}$$

The regime of flow is determined through the wave Reynolds number R_w, and the relative bed roughness *r*:

$$R_w = \frac{U_b A}{\nu}, r = \frac{A}{k_s} \tag{2.46}$$

in which A is the semi-orbital excursion, k_s the Nikuradse equivalent sand grain roughness, and U_b is the wave maximum orbital velocity near the bed.

$$A = \frac{H_s}{2\sinh(kH)} \tag{2.47}$$

$$U_b = A\omega = \frac{\omega H_s}{2\sinh(kH)} \tag{2.48}$$

The bottom friction, the bed forms (such as ripples) and the characteristics of bed materials are strongly interdependent in case of wave actions. The friction coefficient due to waves according to Swart (1974) is

$$f_w = \begin{cases} \exp\left(5.21r^{-0.19} - 6.0\right) & r > 1.57 \\ 0.3 & r \le 1.57 \end{cases} \tag{2.49}$$

2.2.5. Local Wind-Generated Waves

In general, the influence of wind on the field of flow velocity is considerable for the problems of simulations of the hydrodynamic regime, sediment transport for lakes, estuarine and coastal areas with strong wind conditions. Wind effects can not only induce the flow current through the vertical boundary condition at water surface, but also generate surface waves with a wave height of up to several meters. Consequently, the calculation of the total bed shear stress must take the wave factor into account.

In fact, waves on the ocean surface with periods of 3 to 25 seconds are primarily caused by winds. Therefore, it can be seen that wind–generated waves plays an important role in hydrodynamic modeling. The advantage of this Wind-wave sub-model is that it can be easily incorporated into the source code of a hydrodynamic model instead of running a separate wave model. This means that the changes in hydrodynamic parameters are immediately updated in the wave calculations. At the same time it does not take a lot of time for calculation as is the case with other wave models.

In this report the theoretical basis and tests of the wind wave sub-model that is incorporated into EFDC is presented in detail. The mathematical formulae are empirical equations called the SMB model (Sverdrup, Munk and Bretschneider, see Zhen-Gang Ji, 2008). The model calibration is based on the experiment of Cox et al. (1996) for a wave flume. Another verification of the sub-model is the comparison of wave heights computed by the model with those generated by SWAN for the same wind condition in Caloosahatchee Estuary.

The basic assumptions of the SMB model for wind-generated waves include: a) the duration if wind blowing along one direction is long enough to attain the equilibrium condition, b) the wind speed and water depth are spatially uniform over the fetch. The main wave parameters can be determined include wave height, wave direction and wave period. For the SMB model, the wave direction is the same as the wind direction. This means that the effects of refraction, diffraction and reflection are not taken into account. Wave height and period are as follows:

$$H_s = 0.283\alpha \frac{W_s^2}{g} \tanh\left(\frac{0.0125}{\alpha}\left(\frac{gF}{W_s^2}\right)^{0.42}\right) \tag{2.50}$$

$$T_p = 7.54\beta \frac{W_s}{g} \tanh\left(\frac{0.077}{\beta} \left(\frac{gF}{W_s^2} \right)^{0.25} \right) \tag{2.51}$$

where

$$\alpha = \tanh\left\{ 0.53 \left(\frac{gH}{W_s^2} \right)^{0.75} \right\} \tag{2.52}$$

$$\beta = \tanh\left\{ 0.833 \left(\frac{gH}{W_s^2} \right)^{0.375} \right\} \tag{2.53}$$

in which

H_s is the wave height (m),

T_p is the wave period (s),

H is the water depth (m),

W_s is the wind velocity (m/s), and

F is the fetch length from the land boundary to the cell in the upwind direction (m). F is calculated for 16 directions.

2.2.6. Harmonic Forcings

The open boundary conditions in EFDC support a combination of forcing defined as a time series and harmonic forcings. This allow to model situations in estuaries or coastal areas where the influences of tides and river flows or storm surges may happen.

The harmonic repsentation of a time series ζ(t) can be approximated as a combination of sine and cosine functions

$$\zeta(t) = \zeta_0(t) + a_0 + \sum_{k=1}^{N} \left[a_k \cos\left(\omega_k t\right) + b_k \sin\left(\omega_k t\right) \right] \tag{2.54}$$

where

t is the time (s),

$\zeta_0(t)$ is the residual signal other than the periodic components (m),

a_0 is the mean value of the periodic components (m),

N is the number of the harmonic contituents,

a_k, b_k are the harmonic constant of the constituent k (m),

ω_k is the angular speed of constituent k (radians/s),

$$\omega_k = \frac{2\pi}{T_k} \tag{2.55}$$

T_k is the period constituent k (s),

The Eq. (2.60) can be also rewritten in another common form as

$$\zeta(t) = \zeta_0(t) + a_0 + \sum_{k=1}^{N} A_k \cos\left(\omega_k t - \varphi_k\right) \tag{2.56}$$

where

A_k is the amplitude of the harmonic constituent k (m),

$$A_k = \sqrt{a_k^2 + b_k^2} \tag{2.57}$$

φ_k is the phase lag the harmonic constituent k (radians),

$$\varphi_k = \text{atan}\left(\frac{b_k}{a_k}\right) \tag{2.58}$$

The name and period of some major tidal constituents are listed in Table 2.2.

Table 2.2: List of commonly used tidal constituents

No.	Name	Angular Speed (deg,/hour)	Period (hours)	Desctiption
1	O_1	13.943035	25.819	Principal Lunar Declinational diurnal constituent
2	K_1	15.041069	23.934	Luni-solar Declinational diurnal constituent
3	Oo_1	16.139101	22.306	Lunar diurnal constituent
4	S_1	15.000000	24.000	Solar diurnal constituent
5	P_1	14.958931	24.066	Principal Solar Declinational diurnal constituent
6	M_1	14.496694	24.833	Smaller lunar elliptic diurnal constituent
7	J_1	15.585444	23.098	Smaller lunar elliptic diurnal constituent
8	ρ	13.471515	26.723	Larger lunar evectional diurnal constituent
9	Q_1	13.398661	26.868	Larger lunar elliptic diurnal constituent
10	$2Q_1$	12.854286	28.006	Larger elliptic diurnal
11	M_2	28.984104	12.421	Principal Lunar semidiurnal constituent
12	S_2	30.000000	12.000	Principal Solar semidiurnal constituent
13	N_2	28.439730	12.658	Larger Lunar Elliptic semidiurnal constituent
14	ν_2	28.512583	12.626	Larger lunar evectional constituent
15	μ_2	27.968208	12.872	Variational constituent
16	$2N_2$	27.895355	12.905	Lunar elliptical semidiurnal second-order constituent
17	λ_2	29.455626	12.222	Smaller lunar evectional constituent
18	T_2	29.958933	12.016	Larger solar elliptic constituent
19	R_2	30.041067	11.984	Smaller solar elliptic constituent
20	L_2	29.528479	12.192	Smaller Lunar Elliptic semidiurnal constituent
21	K_2	30.082138	11.967	Smaller Lunar Elliptic semi diurnal constituent
22	M_3	43.476160	8.280	Lunar terdiurnal constituent
23	$2MK_3$	42.927140	8.386	Shallow water terdiurnal constituent
24	M_4	57.968210	6.210	Shallow water overtides of principal lunar constituent

25	M_6	86.952320	4.140	Shallow water overtides of principal lunar constituent
26	MK_3	44.025173	8.177	Shallow water terdiurnal
27	S_4	60.000000	6.000	Shallow water overtides of principal solar constituent
28	MN_4	57.423832	6.269	Shallow water quarter diurnal constituent
29	S_6	90.000000	4.000	Shallow water overtides of principal solar constituent
30	$2SM_2$	31.015896	11.607	Shallow water semidiurnal constituent
31	M_8	115.936420	3.105	Shallow water eighth diurnal constituent
32	MS_4	58.984104	6.103	Shallow water quarter diurnal constituent
33	Msf	1.015896	354.367	Lunisolar synodic fortnightly constituent
34	Mf	1.098033	327.859	Lunisolar fortnightly constituent
35	Mm	0.544375	661.309	Lunar monthly constituent
36	Ssa	0.082137	4382.905	Solar semiannual constituent
37	Sa	0.041069	8765.821	Solar annual constituent

2.2.7. Hydraulic Structures

Hydraulic structures can be modeled in EFDC either by rating curves or hydraulic equations. A rating curve is a lookup table which presents a relationship between the flow rate through a structure and the water heads. Depending on the actual water heads of the structure at a curtain time step, the flow rate is determined using the lookup table. EFDC allows a variety of rating curves which the flow discharge can be determined from: a) upstream water depth, b) the head difference (between upstream and downstream), c) the head difference and flow accelerations, d) the upstream and downstream water surface elevations, e) upstream water depth for a low chord structure, and f) head difference for a low chord structure.

The last two types of rating curves use for low chord structures such as bridges. With the low chord structures such as bridges, when flows are below the deck, they may be bi-directional, i.e. flows can be going upstream or downstream. Howerver once the bridge is overtopped, flow is only from upstream to downstream.

2.2.7.1 Rating Curves

If the flow through a structure is uni-directional, i.e., the flow direction is from upstream to downstream of the structures only. The rating curve is a lookup table which composes of a single column for water head and a corresponding single column for flow rate.

If the flow through a structure is bi-directional, the rating curve is a two-dimensional lookup table where the flow rates can be determined based on both upstream and downstream water surface elevations.

Beside of using lookup tables, EFDC can also simulate internally different types of hydraulic structures. This allows the user can model hydraulics structure easily and quickly with EFDC. The built in modeling codes for hydraulic structures inside EFDC include culverts, weirs, sluice gates and orifices.

2.2.7.2 Culverts

Flow rate through a culvert or sluice date is calculated in EFDC based on the water levels at both sides of the structure at its configuration. In a tidal region, the water levels at two sides of a structure are constantly changing that can cause bi-directional flows in which water flows through the structure from the side with a higher water surface elevation to the side with lower water surface elevation.

The characteristics of flow through a culvert are complicated and are determined by the inlet geometry, slope, shape, size, roughness, approach, and headwater and tailwater conditions. Dill (2000) described six different types of culvert flow based on the location of the control section within the culvert and the relative elevations of the headwater, tailwater, and culvert invert and crown elevations. The discharge through a culvert can be expressed as:

$$Q = AV = AC\sqrt{RS} = K\sqrt{S} \tag{2.59}$$

where Q is the flow discharge (m^3/s), A is the cross-sectional flow area (m^2), R is the hydraulic radius (m), S is the culvert slope, K is the conveyance (m^3/s), and C is the Chézy coefficient ($m^{0.5}/s$) which can be caculated by using the Manning's formula

$$C = \frac{1}{n}R^{\frac{1}{6}} \tag{2.60}$$

where n is Manning's roughness coefficient.

a) Initially the tailwater and the headwater is compared to the height of the culvert. If the tailwater is greater than the culvert height or the headwater is greater than 1.5 times the culvert height, the culvert outlet is submerged and the culvert is assumed to flow full. The slope is estimated as the difference in headwater and tailwater elevation divided by the culvert length, L (m), the conveyance is determined for the full culvert, and discharge is calculated by Eq. (2.51).

b) If both the inlet and outlet are not submerged the critical depth, y_c, is computed assuming free flow through the culvert inlet. In this case, it is assumed the approach velocity is negligible so that total energy at the culvert inlet is equal to the headwater. Thus

$$H_{HW} = y_c + \frac{V_c^2}{2g} \tag{2.61}$$

where H_{HW} is the headwater (m), V_c is the critical velocity (m/s), y_c is the critical depth (m), and g is acceleration due to gravity. For critical flow,

$$\frac{V_c^2}{2g} = \frac{D}{2} \tag{2.62}$$

where D is the hydraulic depth (m),

$$D = \frac{A}{T} \tag{2.63}$$

and T is the flow top width (m). Combining Eqs (2.53) and (2.54) yields an expression for the critical depth.

$$y_c = H_{HW} - \frac{D}{2} \tag{2.64}$$

where H_{TW} is the tailwater (m).

Once the critical depth has been determined, the critical velocity V_c, critical discharge Q_{cr}, and critical slope S_{cr} are known as well. The critical discharge represents the maximum possible flow through the culvert for the given headwater.

$$Q_{cr} = V_c A \tag{2.65}$$

If the culvert slope is greater than the critical slope, the culvert can convey more flow than the inlet will allow. As such, the inlet controls the flow and the discharge is assumed to be equal to the critical discharge Q_{cr} calculated as Eq (2.57).

c) If the culvert slope is less than the critical slope, the control section may be at the culvert outlet or downstream of the culvert. The critical depth is then compared to the tailwater, and if the tailwater is greater than the critical depth, the tailwater elevation is used to determine the flow area and hydraulic radius and the flow through the culvert is calculated as Eq. (2.51).

d) If the tailwater depth is less than the critical depth, but the slope is less than the critical slope, it is assumed that uniform flow will occur within the culvert. In this case potential energy is balanced by head loss due to friction in the culvert and conservation of energy between the control section and inlet can be expressed as

$$H_{HW} = y_n + \frac{V^2}{2g} \tag{2.66}$$

where y_n is the normal depth (m) in the culvert and V is the average velocity at the control section. It is also assumed that the approach velocity is negligible, and the slope is small such that the normal depth is approximately equal to the vertical depth.

Equation (2.51) can be re-written as an expression of the velocity head at the control section as:

$$V = \frac{1}{n} R^{\frac{2}{3}} \sqrt{S} \tag{2.67}$$

Combining Eqs (2.58) and (2.59) yields an equation for the normal depth

$$y_n = H_{HW} - \frac{1}{2g}\frac{1}{n^2}R^{\frac{4}{3}}S \tag{2.68}$$

Because R is a function of depth, an adaptive procedure is employed to determine the normal depth. In culverts that experience bi-directional flow, the slope may be adverse or zero. In either case the assumption of uniform flow is problematic because the water surface slope cannot be equal to the culvert slope. In this case, the water surface slope, as determined from the difference in headwater and tailwater elevations, is used in Eq (2.60).

2.2.7.3 Weirs

A general formula for free flow through a weir can be expressed as

$$Q = C_d W\sqrt{2gH_{HW}^3} \tag{2.69}$$

where W is the width of weir (m) and C_d is weir discharge coefficient which depends on the type of weir (broad crested or sharp-/narrow-crested, ogee), shape of opening (rectangular, triangular, trapezoidal), and other weir paramaters.

For submerged flow through a weir, an adjustment factor is applied to Eq. (2.61) to account for the submergence (Villemonte, 1947)

$$Q = \left(1 - \frac{H_{TW}}{H_{HW}}\right)^{0.385} C_d W\sqrt{2gH_{HW}^3} \tag{2.70}$$

2.2.7.4 Sluice Gates

Flow through a sluice gate can be characterized by two basic parameters: the tranquility of the flow (i.e., subcritical or supercritical flow) and the water depth (i.e., gate is submerged or not).

For super-critical weir flow

$$Q = C_1 W\sqrt{g\left(\frac{2}{3}H_{HW}\right)^3} \tag{2.71}$$

For sub-critical weir flow

$$Q = C_2 W H_{TW}\sqrt{2g\left(H_{HW} - H_{TW}\right)} \tag{2.72}$$

where C_1 and C_2 are the super-critical and sub-critical discharge coefficients respectively; H_{HW} and H_{TW} are the headwater and tailwater (m), respectively; W is the width of the gate (m), and g is acceleration due to gravity.

a) When the water surface is determined to be below the top of the gate, the gate is modeled as a broad crested weir and Eq (2.61) is used.

b) When the gate is submerged, the appropriate equation for either free sluice flow (super-critical) or submerged orifice flow (sub-critical) is applied. To determine the flow through the sluice gate at a given model time step, the headwater is compared to the tailwater.

For free sluice flow

$$Q = C_3 A\sqrt{2gH_{HW}} \tag{2.73}$$

For submerged orifice flow

$$Q = C_4 A\sqrt{2g\left(H_{HW} - H_{TW}\right)} \tag{2.74}$$

where C_3 and C_4 are discharge coefficients for free sluice flow and submerged orifice flow, respectively; A is the gate opening (m²).

If the tailwater is less than 0.64 times the headwater, the appropriate Eq. (2.62) for super-critical flow is applied.

If the ratio of tailwater to headwater is greater than 0.68, the appropriate formula Eq. (2.63) for subcritical flow is applied. This is either a free sluice for super-critical flow, or a submerged orifice for sub-critical flow.

In cases when the tailwater to headwater ratio is between 0.64 and 0.88 both discharges are computed and a weighted average of the two is used.

2.2.7.5 Orifices

If the headwater is lower than the opening of a orifice, the flow through the orifice is treated as weir flow, so Eq. (2.61) is used.

If the tailwater is higher than the opening of a orifice, the appropriate Eq. (2.66) submerged flow through the orifice is applied.

If the headwater is higher than the opening of a orifice but the tailwater is lower than the opening of a orifice, the free jet flow through the orifice is calculated as

$$Q = C_2 A\sqrt{2g\left(H_{HW} + 0.5B\right)} \tag{2.75}$$

where B is the height of orifice opening (m), and H_{HW} is calculated based on the center line of the orifice.

2.3 Numerical Solution for the Equations of Motion

The equations of motion (2.2) and (2.3) will be solved in a region subdivided into six faced cells. The projection of the vertical cell boundaries to a horizontal plane forms a curvilinear, orthogonal grid in the orthogonal coordinate system (x,y). In a vertical (x,z) or (y,z) plane, the cells bounded by the same constant z surfaces will be referred to as cell layers or layers. The equations will be solved using a combination of finite volume and finite difference techniques, with the variable locations shown in Figure 2-2.

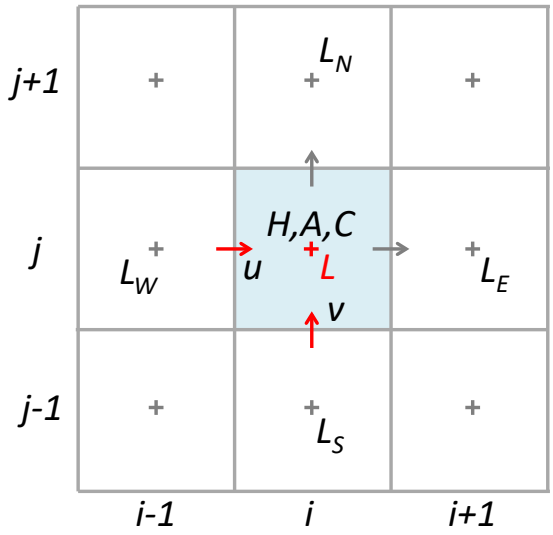


Figure 2-2: Free surface displacement centered horizontal grid

The staggered grid location of variables is often referred to as the Arakawa C grid (Arakawa and Lamb, 1977) or the MAC grid (Peyret and Taylor, 1983). To proceed, it is convenient to modify Eqs. (2.2) and (2.3) by eliminating the vertical pressure gradients using Eq. (2.4). After some manipulation, the horizontal momentum equations become:

$$\begin{aligned}&\frac{\partial}{\partial t}\left(m_x m_y Hu\right)+\frac{\partial}{\partial x}\left(m_y Huu\right)+\frac{\partial}{\partial y}\left(m_x Hvu\right)+\frac{\partial}{\partial z}\left(m_x m_y wu\right)\\&-\left(v\frac{\partial m_y}{\partial x}-u\frac{\partial m_x}{\partial y}\right)Hv-m_x m_y fHv=-m_y H\frac{\partial p}{\partial x}-m_y Hg\frac{\partial \zeta}{\partial x}+m_y Hgb\frac{\partial h}{\partial x}\\&-m_y Hgbz\frac{\partial H}{\partial x}+\frac{\partial}{\partial z}\left(\frac{m_x m_y A_v}{H}\frac{\partial u}{\partial z}\right)+S_u\end{aligned} \tag{2.76}$$

$$\frac{\partial}{\partial t}\left(m_x m_y Hv\right)+\frac{\partial}{\partial x}\left(m_y Huv\right)+\frac{\partial}{\partial y}\left(m_x Hvv\right)+\frac{\partial}{\partial z}\left(m_x m_y wv\right)$$
$$+\left(v\frac{\partial m_y}{\partial x}-u\frac{\partial m_x}{\partial y}\right)Hu+m_x m_y fHu=-m_x H\frac{\partial p}{\partial y}-m_x Hg\frac{\partial \zeta}{\partial y}+m_x Hgb\frac{\partial h}{\partial y} \quad (2.77)$$
$$-m_x Hgbz\frac{\partial H}{\partial y}+\frac{\partial}{\partial z}\left(\frac{m_x m_y A_v}{H}\frac{\partial v}{\partial z}\right)+S_v$$

The vertical discretization of Eqs. (2.76) and (2.77) is considered first. The equations are integrated with respect to z over a cell layer assuming that variables defined vertically at the cell or layer centers are constant and that variables defined vertically at the cell layer interfaces or boundaries vary linearly over the cell. Using the notations for mass fluxes,

$$P_k=m_y Hu_k,\quad Q_k=m_x Hv_k \quad (2.78)$$

the equations reads:

$$\frac{\partial}{\partial t}\left(m_x P_k\Delta_k\right)+\frac{\partial}{\partial x}\left(P_k u_k\Delta_k\right)+\frac{\partial}{\partial y}\left(Q_k u_k\Delta_k\right)+m\left[\left(wu\right)_k-\left(wu\right)_{k-1}\right]$$
$$-\left(v_k\frac{\partial m_y}{\partial x}-u_k\frac{\partial m_x}{\partial y}\right)Hv_k\Delta_k-m_y fQ_k\Delta_k=-\frac{1}{2}m_y H\Delta_k\frac{\partial}{\partial x}\left(p_k+p_{k-1}\right)-m_y H\Delta_k g\frac{\partial \zeta}{\partial x} \quad (2.79)$$
$$+m_y H\Delta_k gb_k\frac{\partial h}{\partial x}-0.5m_y H\Delta_k gb_k\left(z_k+z_{k-1}\right)\frac{\partial H}{\partial x}+m\left[\left(\tau_{xz}\right)_k-\left(\tau_{xz}\right)_{k-1}\right]+\left(S_u\Delta\right)_k$$

$$\frac{\partial}{\partial t}\left(m_y Q_k\Delta_k\right)+\frac{\partial}{\partial x}\left(P_k v_k\Delta_k\right)+\frac{\partial}{\partial y}\left(Q_k v_k\Delta_k\right)+m\left[\left(wv\right)_k-\left(wv\right)_{k-1}\right]$$
$$+\left(v_k\frac{\partial m_y}{\partial x}-u_k\frac{\partial m_x}{\partial y}\right)Hu_k\Delta_k+m_x fP_k\Delta_k=-\frac{1}{2}m_x H\Delta_k\frac{\partial}{\partial y}\left(p_k+p_{k-1}\right)-m_x H\Delta_k g\frac{\partial \zeta}{\partial y} \quad (2.80)$$
$$+m_x H\Delta_k gb_k\frac{\partial h}{\partial y}-0.5m_x H\Delta_k gb_k\left(z_k+z_{k-1}\right)\frac{\partial H}{\partial y}+m\left[\left(\tau_{yz}\right)_k-m\left(\tau_{yz}\right)_{k-1}\right]+\left(S_v\Delta\right)_k$$

where Δ_k is the vertical cell or layer thickness, and the turbulent shear stresses at the cell layer interfaces are defined by:

$$\left(\tau_{xz}\right)_k=\frac{2}{H}\left(A_v\right)_k\frac{u_{k+1}-u_k}{\Delta_{k+1}+\Delta_k} \quad (2.81)$$

$$\left(\tau_{yz}\right)_k=\frac{2}{H}\left(A_v\right)_k\frac{v_{k+1}-v_k}{\Delta_{k+1}+\Delta_k} \quad (2.82)$$

If there are K cells in the z direction, the hydrostatic equation can be integrated from a cell layer interface to the surface to give:

$$p_k = gH\left(\sum_{j=k}^{K} b_j\Delta_j - b_k\Delta_k\right) + p_s \tag{2.83}$$

where p_s is the physical pressure at the free surface or under the rigid lid divided by the reference density. The continuity equation (2.5) is also integrated with respect to z over a cell or layer to give:

$$\frac{\partial}{\partial t}\left(m\zeta\Delta_k\right) + \frac{\partial}{\partial x}\left(P_k\Delta_k\right) + \frac{\partial}{\partial y}\left(Q_k\Delta_k\right) + m\left(w_k + w_{k-1}\right) = S_h \tag{2.84}$$

The numerical solution of the vertically discrete momentum equations (2.79) and (2.80) now proceeds by splitting the external depth integrated mode associated with external long surface gravity waves from the internal mode associated with vertical current structure.

The external mode equations are obtained by summing equations equations (2.79) and (2.80) over K cells or layers in the vertical utilizing equation (2.83), and are given by:

$$\begin{aligned}
&\frac{\partial}{\partial t}\left(m_x\bar{P}\right) + \sum_{k=1}^{K}\left[\frac{\partial}{\partial x}\left(P_k u_k\Delta_k\right) + \frac{\partial}{\partial y}\left(Q_k u_k\Delta_k\right) - \left(v_k\frac{\partial m_y}{\partial x} - u_k\frac{\partial m_x}{\partial y}\right)Hv_k\Delta_k - m_y fQ_k\Delta_k\right] \\
&= -m_y Hg\frac{\partial\zeta}{\partial x} - m_y H\frac{\partial p_s}{\partial x} + m_y Hg\bar{b}\frac{\partial h}{\partial x} - m_y Hg\left[\sum_{k=1}^{K}\left(\beta_k\Delta_k + \frac{1}{2}\left(z_k + z_{k-1}\right)b_k\Delta_k\right)\right]\frac{\partial H}{\partial x} \\
&-m_y gH^2\frac{\partial}{\partial x}\left(\sum_{k=1}^{K}\beta_k\Delta_k\right) + m\left[\left(\tau_{xz}\right)_K - \left(\tau_{xz}\right)_0\right] + \bar{S}_u
\end{aligned} \tag{2.85}$$

$$\begin{aligned}
&\frac{\partial}{\partial t}\left(m_y\bar{Q}\right) + \sum_{k=1}^{K}\left[\frac{\partial}{\partial x}\left(P_k v_k\Delta_k\right) + \frac{\partial}{\partial y}\left(Q_k v_k\Delta_k\right) + \left(v_k\frac{\partial m_y}{\partial x} - u_k\frac{\partial m_x}{\partial y}\right)Hu_k\Delta_k + m_x fP_k\Delta_k\right] \\
&= -m_x Hg\frac{\partial\zeta}{\partial y} - m_x H\frac{\partial p_s}{\partial y} + m_x Hg\bar{b}\frac{\partial h}{\partial y} - m_x Hg\left[\sum_{k=1}^{K}\left(\beta_k\Delta_k + \frac{1}{2}\left(z_k + z_{k-1}\right)b_k\Delta_k\right)\right]\frac{\partial H}{\partial y} \\
&-m_x gH^2\frac{\partial}{\partial y}\left(\sum_{k=1}^{K}\beta_k\Delta_k\right) + m\left[\left(\tau_{yz}\right)_K - \left(\tau_{yz}\right)_0\right] + \bar{S}_v
\end{aligned} \tag{2.86}$$

$$\frac{\partial}{\partial t}\left(m\zeta\right) + \frac{\partial}{\partial x}\bar{P} + \frac{\partial}{\partial y}\bar{Q} = S_h \tag{2.87}$$

where the over bar indicates an average over the depth.

$$\bar{P} = m_y H\bar{u}, \qquad \bar{Q} = m_x H\bar{v} \tag{2.88}$$

$$\beta_k = \sum_{j=k}^{K} b_j\Delta_j - \frac{1}{2}b_k\Delta_k \tag{2.89}$$

The depth integrated continuity equation (2.87) follows from equation (2.6) and provides the continuity constraint for the external mode. Consistent with the form of equation (2.87) the

external mode variables will be chosen to be the free surface displacement, ζ, and the volumetric transports $P = m_y H u$ and $Q = m_x H v$. Details of the solution of the external mode equations (2.85) to (2.87) are presented in Section 2.4.

A number of formulations are possible for the internal mode equations. Equations (2.79) and (2.80) have K degrees of freedom for each of the horizontal velocity components. However, the summation of these equations over K cells or layers in the vertical to form the external mode equations (2.85) and (2.86) effectively removes a degree of freedom since the constraints:

$$\sum_{k=1}^{K} u_k \Delta_k = \overline{u} \tag{2.90}$$

$$\sum_{k=1}^{K} v_k \Delta_k = \overline{v} \tag{2.91}$$

must be satisfied. One approach to the internal mode is to solve equations (2.79) and (2.80) using the free surface slopes, or the surface pressure gradients in the rigid lid case, from the external solution and distribute the error such that equations (2.90) and (2.91) are satisfied. A second approach is to form equations for the deviations of the velocity components from their vertical means by subtracting the external equations (2.85) and (2.86) from the layer integrated equations (2.79) and (2.80). However, it will still be necessary to satisfy the constraints (2.90) and (2.91). The approach proposed herein is to reduce the systems of K layer averaged equations (2.79) and (2.80) to systems of K-1 equations and use equations (2.90) and (2.91) to provide the Kth equation consistent with the actual degrees of freedom.

The internal mode equations are formed by dividing equations (2.79) and (2.80) by the cell layer thickness, Δ_k, subtracting the equations for cell layer k from the equations for cell layer k+1, and then dividing the results by the average thickness of the two cell layers to give:

$$\begin{aligned}
&\frac{\partial}{\partial t}\left(m_x \frac{P_{k+1}-P_k}{\Delta_{k+1,k}} \right) + \frac{\partial}{\partial x}\left(\frac{P_{k+1}u_{k+1}-P_k u_k}{\Delta_{k+1,k}} \right) + \frac{\partial}{\partial y}\left(\frac{Q_{k+1}u_{k+1}-Q_k u_k}{\Delta_{k+1,k}} \right) \\
&+ \frac{m}{\Delta_{k+1,k}}\left[\frac{(wu)_{k+1}-(wu)_k}{\Delta_{k+1}} - \frac{(wu)_k-(wu)_{k-1}}{\Delta_k} \right] - m_y f \frac{Q_{k+1}-Q_k}{\Delta_{k+1,k}} \\
&- \frac{1}{\Delta_{k+1,k}}\left[\left(v_{k+1}\frac{\partial m_y}{\partial x} - u_{k+1}\frac{\partial m_x}{\partial y} \right) H v_{k+1} - \left(v_k \frac{\partial m_y}{\partial x} - u_k \frac{\partial m_x}{\partial y} \right) H v_k \right] \\
&= m_y H \frac{b_{k+1}-b_k}{\Delta_{k+1,k}} g\left(\frac{\partial h}{\partial x} - z_k \frac{\partial H}{\partial x} \right) + \frac{1}{2}\frac{m_y H^2}{\Delta_{k+1,k}} g\left(\Delta_{k+1}\frac{\partial b_{k+1}}{\partial x} + \Delta_k \frac{\partial b_k}{\partial x} \right) \\
&+ \frac{m}{\Delta_{k+1,k}}\left[\frac{(\tau_{xz})_{k+1}-(\tau_{xz})_k}{\Delta_{k+1}} - \frac{(\tau_{xz})_k-(\tau_{xz})_{k-1}}{\Delta_k} \right] + \frac{(S_u)_{k+1}-(S_u)_k}{\Delta_{k+1,k}}
\end{aligned} \tag{2.92}$$

$$\frac{\partial}{\partial t}\left(m_y \frac{Q_{k+1}-Q_k}{\Delta_{k+1,k}} \right)+\frac{\partial}{\partial x}\left(\frac{P_{k+1}v_{k+1}-P_k v_k}{\Delta_{k+1,k}} \right)+\frac{\partial}{\partial y}\left(\frac{Q_{k+1}v_{k+1}-Q_k v_k}{\Delta_{k+1,k}} \right)$$
$$+\frac{m}{\Delta_{k+1,k}}\left[\frac{(wv)_{k+1}-(wv)_k}{\Delta_{k+1}}-\frac{(wv)_k-(wv)_{k-1}}{\Delta_k} \right]+m_x f\frac{P_{k+1}-P_k}{\Delta_{k+1,k}}$$
$$+\frac{1}{\Delta_{k+1,k}}\left[\left(v_{k+1}\frac{\partial m_y}{\partial x}-u_{k+1}\frac{\partial m_x}{\partial y} \right)Hu_{k+1}-\left(v_k\frac{\partial m_y}{\partial x}-u_k\frac{\partial m_x}{\partial y} \right)Hu_k \right] \tag{2.93}$$
$$=m_x H\frac{b_{k+1}-b_k}{\Delta_{k+1,k}}g\left(\frac{\partial h}{\partial y}-z_k\frac{\partial H}{\partial y} \right)+\frac{1}{2}\frac{m_x H^2}{\Delta_{k+1,k}}g\left(\Delta_{k+1}\frac{\partial b_{k+1}}{\partial y}+\Delta_k\frac{\partial b_k}{\partial y} \right)$$
$$+\frac{m}{\Delta_{k+1,k}}\left[\frac{(\tau_{yz})_{k+1}-(\tau_{yz})_k}{\Delta_{k+1}}-\frac{(\tau_{yz})_k-(\tau_{yz})_{k-1}}{\Delta_k} \right]+\frac{(S_v)_{k+1}-(S_v)_k}{\Delta_{k+1,k}}$$

$$\Delta_{k+1,k}=\frac{1}{2}\left(\Delta_{k+1}+\Delta_k \right) \tag{2.94}$$

Inspection of equations (2.92)-(2.93) reveals that they could have also been obtained by differentiating the horizontal momentum equations (2.76)-(2.77) with respect to z and introducing a finite difference discretion in z. Using equations (2.81)-(2.82) to relate the shear stresses to the velocity differences across the interior interfaces suggest that equations (2.92) - (2.93) be interpreted as a system of K-1 equations for either the K-1 interfacial velocity differences or the K-1 interior interfacial shear stresses. Details of the solution of the internal mode equations (2.92) - (2.93) will be presented in Section 2.5.

The solution of the vertical velocity, w, employs the continuity equations. Dividing equation (2.84) by Δ_k, and subtracting equation (2.86) gives:

$$w_k=w_{k-1}-\frac{\Delta_k}{m}\left[\frac{\partial}{\partial x}\left(P_k-\overline{P} \right)+\frac{\partial}{\partial y}\left(Q_k-\overline{Q} \right) \right] \tag{2.95}$$

Since $w_o = 0$, the solution proceeds from the first cell layer to the surface. Provided the constraints (equations (2.90)-(2.91)) are satisfied, the surface velocity at $k = K$ will be zero and satisfy the boundary condition.

2.4 Computational Aspects of the Three Time Level External Mode Solution

The formulation of a computational algorithm for the numerical solution of the external mode equations (2.85) to (2.87) begins by introducing modified variables and reorganizing the equations to give:

$$\begin{aligned}
&\frac{\partial \overline{P}}{\partial t} = -\frac{m_y}{m_x} Hg \frac{\partial \zeta}{\partial x} - \frac{m_y}{m_x} H \frac{\partial p_s}{\partial x} + \frac{m_y}{m_x} Hg \left(\overline{b} \frac{\partial h}{\partial x} - \overline{B} \frac{\partial H}{\partial x} - H \frac{\partial \overline{\beta}}{\partial x} \right) \\
&- \frac{1}{m_x} \sum_{k=1}^{K} \Delta_k \left(\frac{\partial}{\partial x}\left(P_k u_k\right) + \frac{\partial}{\partial y}\left(Q_k u_k\right) \right) + \frac{1}{m_x} \sum_{k=1}^{K} \Delta_k \left[\left(v_k \frac{\partial m_y}{\partial x} - u_k \frac{\partial m_x}{\partial y} \right) H v_k + m_y f Q_k \right] \\
&+ \frac{1}{m_x} \sum_{k=1}^{K} \left[\frac{\partial}{\partial x} \left(\frac{m_y}{m_x} HA_{Hk} \Delta_k \frac{\partial u_k}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{m_x}{m_y} HA_{Hk} \Delta_k \frac{\partial u_k}{\partial y} \right) \right] \\
&+ m_y \left(\tau_{xz}\right)_K - m_y \left(\tau_{xz}\right)_0 + \frac{1}{m_x} \overline{S}_u
\end{aligned} \tag{2.96}$$

$$\begin{aligned}
&\frac{\partial \overline{Q}}{\partial t} = -\frac{m_x}{m_y} Hg \frac{\partial \zeta}{\partial y} - \frac{m_x}{m_y} H \frac{\partial p_s}{\partial y} + \frac{m_x}{m_y} Hg \left(\overline{b} \frac{\partial h}{\partial y} - \overline{B} \frac{\partial H}{\partial y} - H \frac{\partial \overline{\beta}}{\partial y} \right) \\
&- \frac{1}{m_y} \sum_{k=1}^{K} \Delta_k \left(\frac{\partial}{\partial x}\left(P_k v_k\right) + \frac{\partial}{\partial y}\left(Q_k v_k\right) \right) + \frac{1}{m_y} \sum_{k=1}^{K} \Delta_k \left[\left(v_k \frac{\partial m_y}{\partial x} - u_k \frac{\partial m_x}{\partial y} \right) H u_k + m_x f P_k \right] \\
&+ \frac{1}{m_y} \sum_{k=1}^{K} \left[\frac{\partial}{\partial x} \left(\frac{m_y}{m_x} HA_{Hk} \Delta_k \frac{\partial v_k}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{m_x}{m_y} HA_{Hk} \Delta_k \frac{\partial v_k}{\partial y} \right) \right] \\
&+ m_x \left(\tau_{yz}\right)_K - m_x \left(\tau_{yz}\right)_0 + \frac{1}{m_y} \overline{S}_v
\end{aligned} \tag{2.97}$$

$$\frac{\partial \zeta}{\partial t} + \frac{1}{m} \left(\frac{\partial \overline{P}}{\partial x} + \frac{\partial \overline{Q}}{\partial y} \right) = S_h \tag{2.98}$$

where

$$\overline{\beta} = \sum_{k=1}^{K} \beta_k \Delta_k \tag{2.99}$$

$$\overline{B} = \sum_{k=1}^{K} \left[\beta_k \Delta_k + \frac{1}{2} \left(z_k + z_{k-1} \right) b_k \Delta_k \right] \tag{2.100}$$

Equations (2.96)-(2.97) now equate the time rate of change of the external or depth integrated volumetric transports to the pressure gradients associated with the free surface slope, atmospheric pressure and buoyancy, the advective accelerations, the Coriolis and curvature

accelerations, the free surface and bottom tangential stresses and the general source, sink terms. The staggered location of variables on the computational grid, Figure 2-2, allows most horizontal spatial derivatives in equations (2.96) to (2.98) to be represented by second order accurate central differences and results in conservation of volume, mass, momentum and energy in the limit of exact integration of the equations in time (Simons, 1973; Haltiner and Williams, 1980). When a variable is not located at a point required for implementation of central difference operators, averaging in either or both spatial directions is appropriate. The use of the spatial averaging scheme of Arakawa and Lamb (1977) to represent the Coriolis and curvature accelerations also guarantees energy conservation.

Following the introduction of discrete finite difference and averaging representations in space, equations (2.96) to (2.98), for a horizontal grid of L cells, may be viewed as a system of $3L$ ordinary differential equations in time for the volumetric transport and the free surface displacement. The numerous techniques available to solve these equations generally fall within the two categories of explicit and semi-implicit. The most frequently used explicit scheme is the three time level leapfrog scheme where the time derivatives are approximated between time levels n+1 and n-1, and the remaining terms are evaluated at time level n. Although computationally simple to implement, the maximum time step is restricted by the Courant-Fredrick-Levy condition based on the gravity wave phase speed. An alternate approach allowing larger time steps is the semi-implicit three time level scheme (Madala and Piacsek, 1977), which when implemented for equations (2.96) to (2.98) is

$$\begin{aligned}
\overline{P}^{n+1} &= \overline{P}^{n-1} - \Delta t\left(\frac{m_y}{m_x}H\right)^u g\delta_x^u\left(\zeta^{n+1}+\zeta^{n-1}\right) - 2\Delta t\left(\frac{m_y}{m_x}H\right)^u \delta_x^u p_s \\
&+2\Delta t\left(\frac{m_y}{m_x}H\right)^u g\left(\overline{b}^u\delta_x^u h - \overline{B}^u\delta_x^u H - H^u\delta_x^u\overline{\beta}\right) \\
&-2\Delta t\left(\frac{1}{m_x}\right)^u \sum_{k=1}^{K}\Delta_k\left[\delta_x^u\left(P_k u_k\right)+\delta_y^u\left(Q_k u_k\right)\right] \qquad (2.101)\\
&+2\Delta t\left(\frac{1}{m_x}\right)^u \sum_{k=1}^{K}\Delta_k\left[\left(v_k\frac{\partial m_y}{\partial x}-u_k\frac{\partial m_x}{\partial y}\right)Hv_k + m_y fQ_k\right]^u + 2\Delta t m_y^u\left[\left(\tau_{xz}^{n-1}\right)_K-\left(\tau_{xz}^{n-1}\right)_0\right]^u \\
&+2\Delta t\left(\frac{1}{m_x}\right)^u \sum_{k=1}^{K}\Delta_k\left[\frac{\partial}{\partial x}\left(m_y H\tau_{xx}^{n-1}\right)+\frac{\partial}{\partial y}\left(m_x H\tau_{xy}^{n-1}\right)+\frac{\partial}{\partial y}\left(m_x H\tau_{xy}^{n-1}\right)-\frac{\partial}{\partial x}\left(m_y H\tau_{yy}^{n-1}\right)\right]_k^u
\end{aligned}$$

$$\bar{Q}^{n+1} = \bar{Q}^{n-1} - \Delta t\left(\frac{m_x}{m_y}H\right)^v g\delta_y^u\left(\zeta^{n+1}+\zeta^{n-1}\right) - 2\Delta t\left(\frac{m_x}{m_y}H\right)^v \delta_y^u p_s$$

$$+2\Delta t\left(\frac{m_x}{m_y}H\right)^v g\left(\bar{b}^v\delta_y^v h - \bar{B}^v\delta_y^v H - H^v\delta_y^v\bar{\beta}\right)$$

$$-2\Delta t\left(\frac{1}{m_y}\right)^u \sum_{k=1}^{K}\Delta_k\left[\delta_x^v\left(P_k v_k\right)+\delta_y^v\left(Q_k v_k\right)\right] \qquad (2.102)$$

$$-2\Delta t\left(\frac{1}{m_y}\right)^v \sum_{k=1}^{K}\Delta_k\left[\left(v_k\frac{\partial m_y}{\partial x}-u_k\frac{\partial m_x}{\partial y}\right)Hu_k + m_x fP_k\right]^v + 2\Delta t m_x^v\left[\left(\tau_{yz}^{n-1}\right)_K - \left(\tau_{yz}^{n-1}\right)_0\right]^v$$

$$+2\Delta t\left(\frac{1}{m_y}\right)^v \sum_{k=1}^{K}\Delta_k\left[\frac{\partial}{\partial x}\left(m_y H\tau_{yx}^{n-1}\right)+\frac{\partial}{\partial y}\left(m_x H\tau_{yy}^{n-1}\right)+\frac{\partial}{\partial y}\left(m_x H\tau_{xx}^{n-1}\right)-\frac{\partial}{\partial x}\left(m_y H\tau_{yx}^{n-1}\right)\right]_k^v$$

$$\zeta^{n+1} - \zeta^{n-1} + \Delta t\left(\frac{1}{m}\right)^\zeta\left[\delta_x^\zeta\left(\bar{P}^{n+1}+\bar{P}^{n-1}\right)+\delta_y^\zeta\left(\bar{Q}^{n+1}+\bar{Q}^{n-1}\right)\right] = S_h\Delta t \qquad (2.103)$$

with Δt being the time step. All terms in equations (2.101) to (2.103) are understood to be evaluated at the center time level n except those evaluated at the forward and backward time levels, n+1 and n-1, which are denoted by superscripts. The u, v, and ζ superscripts indicate that a variable is evaluated, or that a spatial derivative is centered, at the corresponding spatial point.

The subscript of the spatial central difference operator, δ, indicates direction. The grid cells are presumed to be bounded in the horizontal by lines of constant integer values of the dimensionless orthogonal coordinates x and y, resulting in the central spatial differences having the forms:

$$\delta_x\left(\phi_{i,j,k}\right) = \frac{1}{\Delta x}\left(\phi_{i+\frac{1}{2},j,k} - \phi_{i-\frac{1}{2},j,k}\right) \qquad (2.104)$$

$$\delta_y\left(\phi_{i,j,k}\right) = \frac{1}{\Delta y}\left(\phi_{i,j+\frac{1}{2},k} - \phi_{i,j-\frac{1}{2},k}\right) \qquad (2.105)$$

Application of these finite difference operators to the advective accelerations is illustrated by:

$$\delta_x^u\left(P_{i,j,k}u_{i,j,k}\right) = \frac{1}{\Delta x}\left(P_{i+\frac{1}{2},j,k}\,u_{i+\frac{1}{2},j,k} - P_{i-\frac{1}{2},j,k}\,u_{i-\frac{1}{2},j,k}\right) \qquad (2.106)$$

where the constant y dependence of the variables is implied. Since the u type variables are located at integer values of x, averaging is necessary to obtain values at the half intervals. Averaging both the transport and the velocity gives:

$$\delta_x^u\left(P_{i,j,k}u_{i,j,k}\right)=\frac{1}{\Delta x}\left(\frac{P_{i+1,j,k}+P_{i,j,k}}{2}\frac{u_{i+1,j,k}+u_{i,j,k}}{2}-\frac{P_{i,j,k}+P_{i-1,j,k}}{2}\frac{u_{i,j,k}+u_{i-1,j,k}}{2}\right) \tag{2.107}$$

which is consistent with a central difference approximation of the non-conservative form of this portion of the advective acceleration. Averaging the transport and allowing the velocity to be advected from the upwind direction gives:

$$\begin{aligned}\delta_x^u\left(P_{i,j,k}u_{i,j,k}\right)=&\frac{1}{\Delta x}\left[\max\left(\frac{P_{i+1,j,k}+P_{i,j,k}}{2},0\right)u_{i,j,k}^{n-1}-\max\left(\frac{P_{i,j,k}+P_{i-1,j,k}}{2},0\right)u_{i-1,j,k}^{n-1}\right]\\&+\frac{1}{\Delta x}\left[\min\left(\frac{P_{i+1,j,k}+P_{i,j,k}}{2},0\right)u_{i+1,j,k}^{n-1}-\min\left(\frac{P_{i,j,k}+P_{i-1,j,k}}{2},0\right)u_{i,j,k}^{n-1}\right]\end{aligned} \tag{2.108}$$

which is consistent with an upwind or backward difference approximation of the non-conservative form of this portion of the advective acceleration. In equation (2.108), the transport is still at time level *n*, while the velocity is at time level *n*-1, for both stability and accuracy (Smolarkiewicz and Clark, 1986). The preference for the use of equation (2.107) or equation (2.108) will generally depend upon the physical situation being simulated. The central difference form introduces no numerical diffusion but may produce solution fields which exhibit cell to cell spatial oscillations. These oscillations can be eliminated by the addition of horizontal diffusion terms to the momentum equations. Specification of the horizontal diffusivity allows the degree of spatial smoothing to be controlled. The upwind difference form introduces numerical diffusion and does not produce spatial oscillations in the solution field. The Coriolis and curvature terms in equations (2.101)-(2.102) are discretized using an energy conserving spatial averaging and differencing (Arakawa and Lamb, 1977; Haltiner and Williams, 1980). For example, the Coriolis and curvature term in equation (2.101) is given by:

$$\left[m_y fQ_k+\left(v_k\frac{\partial m_y}{\partial x}-u_k\frac{\partial m_x}{\partial y}\right)Hv_k\right]^u=\frac{1}{2}\left[\left(RH\right)^{\zeta}_{i+\frac{1}{2},j}v^{\zeta}_{i+\frac{1}{2},j,k}+\left(RH\right)^{\zeta}_{i-\frac{1}{2},j}v^{\zeta}_{i-\frac{1}{2},j,k}\right] \tag{2.109}$$

$$R^{\zeta}_{i+\frac{1}{2},j}=\left(mf\right)_{i+\frac{1}{2},j}+\frac{\left(m_y\right)_{i+1,j}-\left(m_y\right)_{i,j}}{\Delta x}v^{\zeta}_{i+\frac{1}{2},j,k}-\frac{\left(m_x\right)_{i+\frac{1}{2},j+\frac{1}{2}}-\left(m_x\right)_{i+\frac{1}{2},j-\frac{1}{2}}}{\Delta y}u^{\zeta}_{i+\frac{1}{2},j,k} \tag{2.110}$$

$$v^{\zeta}_{i+\frac{1}{2},j,k}=\frac{1}{2}\left(v_{i+\frac{1}{2},j+\frac{1}{2},k}+v_{i+\frac{1}{2},j-\frac{1}{2},k}\right) \tag{2.111}$$

$$u^{\zeta}_{i+\frac{1}{2},j,k}=\frac{1}{2}\left(u_{i+1,j,k}+u_{i,j,k}\right) \tag{2.112}$$

where the variables locations are shown in Figure 2-3

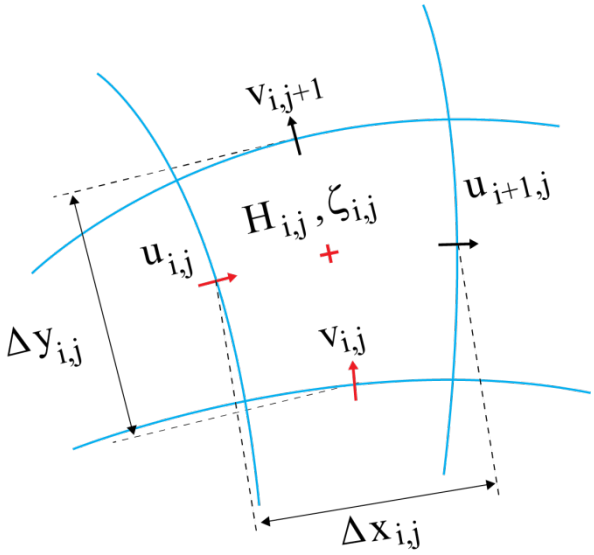


Figure 2-3: U-centered grid in the horizontal (x, y) plane.

Since the bottom tangential stresses in equations (2.101)-(2.102) must be supplied from the internal mode solution which follows the external solution, it is lagged at the backward time level. The general source, sink term has been replaced by horizontal diffusion terms having the form proposed by Mellor and Blumberg (1985). The horizontal stress tensor is taken of the form:

$$\left(\tau_{xx}\right)_k = 2A_H \frac{1}{m_x}\frac{\partial u_k}{\partial x} \tag{2.113}$$

$$\left(\tau_{xy}\right)_k = \left(\tau_{yx}\right)_k = 2A_H \left(\frac{1}{m_x}\frac{\partial v_k}{\partial x} + \frac{1}{m_y}\frac{\partial u_k}{\partial y}\right) \tag{2.114}$$

$$\left(\tau_{yy}\right)_k = 2A_H \frac{1}{m_y}\frac{\partial v_k}{\partial y} \tag{2.115}$$

The horizontal diffusion coefficient, A_H, is often specified as a minimum constant value necessary to smooth cell to cell spatial oscillations in the solution field when the central difference form of the advective acceleration, equation (2.107) is used. When the horizontal turbulent diffusion is used to represent subgrid scale mixing, A_H may be determined as suggested by Smagorinsky (1963).

The solution scheme for equations (2.101) to (2.103) involves first evaluating all terms in the three equations at time levels n and n-1. On boundaries where the transports are specified, the specified values at time level n+1 are inserted into equation (2.103). Equations (2.101)-(2.102) are then used to eliminate the unknown transports at time level n+1, from equation (2.103). The result is a discrete Helmholtz type elliptic equation for the free surface displacement at time level n+1, having the general form:

$$\zeta^{n+1} - g\Delta t^2 \left(\frac{1}{m}\right)^{\zeta}\left[\delta_x^{\zeta}\left(H\frac{m_y}{m_x}\right)^{u}\delta_x^{u}\zeta^{n+1} + \delta_x^{\zeta}\left(H\frac{m_x}{m_y}\right)^{v}\delta_y^{v}\zeta^{n+1}\right] - \phi = 0 \tag{2.116}$$

with the term ϕ containing all of the previously evaluated terms and transport boundary conditions. For cells where the free surface displacement is specified, equation (2.116) is replaced by an equation which enforces the specified boundary condition at time level n+1. For the rigid lid case where the free surface displacement is constant in time and space, equation (2.116) is modified to give an equation for the unknown surface pressure, p_s, by eliminating the first term, replacing $g\zeta$ in the discrete elliptic operator by p_s, and appropriately modifying the last term. In the computer code, the system of equations corresponding to equation (2.116) is solved by a reduced system conjugate gradient scheme with a multicolor or red-black ordering of the cells (Hageman and Young, 1981). The conjugate gradient iterations continue until the sum of the squared residuals is less than a specified value. The free surface displacements or surface pressures are then substituted into equations (2.101)-(2.102) to determine the transports at time level n+1. Since the solution of equation (2.116) is approximate, equation (2.103) may not be identically satisfied upon substitution of the time level n+1 transports and free surface displacement. To insure that equation (2.103) is identically satisfied in the case of a dynamic free surface, it is solved for a revised value of the time level n+1, free surface displacement after introduction of the time level n+1 transports. For the rigid lid case, an external divergence error is calculated and compensated for by adding appropriate volumetric source or sink terms to equation (2.103) during the next time step.

See the Appendix B - Three Time Level Solution for insight into the stability and accuracy of the semi-implicit three time level scheme for solving the external mode equations with a dynamically active free surface with by a Fourier analysis of the linearized discrete equations.

2.5 Computational Aspects of the Three Time Level Internal Mode Solution

The internal mode equations (2.92) and (2.93) are solved using a fractional step scheme (Peyret and Taylor, 1983), with the first step being explicit and the second step being implicit. Figure 2-4 illustrates the location variables in the x, z plane for the x componoent of the internal mode equations.

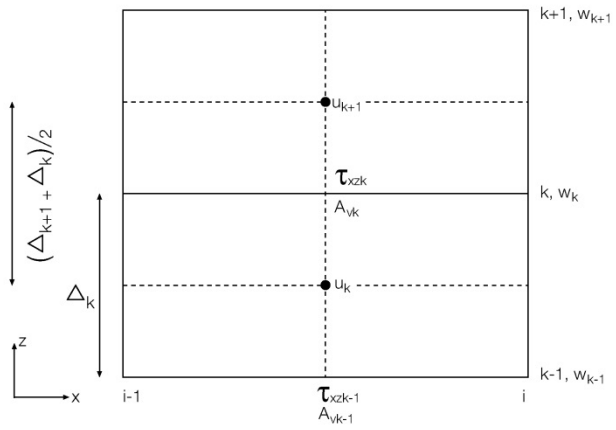


Figure 2-4: U-centered grid in the vertical (x,z) plane

The computational equations for the three time level explicit step are:

$$
\begin{aligned}
&\left(P_{k+1}-P_k\right)^{**}=\left(P_{k+1}-P_k\right)^{n-1}\\
&-2\Delta t\left(\frac{1}{m_x}\right)^u\left[\delta_x^u\left(P_{k+1}u_{k+1}-P_ku_k\right)+\delta_y^u\left(Q_{k+1}u_{k+1}-Q_ku_k\right)\right]\\
&-2\Delta t\left(\frac{1}{m_x}\right)^u\left[\frac{\left(Wu\right)_{k+1}-\left(Wu\right)_k}{\Delta_{k+1}}-\frac{\left(Wu\right)_k-\left(Wu\right)_{k-1}}{\Delta_k}\right]^u\\
&+2\Delta t\left(\frac{1}{m_x}\right)^u\left[m_yfQ_{k+1}-m_yfQ_k\right]^u\\
&+2\Delta t\left(\frac{1}{m_x}\right)^u\left[\left(v_{k+1}\frac{\partial m_y}{\partial x}-u_{k+1}\frac{\partial m_x}{\partial y}\right)Hv_{k+1}-\left(v_k\frac{\partial m_y}{\partial x}-u_k\frac{\partial m_x}{\partial y}\right)Hv_k\right]^u\\
&+2\Delta t\left(\frac{m_y}{m_x}H\right)^u g\left[\left(b_{k+1}-b_k\right)^u\delta_x^u\left(h-z_kH\right)+\frac{1}{2}H^u\delta_x^u\left(b_{k+1}\Delta_{k+1}+b_k\Delta_k\right)\right]\\
&+2\Delta t\left(\frac{1}{m_x}\right)^u\left[\left(S_u\right)_{k+1}-\left(S_u\right)_k\right]^u
\end{aligned}
\tag{2.117}
$$

$$
\begin{aligned}
&\left(Q_{k+1}-Q_k\right)^{**}=\left(Q_{k+1}-Q_k\right)^{n-1}\\
&-2\Delta t\left(\frac{1}{m_y}\right)^v\left[\delta_y^v\left(P_{k+1}v_{k+1}-P_kv_k\right)+\delta_y^v\left(Q_{k+1}v_{k+1}-Q_kv_k\right)\right]\\
&-2\Delta t\left(\frac{1}{m_y}\right)^v\left[\frac{\left(Wv\right)_{k+1}-\left(Wv\right)_k}{\Delta_{k+1}}-\frac{\left(Wv\right)_k-\left(Wv\right)_{k-1}}{\Delta_k}\right]^v\\
&-2\Delta t\left(\frac{1}{m_y}\right)^v\left[m_xfP_{k+1}-m_xfP_k\right]^v\\
&-2\Delta t\left(\frac{1}{m_y}\right)^v\left[\left(v_{k+1}\frac{\partial m_y}{\partial x}-u_{k+1}\frac{\partial m_x}{\partial y}\right)Hu_{k+1}-\left(v_k\frac{\partial m_y}{\partial x}-u_k\frac{\partial m_x}{\partial y}\right)Hu_k\right]^v\\
&+2\Delta t\left(\frac{m_x}{m_y}H\right)^v g\left[\left(b_{k+1}-b_k\right)^v\delta_y^v\left(h-z_kH\right)+\frac{1}{2}H^v\delta_y^v\left(\Delta_{k+1}b_{k+1}+\Delta_kb_k\right)\right]\\
&+2\Delta t\left(\frac{1}{m_y}\right)^v\left[\left(S_v\right)_{k+1}-\left(S_v\right)_k\right]^v
\end{aligned}
\tag{2.118}
$$

$$W\ =\ m_xm_yw\ =\ mw \tag{2.119}$$

where the superscript ** denotes the provisional solution, and all terms not having a specified time level are understood to be at the centered time level *n*. The horizontal volume transports, *P* and *Q* are as defined by equation (2.78) and *W* is the vertical volume transport. The horizontal difference operations on the horizontal advection terms are identical to those presented in

Section 3, equations (2.106) to (2.108). The vertical momentum flux terms may be represented in forms consistent with central or upwind differencing,

$$(Wu)^u_{i,j,k} = \frac{W_{i-\frac{1}{2},j,k} + W_{i+\frac{1}{2},j,k}}{2} \frac{u_{i,j,k} + u_{i,j,k+1}}{2} \tag{2.120}$$

$$(Wu)^u_{i,j,k} = max\left(\frac{W_{i-\frac{1}{2},j,k} + W_{i+\frac{1}{2},j,k}}{2}, 0\right) u^{n-1}_{i,j,k} + min\left(\frac{W_{i-\frac{1}{2},j,k} + W_{i+\frac{1}{2},j,k}}{2}, 0\right) u^{n-1}_{i,j,k+1} \tag{2.121}$$

where the advected velocity in the upwind form, equation (2.121) is evaluated at time level n-1 for stability. The horizontal difference operations on the buoyancy and mean and total depths are central difference operators defined by equations (2.104) and (2.105). The inclusion of horizontal diffusion in the source, sink terms in equations (2.117)-(2.118) would follow from its inclusion in equations (2.101)-(2.102). The Coriolis and curvature terms are averaged and differenced by the energy conserving scheme presented in Section 3, equations (2.109) to (2.112). The stability of the explicit fractional step, equations (2.117)-(2.118) is governed by the stability of the discretization of the horizontal and vertical advective accelerations, which will be discussed in Section 5, and the discretization of the Coriolis and curvature terms. The results of the Fourier stability analysis of the external mode scheme, with respect to the Coriolis acceleration, can be shown to apply to the internal mode scheme as well.

The computational equations for the second step of the three time level scheme are:

$$\frac{(P_{k+1} - P_k)^{n+1}}{m^u_y \Delta_{k+1,k}} = \frac{(P_{k+1} - P_k)^{**}}{m^u_y \Delta_{k+1,k}} + 2\Delta t \left[\frac{(\tau_{xz})_{k+1} - (\tau_{xz})_k}{\Delta_{k+1}\Delta_{k+1,k}} - \frac{(\tau_{xz})_k - (\tau_{xz})_{k-1}}{\Delta_k \Delta_{k+1,k}}\right]^{n+1} \tag{2.122}$$

$$\frac{(Q_{k+1} - Q_k)^{n+1}}{m^v_x \Delta_{k+1,k}} = \frac{(Q_{k+1} - Q_k)^{**}}{m^v_x \Delta_{k+1,k}} + 2\Delta t \left[\frac{(\tau_{yz})_{k+1} - (\tau_{yz})_k}{\Delta_{k+1}\Delta_{k+1,k}} - \frac{(\tau_{yz})_k - (\tau_{yz})_{k-1}}{\Delta_k \Delta_{k+1,k}}\right]^{n+1} \tag{2.123}$$

Using equations (2.81)-(2.82), the turbulent shear stresses are related to the horizontal transports by:

$$(\tau_{xz})^{n+1}_k = \left(\frac{A^u_v}{H^u}\right)^n_k \left(\frac{1}{m^u_y H^u} \frac{P_{k+1} - P_k}{\Delta_{k+1,k}}\right)^{n+1} \tag{2.124}$$

$$(\tau_{yz})^{n+1}_k = \left(\frac{A^v_v}{H^v}\right)^n_k \left(\frac{1}{m^v_x H^v} \frac{Q_{k+1} - Q_k}{\Delta_{k+1,k}}\right)^{n+1} \tag{2.125}$$

Equations (2.124)-(2.125) could be used to eliminate the turbulent shear stresses from equations (2.122)-(2.123) to give a pair of K-1 systems of equations for the transport differences between layers, however, the resulting equations are poorly conditioned. Instead, equations (2.124)-(2.125) are used to eliminate the horizontal transport differences at time level n+1 from equations (2.122)-(2.123) to give a pair of K-1 equations for the turbulent shear stresses

$$-\frac{1}{\Delta_k\Delta_{k+1,k}}\left(\tau_{xz}\right)_{k-1}^{n+1}+\left[\frac{1}{\Delta_k\Delta_{k+1,k}}+\frac{\left(H^u\right)^{n+1}}{2\Delta t}\left(\frac{H^u}{A_v^u}\right)_k^n+\frac{1}{\Delta_{k+1}\Delta_{k+1,k}}\right]\left(\tau_{xz}\right)_k^{n+1}$$
$$-\frac{1}{\Delta_{k+1}\Delta_{k+1,k}}\left(\tau_{xz}\right)_{k+1}^{n+1}=\frac{1}{2\Delta t m_y^u}\frac{\left(P_{k+1}-P_k\right)^{**}}{\Delta_{k+1,k}} \tag{2.126}$$

$$-\frac{1}{\Delta_k\Delta_{k+1,k}}\left(\tau_{yz}\right)_{k-1}^{n+1}+\left[\frac{1}{\Delta_k\Delta_{k+1,k}}+\frac{\left(H^v\right)^{n+1}}{2\Delta t}\left(\frac{H^v}{A_v^v}\right)_k^n+\frac{1}{\Delta_{k+1}\Delta_{k+1,k}}\right]\left(\tau_{yz}\right)_k^{n+1}$$
$$-\frac{1}{\Delta_{k+1}\Delta_{k+1,k}}\left(\tau_{yz}\right)_{k+1}^{n+1}=\frac{1}{2\Delta t m_x^v}\frac{\left(Q_{k+1}-Q_k\right)^{**}}{\Delta_{k+1,k}} \tag{2.127}$$

These equations are diagonally dominant and well conditioned, and can be solved independently at each of the horizontal velocity locations. Since equations (2.126)-(2.127) represent fully implicit, backward difference in time, schemes for one dimensional parabolic diffusion equations, the solutions are unconditionally stable (Fletcher, 1988). Given the solutions of (2.126)-(2.127) the shear stresses, the K-1 transport differences, P_{k+1}-P_k and Q_{k+1}-Q_k, are determined from equations (2.124)-(2.125) and combined with the continuity constraints, equations (2.90)-(2.91), to form a pair of K equations for the horizontal transports in each cell layer. To illustrate, the horizontal transports in the surface cell layer are determined analytically and given by:

$$P_k=\overline{P}+\sum_{k=1}^{K-1}\left(\sum_{j=1}^{k}\Delta_j\right)\left(P_{k+1}-P_k\right) \tag{2.128}$$

and a similar expression for Q_K. Working down from the surface using the K-1 transport differences allows the remaining transports to be determined. It is noted for later use that the bottom cell layer transports can be expressed in terms of the depth integrated transports and the transport differences using:

$$P_1=\overline{P}-\sum_{k=1}^{K-1}\left(1-\sum_{j=1}^{k}\Delta_j\right)\left(P_{k+1}-P_k\right) \tag{2.129}$$

and an identical equation for Q_1.

Solution of equations (2.126)-(2.127) requires specification of bottom and surface stresses at k=0 and k=K, respectively. On the free surface, k=K, the surface wind stress components are specified. On the bottom fluid-solid boundary, k=0, the bottom stress must be specified. The simplest approach to specifying the bottom stress components utilizes the velocity component in the bottom cell layer and the quadratic friction relations:

$$\left(\tau_{xz}\right)_0^{n+1}=C_b\left(\sqrt{\left(u_1\right)^2+\left(v_1^u\right)^2}\right)^n\left(\frac{P_1}{m_y^uH^u}\right)^{n+1} \tag{2.130}$$

$$\left(\tau_{yz}\right)_0^{n+1} = C_b\left(\sqrt{\left(u_1^v\right)^2+\left(v_1\right)^2}\right)^n\left(\frac{Q_1}{m_x^v H^v}\right)^{n+1} \tag{2.131}$$

Assuming a logarithmic velocity profile between the solid bottom and the middle of the bottom cell layer gives the bottom stress coefficient:

$$C_b = \frac{\kappa^2}{\left[\ln\left(\frac{\Delta_1 H}{2z_0^*}\right)\right]^2} \tag{2.132}$$

where z_o^* is the dimensional bottom roughness height. Inserting equation (2.129) and a corresponding equation for Q_1 into equations (2.130)-(2.131), respectively allows the bottom stresses at time level n+1 to be expressed in terms of the depth integrated transport components, known from the external mode solution, and the unknown transport differences at time level n+1. However, the transport differences at time level n+1 are related to the shear stress components by equations (2.124)-(2.125), allowing the bottom stresses to be expressed in terms of the depth integrated transports and the internal shear stresses by:

$$\left(\tau_{xz}\right)_0^{n+1} = C_b\left(\sqrt{\left(u_1\right)^2+\left(v_1^u\right)^2}\right)^n\left[\left(\frac{\bar{P}}{m_y^u H^u}\right)^{n+1} - \sum_{k=1}^{K-1}\left(1-\sum_{j=1}^{k}\Delta_j\right)\frac{\Delta_{k+1,k}\left(\tau_{xz}\right)_k^{n+1}}{\left(\frac{A_v^u}{H^u}\right)_k^n}\right] \tag{2.133}$$

and a similar expression for the y component. Inserting equation (2.133) and the corresponding y component equation for the bottom stress components into the k=1 pair of equations (2.126)-(2.127) results in a nearly tri-diagonal system with a fully populated first row. The systems of equations are still efficiently solved using a tri-diagonal equation solver and the Sherman-Morrison formula (Press *et al.*, 1986).

The internal mode solution is completed by the determination of the vertical velocity using:

$$w_k = w_{k-1} - \frac{\Delta_k}{m^\zeta}\left[\delta_x^\zeta\left(P_k-\bar{P}\right)+\delta_y^\zeta\left(Q_k-\bar{Q}\right)\right] \tag{2.134}$$

which follows from equation (2.95). The solution of equation (2.134), where all variables are at time level n+1, proceeds from k=1 since w_o = 0. A two time level correction step is also periodically inserted into the internal mode time integration on the same time step as the external mode correction. The computational equations follow directly from the three time level equations using the details of the external mode presentated in Section 2.4.

2.6 Vertical Layering Options

This Section summarizes the vertical coordinate options in the Environmental Fluid Dynamics Computer Code (EFDC). It supplements the theoretical and computational description of the basic EFDC hydrodynamic and transport model components. The EFDC model was originally formulated with a sigma stretched vertical coordinate. Later, more efficient vertical layering options, namely

sigma-zed options have been implemented to reduce the error due to the horizontal pressure gradients and to reduce the number of computational cells.

2.6.1. Sigma Coordinate System

A sigma coordinate system is a topographically conformal vertical coordinate system which is widely used in three-dimensional hydrodynamic models. In this vertical coordinate system, the number of vertical levels in the water column is the same everywhere in the domain irrespective of the depth of the water column. This can resolve the water column equally well and equally efficiently in both shallow and deep regions of a computational domain simultaneously and it is suitable for a water body with complicated geometry and large changes in bottom elevation. The transformation of the governing equations using sigma-coordinate in the vertical is described in Section 2.1.

In the sigma coordinate formulation, the number of vertical layers is the same at all horizontal locations in the model grid. Although this formulation is widely accepted, conceptually attractive and adequate for a large range of applications, there are a number of application classes where a traditional z or physical vertical grid is desirable, such as deep reservoirs which rapid and large lateral bathymetric changes. There are also applications where the ability to use a combination of sigma and physical z vertical layering in different regions of the horizontal domain would be desirable. An example would be a deep navigation channel in an otherwise shallow estuary. The sigma stretched vertical grid formulation may also be subject to internal pressure gradient errors (Mellor et al., 1994, 1998) providing another motivation for having the options to the sigma formulation.

2.6.2. Sigma-Zed Approach (SGZ)

The standard sigma grid using for the transformation of the vertical coordinate introduces a well-known error in the horizontal gradient terms including the concentration, velocity and the pressure (Mellor, et.al. 1994). In general, this error is significant only in regions with steeply varying bathymetry. In order to overcome this weakness, two new vertical layering approaches that are computationally efficient have been developed and applied to the EFDC+ model (Craig, et.al, 2014). The vertical layering scheme has been modified to allow the number of layers to vary over the model domain based on the water depth. Consequently, each cell can have a different number of layers. The z coordinate system varies for each cell face, matching the number of active layers to the adjacent cells (face matching of layering is a fundamental difference with the GVC approach). Such a transformation is referred to as the sigma-zed (SGZ) coordinate. The differences in the two optional SGZ approaches relate to the sigma layer thickness computed for each cell. Figure 2-5 shows a schematic demonstrating the layering options. Panel (a) represents a standard sigma stretch grid with 10 layers. Panels (b) and (c) represent SGZ options with varying number of layers per cell but with the relative layer thicknesses varying per cell. It should be noted that in SGZ coordinate the number of vertical layers can be very large, but the computational time is shorter in comparison with a similarly configured sigma (SIG) coordinate model.

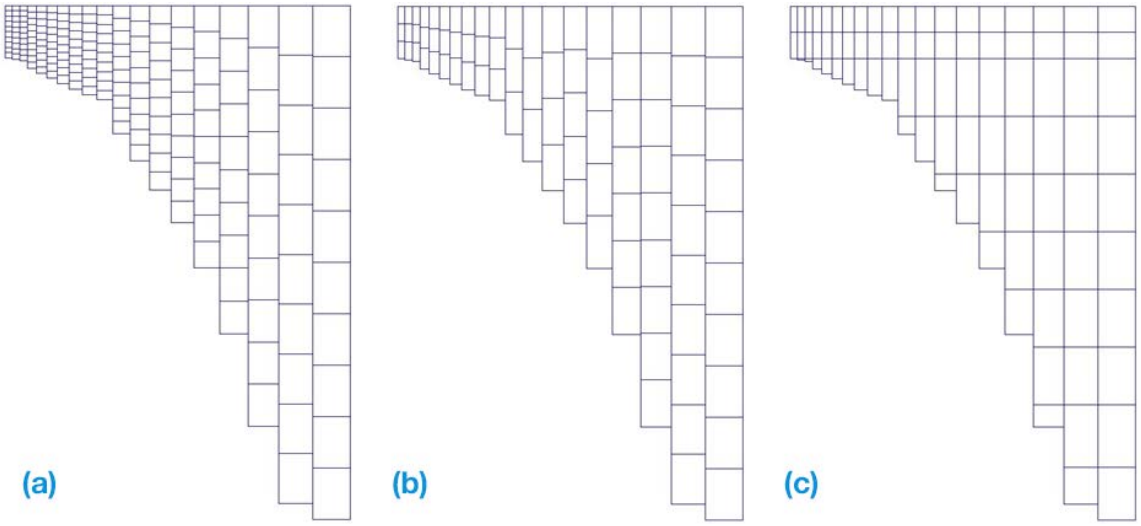


Figure 2-5: An illustration of EFDC+ Layering options for a K = 10

For SGZ transformation the equations are still the same as the standard sigma transformation, however, the number of layers at each cell differs based on a factor determined on the basis of the ratio between bed elevation and the minimum elevation. In addition, the thickness of layers at each cell has to satisfy the following expression:

$$\sum_{k=n}^{KC} \Delta_k = 1 \tag{2.135}$$

in which KC is the maximum number of layers, n the index of bottom layer and Δk the thickness of layer k.

For the original sigma the index of the bed layer always is equal to n = 1 while in the SGZ this value can be varied in the range $1 \le n \le K$ depending on the number of layers due to the rescaling. This requirement improves the accuracy of the horizontal gradient calculation for the variable $C_{i,j,k}$ of the cell $L(i,j)$ at layer k:

$$\frac{\partial C_{i,j,k}}{\partial x} = \frac{C_{i,j,k} - C_{i-1,j,k}}{\Delta x} \tag{2.136}$$

When sediment transport is simulated and bed morphology is taken into account, the determination of the new indices of bottom layers should be implemented at every time step. This is because at this time the ratios between water depths and the maximum are changing due to erosion or deposition in comparison with the initial time. Therefore, an update of layering for the whole domain is important and necessary for SGZ. However, the re-layering is only an optional approach.

The other necessary modification for SGZ coordinate system is the treatment on wet/dry in 3-D calculation of the horizontal gradient when the number of layers at cell $L(i\text{-}1,j)$ is less than that at cell $L(i,j)$. It should be noted that this problem does not appear for SIG coordinates, because the number of layers is the same for every cell. This means that the SIG model requires more calculation time and is therefore one of the weak points of the SIG coordinate system which is overcome with SGZ.

2.6.3. Verification of SGZ Approach

A modification of the vertical coordinate system algorithm in EFDC with SGZ transformation has been successfully implemented. It has been tested for many computation domains with different

hydrodynamic regimes, such as the application to Lake Washington (Seattle, WA, USA), which has steep bottom gradients and sharp thermoclines.

In this test two EFDC options including the original sigma (SIG) and sigma-zed (SGZ), were applied with the same model conditions. Some parameters in the model for the two options are presented in Table 2.3.

Table 2.3: Input parameters of two options

Options	Time (days)	Time step (secconds)	Number of layers	Thermal model
Standard sigma stretched grid	120	5	55	Yes
Sigma-Zed grid	120	5	2 - 55	Yes

In order to investigate the influence of horizontal gradient terms on the vertical profiles of temperature as well as the accuracy between SIG and SGZ models, comparisons the models versus the data were computed for every day during the summer 2008 simulation period. Eight representative vertical profiles, starting at the beginning of the simulation and ending on October 15, are presented in Figure 2-6 for the SIG model and Figure 2-7 for the SGZ model. It clearly shows that the SGZ model produces a much better vertical profiles of the temperature than SIG model.

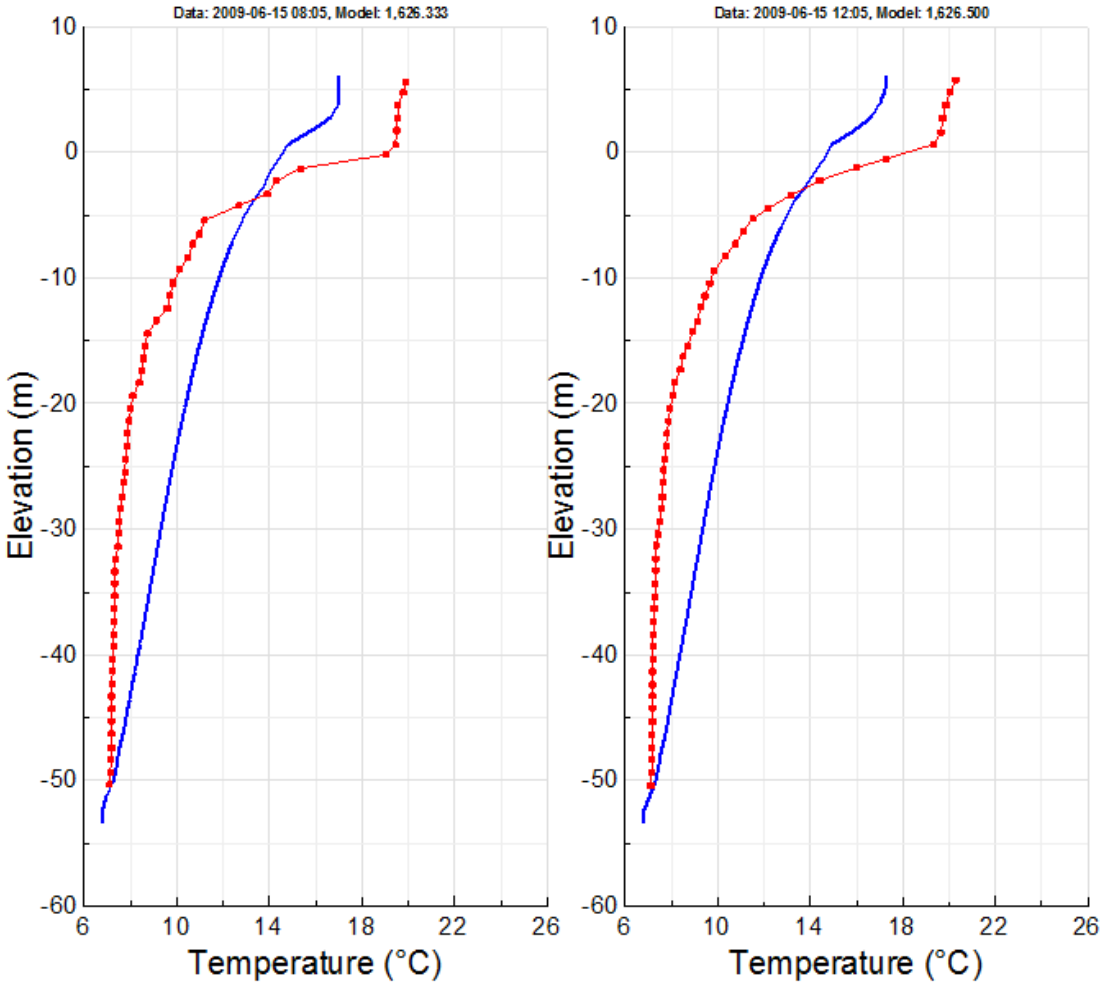


Figure 2-6: Comparison on vertical profile of temperature between data and SIG model

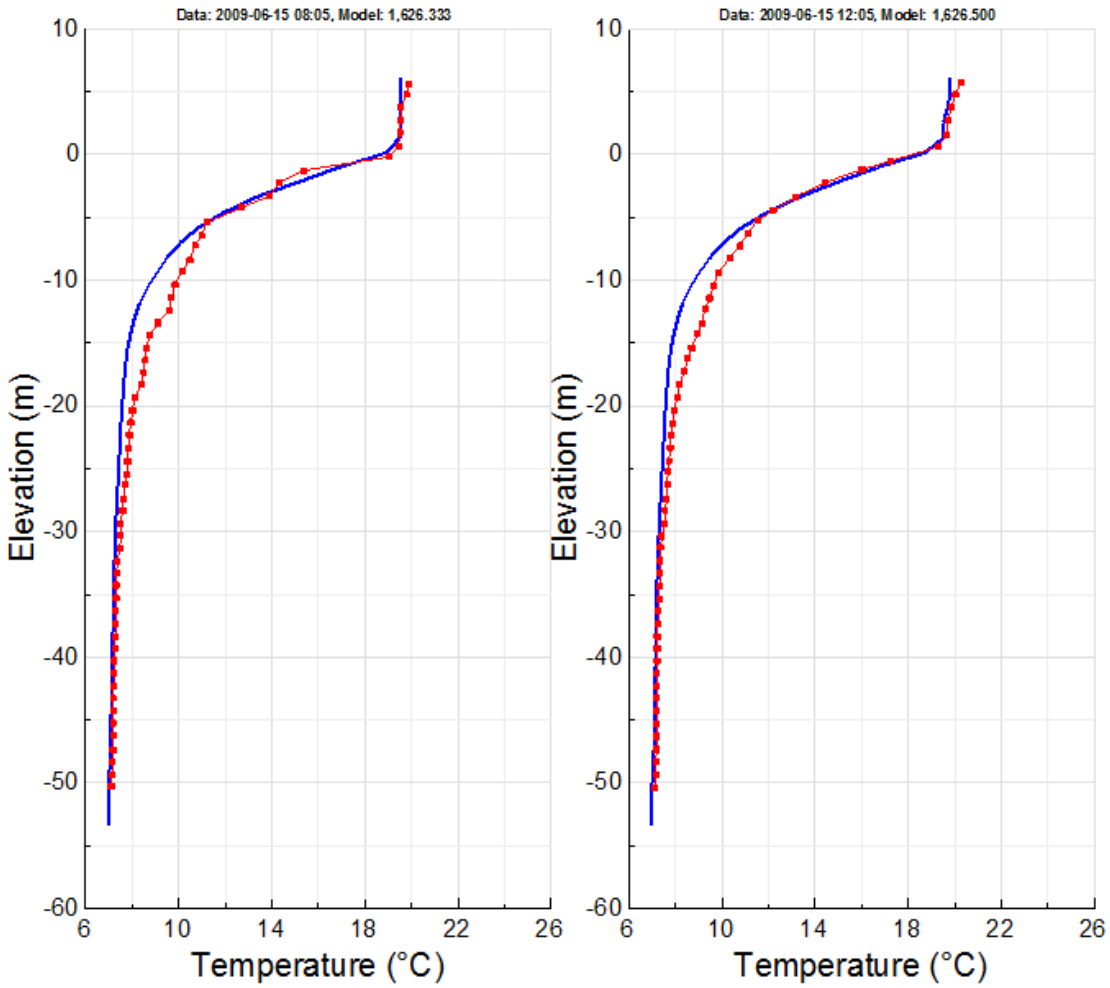


Figure 2-7: Comparison on vertical profile of temperature between data and SGZ model.

2.7 Near-Field Discharge Dilution and Mixing Zone Analysis

The calculation procedure of the jet/plume submodel is mainly based on Winiarski and Frick (1984) and Lee and Cheung (1990). The trajectory of a group of plume particles is traced in time using a Lagrangian formulation. The plume puff gains mass as ambient fluid is entrained and mixed within it, but one entrained the new mass becomes an indistinguishable part of the plume puff. In the simplest version, the plume is assumed to be essentially a cyclindrical segment whose radius grows as mass is entrained.

The initial plume mass is identified as the mass issuing form a diffuser with radius b_0:

$$M_0 = \rho_0 \pi b_0^2 h_0 \tag{2.137}$$

where h_0 is the length of the plume mass and is chosen to be comparable to b_0., for example, $h_0 = r$ and $b_0 = r$ where r is the radius of the diffuser.

$$h_0 = V_0 \Delta t \tag{2.138}$$

The increment in the plume mass at the time step n^{th} is evaluated as the sum of the plume mass increasement due to the shear-induced entrainment and the forced entrainment.

$$\Delta M_n = \Delta M_s + \Delta M_f \tag{2.139}$$

where ΔM_s is the increase in mass due to shear entrainment, and ΔM_f is the increase in mass due to forced entrainment.

A schematic of a rising plume discharged into a water body is show in Figure 2-8 below.

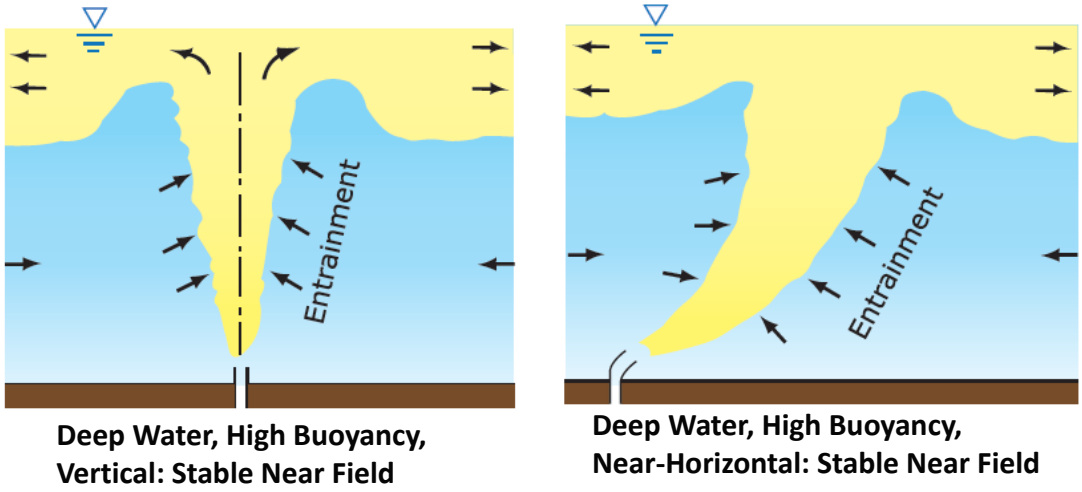


Figure 2-8: Near field Jet Plume mixing

2.7.1. Shear-induced entraimnent

The increase in mass of the plume element is due to turbulent entrainment of the ambient flow. Close to the discharge point, or in a very weak current, shear-induced entrainment dominates. In general, however, the forced entrainment of the cross flow dominates except very close to the source. In the model, assuming the total entrainment is a function of the horizontal currents and a shearing action of the plume relative to the currents, the increase in mass due to shear entrainment, ΔM_s, is written as

$$\Delta M_s = \rho_a 2\pi b_n h_n E \left| V_n - u_a \cos\phi_n \cos\theta_n \right| \Delta t \tag{2.140}$$

In which the subscript n denotes the value of plume element at n^{th} step of calculation, the subscript a denotes the local ambient value, E is the entraiment coefficient which is dependent on the local densimetric Froude number F_1 and jet orientation

$$E = \sqrt{2}\,\frac{0.057 - 0.554\dfrac{\sin\phi_n}{F_1^2}}{1 + 5\dfrac{u_a \cos\phi_n \cos\theta_n}{\left| V_n - u_a \cos\phi_n \cos\theta_n \right|}} \tag{2.141}$$

where F_1 is the local densimetric Froude number

$$F_1 = \alpha \frac{\left|V_n - u_a \cos\phi_n \cos\theta_n\right|}{\sqrt{g \frac{\Delta\rho_n}{\rho_a} b_n}} \tag{2.142}$$

and a is a proportionality constant.

2.7.2. Forced entrainment

Experimental observations by Chu and Goldberg (1984) and Chu (1975) have shown that the transfer of horizontal momentum is complete beyond a few jet diameters. We assume that all the ambient flow on the downdrift side of the plume is entrained into the plume element. This forced entrainment of the ambient flow into an arbitrarily inclined plume element can be formulated as

$$\Delta M_f = \rho_a u_a \left[2b\Delta s\sqrt{1-\cos^2\phi\cos^2\theta} + \pi b\Delta b \cos\phi\cos\theta + \frac{1}{2}\pi b^2 \Delta\left(\cos\phi\cos\theta\right) \right] \tag{2.143}$$

In which the first term represents the forced entrainment due to the projected plume area normal to the corss flow; the second term is a correction due to the growth of plume radius; and the third term is a correction fue to the curvature of the trajectory.

An intial estimate of ΔM_f can be ontained as

$$\begin{aligned} \Delta M_f = \rho_a u_a h_n b_n [2\sqrt{\left(\sin^2\phi + \sin^2\theta - \sin^2\phi\sin^2\theta\right)_n} + \\ \pi\left(\frac{\Delta b}{\Delta s}\cos\phi\cos\theta\right)_n + \frac{\pi b_n}{2}\frac{\left(\cos\phi\cos\theta\right)_n - \left(\cos\phi\cos\theta\right)_{n-1}}{\Delta s_n}]\Delta t \end{aligned} \tag{2.144}$$

2.7.3. Model implementation

At the nth step consider a plume element located at (x_n, y_n, z_n) with the velocity (u_n, v_n, w_n) and its magnitude V_n. The jet axis makes and angle of ϕ_n with the horizontal plane, and θ_n is the angle between the x-axis and the projection of the jet axis on the horizontal plane. The half-width or radius of the plume element is b_n; h_n is the thickness, defined as proportional to the magnitude of the local jet velocity, $h_n = V_n\Delta t$. The mass of the plume element is then given by

$$M_n = \rho_n \pi b_n^2 h_n \tag{2.145}$$

Given the increase in mass due to turbulemt entrainment, ΔM_n, the plume element characteristics at the next step are obtained by applying conservation of mass, horizontal and vertical momentum, energy, and tracer mass to the discrete element. For completeness, the self-explanatory euqations of the generalized Lagraangian model, essentially similar to its original two-dimensional counterpart (Winiarski and Frick, 1976), as summarized as follows:

Mass conservation

$$M_{n+1} = M_n + \Delta M_n \tag{2.146}$$

$$M_{n+1} = \rho_{n+1}\pi b_{n+1}^2 h_{n+1} \tag{2.147}$$

The concentration of the tracer, salinity, temperature and water density

$$C_{n+1} = \frac{M_n C_n + \Delta M_n C_a}{M_{n+1}} \tag{2.148}$$

$$S_{n+1} = \frac{M_n S_n + \Delta M_n S_a}{M_{n+1}} \tag{2.149}$$

$$T_{n+1} = \frac{M_n T_n + \Delta M_n T_a}{M_{n+1}} \tag{2.150}$$

$$\rho_{n+1} = \rho\left(S_{n+1}, T_{n+1}\right) \tag{2.151}$$

The horizonral momentum

$$u_{n+1} = \frac{M_n u_n + \Delta M_n u_a}{M_{n+1}} \tag{2.152}$$

$$v_{n+1} = \frac{M_n v_n}{M_{n+1}} \tag{2.153}$$

The vertical momentum

$$w_{n+1} = \frac{M_n w_n + \Delta M_{n+1}\left(\frac{\Delta\rho}{\rho}\right)_{n+1} g\Delta t}{M_{n+1}} \tag{2.154}$$

$$\left(Mw\right)_{n+1} = \left(Mw\right)_n + \left(\Delta\rho V\right)_{n+1} g\Delta t \tag{2.155}$$

where

$$V_{n+1} = \sqrt{u_{n+1}^2 + v_{n+1}^2 + w_{n+1}^2} \tag{2.156}$$

$$U_{n+1} = \sqrt{u_{n+1}^2 + v_{n+1}^2} \tag{2.157}$$

The new thickness and radius of the plume element

$$h_{n+1} = \frac{V_{n+1}}{V_n} h_n \tag{2.158}$$

$$b_{n+1} = \sqrt{\frac{M_{n+1}}{\pi \rho_{n+1} h_{n+1}}} \tag{2.159}$$

The jet orientation

$$\phi_{n+1} = \arctan\left(\frac{w_{n+1}}{U_{n+1}}\right) \tag{2.160}$$

$$\theta_{n+1} = \arctan\left(\frac{v_{n+1}}{u_{n+1}}\right) \tag{2.161}$$

The new location of the plume element

$$x_{n+1} = x_n + u_{n+1}\Delta t \tag{2.162}$$

$$y_{n+1} = y_n + v_{n+1}\Delta t \tag{2.163}$$

$$z_{n+1} = z_n + w_{n+1}\Delta t \tag{2.164}$$

The distance along the trajectory

$$\Delta s_{n+1} = V_{n+1}\Delta t \tag{2.165}$$

The time step Δt can be fixed or variable; its is chosen via a "prediction-correction" procedure to attain a prescribed fractional change in mass (typically of the order of 1%) at each step.

3 GENERAL TRANSPORT MODEL

3.1 Introduction

This section summarizes theoretical and computational aspects of the transport formulations used in the EFDC model. Theoretical and computational aspects for the basic EFDC hydrodynamic and generic transport model components are presented in Hamrick (1992).

3.2 Basic Equation of Advection-Diffusion Transport

The generic transport equation for a dissolved or suspended material is shown in Eq. (3.1):

$$\begin{aligned}&\frac{\partial}{\partial t}\left(m_x m_y HC\right)+\frac{\partial}{\partial x}\left(m_y HuC\right)+\frac{\partial}{\partial y}\left(m_x HvC\right)+\frac{\partial}{\partial z}\left(m_x m_y wC\right)-\frac{\partial}{\partial z}\left(m_x m_y w_{sc}C\right)\\&=\frac{\partial}{\partial x}\left(\frac{m_y}{m_x}HA_H\frac{\partial C}{\partial x}\right)+\frac{\partial}{\partial y}\left(\frac{m_x}{m_y}HA_H\frac{\partial C}{\partial y}\right)+\frac{\partial}{\partial z}\left(\frac{m_x m_y}{H}A_b\frac{\partial C}{\partial z}\right)+S_C\end{aligned} \tag{3.1}$$

where:

x, y are the orthogonal curvilinear coordinates in the horizontal direction (m),

z is the sigma coordinate (dimensionless),

t is time (s),

m_x, m_y are the square roots of the diagonal components of the metric tensor (m),

m is the Jacobian, $m = m_x m_y$ (m²),

C is the concentration or intensity of transport constituent (g/m³ for concentration of dissolved/suspended material, °C for temperature, ppt for salinity),

H is the total water depth (m),

u, v are the horizontal velocity components in the curvilinear coordinates (m/s),

w is the vertical velocity component (m/s),

A_H is the horizontal turbulent eddy diffusivity (m²/s),

A_b is the vertical turbulent eddy diffusivity (m²/s), and

w_{sc} is a positive settling velocity when C represents a suspended material.

S_c is the source/sink term for the constituent that includes subgrid scale horizontal diffusion and thermal sources and sinks.

A simple example is salinity transport equation:

$$\frac{\partial}{\partial t}\left(mHC\right)+\frac{\partial}{\partial x}\left(PC\right)+\frac{\partial}{\partial y}\left(QC\right)+\frac{\partial}{\partial z}\left(mwC\right)=\frac{\partial}{\partial z}\left(\frac{m}{H}A_b\frac{\partial C}{\partial z}\right)+S_C \tag{3.2}$$

where:

C is the salinity concentration (g/m³); P and Q are mass fluxes per unit length as defined in (2.9).

3.3 Numerical Solution for Transport Equations

In this section, solutions techniques for the transport equations for salinity, temperature, turbulence intensity and turbulence length scale are presented. Stability and accuracy aspects of the advection schemes common to the transport equations and the external and internal horizontal momentum equations are also discussed. The salinity transport equation (3.2) is used as a generic example and the location of variables is shown in Figure 3-1.

The salinity transport equation (3.2) is integrated over a cell layer to give:

$$\frac{\partial}{\partial t}\left(mHC_k\right)+\frac{\partial}{\partial x}\left(P_kC_k\right)+\frac{\partial}{\partial y}\left(Q_kC_k\right)+\frac{\left(WC\right)_k-\left(WC\right)_{k-1}}{\Delta_k}$$
$$-\frac{m}{\Delta_k}\left[\left(\frac{A_b}{H}\frac{\partial C}{\partial z}\right)_k-\left(\frac{A_b}{H}\frac{\partial C}{\partial z}\right)_{k-1}\right]-\left(S_C\right)_k=0 \tag{3.3}$$

where P_k, Q_k, and W_k are defined by Eqs. (2.78) and (2.119). The source, sink, advection, and vertical diffusion portions of equation (3.3) are treated in separate fractional steps, as was done for the internal mode momentum equations in Section 2.5.

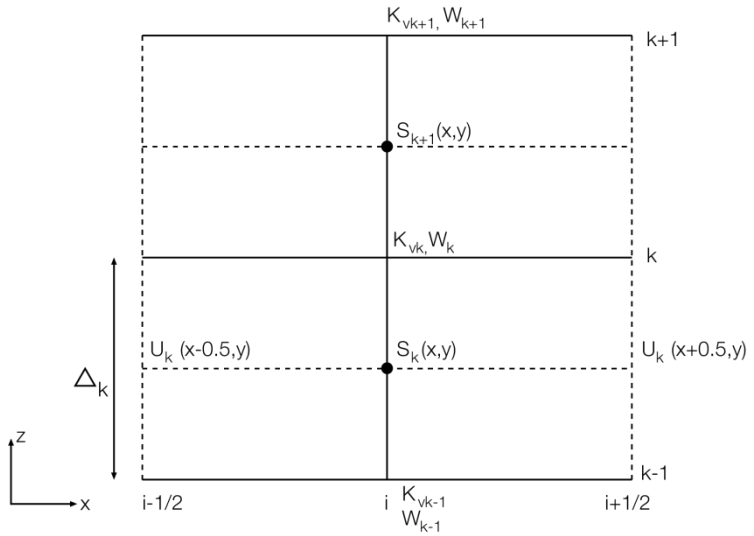


Figure 3-1: S-centered grid in the vertical (x,z) plane

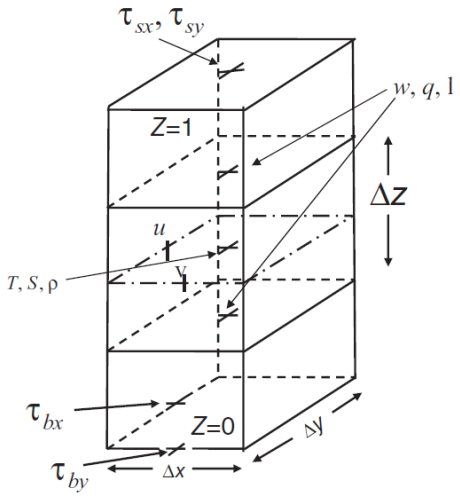


Figure 3-2: Sigma coordinate and variable center (from Ji, 2008)

The three time level fractional step sequence is given by:

$$C_k^* = C_k^{n-1} + \frac{2\Delta t}{mH^{n-1}}\left(S_C\right)_k^{n-1} \tag{3.4}$$

$$\left(mH\right)^{n+1} C_k^{**} = \left(mH\right)^{n-1} C_k^* - 2\Delta t\left[\delta_x^\zeta\left(P_k C_k\right) + \delta_y^\zeta\left(Q_k C_k\right) + \frac{\left(WC\right)_k - \left(WC\right)_{k-1}}{\Delta_k}\right] \tag{3.5}$$

$$\left(HC_k\right)^{n+1} - 2\Delta t\left\{\left[\left(\frac{A_b}{H}\right)_k^n \frac{\left(C_{k+1} - C_k\right)^{n+1}}{\Delta_k \Delta_{k+1,k}}\right] - \left[\left(\frac{A_b}{H}\right)_{k-1}^n \frac{\left(C_k - C_{k-1}\right)^{n+1}}{\Delta_k \Delta_{k,k-1}}\right]\right\} = H^{n+1} C_k^{**} \tag{3.6}$$

The source, sink step (see Eq. (3.4)) is explicit and involves no changes in cell volumes. When the source, sink term represents horizontal turbulent diffusion, it is evaluated at time level *n*-1, for stability (Fletcher, 1988). The advection step, Eq. (3.5), is explicit and involves changes in cell volumes. The vertical diffusion step, Eq. (3.6), which involves no changes in cell volumes, is fully implicit and unconditionally stable (Fletcher, 1988).

Rearranging equation (3.6), the vertical diffusion step, gives:

$$\begin{aligned} &-\frac{2\Delta t}{\Delta_k \Delta_{k,k-1}}\left(\frac{A_b}{H}\right)_{k-1}^n C_{k-1}^{n+1} + \left[\frac{2\Delta t}{\Delta_k \Delta_{k,k-1}}\left(\frac{A_b}{H}\right)_{k-1}^n + H^{n+1} + \frac{2\Delta t}{\Delta_k \Delta_{k+1,k}}\left(\frac{A_b}{H}\right)_k^n\right] C_k^{n+1} \\ &-\frac{2\Delta t}{\Delta_k \Delta_{k+1,k}}\left(\frac{A_b}{H}\right)_k^n C_{k+1}^{n+1} = H^{n+1} C_k^{**} \end{aligned} \tag{3.7}$$

For salinity, temperature, and suspended sediment concentration, the generic variable C is defined vertically at cell layer centers, and the diffusivity is defined at cell layer interfaces. Equation (3.7) then represents a system of *K* equations and the boundary conditions are generally of the specified flux type. Specified surface and bottom flux boundary conditions are most conveniently incorporated in the surface and bottom cell layer source and sink terms allowing A_b at the bottom boundary, k = 0, and the surface boundary, k = K+1, to be set to zero making Eq. (3.7) tri-diagonal.

For turbulence intensity and turbulence length scale, Eqs (2.23) and (2.24), the generic variable *C* is defined vertically at cell layer interfaces and the diffusivity is defined at cell layer centers. Equation (3.7) then represents a system of *K*-1 equations for the variables at internal interfaces with the variable values at the free surface and bottom being provided as boundary conditions. For the turbulence intensity and length scale, the boundary conditions are:

$$q_0^2 = B_1^{2/3}\sqrt{\tau_{bx}^2+\tau_{by}^2}, \quad l_0 = 0, \quad \text{at } z = 0 \tag{3.8}$$

$$q_K^2 = B_1^{2/3}\sqrt{\tau_{sx}^2+\tau_{sy}^2}, \quad l_K = 0, \quad \text{at } z = 1 \tag{3.9}$$

where $\boldsymbol{\tau}_b$ and $\boldsymbol{\tau}_s$ are the bottom and surface stress vectors respectively. Insertion of these boundary conditions results in Eq. (3.7) representing tri-diagonal systems of *K*-1 equations for the turbulence intensity and length scale.

Without loss of generality, the notation used in analyzing the three time level advection step, Eq. (3.5), is simplified by replacing the double and single asterisk intermediate time level indicators by *n*+1 and *n*-1, respectively to give:

$$\begin{aligned}\left(mHC_k\right)^{n+1} &= \left(mHC_k\right)^{n-1} - 2\frac{\Delta t}{\Delta x}\left[\left(PC\right)_{i+\frac{1}{2},j,k} - \left(PC\right)_{i-\frac{1}{2},j,k}\right] \\ &-2\frac{\Delta t}{\Delta y}\left[\left(QC\right)_{i,j+\frac{1}{2},k} - \left(QC\right)_{i,j-\frac{1}{2},k}\right] - 2\frac{\Delta t}{\Delta_k}\left[\left(WC\right)_k - \left(WC\right)_{k-1}\right]\end{aligned} \tag{3.10}$$

where the horizontal central difference operators have been expanded about the cell volume centroid (*x*,*y*), according to Eqs. (2.104) and (2.105). The cell face fluxes can be represented consistent with centered in time and space differencing as was illustrated by Eqs. (2.106), (2.107) and (2.120) or forward in time and backward or upwind in space as was illustrated by Eqs. (2.108) and (2.121) for the *x* momentum fluxes. For the centered in time and space form, Eq. (3.10) becomes:

$$\begin{aligned}\left(mHC_k\right)^{n+1} &= \left(mHC_k\right)^{n-1} \\ &-\frac{\Delta t}{\Delta x}\left[\tilde{P}_{i+\frac{1}{2},j,k}\left(C_{i+1,j,k}+C_{i,j,k}\right) - \tilde{P}_{i-\frac{1}{2},j,k}\left(C_{i,j,k}+C_{i-1,j,k}\right)\right] \\ &-\frac{\Delta t}{\Delta y}\left[\tilde{Q}_{i,j+\frac{1}{2},k}\left(C_{i,j+1,k}+C_{i,j,k}\right) - \tilde{Q}_{i,j-\frac{1}{2},k}\left(C_{i,j,k}+C_{i,j-1,k}\right)\right] \\ &-\frac{\Delta t}{\Delta_k}\left[\tilde{W}_{i,j,k}\left(C_{i,j,k+1}+C_{i,j,k}\right) - \tilde{W}_{i,j,k-1}\left(C_{i,j,k}+C_{i,j,k-1}\right)\right]\end{aligned} \tag{3.11}$$

The transports in equation (3.11) are evaluated at the centered time level when used in the external and internal momentum equations, and are averaged to the centered time level using

$$\tilde{P}_k = \frac{1}{2}\left(P_k^{n+1} + P_k^{n-1}\right) \tag{3.12}$$

when used in the transport equations for scalar variables.

3.4 Dye Constituent

The dye constituent in EFDC+ represents a dilute substance in the water column that does not impact the hydrodynamics (i.e. density impacts) or any other water column process (e.g. light extinction). This constituent can be used in one of two ways:

- Tracer, with or without decay, or
- Age of Water where the dye constituent is used to compute the age of water in days.

The dye is transported in the water column following Eq. 3.1.

3.4.1. Decay

The dye constituent can optionally be configured with a first order decay rate

$$\frac{dC}{dt} = -KC \tag{3.13}$$

where C is the dye concentration in g/m^3, K is the first oder decay rate in 1/s and t is time is seconds.

3.4.2. Age of Water

The dye constituent can be used to calculate the age of water in days. With this option a zero order kinetic rate approach is used

$$\frac{dC}{dt} = -K \tag{3.14}$$

where C is "age" in days, t is in days and K is equal to -1/day. By averaging cell ages over all or parts of the model domain, residence times can be computed. If the model is run sufficiently long enough to achieve a dynamic steady state, the hydraulic residence time can be computed. See the Age of Water guidance document on usage of this option.

4 TEMPERATURE AND HEAT TRANSFER

4.1 Introduction

This chapter presents the basic of water temperature and heat transfer sub models. This includes the sub-models for water column temperature, surface and bed heat exchanges, thermal power plant cooling water and ice formation and melt.

4.2 Basic Equation of Heat Transfer

The basic equation for water temperature and heat transfer in curvilinear and sigma coordinates is given as follows (Ji, 2008):

$$\frac{\partial}{\partial t}(mHT)+\frac{\partial}{\partial x}(PT)+\frac{\partial}{\partial y}(QT)+\frac{\partial}{\partial z}(mwT)=\frac{\partial}{\partial z}\left(\frac{m}{H}A_b\frac{\partial T}{\partial z}\right)=\frac{\partial I}{\partial z}+S_T \tag{4.1}$$

where

x, y are the orthogonal curvilinear coordinates in the horizontal direction (m),

z is the sigma coordinate (dimensionless),

t is time (s),

m_x, m_y are the square roots of the diagonal components of the metric tensor (m),

m is the Jacobian, $m = m_x m_y$ (m²),

T is temperature (°C),

H is the total water depth (m),

P and Q are the mass flux components in the x and y directions, respectively (m²/s) which defined in (2.9),

u, v are the horizontal velocity components in the curvilinear coordinates (m/s),

w is the vertical velocity component (m/s),

A_v is the vertical turbulent eddy viscosity (m²/s), and

I is the short wave solar radiation (W/m²), and

S_T is the source/sink term for heat exchanges (J/s),

To solve the above equation (4.1), the hydrodynamic transport, (u, v, w), and the turbulent mixing, A_b, are provided by the hydrodynamic module in EFDC model. The depth distribution of the solar radiation heating is exponential and can be expressed in simple form as Beer's Law:

$$I(D)=I_s\exp(-K_e D) \tag{4.2}$$

where

$I(D)$ is solar radiation at depth D below the surface (W/m²),

I_S is solar radiation at the surface (D = 0) (W/m²),

$D = H(1\text{-}z)$ is the depth below water surface (m),

K_e is the light extinction coefficient (1/m),

Solar radiation that penetrates the surface of water is absorbed by water. The absorption heats the water column and can take place over a considerable depth, depending on the light extinction coefficient (K_e). The light extinction coefficient (also referred to as the light attenuation coefficient) is the measure for the reduction (absorption) of light intensity within a water column. The solar radiation at the surface I_S a function of location, time of the year, time of day, meteorological conditions, and other factors.

4.3 Surface Heat Exchange

4.3.1. Equilibrium Temperature

A computed equilibrium temperature can be used for the surface heat exchange in EFDC+. The approach used here is based on the equilibrium temperature computation approach in the CE-QUAL-W2 (Wells & Cole, 2000).

Because some of the terms in the term-by-term heat balance equation are surface temperature dependent and others are measurable or computable input variables, the most direct route to simplify computation is to define an equilibrium temperature, T_e, as the temperature at which the net rate of surface heat exchange is zero.

Linearization of the term-by-term heat balance along with the definition of equilibrium temperature allows for expression of the net rate of surface heat exchange, H_n, as:

$$H_{aw} = -K_{aw} \left(T_w - T_e \right) \tag{4.3}$$

where

H_{aw} is the rate of surface heat exchange (W/m²),

K_{aw} is the coefficient of surface heat exchange (W/m²/°C),

T_w is the water surface temperature (°C), and

T_e is the equilibrium temperature (°C).

Seven separate heat exchange processes are summarized in the coefficient of surface heat exchange and equilibrium temperature.

The definition of the coefficient of surface heat exchange can be shown to be the first term of a Taylor series expansion by considering the above equation as:

$$H_{aw} = \frac{dK_{aw}}{dT_s} \left(T_s - T_e \right) \tag{4.4}$$

where the derivative of H_{aw} with respect to surface temperature is evaluated from Eq. (4.3) to give K_{aw}, the coefficient of surface heat exchange. All approximations of the individual surface heat exchange terms enter into the evaluation of the coefficient of surface heat exchange and the equilibrium temperature.

Equilibrium temperature submodel is fully linked with ice submodel that incorporates ice growth and ice melt processes.

4.3.2. Full Heat Balance

At the water surface (z = 1), the boundary condition for the temperature transport Eq. (4.1) is:

$$-\frac{\rho c_p A_b}{H}\frac{\partial T}{\partial z} = H_L + H_E + H_C \tag{4.5}$$

where

ρ is the water density (kg/m³),

c_p is the water specific heat,

A_b is the vertical turbulent mass mixing coefficient (m²/s),

H is the water depth (m),

H_L is the surface heat exchange due to long wave back radiation (W/m²),

H_E is the surface heat exchange due to evaporation/condensation (W/m²),

H_C is the surface heat exchange flux due to convection (W/m²),

Based on approach proposed by Rosati and Miyakoda (1988), Hamrick (1992) applied the following to the temperature boundary condition at water surface:

$$H_L = \varepsilon\sigma T_s^4\left(0.39 - 0.05\sqrt{e_a}\right)\left(1 + B_c C\right) + 4\varepsilon\sigma T_s^3\left(T_s - T_a\right) \tag{4.6}$$

$$H_E = c_e \rho_a L_E W_s \left(e_s - e_a\right)\frac{0.622}{P_a} \tag{4.7}$$

$$H_C = c_h \rho_a c_{pa} W_s \left(T_s - T_a\right) \tag{4.8}$$

where

ε is the emissivity of the waterbody (ε = 0.97),

σ is the Stefan–Boltzmann constant ($\sigma = 5.67\times10^{-8}$ W/m²/K⁴),

e_a is the actual vapor pressure (mb),

C is the cloud fraction (C =0: cloudless, C =1: full cloud coverage),

B_c is the an empirical constant (B_c = 0.8),

T_s is the water surface temperature (°C),

T_a is the air temperature (°C),

$c_e = c_h$ is the turbulent exchange coefficients ($c_e = 1.1\times10^{-3}$),

ρ_a is the atmospheric density (ρ_a = 1.2 kg/m³),

c_{pa} is the specific heat of air (c_{pa} = 1005 J/kg/K),

L_E is the latent heat of evaporation ($L_E = 2.501\times10^6$ J/kg),

W_s is the wind speed (m/s),

e_s is the saturation vapor pressure at surface water temperature (mb),

P_a is the atmospheric pressure (mb).

4.3.3. Solar Radiation

Short wave solar radiation is adsorbed into the water as it passes down through the water column. The water temperature rises due to this increase in energy. This process is often referred to as light extinction. The light extinction coefficient (also called light attenuation coefficient) is the measure for the reduction (absorption) of light intensity within a water column. The light field in the water column is governed by:

$$\frac{\partial I}{\partial z^*} = -K_{ess} I \tag{4.9}$$

where

I is the short wave solar radiation or light intensity (W/m²),

K_{ess} is the light extinction coefficient (1/m), and

z^* is the depth below the water surface (m).

Integration of Eq. (4.9) gives:

$$I = I_{ws} \exp\left(-\int_0^{z^*} K_{ess} dz^* \right) \tag{4.10}$$

The solar radiation at the surface I_0 is a function of location, time of the year, time of day, meteorological conditions, and other factors. The light intensity at the water surface, I_{ws}, is given by:

$$I_{sw} = I_0 S_f \min\left\{\exp\left[-K_{e,me}\left(H_{rps} - H\right)\right], 1\right\} \min\left\{\exp\left[-K_{e,ice} H_{ice}\right], 1\right\} \tag{4.11}$$

where

I_0 is the solar radiation at the Earth's surface (W/m²),

S_f is the tree canopy and/or terrain shading factor (dimensionless),

H_{ice} is the ice thickness (m),

$K_{e,ice}$ is the light extinction coefficient for ice cover (1/m),

$K_{e,me}$ is the light extinction coefficient for emergent shoots (1/m),

H_{rps} is the rooted plant shoot height (m),

H is the water column depth (m).

In EFDC+, depending on which heat exchange option has been selected, light extinction is treated differently.

Term by Term Heat Exchange: EFDC+ computes the net extinction coefficient for each cell and layer at every time step.

Term by Term Heat Exchange (Legacy): The extinction coefficient is constant spatially and temporally. This is consistent with previous implementations of EFDC.

Equilibrium Temperature: EFDC+ computes the net extinction coefficient for each cell and layer at every time step.

4.3.3.1 *Term by Term Heat Exchange*

EFDC+ computes the net extinction coefficient for each cell and layer at every time step using Eq. (4.10).

4.3.3.2 *Term by Term Heat Exchange (Legacy)*

The extinction coefficient is constant spatially and temporally. This is consistent with previous implementations of EFDC. To solve the above Eq. (4.1), the hydrodynamic transport, (u, v, w), and the turbulent mixing, A_b, are provided by EFDC model. The depth distribution of the solar radiation heating is an exponential function and is expressed as :

$$I = rI_{sw}\exp\left[-\beta_f H\left(1-z\right)\right]+\left(1-r\right)I_{sw}\exp\left[-\beta_s H\left(1-z\right)\right] \tag{4.12}$$

where

I is the solar radiation at water depth z (W/m²),

I_{sw} is the incident solar radiation at water surface (z = 1) (W/m²),

β_f is the fast scale attenuation coefficients (1/m),

β_s is the slow scale attenuation coefficients (1/m),

r is the distribution fraction between 0 and 1.

4.3.3.3 *Equilibrium Temperature*

EFDC+ computes the net extinction coefficient for each cell and layer at every time step. If the Equilibrium Temperature heat exchange option has been selected the light extinction is handled the same as with the Full Heat Balance option with the exception of the addition of a constant fraction of the solar radiation is always adsorbed in the top layer, regardless of how thick it is or what the extinction coefficient is. This is described by Beer's law with the addition term β:

$$H_s(z) = \left(1-\beta\right)H_s\exp\left(-\eta z\right) \tag{4.13}$$

where:

$H_s(z)$ is the short wave radiation at depth z (W/m²),

β is the fraction absorbed at the water surface (dimensionless),

K_e is the extinction coefficient (1/m),

H_s is the short wave radiation reaching the water surface (W/m²).

The extinction coefficient can vary spatially and temporally as described in the Full Heat Exchange option.

4.3.4. Light Extinction Factors

The standard EFDC+ term by term full heat balance surface heat exchange processes are same as the full heat balance (legacy) option. The major difference between these two options is that standard EFDC+ full heat balance uses variable light extinction factors. Total light extinction in the model is given by:

$$K_{e,SS} = K_{e,b} + K_{e,TSS}TSS + K_{e,POC}POC + K_{e,DOC}DOC + K_{e,Chl}\sum Chl + K_{e,RPS}RPS \qquad (4.14)$$

If the legacy version of the surface heat exchange option is not selected, EFDC+ will compute the light extinction coefficient in the water column by:

$$K_{e,SS} = K_{e,b} + K_{e,TSS}SED + K_{e,VSS}VSS + K_{e,Chl}\sum_{m=1}^{M}\frac{B_m}{CChl_m} + K_{e,RPS}RPS \qquad (4.15)$$

where

$K_{e,SS}$ is the total light extinction coefficient (1/m),

$K_{e,TSS}$ is the light extinction coefficient for total inorganic suspended solid (1/m per g/m^3),

$K_{e,b}$ is the background light extinction (1/m),

$K_{e,TSS}$ is the light extinction coefficient for total inorganic suspended solid (1/m per g/m^3),

TSS is the inorganic suspended solid concentration (g/m^3) provided from the sediment transport model,

POC is the total Particulate Organic Carbon concentration (Labile and Refractory) (g/m^3) provided from the water quality model,

$K_{e,POC}$ is the light extinction factor as a function of POC concentrations (1/m per g/m^3),

DOC is the Dissolved Organic Carbon concentration (Labile and Refractory) (g/m^3) provided from the water quality model,

$K_{e,DOC}$ is the light extinction factor as a function of DOC concentrations (1/m per g/m^3),

$CChl_{RPE}$ is the carbon-to-chlorophyll ratio for epiphytes (g C per mg Chl),

$K_{e,Chl}$ is the light extinction coefficient for algae chlorophyll (1/m per mg Chl per m^3),

B_m is the concentration of algae group m (g C per ml),

$CChl_m$ is the carbon-to-chlorophyll ratio in algal group m (g C per mg Chl),

$K_{e,RPS}$ is the light extinction coefficient for rooted plant shoots (1/m per gm C per m^2),

RPS is the concentration of plant shoots (g C per m^2).

If only hydrodynamics and temperature is simulated in EFDC model, then background light extinction coefficient will be used. If hydrodynamics, temperature and TSS is simulated, then the total light extinction coefficient is a function of background extinction coefficient and light extinction coefficient due to TSS. If a full water quality model is simulated with TSS, then the total light extinction coefficient is the function of background extinction, TSS, POC, DOC and Chl-a.

Full heat balance with variable light extinction sub-model is fully coupled with ice sub-model and accounts for ice melt and ice growth. Finally, the surface heat exchange coefficients for evaporative and conductive exchange can be spatially variable.

4.4 Bed Thermal Model

Sediment-water heat exchange with water is generally small compared to surface heat exchange and is frequently neglected. However, including sediment bed heat exchange can improve the

simulation of temperature in deep lakes and reservoirs. This section describes the EFDC+ bed thermal model.

The heat exchange between the sediment bed and the bottom layer of the water column can be described as

$$H_b = -\left(K_{b,v}U + K_{b,c}\right)\left(T_w - T_b\right) \tag{4.16}$$

$$U = \sqrt{u_1^2 + v_1^2} \tag{4.17}$$

where

H_b is the sediment bed-water heat exchange (W/m²),

$K_{b,v}$ is the convective heat exchange coefficient (W-s/m³-°C),

$K_{b,c}$ is the conductive heat exchange coefficient (W/m²-°C),

u_1 is the u component water velocity in layer 1 (m/s),

v_1 is the v component water velocity in layer 1 (m/s),

T_w is the water temperature in layer 1 (°C), and

T_b is the water temperature in layer 1 (°C).

Typical applications have used a value of 0.3 W/m²-°C for $K_{b,c}$ that is approximately two orders of magnitude smaller than the surface heat exchange coefficient. $K_{b,c}$ is often not used (i.e. equal to zero) but can be in the range of 0 to 10. Average yearly air temperature is a good initial estimate of T_b.

Optionally, the bed temperature (T_b) can change with time due to the heat exchange.

$$\frac{\delta\left(D_b T_b\right)}{\delta t} = -\left(K_{b,v}U + K_{b,c}\right)\left(T_b - T_w\right) \tag{4.18}$$

where D_b is the sediment bed-thermal thickness (m). Selection of the thermal thickness is subject to initial approximation and subsequent calibration. The larger the thermal thickness is the slower the bed temperature will change.

4.5 Thermal Power Plant Cooling Water

The EFDC model has features specifically designed for thermal simulation of a power plant cooling systems. In EFDC, the cooling water withdrawal from a nearby river or lake and discharge from the plant are represented by the withdrawal and return boundaries and can be represented by specifying the flow rate Q at the withdrawal and return cells (Hamrick and Mills 2000):

Heat energy added to or removed from a receiving waterbody can be estimated using:

$$H_T = Q_c \rho c_p \Delta T \tag{4.19}$$

where:

H_{WR} is the rate of heat energy exchange due to temperature rise/fall (J/s),

Q_c is the withdrawal flow rate (m³/s),

ρ is the water density (kg/m^3),

c_p is the specific heat of water (c_p = 2400 J/kg/°C), and

ΔT is the temperature difference between the discharged water and the receiving water (°C).

So, for any industrial process that uses once-through cooling water we can define the following:

Q_{cool} is the process water flow rate that is being pumped from the water body. The sign convention is positive for a standard withdrawal. The thermal sinks and sources H_T are specified in the transport equation for heat:

$$H_{intake} = -Q_{cool}\rho c_p T_1 \tag{4.20}$$

$$H_{discharge} = Q_{cool}\rho c_p \left(T_1 + \Delta T\right) \tag{4.21}$$

where

H_{intake} is the heat removed from the withdrawal cell or intake,

$H_{discharge}$ is the heat added back to the water body at the power plant discharge location,

T_1 is the ambient intake temperature, and

ΔT is the temperature rise of a power plant.

4.6 Thermal Power Plant Forced Evaporation

The Forced Evaporation (FE) Analysis capability has been developed to quantify increased evaporation induced by increased water temperatures due to releases from thermoelectric power plants. These power plants withdraw cooling waters, which once run through the plant and are returned/discharged to rivers or lakes at a higher temperature than the ambient water temperature. This higher temperature water causes additional evaporation (forced evaporation) from the river or lake. This additional evaporation is counted as water consumption by regulators as it is no longer available to downstream users.

Evaporation is dependent on wind speed, atmospheric humidity, and water temperature. There are a number of methods to compute FE using different wind functions as listed in Table 4 1.

Using these various evaporation methods, the model is first run with the power plant, and then run again without the power plant. EE then subtracts the output from two models and displays the difference which is the consumption of water from the power plant. Once temperature is activated and the correct Surface Heat Exchange Sub-Model option has been selected the user can choose which evaporation approach is desired. Even if evaporative losses are not a major concern, the evaporative mass fluxes should normally be activated for most models.

Heat flux due to evaporation is always included for the Full Heat and the Equilibrium Temperature (W2) options.

EFDCPlus/EFDC_Explorer Forced Evaporation (FE) toolset results have been compared to the Electric Power Research Institute's (EPRI) FE estimates. EPRI's once through cooling FE analysis for river discharges is based on a USGS report on water consumption by thermoelectric power plants (USGS, 2013; EPRI, 2013).

Table 4.1: List of Evaporation Calculation Methods

IEVAP	Evaporation Approach	General Usage
0	Do Not Include Evaporation	
1	Use Evaporation from ASER	Measured or Externally Estimated
2	EFDC Original	
3	Ward, 1980	Cooling Lake
4	Harbeck, 1964	Cooling Lake
5	Brady et al, 1969	Cooling Pond
6	Anderson, 1954	Large Lake
7	Webster-Sherman, 1995	Lakes
8	Fulford-Sturm, 1984	Rivers
9	Gulliver-Stefan, 1986	Streams
10	Edinger et al., 1974	Lakes/Rivers
11	Ryan-Harleman 1974	Lakes/Rivers

4.7 Ice Sub Model

4.7.1. Introduction

A robust ice sub-model has been implemented in EFDC+ which is based on the CE-QUAL-W2 (Wells and Cole, 2000) ice module. With this model:

- Ice formation and melt is simulated by EFDC+ using a coupled heat model.
- Ice dynamics (i.e. movement of ice block/chunks) have not yet been implemented.

4.7.2. Heat Balance

The heat balance for the water-to-ice air system is given by:

$$\rho_i L_f \frac{dh}{dt} = h_{ai}\left(T_i - T_e\right) - h_{wi}\left(T_w - T_m\right) \tag{4.22}$$

where

ρ_i is the density of ice (kg/m³),

L_f is the latent heat of fusion of ice (J/kg),

dh/dt is the change in ice thickness (h) with time (t) (m/s),

h_{ai} is the coefficient of ice-to-air heat exchange (W/m²/°C),

h_{wi} is the coefficient of water-to-ice heat exchange through the melt layer (W/m²/°C),

T_i is the ice temperature (°C),

T_{ei} is the equilibrium temperature of ice to air heat exchange (°C),

T_w is the water temperature below ice (°C), and

T_m is the melt temperature (°C).

4.7.3. Ice Formation

Formation of ice requires lowering the surface water temperature to the freezing point by normal surface heat exchange processes. With further heat removal, ice begins to form on the water surface. This is indicated by a negative water surface temperature. The negative water surface temperature is then converted to equivalent ice thickness and equivalent heat is added to the heat source and sink term for water.

$$\theta_0 = -\frac{T_{wn}\rho_w c_{pw} h}{\rho_i L_f} \tag{4.23}$$

where:

θ_0 is the thickness of initial ice formation during a timestep (m),

T_{wn} is the local temporary negative water temperature (°C),

h is the layer thickness (m),

ρ_w is the density of water (kg/m³),

c_{pw} is the specific heat of water (J/kg/°C),

ρ_i is the density of ice (kg/m³), and

L_f is the latent heat of fusion (J/kg).

4.7.4. Ice Surface Temperature

The ice surface temperature is given by the equations:

$$T_s^n \approx \frac{\theta^{n-1}}{K_i}\left[H_{sn}^n + H_{an}^n - H_{br}\left(T_s^n\right) - H_c\left(T_s^n\right)\right] \tag{4.24}$$

$$H_{sn} + H_{an} - H_{br} - H_e - H_c + q_i = \rho_i L_f \frac{d\theta_{ai}}{dt} \text{ for } T_s = 0°C \tag{4.25}$$

$$q_i = K_i \frac{T_f - T_s(t)}{\theta(t)} \tag{4.26}$$

where

K_i is the thermal conductivity of ice (W/m/°C),

T_f is the freezing point temperature (°C),

n is the time level,

q_i is the heat flux through ice (W/m²),

H_n is the net rate of heat exchange across the water surface (W/m²),

H_s is the incident short wave solar radiation (W/m²),

H_a is the incident long wave radiation (W/m²),

H_{sr} is the reflected short wave solar radiation (W/m²),

H_{ar} is the reflected long wave radiation (W/m²),

H_{br} is the back radiation from the water surface (W/m²),

H_e is the evaporative heat loss (W/m²), and

H_c is the heat conduction (W/m²).

4.7.5. Freezing Temperature

The freezing temperature relationship is described as below.

$$T_f = \begin{cases} -0.0545\,TDS, & TDS < 35 \text{ ppt} \\ -0.3146 - 0.0417\,TDS - 0.000166\,TDS^2, & TDS > 35 \text{ ppt} \end{cases} \tag{4.27}$$

where

T_f is the freezing point temperature (°C),

TDS is the total dissolved solids (ppt).

4.7.6. Ice Melt at Air/Water Interface

The ice melt at the air/water interface is described by the equation below:

$$\rho_i c_{pi} \frac{T_s(t)}{2} \theta(t) = \rho_i L_f \Delta\theta_{ai} \tag{4.28}$$

where

c_{pi} is the specific heat of ice (J/kg/°C), and

θ_{ai} is the ice melt at the air-ice interface (1/m).

4.7.7. Ice Growth/Melt at Bottom of Ice

The ice growth/melt at the bottom of the ice is described by the equation below:

$$q_i - q_{iw} = \rho_i L_f \frac{d\theta_{iw}}{dt} \tag{4.29}$$

where

q_i is the heat flux through the ice (W/m²),

q_{iw} is the heat flux at the ice/water interface (W/m²),

θ_{iw} is the ice growth/melt at the ice-water interface.

$$\Delta\theta_{iw}^{n} = \frac{1}{\rho_i L_f}\left[K_i \frac{T_f - T_s^n}{\theta^{n-1}} - h_{wi}\left(T_w^n - T_f\right)\right] \tag{4.30}$$

4.7.8. Solar Radiation at Bottom of Ice

Solar radiation at the bottom of the ice is given by the equation below:

$$H_{ps} = H_s\left(1-\alpha_i\right)\left(1-\beta_i\right)\exp\left[-\gamma_i\theta(t)\right] \tag{4.31}$$

where

H_{ps} is the solar radiation absorbed by water under ice cover (W/m²),

H_s is the incident solar radiation (W/m²),

a_i is the ice albedo,

β_i is the fraction of the incoming solar radiation absorbed in the ice surface,

γ_i is the ice extinction coefficient (1/m).

5 SEDIMENT TRANSPORT

5.1 Introduction

This section describes the solution of the transport equations for suspended sediment. The general procedure follows that for the salinity transport equation, which uses a high order upwind difference solution scheme for the advective terms, described in Hamrick (1992).

This document provides the basic theory for sediment transport using two sediment transport options included in EFDC+. They are:

- "Original" sediment transport module. This module is based on the Hamrick's work (Tetra Tech, 2007).
- SEDZLJ sediment transport module. The module is based on the SEDiment dynamics work by Ziegler, Lick, and Jones (SEDZLJ) (Ziegler and Lick 1986, 1988, Jones and Lick 1999, 2001, Jones 2001).

5.2 Basic Equations for Suspended Sediment Transport

5.2.1. Suspended Sediment Transport

The water column equation for suspended sediment transport is derived from the generic transport equation (3.1) for a dissolved or suspended material. Although the advection scheme is designed to minimize numerical diffusion, a small amount of horizontal diffusion remains inherent in the upwind finite difference scheme. Due the small inherent numerical diffusion, the physical horizontal diffusion terms in (3.1) are omitted as to give:

$$\begin{aligned}&\frac{\partial}{\partial t}\left(mHC_j\right)+\frac{\partial}{\partial x}\left(PC_j\right)+\frac{\partial}{\partial y}\left(QC_j\right)+\frac{\partial}{\partial z}\left(mwC_j\right)-\frac{\partial}{\partial z}\left(mw_{s,j}C_j\right)\\&=\frac{\partial}{\partial z}\left(m\frac{A_b}{H}\frac{\partial}{\partial z}C_j\right)+S^E_{s,j}+S^I_{s,j}\end{aligned} \tag{5.1}$$

where C_j represents the concentration of the jth sediment class and the source-sink term has been split into an external part, which would include point and nonpoint source loads, and internal part which could include reactive decay of organic sediments or the exchange of mass between sediment classes if floc formation and destruction were simulated. Vertical boundary conditions for (5.1) are:

$$-\frac{A_b}{H}\frac{\partial}{\partial z}C_j-w_{s,j}C_j=J_{jo} \quad \text{at } z=0 \tag{5.2}$$

$$-\frac{A_b}{H}\frac{\partial}{\partial z}C_j-w_{s,j}C_j=0 \quad \text{at } z=1 \tag{5.3}$$

where J_{j0} is the net water column-bed exchange flux defined as positive into the water column.

5.2.2. Numerical Solution

The numerical solution of (5.1) utilizes a fractional step procedure. The first step advances the concentration due to advection and external sources and sinks having corresponding volume fluxes by

$$H^{n+1}C^{*} = H^{n}C^{n} + \frac{\Delta t}{m}\left(S^{E}_{s,j}\right)^{n+1/2} - \frac{\Delta t}{m}\left[\frac{\partial}{\partial x}\left(P^{n+\frac{1}{2}}C^{n}\right) + \frac{\partial}{\partial y}\left(Q^{n+\frac{1}{2}}C^{n}\right) + \frac{\partial}{\partial z}\left(mw^{n+\frac{1}{2}}C^{n}\right)\right] \quad (5.4)$$

where the superscripts *n* and *n+1* denote the old and new time levels and the superscript * denotes the intermediate fractional step results. The portion of the source and sink term, associated with volumetric sources and sinks is included in the advective step for consistency with the continuity constraint. This source-sink term, as well as the advective field (*u,v,w*), is defined as intermediate in time between the old and new time levels consistent with the temporal discretization of the continuity equation. Note that the sediment class subscripts have been dropped for clarity. The advection step uses the anti-diffusive MPDATA scheme (Smolarkiewicz and Clark, 1986) with optional flux corrected transport (Smolarkiewicz and Grabowski, 1990).

The second fractional step or settling step is given by

$$C^{**} = C^{*} + \frac{\Delta t}{H^{n+1}}\frac{\partial}{\partial z}\left(w_s C^{**}\right) \quad (5.5)$$

which is solved by a fully implicit upwind difference scheme

$$C^{**}_{k,c} = C^{*}_{k,c} + \frac{\Delta t}{\Delta_z H^{n+1}}\left(w_s C^{**}\right)_{k,c} \quad (5.6)$$

$$C^{**}_{k} = C^{*}_{k} + \frac{\Delta t}{\Delta_1 H^{n+1}}\left(w_s C^{**}\right)_{k+1} - \frac{\Delta t}{\Delta_k H^{n+1}}\left(w_s C^{**}\right)_{k} \quad \text{with } 2 \le k \le KC\text{-}1 \quad (5.7)$$

$$C^{**}_{1} = C^{*}_{1} + \frac{\Delta t}{\Delta_z H^{n+1}}\left(w_s C^{**}\right)_{2} \quad (5.8)$$

Where C_k is the concentration in each layer (g/m³), C_1 is the bottom layer concentration and k is the layer index. The solution starts at $k = KC$ and marches downward to the bottom layer. The implicit solution includes an optional anti- diffusion correction across internal water column layer interfaces.

The third fractional step accounts for water column-bed exchange by resuspension and deposition

$$C^{***}_{1} = C^{**}_{1} + \frac{\Delta t}{\Delta_z H^{n+1}} L_0 J^{***}_{0} \quad (5.9)$$

Where L_0 is a flux limiter such that only the current top layer of the bed can be completely resuspended in single time step. The representation of the water column bed exchange by a distinct fractional step is equivalent to a splitting of the bottom boundary condition (5.2) such that the bed flux is imposed intermediate between settling and vertical diffusion.

For resuspension and deposition of suspended non-cohesive sediment, the bed flux is given by

$$J^{***}_{0} = \frac{w_s}{\nu}\left(\mu C_{eq} - C^{***}_{1}\right) \quad (5.10)$$

which will be further discussed in the next section.

For cohesive sediment resuspension, the bed flux is specified as a function of the bed stress and bed geomechanical properties. For cohesive sediment deposition, the bed flux is typically given by

$$J_0^{***} = -P_d w_s C_1^{***} \tag{5.11}$$

where P_d is a probability of deposition.

The remaining step is an implicit vertical turbulent diffusion step corresponding

$$C^{n+1} = C^{***} + \Delta t \frac{\partial}{\partial z}\left[\left(\frac{A_b}{H^2}\right)^{n+1} \frac{\partial}{\partial z} C^{n+1}\right] \tag{5.12}$$

with zero diffusive fluxes at the bed and water surface.

5.2.3. Definitions

Void ratio ε is defined as the ratio of the volume of voids ϕ to the total volume of the soil (dimensionless).

$$\varepsilon = \frac{\phi}{1-\phi} \tag{5.13}$$

Bulk density ρ_b is defined as the dry weight of soil per unit volume of soil (kg/m³). Considering the relationship between dry and bulk density

$$\rho_d = \rho_s \frac{(\rho_b - \rho_w)}{(\rho_s - \rho_w)} \tag{5.14}$$

Where ρ_s is the solids or grain density (kg/m³) and ρ_w is the water density (kg/m³).

5.3 Original EFDC+ Sediment Transport Approach

5.3.1. Non-Cohesive Sediments

5.3.1.1 Settling Velocity

Non-cohesive inorganic sediments settle as discrete particles, with hindered settling and multiphase interactions becoming important in regions of high sediment concentration near the bed. At low concentrations, the settling velocity for the jth non-cohesive sediment class corresponds to the settling velocity of a discrete particle:

$$w_{sj} = w_{soj} \tag{5.15}$$

Useful expressions for the discrete particle settling velocity which depends on the sediment density, effective grain diameter, and fluid kinematic viscosity, provide by van Rijn (1984b) are:

$$w_{soj} = \sqrt{g'd_j}\begin{cases} \dfrac{R_{dj}}{18}, & d \le 100\ \mu\text{m} \\ \dfrac{10}{R_{dj}}\left(\sqrt{1+0.01R_{dj}^2}-1\right), & 100\mu m < d_j \le 1000\ \mu\text{m} \\ 1.1, & d_j > 1000\ \mu\text{m} \end{cases} \tag{5.16}$$

$$g' = g\left(\frac{\rho_{sj}}{\rho_w}-1\right) \tag{5.17}$$

is the reduced gravitational acceleration and

$$R_{dj} = \frac{d_j\sqrt{g'd_j}}{\nu} \tag{5.18}$$

is the sediment grain densimetric Reynolds number.

At higher concentrations and hindering settling conditions, the settling velocity is less than the discrete velocity and can be expressed in the form

$$w_{sj} = \left(1-\sum_i^I \frac{C_i}{\rho_{si}}\right)^n w_{soj} \tag{5.19}$$

where ρ_s is the sediment particle density with values of n ranging from 2 (Cao et al., 1996) to 4 (Van Rijn, 1984). The expression (5.16) is approximated to within 5 percent by

$$w_{sj} = \left(1-n\sum_i^I \frac{C_i}{\rho_{si}}\right) w_{soj} \tag{5.20}$$

for total sediment concentrations up to 200,000 mg/l. For total sediment concentrations less than 25,000 mg/l, neglect of the hindered settling correction results in less than a 5 per cent error

in the settling velocity that is well within the range of uncertainty in parameters used to estimate the discrete particle settling velocity.

5.3.1.2 Deposition, Resuspension and Bedload

Non-cohesive sediment is transported as bedload and suspended load. The initiation of both modes of transport begins with erosion or resuspension of sediment from the bed when the bed stress, τ_b, exceeds a critical stress referred to as the Shield's stress, τ_{cs}. The Shield's stress depends upon the density and diameter of the sediment particles and the kinematic viscosity of the fluid and can be expressed in empirical dimensionless relationships of the form:

$$\theta_{csj} = \frac{\tau_{csj}}{g'd_j} = \frac{u_{*csj}^2}{g'd_j} = f\left(R_{dj}\right) \tag{5.21}$$

Useful numerical expressions of the relationship (5.19), provided by van Rijn (1984b), are:

$$\theta_{csj} = \begin{cases} 0.24\left(R_{dj}^{2/3}\right)^{-1}, & R_{dj}^{2/3} < 4 \\ 0.14\left(R_{dj}^{2/3}\right)^{-0.64}, & 4 \le R_{dj}^{2/3} < 10 \\ 0.04\left(R_{dj}^{2/3}\right)^{-0.1}, & 10 \le R_{dj}^{2/3} < 20 \\ 0.013\left(R_{dj}^{2/3}\right)^{0.29}, & 20 \le R_{dj}^{2/3} < 150 \\ 0.055, & R_{dj}^{2/3} \ge 150 \end{cases} \tag{5.22}$$

A number of approaches have been used to distinguish whether a particular sediment size class is transported as bedload or suspended load under specific local flow conditions characterized by the bed stress or bed shear velocity:

$$u_* = \sqrt{\tau_b} \tag{5.23}$$

The approach proposed by van Rijn (1984a) is adopted in the EFDC+ model and is as follows. When the bed velocity is less than the critical shear velocity

$$u_{*csj} = \sqrt{\tau_{csj}} = \sqrt{g'd_j\theta_{csj}} \tag{5.24}$$

no erosion or resuspension takes place and there is no bedload transport. Sediment in suspension under this condition will deposit to the bed as will be subsequently discussed.

When the bed shear velocity exceeds the critical shear velocity but remains less than the settling velocity,

$$u_{*csj} < u_* < w_{soj} \tag{5.25}$$

sediment will be eroded from the bed and transported as bedload. Sediment in suspension under this condition will deposit to the bed. When the bed shear velocity exceeds both the critical shear velocity and the settling velocity, bedload transport ceases and the eroded or resuspended sediment will be transported as suspended load. These various transport modes are further illustrated by reference to Figure 1, which shows dimensional forms of the settling velocity relationship (5.16) and the critical Shield's shear velocity (5.24) determined using (5.22) for sediment with a specific gravity of 2.65. For grain diameters less than approximately 1.3×10^{-4} m (130 μm) the settling velocity is less than the critical shear velocity and sediment resuspend

from the bed when the bed shear velocity exceeds the critical shear velocity will be transported entirely as suspended load. For grain diameters greater than 1.3×10^{-4} m, eroded sediment be transported by bedload in the region corresponding to (5.25) and then as suspended load when the bed shear velocity exceeds the settling velocity.

In the EFDC+ model, the preceding set of rules are used to determine the mode of transport of multiple size classes of non-cohesive sediment. Bedload transport is determined using a general bedload transport rate formula:

$$\frac{q_B}{\rho_s d\sqrt{g'd}} = \phi\left(\theta, \theta_{cs}\right) \tag{5.26}$$

where q_B is the bedload transport rate (mass per unit time per unit width) in the direction of the near bottom horizontal flow velocity vector. The function ϕ depends on the Shield's parameter

$$\theta = \frac{\tau_b}{g'd_j} = \frac{u_*^2}{g'd_j} \tag{5.27}$$

and the critical Shield's parameter defined by (5.21) and (5.22). A number of bedload transport formulas explicitly incorporate the settling velocity. However, since both the critical Shield's parameter and the settling velocity are unique functions of the sediment grain densimetric Reynolds number, the settling velocity can also be expressed as a function of the critical Shield's parameter with (5.26) remaining an appropriate representation.

A number of bedload formulations developed for riverine prediction (Ackers and White, 1973; Laursen, 1958; Yang, 1973; Yang and Molinas, 1982) do not readily conform to (5.26) and were not incorporated as options in the EFDC+ model. Two widely used bedload formulations which do conform to (5.26) are the Meyer-Peter and Muller (1948) and Bagnold (1956) formulas and their derivatives (Raudkivi, 1967; Neilson, 1992; Reid and Frostick, 1994) which have the general form

$$\Phi\left(\theta, \theta_{cs}\right) = \phi\left(\theta - \theta_{cs}\right)^{\alpha}\left(\sqrt{\theta} - \gamma\sqrt{\theta_{cs}}\right)^{\beta} \tag{5.28}$$

where

$$\phi = \phi\left(\theta_{cs}\right) \text{ or } \phi = \phi\left(R_d\right) \tag{5.29}$$

The Meyer-Peter and Muller formulations are typified by

$$\Phi = \phi\left(\theta - \theta_{cs}\right)^{3/2} \tag{5.30}$$

while Bagnold formulations are typified by

$$\Phi = \phi\left(\theta - \theta_{cs}\right)\left(\sqrt{\theta} - \gamma\sqrt{\theta_{cs}}\right) \tag{5.31}$$

with Bagnold's original formula having γ equal to zero. The Meyer-Peter and Muller formulation has been extended to heterogeneous beds by Suzuki *et al.* (1998), while Bagnold's formula has been similarly extended by van Niekerk *et al.* (1992). The bedload formulation by van Rijn (1984a) having the form

$$\Phi = \phi\left(\theta - \theta_{cs}\right)^{2.1}$$

$$\phi = \frac{0.053}{R_d^{1/5}\theta_{cs}^{2.1}} \tag{5.32}$$

has been incorporated into the CH3D-SED model and modified for heterogeneous beds by Spasojevic and Holly (1994). Equation (5.32) can be implemented in the EFDC+ model with an appropriately specified ϕ. A modified formulation of the Einstein bedload function (Einstein, 1950) which conforms to (5.26) and (5.28) has been presented by Rahmeyer (1999) and will be later incorporated into the EFDC+ model.

The procedure for coupling bedload transport with the sediment bed in the EFDC+ model is as follows. First, the magnitude of the bedload mass flux per unit width is calculated according to (5.26) at horizontal model cell centers, denoted by the subscript *c*. The cell center flux is then transformed into cell center vector components using

$$q_{bcx} = \frac{u}{\sqrt{u^2 + v^2}} q_{bc}$$

$$q_{bcy} = \frac{v}{\sqrt{u^2 + v^2}} q_{bc} \tag{5.33}$$

where u and v are the cell center horizontal velocities near the bed. Cell face mass fluxes are determined by downwind projection of the cell center fluxes

$$q_{bfx} = \left(q_{bcx}\right)_{upwind}$$

$$q_{bfy} = \left(q_{bcy}\right)_{upwind} \tag{5.34}$$

where the subscript *upwind* denotes the cell center upwind of the x normal and y normal cell faces. The net removal or accumulation rate of sediment material from the deposited bed underlying a water cell is then given by:

$$m_x m_y J_b = \left(m_y q_{bfx}\right)_e - \left(m_y q_{bfx}\right)_w + \left(m_x q_{bfy}\right)_n - \left(m_x q_{bfy}\right)_s \tag{5.35}$$

where J_b is the net removal rate (gm/m²-sec) from the bed, m_x and m_y are x and y dimensions of the cell, and the compass direction subscripts define the four cell faces. The implementation of (5.33) through (5.35) in the EFDC+ code includes logic to limit the out fluxes (5.34) over a time step, such that the time integrated mass flux from the bed does not exceed bed sediment available for erosion or resuspension.

Under conditions when the bed shear velocity exceeds the settling velocity and critical Shield's shear velocity, non-cohesive sediment will be resuspended and transported as suspended load. When the bed shear velocity falls below both the settling velocity and the critical Shield's shear velocity, suspended sediment will deposit to the bed.

A consistent formulation of these processes can be developed using the concept of a near bed equilibrium sediment concentration. Under steady, uniform flow and sediment loading conditions, an equilibrium distribution of sediment in the water column tends to be established, with the resuspension and deposition fluxes canceling each other. Using a number of simplifying assumptions, the equilibrium sediment concentration distribution in the water column can be expressed analytically in terms of the near bed reference or equilibrium concentration, the

settling velocity and the vertical turbulent diffusivity. For unsteady or spatially varying flow conditions, the water column sediment concentration distribution varies in space and time in response to sediment load variations, changes in hydrodynamic transport, and associated nonzero fluxes across the water column-sediment bed interface. An increase or decrease in the bed stress and the intensity of vertical turbulent mixing will result in net erosion or deposition, respectively, at a particular location or time.

To illustrate how an appropriate suspended non-cohesive sediment bed flux boundary condition can be established, consider the approximation to the sediment transport equation (5.1) for nearly uniform horizontal conditions

$$\frac{\partial}{\partial t}(HC) = \frac{\partial}{\partial z}\left(\frac{A_b}{H}\frac{\partial C}{\partial z} + w_z C\right) \tag{5.36}$$

Integrating (5.36) over the depth of the bottom hydrodynamic model layer gives

$$\frac{\partial}{\partial t}\left(\Delta H\overline{C}\right) = J_0 - J_\Delta \tag{5.37}$$

where the over bar denotes the mean over the dimensionless layer thickness, Δ. Subtracting (5.37) from (5.36) gives

$$\frac{\partial}{\partial t}(HC') = \frac{\partial}{\partial z}\left(\frac{A_b}{H}\frac{\partial C}{\partial z} + w_z C\right) - \left(\frac{J_0 - J_\Delta}{\Delta}\right) \tag{5.38}$$

Assuming that the rate of change of the deviation of the sediment concentration from the mean is small

$$\frac{\partial}{\partial t}(HC') << \frac{\partial}{\partial t}\left(H\overline{C}\right) \tag{5.39}$$

allows (5.38) to be approximated by

$$\frac{\partial}{\partial z}\left(\frac{A_b}{H}\frac{\partial C}{\partial z} + w_z C\right) = \left(\frac{J_0 - J_\Delta}{\Delta}\right) \tag{5.40}$$

Integrating (5.40) once gives

$$\frac{A_b}{H}\frac{\partial C}{\partial z} + w_z C = \left(J_0 - J_\Delta\right)\frac{z}{\Delta} - J_0 \tag{5.41}$$

Very near the bed, (5.41) can be approximated by

$$\frac{A_b}{H}\frac{\partial C}{\partial z} + w_z C = -J_0 \tag{5.42}$$

Neglecting stratification effects and using the results of Section 5.3.2, the near bed diffusivity is approximately

$$\frac{A_b}{H} = K_o q \frac{l}{H} \cong u_* \kappa z \tag{5.43}$$

Introducing (5.43) into (5.42) gives

$$\frac{\partial C}{\partial z} + \frac{R}{z} C = -\frac{R}{z} \frac{J_o}{w_s} \tag{5.44}$$

where

$$R = \frac{w_s}{u_* \kappa} \tag{5.45}$$

is the Rouse parameter. The solution of (5.44) is

$$C = -\frac{J_o}{w_s} + \frac{C_0}{z^R} \tag{5.46}$$

The constant of integration is evaluated using

$$C = C_{eq} \quad \text{at } z = z_{eq} \text{ and}$$
$$J_o = 0 \tag{5.47}$$

which sets the near bed sediment concentration to an equilibrium value, defined just above the bed under no net flux condition. Using (5.47), equation (5.46) becomes

$$C = \left(\frac{z_{eq}}{z} \right)^R C_{eq} - \frac{J_o}{w_s} \tag{5.48}$$

For non-equilibrium conditions, the net flux is given by evaluating (5.48) at the equilibrium level

$$J_o = w_s \left(C_{eq} - C_{ne} \right) \tag{5.49}$$

where C_{ne} is the actual concentration at the reference equilibrium level. Equation (5.49) clearly indicates that when the near bed sediment concentration is less than the equilibrium value a net flux from the bed into the water column occurs. Likewise when the concentration exceeds equilibrium, a net flux to the bed occurs. For this case when C_{ne} is greater than C_e

$$J_o = -w_s C_{ne} \left(1 - \frac{C_{eq}}{C_{ne}} \right) \tag{5.50}$$

and the term inside the brackets can be considered as the deposition factor which does not exceed unity.

For the relationship (5.49) to be useful in a numerical model, the bed flux must be expressed in terms of the model layer mean concentration. For a three-dimensional

$$J_o = w_s\left(\overline{C}_{eq} - \overline{C}\right) \tag{5.51}$$

where

$$\overline{C}_{eq} = \frac{\ln\left(\Delta z_{eq}^{-1}\right)}{\left(\Delta z_{eq}^{-1} - 1\right)} C_{eq}, \quad R = 1$$

$$\overline{C}_{eq} = \frac{\left(\Delta z_{eq}^{-1}\right)^{1-R} - 1}{(1-R)\left(\Delta z_{eq}^{-1} - 1\right)} C_{eq}, \quad R \neq 1 \tag{5.52}$$

defines an equivalent layer mean equilibrium concentration in terms of the near bed equilibrium concentration. The corresponding quantities in the numerical solution bottom boundary condition (5.9) are

$$w_r C_r = w_s \overline{C}_{eq}$$

$$P_d w_s = w_s \tag{5.53}$$

If the dimensionless equilibrium elevation, z_{eq} exceeds the dimensionless layer thickness, (5.33) can be modified to

$$\overline{C}_{eq} = \frac{\ln\left(M\,\Delta z_{eq}^{-1}\right)}{\left(M\,\Delta z_{eq}^{-1} - 1\right)} C_{eq}, \quad R = 1$$

$$\overline{C}_{eq} = \frac{\left(M\,\Delta z_{eq}^{-1}\right)^{1-R} - 1}{(1-R)\left(M\,\Delta z_{eq}^{-1} - 1\right)} C_{eq}, \quad R \neq 1 \tag{5.54}$$

where the over bars in (5.51) and (5.53) implying an average of the first *M* layers above the bed.

For two-dimensional, depth averaged model application, a number of additional considerations are necessary. For depth average modeling, the equivalent of (5.41) is

$$\frac{A_b}{H}\frac{\partial C}{\partial z} + w_s C = -J_o\left(1 - z\right) \tag{5.55}$$

Neglecting stratification effects and using the results of the sediment boundary layers, the diffusivity is

$$\frac{A_b}{H} = K_o q \frac{1}{H} \cong u_* \kappa z \left(1 - z\right)^{\lambda} \tag{5.56}$$

Introducing (5.56) into (5.55) gives

$$\frac{\partial C}{\partial z}+\frac{R}{z\left(1-z\right)^{\lambda}}C=-\frac{R\left(1-z\right)^{1-\lambda}}{z}\frac{J_o}{w_s} \tag{5.57}$$

A close form solution of (5.57) is possible for λ equal to zero. Although the resulting diffusivity is not as reasonable as the choice of λ equal to one, the resulting vertical distribution of sediment is much more sensitive to the near bed diffusivity distribution than the distribution in the upper portions of the water column. For λ equal to zero, the solution of (5.57) is

$$C=-\left(1-\frac{Rz}{\left(1+R\right)}\right)\frac{J_o}{w_s}+\frac{C_0}{z^R} \tag{5.58}$$

Evaluating the constant of integration using (5.56) gives

$$C=\left(\frac{z_{eq}}{z}\right)^R C_{eq}-\left(1-\frac{Rz}{\left(1+R\right)}\right)\frac{J_o}{w_s} \tag{5.59}$$

For non-equilibrium conditions, the net flux is given by evaluating (5.59) at the equilibrium level

$$J_o=w_s\left(\frac{1+R}{1+R\left(1-z_{eq}\right)}\right)\left(C_{eq}-C_{ne}\right) \tag{5.60}$$

where C_{ne} is the actual concentration at the reference equilibrium level. Since z_{eq} is on the order of the sediment grain diameter divided by the depth of the water column, (5.60) is essentially equivalent(5.49). To obtain an expression for the bed flux in terms of the depth average sediment concentration, (5.59) is integrated over the depth to give

$$J_o=w_s\left(\frac{2\left(1+R\right)}{2+R\left(1-z_{eq}\right)}\right)\left(\overline{C}_{eq}-\overline{C}\right) \tag{5.61}$$

where

$$\begin{aligned}\overline{C}_{eq}&=\frac{\ln\left(z_{eq}^{-1}\right)}{\left(z_{eq}^{-1}-1\right)}C_{eq},\quad R=1\\ \overline{C}_{eq}&=\frac{\left(z_{eq}^{R-1}-1\right)}{\left(1-R\right)\left(z_{eq}^{-1}-1\right)}C_{eq},\quad R\neq 1\end{aligned} \tag{5.62}$$

The corresponding quantities in the numerical solution bottom boundary condition (5.9) are

$$w_r s_r = w_s \left(\frac{2(1+R)}{2+R(1-z_{eq})} \right) \bar{C}_{eq}$$
$$P_d w_s = \left(\frac{2(1+R)}{2+R(1-z_{eq})} \right) w_s \tag{5.63}$$

When multiple sediment size classes are simulated, the equilibrium concentrations given by (5.52), (5.54), and (5.62) are adjusted by multiplying by their respective sediment volume fractions in the surface layer of the bed.

The specification of the water column-bed flux of non-cohesive sediment has been reduced to specification of the near bed equilibrium concentration and its corresponding reference distance above the bed. Garcia and Parker (1991) evaluated seven relationships, derived by combinations of analysis and experiment correlation, for determining the near bed equilibrium concentration as well as proposing a new relationship. All of the relationships essential specify the equilibrium concentration in terms of hydrodynamic and sediment physical parameters

$$C_{eq} = C_{eq}(d, \rho_s, \rho_w, w_s, u_*, \nu) \tag{5.64}$$

including the sediment particle diameter, the sediment and water densities, the sediment settling velocity, the bed shear velocity, and the kinematic molecular viscosity of water. Garcia and Parker concluded that the representations of Smith and McLean (1977) and Van Rijn (1984b) as well as their own proposed representation perform acceptably when tested against experimental and field observations.

Smith and McLean's formula for the equilibrium concentration is

$$C_{eq} = \rho_s \frac{0.65\gamma_o T}{1+\gamma_o T} \tag{5.65}$$

where γ_o is a constant equal to 2.4×10^{-3} and T is given by

$$T = \frac{\tau_b - \tau_{cs}}{\tau_{cs}} = \frac{u_*^2 - u_{*cs}^2}{u_{*cs}^2} \tag{5.66}$$

where τ_b is the bed stress and τ_{cs} is the critical Shields stress. The use of Smith and McLean's formulation requires that the critical Shields stress be specified for each sediment size class. Van Rijn's formula is

$$C_{eq} = 0.015\rho_s \frac{d}{z_{eq}^*} T^{3/2} R_d^{-1/5} \tag{5.67}$$

where $z_{eq}^* = Hz_{eq}$ is the dimensional reference height and R_d is a sediment grain Reynolds number. When van Rijn's formula is select for use in EFDC+, the critical Shields stress in

internally calculated using relationships from van Rijn (1984b). van Rijn suggested setting the dimensional reference height to three grain diameters. In the EFDC+ model, the user specifies the reference height as a multiple of the largest non-cohesive sediment size class diameter.

Garcia and Parker's general formula for multiple sediment size classes is

$$C_{jeq} = \rho_s \frac{A\left(\lambda Z_j\right)^5}{\left(1+3.33A\left(\lambda Z\right)^5\right)} \tag{5.68}$$

$$Z_j = \frac{u_*}{w_{sj}} R_{dj}^{3/5} F_H \tag{5.69}$$

$$F_H = \left(\frac{d_j}{d_{50}}\right)^{1/5} \tag{5.70}$$

$$\lambda = 1+\frac{\sigma_\phi}{\sigma_{\phi o}}\left(\lambda_o - 1\right) \tag{5.71}$$

where A is a constant equal to 1.3×10^{-7}, d_{50} is the median grain diameter based on all sediment classes, λ is a straining factor, F_H is a hiding factor and σ_ϕ is the standard deviation of the sedimentological phi scale of sediment size distribution. Garcia and Parker's formulation is unique in that it can account for armoring effects when multiple sediment classes are simulated. For simulation of a single non-cohesive size class, the straining factor and the hiding factor are set to one. The EFDC+ model has the option to simulate armoring with Garcia and Parker's formulation. For armoring simulation, the current surface layer of the sediment bed is restricted to a thickness equal to the dimensional reference height.

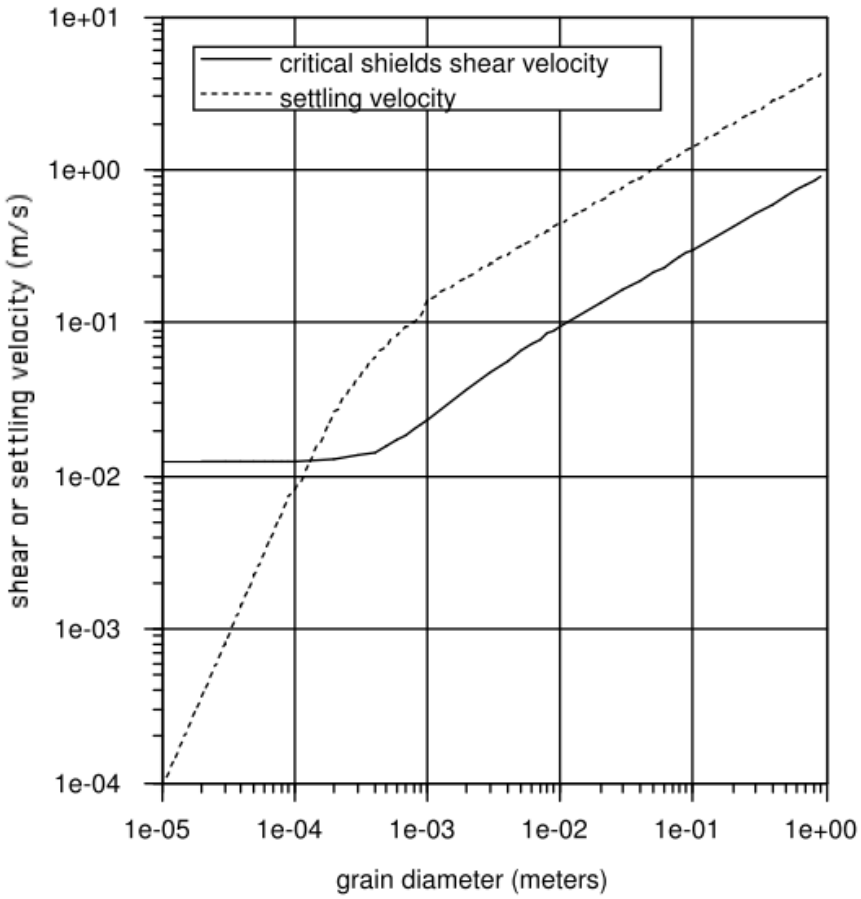


Figure 5-1: Critical Shield's shear velocity and settling velocity as a function of sediment grain size

5.3.2. Cohesive Sediments

5.3.2.1 Settling Velocities

The settling of cohesive inorganic sediment and organic particulate material is an extremely complex process. Inherent in the process of gravitational settling is the process of flocculation, where individual cohesive sediment particles and particulate organic particles aggregate to form larger groupings or flocs having settling characteristics significantly different from those of the component particles (Burban et al., 1989; 1990; Gibbs, 1985; Mehta *et al*, 1989). Floc formation is dependent upon the type and concentration of the suspended material, the ionic characteristics of the environment, and the fluid shear and turbulence intensity of the flow environment. Progress has been made in first principles mathematical modeling of floc formation or aggregation, and disaggregation by intense flow shear (Lick and Lick, 1988; Tsai *et al.*, 1987). However, the computational intensity of such approaches precludes direct simulation of flocculation in operational cohesive sediment transport models for the immediate future.

An alternative approach, which has met with reasonable success, is the parameterization of the settling velocity of flocs in terms of cohesive and organic material fundamental particle size, d; concentration, S; and flow characteristics such as vertical shear of the horizontal velocity, du/dz, shear stress, $A_v du/dz$, or turbulence intensity in the water column or near the sediment bed, q. This has allowed semi-empirical expressions having the functional form

$$w_{se} = w_{se}\left(d, C, \frac{du}{dz}, q\right) \tag{5.72}$$

to be developed to represent the effective settling velocity. The following settling approaches are available in EFDC+.

(1) Option 1

Hwang and Mehta (1989) proposed

$$w_s = \frac{aC''}{\left(C^2 + b^2\right)^m} \tag{5.73}$$

based on observations of settling at six sites in Lake Okeechobee. This equation has a general parabolic shape with the settling velocity decreasing with decreasing concentration at low concentrations and decreasing with increasing concentration at high concentration. Least squares analysis for the parameters, a, m, and n, in (5.73) was shown to agree well with observational data. Equation (5.73) does not have a dependence on flow characteristics, but is based on data from an energetic field condition having both currents and high frequency surface waves.

(2) Option 2

The Shrestha and Orlob (1996) as modified by Mehta (1989) formulation in EFDC+ has the form

$$\begin{aligned} w_s &= C^{\alpha} \exp\left(-4.21 + 0.147G\right) \\ \alpha &= 1.11075 + 0.0386G \end{aligned} \tag{5.74}$$

where

$$G = \sqrt{\left(\frac{\partial u}{\partial z}\right)^2 + \left(\frac{\partial v}{\partial z}\right)^2} \tag{5.75}$$

is the magnitude of the vertical shear of the horizontal velocity. It is noted that all of these formulations are based on specific dimensional units for input parameters and predicted settling velocities and that appropriate unit conversion are made internally in their implementation in the EFDC+ model.

(3) Option 3

Ziegler and Nisbet (1994, 1995) proposed a formulation to express the effective settling as a function of the floc diameter, d_f

$$w_s = a d_f^b \tag{5.76}$$

with the floc diameter given by:

$$d_f = \sqrt{\frac{\alpha_f}{C\sqrt{\tau_{xz}^2 + \tau_{yz}^2}}} \tag{5.77}$$

where C is the sediment concentration, a_f is an experimentally determined constant and τ_{xz} and τ_{yz} are the x and y components of the turbulent shear stresses at a given position in the water column. Other quantities in (5.77) have been experimentally determined to fit the relationships:

$$a = B_1\left(C\sqrt{\tau_{xz}^2 + \tau_{yz}^2}\right)^{-0.85} \tag{5.78}$$

$$b = -0.8 - 0.5\log\left(C\sqrt{\tau_{xz}^2 + \tau_{yz}^2} - B_2\right) \tag{5.79}$$

where B_1 and B_2 are experimental constants.

(4) Option 4

Generalized approach to compute settling velocities based on shear stress is as follows:

$$w_s = \begin{cases} 1.510\times10^{-5}(C')^{0.45}, & C' < 40 \\ 8\times10^{-5}, & 40 \le C' \le 400 \\ 0.893\times10^{-6}(C')^{0.75}, & C' > 400 \end{cases} \tag{5.80}$$

$$C' = \tau C \tag{5.81}$$

where τ is shear stress (cm²/s²) and C is total cohesive concentration (g/m³).

(5) Option 5

The following approach for settling velocities are based on the Housatonic River approach combining Burban & Lick with site specific regressions.

$$w_s = \begin{cases} \dfrac{1.270}{86400(C')^{0.79}}, & C' < 3.8 \\ \dfrac{3.024}{86400(C')^{0.14}}, & C' \geq 3.8 \end{cases} \tag{5.82}$$

$$C' = \tau C \tag{5.83}$$

where τ is shear stress (cm^2/s^2) and C is total cohesive concentration (g/m^3).

(6) Option 6

Generalized approach to compute settling velocities based on shear stress is as follows:

$$w_s = \begin{cases} 2.32\times10^{-5}(C')^{0.5}, & C' < 100 \\ 3.68\times10^{-5}(C')^{0.4}, & C' \geq 100 \end{cases} \tag{5.84}$$

$$C' = \tau C \tag{5.85}$$

where τ is shear stress (cm^2/s^2) and C is total cohesive concentration (g/m^3).

(7) Option 7

Generalized approach to compute settling velocities based on shear stress is as follows:

$$w_s = 0.0052(C')^{0.470138} \tag{5.86}$$

$$C' = \tau C \tag{5.87}$$

where τ is shear stress (m^2/s^2) and C is total cohesive concentration (g/m^3).

(8) Option 8

A modified Shrestha approach to compute settling velocities based on shear stress is as follows:

$$w_s = a\left(\frac{C}{2650}\right)^{0.3333} \tag{5.88}$$

$$a = \begin{cases} \dfrac{0.06}{\sqrt{\tau}}, & \tau > 0.1 \\ 0.1, & \tau \leq 0.1 \end{cases} \tag{5.89}$$

where τ is shear stress (m^2/s^2) and C is total cohesive concentration (g/l).

5.3.2.2 Deposition

Water column-sediment bed exchange of cohesive sediments and organic solids is controlled by the near bed flow environment and the geomechanics of the deposited bed. Net deposition to the bed occurs as the flow-induced bed surface stress decreases. The most widely used expression for the depositional flux is:

$$J_o^d = \begin{cases} -w_s C_d \left(\dfrac{\tau_{cd} - \tau_b}{\tau_{cd}} \right) = -w_s P_d C_d , & \tau_b \le \tau_{cd} \\ 0, & \tau_b \ge \tau_{cd} \end{cases} \tag{5.90}$$

where τ_b is the stress exerted by the flow on the bed, τ_{cd} is a critical stress for deposition which depends on sediment material and floc physiochemical properties (Mehta *et al.*, 1989) and C_d is the near bed depositing sediment concentration. The probability of deposition P_d is based on the linear term $(\tau_{cd}-\tau_b)/\tau_{cd}$. The critical deposition stress is generally determined from laboratory or *in situ* field observations and values ranging from 0.06 to 1.1 N/m² have been reported in the literature. Given this wide range of reported values, in the absence of site specific data the depositional stress and is generally treated as a calibration parameter. The depositional stress is an input parameter in the EFDC+ model.

5.3.2.3 Erosion

Cohesive bed erosion occurs in two distinct modes, mass erosion and surface erosion. Mass erosion occurs rapidly when the bed stress exerted by the flow exceeds the depth varying shear strength, τ_s, of the bed at a depth, H_{me}, below the bed surface. Surface erosion occurs gradually when the flow-exerted bed stress is less than the bed shear strength near the surface but greater than a critical erosion or resuspension stress, τ_{ce}, which is dependent on the shear strength and density of the bed. A typical scenario under conditions of accelerating flow and increasing bed stress would involve first the occurrence of gradual surface erosion, followed by a rapid interval of mass erosion, followed by another interval of surface erosion. Alternately, if the bed is well consolidated with a sufficiently high shear strength profile, only gradual surface erosion would occur. Transport into the water column by mass or bulk erosion can be expressed in the form

$$J_o^r = w_r C_r = \frac{m_{me}\left(\tau_s \le \tau_b\right)}{T_{me}} \tag{5.91}$$

where J_o is the erosion flux, the product $w_r C_r$ represents the numerical boundary condition(5.9), m_{me} is the dry sediment mass per unit area of the bed having a shear strength, τ_s, less than the flow-induced bed stress, τ_b, and T_{me} is a somewhat arbitrary time scale for the bulk mass transfer. The time scale can be taken as the numerical model integration time step (Shrestha and Orlob, 1996). Observations by Hwang and Mehta (1989) have indicated that the maximum rate of mass erosion is on the order of 0.6 gm/s-m² which provides a means of estimating the transfer time scale in Eq. (5.91). The shear strength of the cohesive sediment bed is generally agreed to be a linear function of the bed bulk density (Mehta *et al.*, 1982; Villaret and Paulic, 1986; Hwang and Mehta, 1989).

$$\tau_s = a_s \rho_b + b_s \tag{5.92}$$

For the shear strength in N/m² and the bulk density in gm/cm³, Hwang and Mehta (1989) give a_S and b_S values of 9.808 and -9.934 for bulk density greater than 1.065 gm/cm³. The EFDC+ model currently implements Hwang and Mehta's relationship, but can be readily modified to incorporate other functional relationships.

Surface erosion is generally represented by relationships of the form

$$J_o^r = w_r C_r = \frac{dm_e}{dt}\left(\frac{\tau_b - \tau_{ce}}{\tau_{ce}}\right)^{\alpha}, \quad \tau_b \geq \tau_{ce} \tag{5.93}$$

or

$$J_o^r = w_r C_r = \frac{dm_e}{dt}\exp\left(-\beta\left(\frac{\tau_b - \tau_{ce}}{\tau_{ce}}\right)^{\gamma}\right), \quad \tau_b \geq \tau_{ce} \tag{5.94}$$

where dm_e/dt is the surface erosion rate per unit surface area of the bed and τ_{ce} is the critical stress for surface erosion or resuspension. The critical erosion rate and stress and the parameters α, β, and γ are generally determined from laboratory or *in situ* field experimental observations. Equation (5.93) is more appropriate for consolidated beds, while (5.94) is appropriate for soft partially consolidated beds. The base erosion rate and the critical stress for erosion depend upon the type of sediment, the bed water content, total salt content, ionic species in the water, pH and temperature (Mehta *et al.*, 1989) and can be measured in laboratory and sea bed flumes.

Surface erosion rates ranging from 0.005 to 0.1 gm/s-m² have been reported in the literature, and it is generally accepted that the surface erosion rate decreases with increasing bulk density. The critical erosion stress is related to but generally less than the shear strength of the bed, which in turn depends upon the sediment type and the state of consolidation of the bed. Experimentally determined relationships between the critical surface erosion stress and the dry density of the bed of the form

$$\tau_{ce} = c\rho_s^d \tag{5.95}$$

have been presented (Mehta *et al.*, 1989).

The EFDC+ model allows for a user defined constant critical stress for surface erosion or the use of a computed τ_{ce} based on one of the following options.

(1) Option 1

Hwang and Mehta (1989) proposed the relationship

$$\tau_{ce} = \begin{cases} a\left(\rho_b - \rho_l\right)^b + c, & \rho_b > 1.065 \\ 0, & \rho_b \leq 1.065 \end{cases} \tag{5.96}$$

between the critical surface erosion stress and the bed bulk density with a = 0.883, b = 0.2, c = 0.05, and ρ_1 = 1.065 for the stress in N/m² and the bulk density in gm/cm³.

(2) Option 2 and 3

Sanford and Maa (2001) proposed the relationship

$$\tau_{ce} = \tau_{ci} \frac{(1+VR_r)}{(1+VR_b)} \tag{5.97}$$

where τ_{ci} is the reference critical surface erosion rate (m^2/s^2) and VR_r and VR_b are the void ratios of the reference and the bed, respectively (dimensionless).

(3) Option 4 and 5

$$\tau_{ce} = \tau_{ci} \tag{5.98}$$

where τ_{ci} is the reference critical surface erosion rate (m^2/s^2).

(4) Option 99

The Housatonic River relationship is

$$\tau_{ce} = \begin{cases} \dfrac{0.2}{1000}, & L \le 265 \\ \dfrac{0.4}{1000}, & L > 265 \end{cases} \tag{5.99}$$

Where L is the EFDC+ linear cell index. This is a hardwired option for the Housatonic River.

5.3.3. Consolidation of Mixed Cohesive and Non-Cohesive Sediment Beds

This section presents a methodology for representing consolidation of sediment beds containing both cohesive and non-cohesive sediments. The methodology allows for both cohesive and non-cohesive sediment in any bed layer and is based on the following assumptions. First, it is assumed that during the consolidation step, a fraction of the bed pore water volume per unit horizontal area is associated with each sediment type or

$$\left(\frac{\varepsilon H_{bed}}{1+\varepsilon}\right) = (\psi_{wc} + \psi_{wn}) H_{bed} \tag{5.100}$$

where H_{bed} is the bed thickness (m) and ψ with the subscripts wc and wn denote volume fraction of water associated with cohesive and non-cohesive sediment, respectively. Likewise the volume of sediment per unit horizontal area can be fractionally partitioned between cohesive and non-cohesive

$$\left(\frac{H_{bed}}{1+\varepsilon}\right) = (\psi_{sc} + \psi_{sn}) H_{bed} \tag{5.101}$$

Following the Lagrangian formulation of the previous section, the total volume of sediment and the fractional sediment volume in a bed layer remain constant during a consolidation step.

$$\frac{\partial}{\partial}(H_{bed}\psi_{sc}) = \frac{\partial}{\partial}(H_{bed}\psi_{sn}) = 0 \tag{5.102}$$

Fractional void ratios can also be defined

$$\varepsilon_c = \frac{\psi_{wc}}{\psi_{sc}} \tag{5.103}$$

$$\varepsilon_n = \frac{\psi_{wn}}{\psi_{sn}} \tag{5.104}$$

and using (5.100) and (5.101), the void ratio of the mixture is

$$\varepsilon = \frac{\psi_{sc}\varepsilon_c + \psi_{sn}\varepsilon_n}{\psi_{sc} + \psi_{sn}} \tag{5.105}$$

which is the sediment volume weighted average of the void ratios of the two sediment types.

The second assumption is that during the consolidation time step, the fraction of water associated with non-cohesive sediment remains constant, as does the fractional void ratio. This is equivalent to the assuming that the portion of the bed layer associated with non-cohesive sediment is incompressible, and that the pore water associated the non-cohesive sediment is specified by ε_n.

Consistent with the preceding assumptions, the thickness of the bed layer can be divided into cohesive and non-cohesive fractions, $H_{bed,c}$ and $H_{bed,n}$, respectively.

$$\begin{aligned} H_{bed,c} &= \left(\psi_{wc} + \psi_{sc}\right) H_{bed} = \left(1+\varepsilon_c\right)\psi_{sc} H_{bed} \\ H_{bed,n} &= \left(\psi_{wn} + \psi_{sn}\right) H_{bed} = \left(1+\varepsilon_n\right)\psi_{sn} H_{bed} \end{aligned} \tag{5.106}$$

The hydraulic conductivity of the layer can be expressed by

$$K = \frac{\left(H_{bed,c} + H_{bed,n}\right)}{\left(\frac{H_{bed,c}}{K_c} + \frac{H_{bed,n}}{K_n}\right)} \tag{5.107}$$

which is equivalent to an infinite number of alternating infinitesimal cohesive and non-cohesive sublayers of proportional thickness comprising the mixed bed layer. Equation (5.107) can be written as

$$\frac{K}{\left(1+\varepsilon\right)} = \frac{1}{\left(f_{sc}\frac{\left(1+\varepsilon_c\right)}{K_c} + f_{sn}\frac{\left(1+\varepsilon_n\right)}{K_n}\right)} \tag{5.108}$$

where

$$f_{sc} = \frac{\psi_{sc}}{\left(\psi_{sc} + \psi_{sn}\right)} \tag{5.109}$$

$$f_{sn} = \frac{\psi_{sn}}{(\psi_{sc} + \psi_{sn})}$$

are the time invariant total cohesive and non-cohesive sediment fractions in the bed layer. Likewise, (5.105) can be written as

$$\varepsilon = f_{sc}\varepsilon_c + f_{sn}\varepsilon_n \tag{5.110}$$

The final assumption for the mixed material consolidation formulation is that changes in effective stress are due entirely to changes in the cohesive void ratio. Under this assumption, the specific discharge can be written as

$$q = -\left(\frac{K}{1+\varepsilon}\right)_{k+\frac{1}{2}} \frac{2\lambda_{k+\frac{1}{2}}}{(\Delta_{k+1}+\Delta_k)}\left[(f_{sc}\varepsilon_c)_{k+1} - (f_{sc}\varepsilon_c)_k\right] + \left(\frac{K}{1+\varepsilon}\right)_{k+\frac{1}{2}}\left(\frac{\overline{\rho}_s}{\rho_w} - 1\right)_{k+\frac{1}{2}} \tag{5.111}$$

and

$$\lambda_{k+\frac{1}{2}} = -\frac{1}{g\rho_w}\left(\frac{\sigma_{e,k+1} - \sigma_{e,k}}{(f_{sc}\varepsilon_c)_{k+1} - (f_{sc}\varepsilon_c)_k}\right) \tag{5.112}$$

When the depositional void ratio is specified for the surface layer specific discharge becomes

$$q_{w:kt+} = -\left(\frac{2\lambda_{kt+}}{\Delta_{kt}}\right)\left(\frac{K}{1+\varepsilon}\right)_{kt+}\left[(\varepsilon_c)_{dep} - (\varepsilon_c)_k\right] + \left(\frac{\overline{\rho}_s}{\rho_w} - 1\right)_{kt+}\left(\frac{K}{1+\varepsilon}\right)_{kt+} \tag{5.113}$$

When the zero excess pore pressure boundary condition at the bed surface is used

$$\begin{aligned} q_{w:kt+} &= \left(\frac{K}{1+\varepsilon}\right)_{Kt}\frac{2}{\Delta_{Kt}}\left(\lambda^* f_{sc}\varepsilon_c^{n+1}\right)_{Kt} \\ &+ \left(\frac{K}{1+\varepsilon}\right)_{Kt}\left(\frac{\overline{\rho}_s}{\rho_w} - 1\right)_{Kt} - \left(\frac{K}{1+\varepsilon}\right)_{Kt}\frac{2}{\Delta_{Kt}}\left(\frac{\sigma_e^n}{g\rho_w} + \lambda^* f_{sc}\varepsilon^*\right)_{Kt} \end{aligned} \tag{5.114}$$

The equation for updating the void ratio is modified using (5.110) to give

$$(f_{sc}\varepsilon_c)_k^{**} = (f_{sc}\varepsilon_c)_k^{*} + \frac{\Delta t}{2}\left(\frac{1+\varepsilon}{H_{bed}}\right)_k^{**}(q_{w:k-} - q_{w:k+}) \tag{5.115}$$

Thus the mixed bed layer consolidation formulation essentially solves of the space and time evolution of $f_{sc}\varepsilon_c$ with the continuum constitutive relationship for λ given by

$$\lambda = -\frac{1}{f_{sc}}\frac{\partial}{\partial \varepsilon}\left(\frac{\sigma}{g\rho_w}\right) \tag{5.116}$$

The formulation has the desirable characteristic of reducing to the well established cohesive formulation in the absence of non-cohesive material. The solution for $f_{SC}\varepsilon_C$ proceeds by introducing (5.101) and (5.103) or (5.104) into (5.105) and solving the resulting tri-diagonal system of equations. The new specific discharges are then directly calculated using (5.101) and (5.103) or (5.104) and used to update the layer thickness

$$H_{bed,k}^{n+1} = H_{bed,k}^{*} + \Delta t\left(q_{w:k-} - q_{w:k+}\right) \tag{5.117}$$

The ratio $H_{bed}/(1+\varepsilon)$ can then be updated

$$\left(\frac{H_{bed}}{1+\varepsilon}\right)_k^{n+1} = \left(\frac{H_{bed}}{1+\varepsilon}\right)_k^{*} \tag{5.118}$$

Followed by the solution of(5.110) for the cohesive void ratio

$$\varepsilon_c = \frac{\varepsilon - f_{sn}\varepsilon_n}{f_{sc}} \tag{5.119}$$

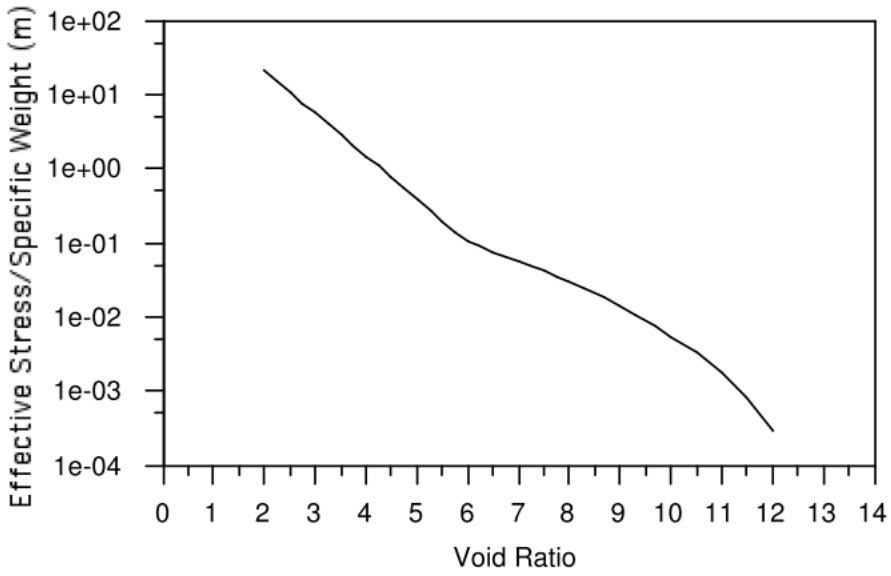


Figure 5-2: Specific Weight Normalized Effective Stress versus Void Ratio.

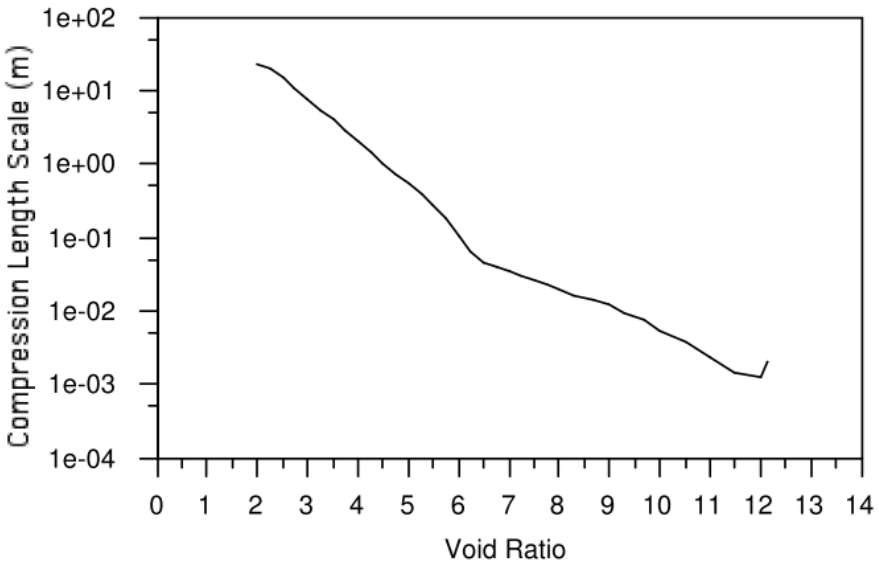


Figure 5-3: Compress Length Scale, $(\boldsymbol{g\rho_w})^{-1}$ $(\boldsymbol{d\sigma_e/d\varepsilon})$, versus Void Ratio.

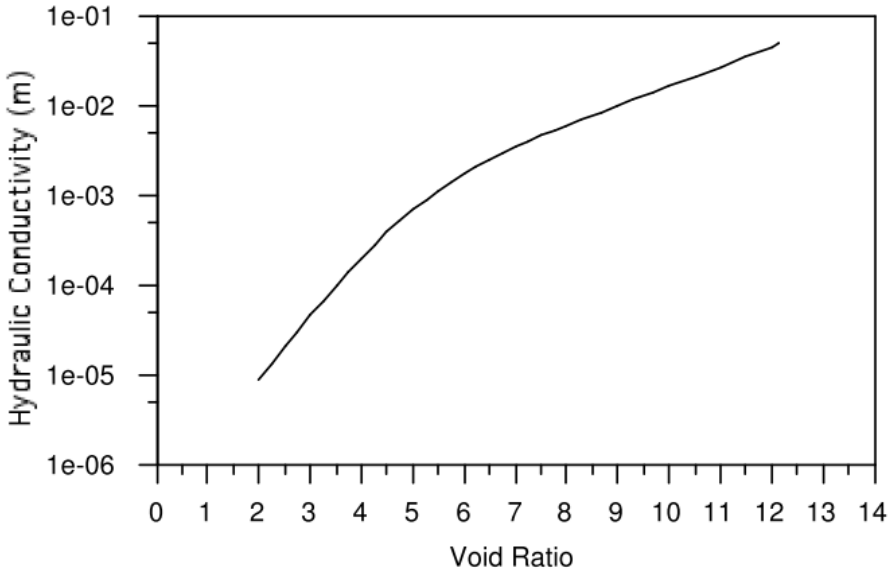


Figure 5-4: Hydraulic Conductivity versus Void Ratio.

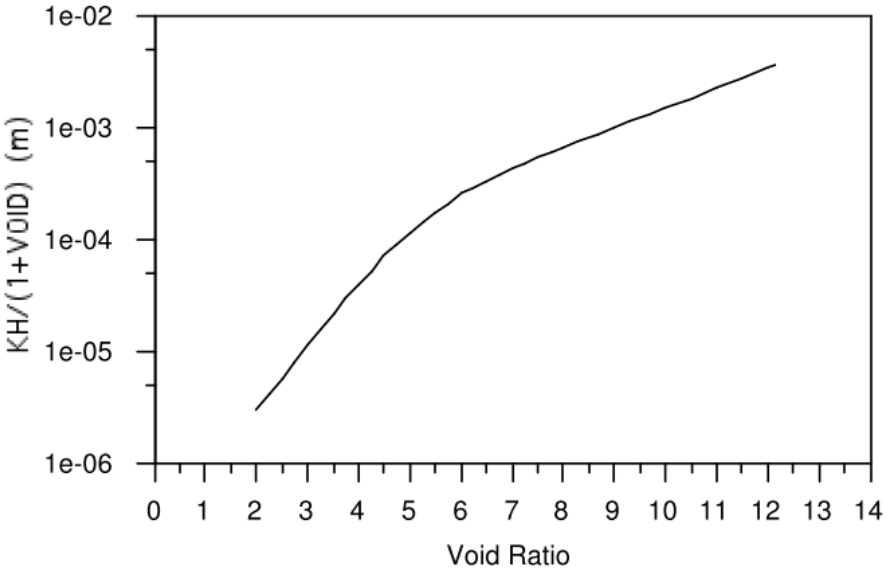


Figure 5-5: Hydraulic Conductivity/(1 + Void Ratio) versus Void Ratio.

5.4 SEDZLJ Sediment Transport

5.4.1. Overview

The EFDC+ model was extended (James, et.al. 2010) to include measured erosion rates from an erosion rate measurement flume referred to as SEDFlume (Jones and Lick, May 2001). The mathematical framework for the unified treatment of erosion, deposition and bedload transport is referred to as the SEDZLJ model. This section of the EFDC+ theory document provides a summary of the theory of SEDZLJ and larger is taken from Jones and Lick (2001), Thanh, et.al. (2008) and James, et.al. (2010).

A SEDFlume consists of a straight flume with an open bottom through which a rectangular cross-section core tube containing sediment can be inserted. The main components of the flume are the core tube and sediment, the test section, the inlet section for uniform, fully-developed, turbulent flow, the flow exit section, the water storage tank, and the pump (which forces water through the system). A schematic of the SEDFlume is shown in Figure 5-6. Data produced from these tests produce erosion rates, critical shear stress, and bulk density by depth in a core.

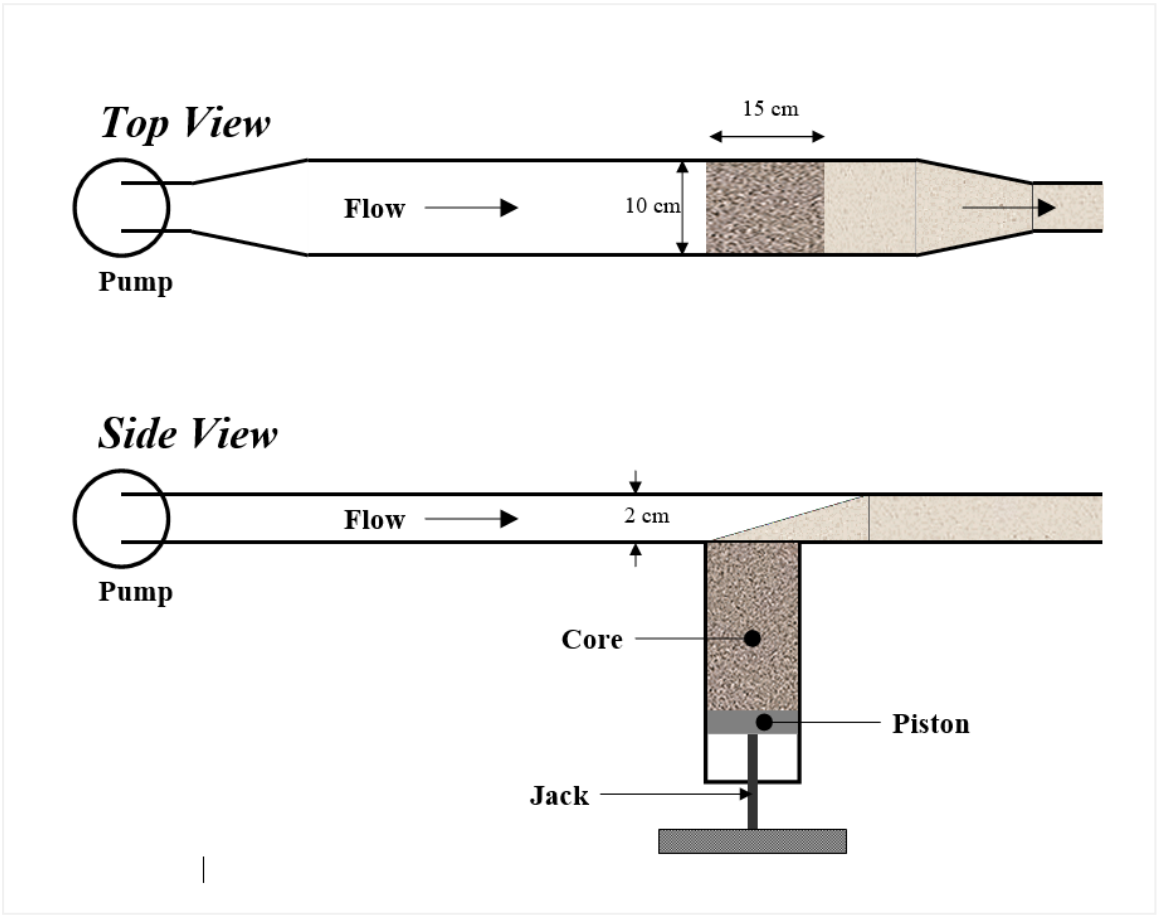


Figure 5-6: Schematic of the SEDFlume apparatus.

5.4.2. Introduction

Most models are calibrated using hind casting techniques which can have limitations when extending the simulation to future conditions. The most typically available sediment transport indicator measured in aquatic systems is the suspended sediment concentration. Unfortunately, many different combinations of erosion and deposition rates can be used to reach the same suspended sediment concentration. This can be illustrated as follows. In the steady state, an equilibrium exists between erosion and deposition. Deposition is generally described as $D = Pw_sC$ where P is a probability of deposition, w_s is the settling speed of the sediment particles, and C is the sediment concentration in the water. Equilibrium then gives

$$E - Pw_sC = 0 \tag{5.120}$$

This can be solved for the sediment concentration, which is

$$C = \frac{E}{Pw_s} \tag{5.121}$$

From this equation, it is seen that any suspended sediment concentration can be matched with an infinite number of erosion rates and deposition parameters by adjusting both accordingly. For example, the observed value of C can be obtained by high values of E and high values of Pw_s or by low values of E and low values of Pw_s. In other words, measurements of suspended sediment concentrations are not sufficient to determine erosion and/or deposition, parameters essential for predicting sediment and contaminant transport and fate. Historically, the erosion was constrained by theoretical relationships between shear stress and grain sizes, however there was still a range of parameters that would produce the same suspended concentrations. In order to predict erosion and deposition accurately, these quantities should be determined as functions of sediment characteristics and hydrodynamic variables by means of experiments or theory based on experiments.

5.4.3. SEDZLJ Model

5.4.3.1 Bed Shear Stress

The bottom stress is represented by τ_b whose components are given by

$$\tau_b = c_f V \tag{5.122}$$

where V is the velocity magnitude (cm/s) and c_f is a bottom shear stress friction factor calculated using a log-layer distribution of velocity; c_f is given by

$$c_f = \max\left[\frac{\kappa^2}{\left(\ln\frac{h}{2z_b}\right)^2}, c_{f,min}\right] \tag{5.123}$$

where κ is von Karman's constant (0.42), z_b is bottom skin friction based on the d_{50} at the sediment surface, and $c_{f,min}$ is the minimum value of the friction factor. The bottom roughness is assumed to be equal to the average particle diameter of the surface of the sediment bed at any given location. As the bottom roughness increases or as the depth of water decreases, c_f increases.

5.4.3.2 Erosion Rates

Results of a typical application of SEDFlume are shown in Figure 5-7, where erosion rates, *E*, in units of cm/s are plotted as a function of depth (cm) with shear stress, τ (N/m²). Erosion rates are generally highest at the surface and decrease with depth; they also increase with shear stress. In general, information of this type for sediments throughout the system is necessary for accurate predictions of sediment transport (Jones and Lick, 2000). Availability of this type of data is assumed and is used in the present model.

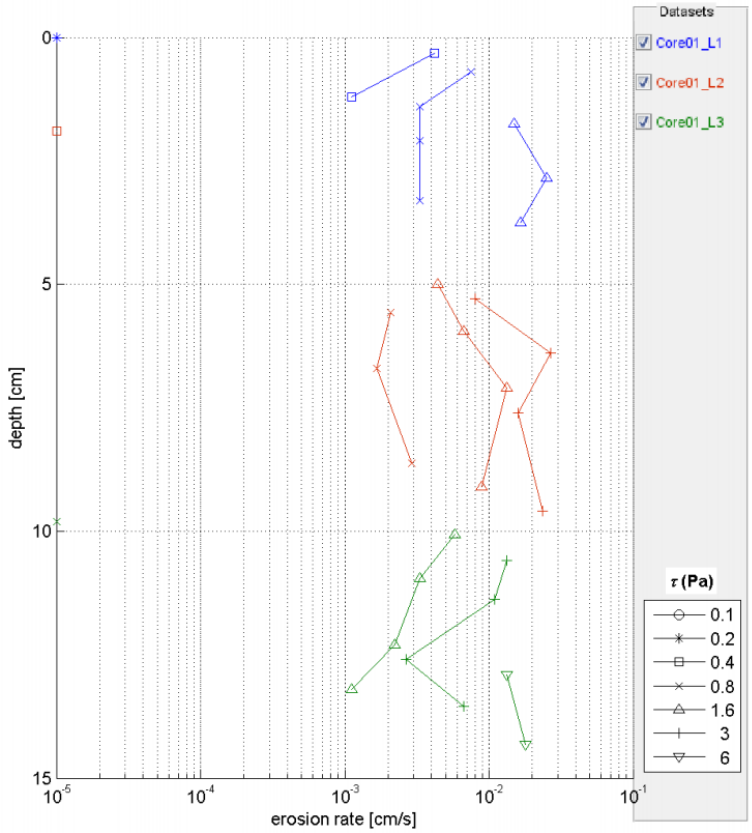


Figure 5-7: SEDFlume data for Conowingo Reservoir (DNR Maryland).

Information on erosion rates is generally reported in units of cm/s. In order to convert this to a mass flux in units of g/cm²/s which is needed in the modeling, the mass of solids within a sediment volume is needed. This quantity, for a sediment consisting of solids and water only (i.e. no gas), can be determined in terms of the bulk density of the sediments, ρ, as follows:

$$\rho = \rho_s x_s + \rho_w x_w = \rho_s x_s + \rho_w (1 - x_s) \tag{5.124}$$

Where ρ_s is the density of solids (g/cm³), x_s is the volume fraction of the solids, ρ_w is the density of water (g/cm³) and x_w is the volume fraction of water. Since $x_w = 1 - x_s$. The mass of solids per unit volume is $x_s\rho_s$ and can be determined from the above equation as

$$x_s\rho_s = \frac{\rho_s(\rho - \rho_w)}{\rho_s - \rho_w} = \frac{2.6}{1.6}(\rho - 1) \tag{5.125}$$

where it has been assumed that ρ_s = 2.6 g/cm³ and ρ_w = 1.0 g/cm³. Once the bulk density of the sediments is known, the erosion rate in units of g/cm²·s can be determined by multiplying the erosion rate in units of cm/s by $x_s\rho_s$.

As indicated above, erosion rates change as a function of depth. This variation is incorporated into the sediment bed model through a discrete layering system where the erosion rate is defined at each layer interface

and the particle size distribution and bulk density are defined as constant throughout the layer. Any number and thickness of layers required to approximate the variation of sediment properties with depth can be introduced as necessitated by field data.

The sediment transport model developed herein can incorporate erosion rate data collected in the field that are typically spatially discrete and at specific depths, but can be interpolated where no direct data are available. The total erosion rate is interpolated across sediment layer thicknesses and shear stresses. Linear interpolation is used to calculate an erosion rate at a specified shear stress τ as

$$E(\tau) = \left(\frac{\tau_{i+1} - \tau}{\tau_{i+1} - \tau_i}\right) E_i + \left(\frac{\tau - \tau_i}{\tau_{i+1} - \tau_i}\right) E_{i+1} \tag{5.126}$$

where subscript i denotes data for a shear stress less than τ and $i + 1$ denotes measured data for a shear stress greater than τ, with $\tau_i < \tau < \tau_{i+1}$.

Because E often changes rapidly with depth, the logarithmic interpolation between data points best represents erosion rates as a function of depth

$$\ln[E(T)] = \left(\frac{T_0 - T}{T_0}\right) \ln(E^j) + \frac{T}{T_0} \ln(E^{j+1}) \tag{5.127}$$

where T is the actual layer thickness, T_0 is the initial layer thickness, superscripts j and $j + 1$ denote data for the interface at the top and the bottom of the specific layer where the erosion rate is required, respectively. Equations (5.126) and (5.127) are combined so that the erosion rates may be calculated as a function of shear stress and depth.

5.4.3.3 Critical Shear Stress for Erosion

In addition to erosion rates, another parameter of significance in modeling is the critical stress for erosion, τ_{ce} This quantity can be understood and quantified as follows. Consider the flow of water over a sediment bed. As the rate of flow is increased starting from rest, there is a range of velocities (or shear stresses) at which the movement of the easiest-to-move particles (generally the sm allest) is first noticeable to an observer. These eroded particles then travel a relatively short distance until they come to rest in a new location. This initial motion tends to occur only at a few isolated spots. As the flow · velocity and shear stress increase further, more particles participate in this process of erosion, transport, and deposition, and the movement of the particles becomes more sustained.

Because of this gradual increase in sediment erosion as the shear stress increases, it is difficult to precisely define a critical velocity or critical shear stress at which sediment erosion is first initiated. More quantitatively and with less ambiguity, a critical shear stress for erosion can be defined as the shear stress at which a small, but accurately measurable, rate of erosion occurs. Roberts et al. (1998) defined this rate as 10^{-6} m/s, represented by approximately 1 mm of erosion in 15 minutes, though different rates have been used to define τ_{ce}.

Critical shear stresses for erosion as a function of particle diameter, d, are shown in Figure 5-8. For d >200 µm, the sediments behave in a non-cohesive manner, i.e., they consolidate rapidly and they erode particle by particle. For d < 200 µm, cohesive effects between particles become significant. The sediments consolidate relatively slowly with time, and the critical stresses depend not only on particle diameter but also on the bulk density of the sediments. For these cohesive sediments, τ_{ce} increases as d decreases and as bulk density increases.

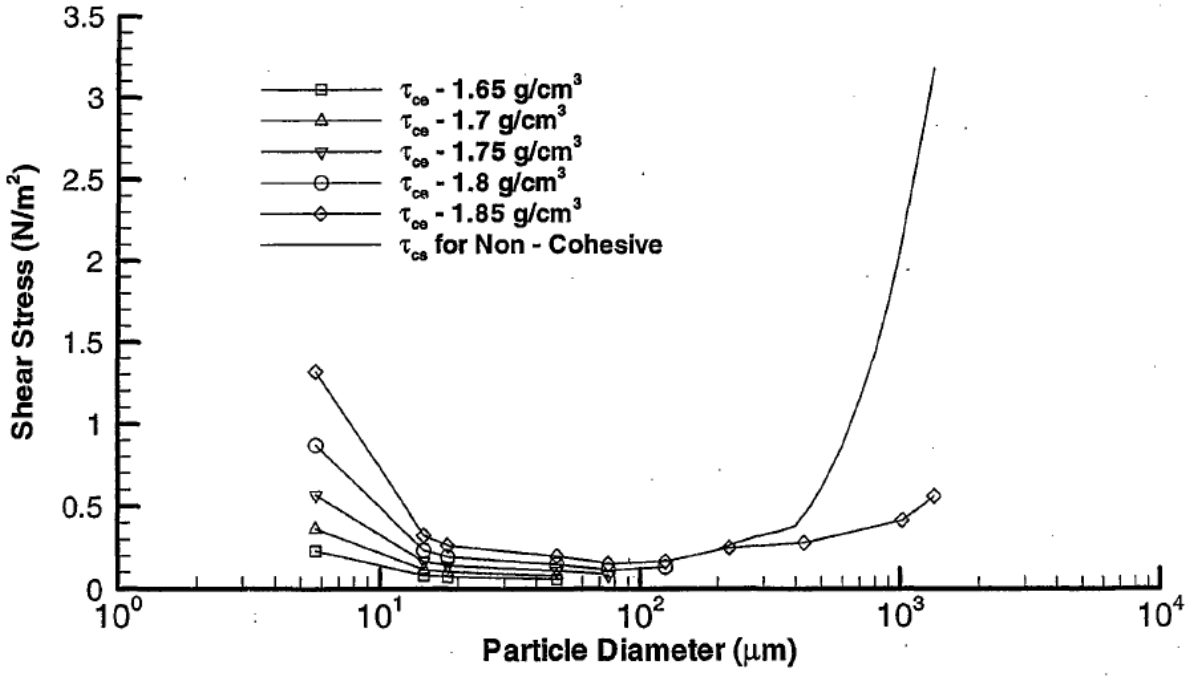


Figure 5-8: Critical shear stresses for erosion and suspension of quartz particles.

For non-cohesive sediment beds, the curve of Shields (1936), or any approximation thereof (van Rijn 1984a), could be used to define the critical shear stress for erosion. Soulsby (1997) approximated the critical shear for erosion as

$$\tau_{ce} = \rho g d\theta = \rho g d\left\{\frac{0.3}{1 + 1.2d_*} + 0.055[1 - \exp(-0.02d_*)]\right\} \tag{5.128}$$

$$d_* = d\left[\left(\rho_{sd}/\rho_w - 1\right)g/v^2\right]^{1/3} \tag{5.129}$$

where g is the acceleration due to gravity, d the sediment particle diameter, d_* is the non-dimensional particle diameter, v the kinematic fluid viscosity, and θ the critical Shields' parameter, represented by the algebraic fit shown in the brackets in Eq. (5.128)

5.4.3.4 Erosion into Suspended Load versus Bedload

As bottom sediments are eroded, a fraction of the sediments are suspended into the overlying water and are transported as suspended load; the rest of the eroded · sediments move by rolling and/or saltation in a thin layer near the bed in what is called bedload. The fraction in each of the transport modes depends on particle size and shear stress.

For fine-grained particles (which are generally cohesive), erosion occurs both as individual particles and in the form of chunks or small aggregates of particles. The individual particles move as suspended load. The aggregates tend to move downstream near the bed but generally seem to disintegrate into small particles in the high stress boundary layer near the bed as they move downstream. These disaggregated particles then move as suspended load. For this reason, it is assumed here that fine-grained sediments less than about 200 μm are completely transported as suspended load.

Coarser, non-cohesive particles (defined here as those particles with diameters greater than about 200 μm) can be transported both as suspended load and bedload, with the fraction in each dependent on

particle diameter and shear stress. For particles of a particular size, the shear stress at which suspended load (or sediment suspension) is initiated is defined as τ_{cs} (N/m²). This shear stress,τ_{cs}, can be defined from van Rijn's (1984b) formulations as

$$\tau_{cs} = \begin{cases} \frac{1}{\rho_w}\left(\frac{4w_s}{d_*}\right)^2 \text{ for } d \le 400\ \mu m \\ \frac{1}{\rho_w}(0.4w_s)^2 \text{for } d > 400\ \mu m \end{cases} \quad (5.130)$$

where d_* is the non-dimensional particle diameter calculated from $d_* = d\left[\left(\rho_s - 1\right)g/v^2\right]^{1/3}$ where d is the particle diameter (cm), and w_s is the particle settling speed (cm/s). For $\tau^b > \tau_{cs}$, sediments are transported both as bed load and suspended load with the fraction in suspended load, f, increasing with τ^b from $f = 0$ until f reaches 1. For τ^b greater than this, sediments are transported completely as suspended load.

The settling speed can be determined from Cheng (1997), van Rijn (1984) or user specified.

Cheng's formula is

$$w_s = \frac{v}{d}\left(\sqrt{25 + 1.2d_*^2} - 5\right)^{1.5} \quad (5.131)$$

where v is the kinematic fluid viscosity (cm²/s). Since Cheng's formula is based on observations of the settling of real sediment particles, it produces settling speeds lower than Stoke's law. This is because real sediments are often irregular in shape and have a greater hydrodynamic resistance to settling than perfect spheres as in Stoke's law.

Van Rijn computes the settling velocity as the following

$$w_s = \begin{cases} \frac{1}{18}\left[\frac{\left(s-1\right)gD_s^2}{v}\right], & D_s < 100\,\mu\text{m} \\ 10\frac{v}{D_s}\left\{\left[1+\frac{0.01\left(s-1\right)gD_s^3}{v^2}\right]^{0.5} - 1\right\}, & 100\,\mu\text{m} \le D_s < 1000\,\mu\text{m} \\ 1.1\left[\left(s-1\right)gD_s\right]^{0.5}, & D_s \ge 1000\,\mu\text{m} \end{cases} \quad (5.132)$$

where D_s is the representative particle size in m, s is the specific density; g is the acceleration of gravity; and v is the kinematic viscosity coefficient and w_s is in m/s.

Guy et al. (1966) have performed detailed flume measurements of suspended load and bedload transport for sediments ranging in median diameter, d_{50}, from 190 µm to 930 µm. They found that, as the ratio of shear velocity (defined as $u_* = \sqrt{\tau^b/\rho_w}$) to settling velocity increases, the proportion of suspended load to total load transport, q_s/q_t increases. An approximation of their data can be made with the following function:

$$\frac{q_s}{q_t}=\begin{cases}0, & \tau^b<\tau_{cs}\\ \dfrac{\ln\left(u_*/w_s\right)-\ln\left(\sqrt{\tau_{cs}/\rho_w}/w_s\right)}{\ln(4)-\ln\left(\sqrt{\tau_{cs}/\rho_w}/w_s\right)}, & \tau^b>\tau_{cs}\ \text{and}\ \dfrac{u_*}{w_s}<4\\ 1, & \dfrac{u_*}{w_s}>4\end{cases} \tag{5.133}$$

This approximation is used here. The original data is shown with the result given by the above equation in Figure 5-9.

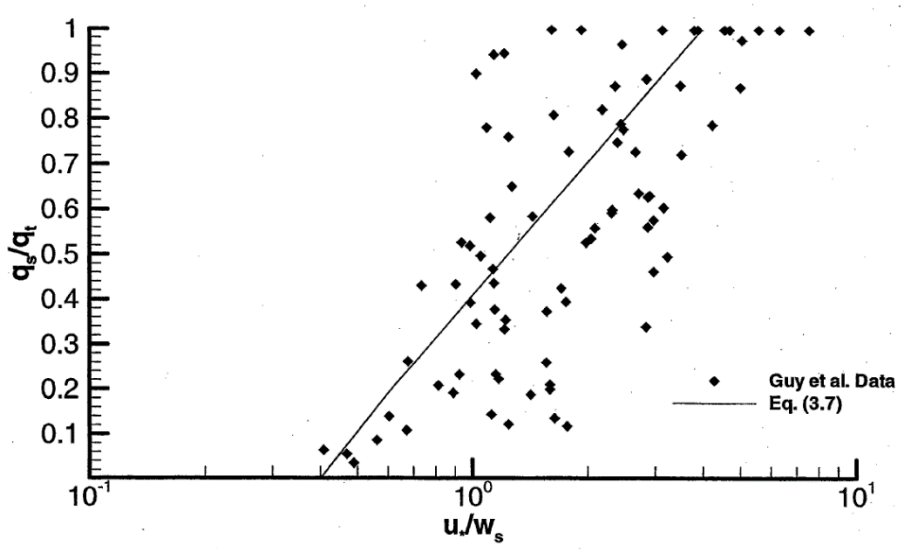


Figure 5-9: Results from flume measurements of suspended load and bed load (Guy, 1966).

Although sediments in nature have a continuous size distribution, physical quantities in numerical models are inherently discrete; hence, sediment particle sizes are discretized. The discretization of particle size classes j is done by measuring the different sediment sizes in a site-specific sediment core and grouping into appropriate size classes. The sediment bed in the model is described as the product of the particle size class and the corresponding mass fraction. By multiplying the total erosion flux of a particular size class j by q_s/q_t, the erosion flux of that class into suspended load $E_{s,j}$ can be calculated. The corresponding erosion flux into bedload $E_{b,j}$ is also calculated by multiplying the total erosion flux of the size class by the factor$(1-q_s/q_t)$. Thus, the erosion flux for any size class j is

$$E_{s,j}=\begin{cases}0, & \tau^b<\tau_{\text{ce}}\\ \dfrac{q_s}{q_t}f_jE, & \tau^b\geq\tau_{\text{ce}}\end{cases} \tag{5.134}$$

$$E_{b,j}=\begin{cases}0, & \tau^b<\tau_{\text{ce}}\\ \left(1-\dfrac{q_s}{q_t}\right)f_jE, & \tau^b\geq\tau_{\text{ce}}\end{cases} \tag{5.135}$$

where f_j is the mass fraction of the jth sediment size class.

5.4.3.5 *Suspended Load*

For suspended sediments, the three-dimensional, time dependent transport equation in the water over the bed is shown in Eq. (5.1). The net sediment flux into suspension, Q_s, is calculated as the total erosion flux into suspended load, $E_{s,j}$, minus the deposition flux from suspended load, $D_{s,j}$, for each sediment size class j

$$Q_{s,j} = E_{s,j} - D_{s,j} \tag{5.136}$$

where

$$Q_s = \sum_j Q_{s,j} \tag{5.137}$$

In a quiescent fluid where no shear stress is present, the deposition flux for suspended sediments can be described as the product of the settling speed of the sediment and the concentration of the sediment in the overlying water. However, in flowing water, the deposition is affected by the fluid turbulence, quantified as shear stress. In this case, a probability of deposition for each size class j, P_k, can be included in the formulation to account for the effects of the shear stress to yield

$$D_{sj} = P_j w_{sj} C_{sj} \tag{5.138}$$

This probability would be unity in the case of quiescent flow and decrease as the flow, turbulence, and shear stress increase. The probability for suspended load deposition seems to differ for cohesive and non-cohesive particle sizes. For cohesive particles, size classes with effective diameters less than 200 µm, Krone (1962) found that the probability of deposition varied approximately as

$$P_j = \begin{cases} 0 \text{ for } \tau^b > \tau_{cs,j} \\ \left(1 - \dfrac{\tau^b}{\tau_{cs,j}}\right) \text{ for } \tau^b > \tau_{cs,j} \end{cases} \tag{5.139}$$

For larger non-cohesive particles, size classes with an effective diameter greater than 200 µm, Gessler (1967) showed that the probability of deposition could be described with a Gaussian distribution, or error function given by

$$P_j(Y) = \text{erf}\left(\frac{Y}{2}\right) = \frac{2}{\sqrt{\pi}} \int_0^{Y/2} \exp(-\xi^2)\, \text{d}\xi \tag{5.140}$$

where

$$Y = \frac{1}{\sigma}\left(\frac{\tau_{cs,j}}{\tau^b} - 1\right) \tag{5.141}$$

and $\tau_{cs,j}$ is the critical shear stress for suspension for size class j and σ is the standard deviation for shear stress variation, which Gessler determined to be about 0.57.

An approximation to this function for $Y > 0$ with an error of less than 0.001% is found to be (Dwight, 1961; Abramowitz and Stegun, 1972; HydroQual, 1997)

$$P_j = 1 - F(Y)(0.4632X - 0.1202X^2 + 0.9373X^3) \tag{5.142}$$

where

$$F(Y) = \frac{1}{(2\pi)^{1/2}} e^{-\frac{1}{2}Y^2} \tag{5.143}$$

$$X = \frac{1}{(1 + 0.235235Y)} \tag{5.144}$$

when $Y < 0$

$$P_j = 1 - P(|Y|) \tag{5.145}$$

Figure 5-10 shows sample probability distributions using the formulations for cohesive and non-cohesive particles.

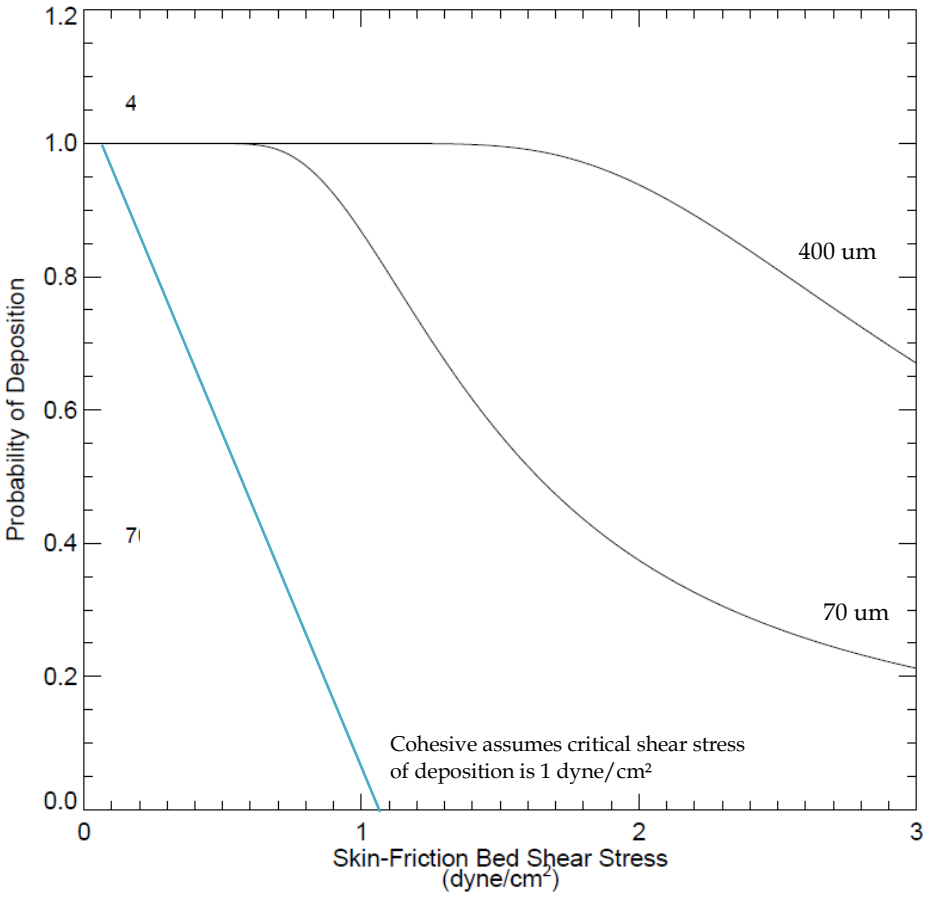


Figure 5-10: Sample probability distributions for cohesive and non-cohesive particles.

5.4.3.6 Bedload

For the description of bedload transport, van Rijn's (1984a) approach is used. To calculate the concentration of particles moving in bedload, a mass balance equation can be written as

$$\frac{\partial(mC_b)}{\partial t} = \frac{\partial(mq_{bx})}{\partial x} + \frac{\partial(mq_{by})}{\partial y} + Q_b \tag{5.146}$$

where C_b is the bedload concentration (g/cm²), q_b is the horizontal bedload flux in the x or y directions (g/s/cm), m is the cell area (cm²) and Q_b is the net vertical flux of sediments between the sediment bed and bedload (g/s). This equation is solved using a central difference approximation for the fluxes in the x and y directions. The horizontal bedload flux in general is calculated as

$$q_b = u_b C_b \tag{5.147}$$

where u_b is the bedload velocity (cm/s) in the direction of interest. The bedload velocity and thickness can be calculated from van Rijn's (1984a) formulations as

$$u_b = 1.5T^{0.6}[(\rho_s - 1)gd]^{0.5} \tag{5.148}$$

$$h_b = 3dd_*^{0.6}T^{0.9} \tag{5.149}$$

The transport parameter, T, is calculated as

$$T = \frac{\tau^b - \tau_{ce}}{\tau_{ce}} \tag{5.150}$$

The flux of sediments between the bottom sediments and bedload, Q_b, is calculated as the erosion of sediments into bedload, E_b, minus the deposition of sediments from bedload, D_b, and is

$$Q_b = E_b - D_b \tag{5.151}$$

where D_b is given by

$$D_b = Pw_s C_b \tag{5.152}$$

In steady state equilibrium, the concentration of sediments in bedload, C_e, is due to a dynamic equilibrium between erosion and deposition, i.e.,

$$E_b = Pw_s C_e \tag{5.153}$$

From this, the probability of deposition can be written as

$$P = \frac{E_b}{w_s C_e} \tag{5.154}$$

The erosion rate can be determined from SEDFlume, while the settling speed can be calculated from Eq. (5.132). The equilibrium concentration, C_e, has been investigated by several authors; the formulation by van Rijn (1984a) will be used here and is calculated as

$$C_e = 0.117\frac{\rho_s T}{d_*} \tag{5.155}$$

Once E_b, w_s, and C_e are known as a function of particle diameter and shear stress, P can be calculated from Eq. (5.154). It is then assumed that this probability is also valid for the non-steady case so that the deposition rate can be calculated in this case also.

The equilibrium concentration, C_e, is based on experiments with uniform sediments. In general, the sediment bed contains and must be represented by more than one size class. In this case; the erosion rate for a particular size class is given by $f_j E_b$, it follows that the probability of deposition for size class j is then given by

$$P_j = \frac{f_j E_b}{w_{sj} f_j C_{ej}} = \frac{E_b}{w_{sj} C_{ej}} \tag{5.156}$$

In this equation, it is implicitly assumed that there is a dynamic equilibrium between erosion and deposition for each size class j.

5.4.3.7 Bed Armoring

A decrease in sediment erosion rates with time, or bed armoring, can occur due to (1) the consolidation of cohesive sediments with depth and time, (2) the deposition of coarser sediments on the sediment bed during a flow event, and (3) the erosion of finer sediments from the surficial sediment, leaving coarser sediments behind, again during a flow event. The consolidation of sediments and subsequent change in erosion rates with of depth can be determined by SEDFlume in situ measurements. The consolidation of sediment and increase of erosion rates with time can be determined approximately from consolidation studies, again by means of SEDFlume.

Here we are concerned about bed armoring due to processes (2) and (3). In order to describe these processes, it is assumed in the present model that a thin mixing layer, or active layer, is formed at the surface of the bed. The existence and properties of this have been discussed by previous researchers (Borah et al., 1983; van Niekerk et al., 1992; Parker et al., 2000). The presence of this active layer permits the interaction of depositing and eroding sediments to occur in a discrete layer without allowing deposited sediments to affect the undisturbed sediments below. Van Niekerk et al. (1992) have suggested that the thickness, Ta, can be approximated by

$$T_a = 2d_{50}\frac{\tau^b}{\tau_{ce}} \tag{5.157}$$

This formulation takes into account the deeper penetration of turbulence into the bed with increasing shear stress. In the present calculations, d50 is approximated by the average diameter in the interest of computational efficiency.

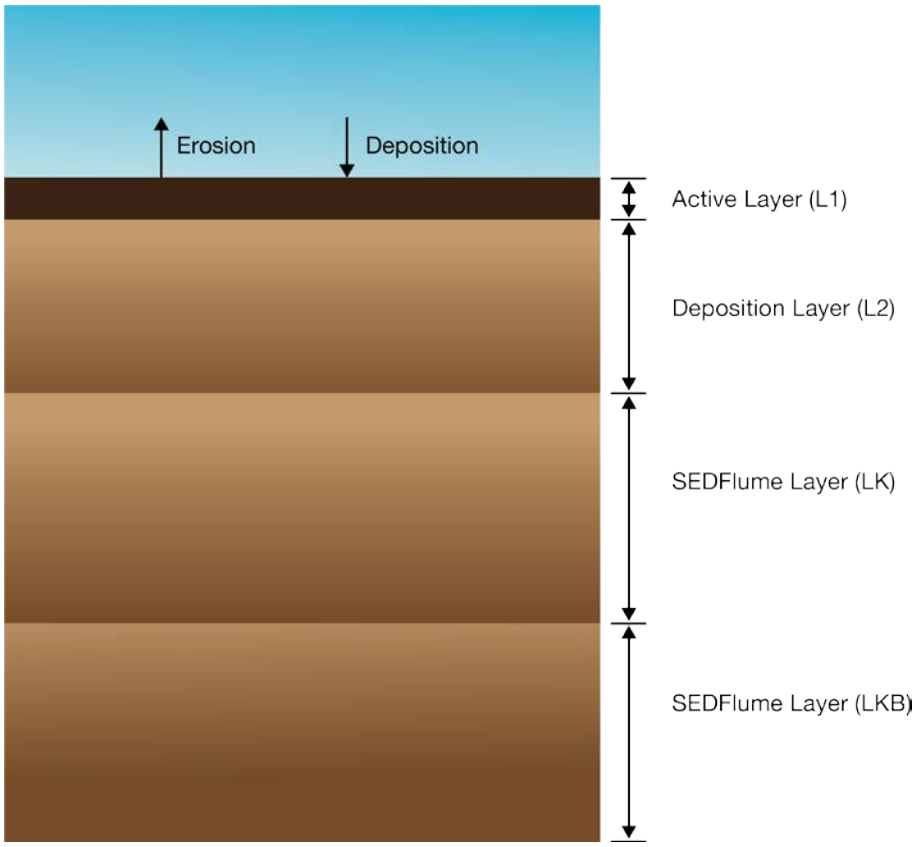


Figure 5-11: Diagram of SEDflume layering system.

Since the active layer is kept at a constant thickness, T_a, three possible states of the active layer must be considered. The first state is a net erosion of the active layer, where there may be deposition occurring but the net flux is erosional. If the thickness of the active layer after this net erosion is T, then a thickness of material equal to T_a-T is added to the active layer so that a thickness of T_a can be maintained. This material is added from the layer below in size class proportions equivalent to that in the layer below. The second possible state of the active layer is a net depositional state where the thickness of the active layer exceeds T_a. In this case, the excess material, T-T_a, is put into a new deposited material layer just below the active layer, but above the parent bed. This material is added to the deposited layer in size class proportions equal to the active layer. The third state of the active layer is where T is equal in thickness to T_a: In this case, no action is taken. Figure 5-11 shows a diagram of the layering system.

The erosion rates for this active layer are dependent on its average particle size. Figure 5-12 shows the erosion rate vs. particle diameter for quartz sediment. It is seen that as the particle diameter increases beyond 200 µm, the erosion rate decreases. This demonstrates how bed coarsening affects erosion rates. A dataset of this type can be constructed utilizing laboratory and field cores to determine erosion rates as a function of particle size for any particular site. The erosion rate for an active or deposited layer can then be calculated from the average particle size of the layer with an interpolation similar to Eq. (5.126) with particle size in place of thickness.

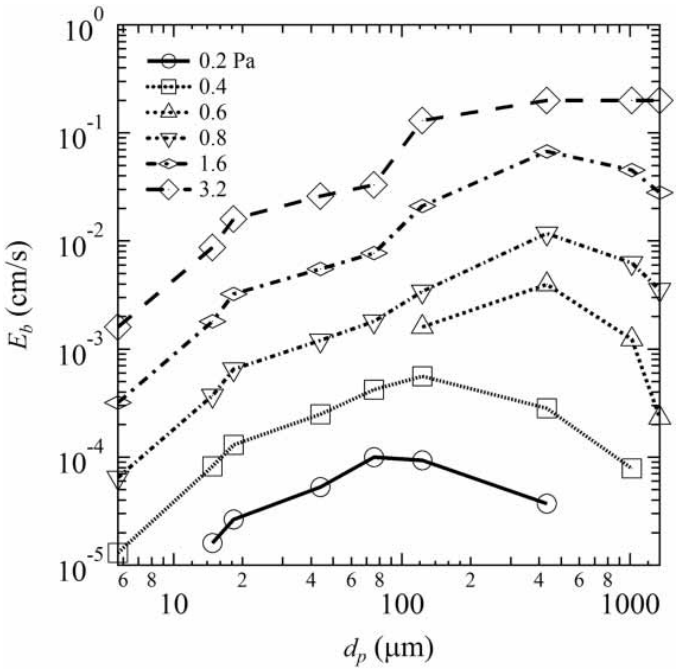


Figure 5-12: Erosion rates versus particle size and shear stress for a bulk density of 1.9 g/cm², adapted from Roberts et al. (1998) by James (2010). Model uses interpolated data to estimate erosion rates at 1 Pa.

6 TOXIC CONTAMINANT TRANSPORT AND FATE

6.1 Introduction

This chapter presents processes associated with toxics and their mathematical modeling. It starts with the basic equations and their numerical aspects then follows by characteristics of toxic organic chemicals and metals, and the sorption and de sorption processes.

6.2 Basic Equations

The transport of a sorptive contaminant in the water column is governed by transport equations for the contaminant dissolved in the water phase, for the contaminant sorbed to material effectively dissolved in the water phase, and for the contaminant sorbed to suspended particles. For the portion of the contaminant dissolved directly in the water phase

$$\begin{aligned}
&\frac{\partial}{\partial t}\left(mHC_w\right)+\frac{\partial}{\partial x}\left(m_yHuC_w\right)+\frac{\partial}{\partial y}\left(m_xHvC_w\right)+\frac{\partial}{\partial z}\left(mwC_w\right)\\
&=\frac{\partial}{\partial z}\left(m\frac{A_b}{H}\partial_zC_w\right)+mH\left(\sum_i\left(K_{dS}^iS^i\chi_S^i\right)+\sum_j\left(K_{dD}^jD^j\chi_D^j\right)\right)\\
&-mH\sum_i\left(K_{aS}^iS^i\right)\left(\psi_w\frac{C_w}{\phi}\right)\left(\hat{\chi}_S^i-\chi_S^i\right)\\
&-mH\sum_j\left(K_{aD}^jD^j\right)\left(\psi_w\frac{C_w}{\phi}\right)\left(\hat{\chi}_D^j-\chi_D^j\right)+S_C^E+S_C^I
\end{aligned}\tag{6.1}$$

where

C_w is the mass of water dissolved contaminant per unit total volume (µg/l),

S^i is the mass of sediment class i (g/m^3),

D^j is the mass of the dissolved substance (i.e. DOC) (g/m^3),

χ_S is the mass of contaminant sorbed to sediment class *i* per mass of sediment (mg/g),

χ_D is the mass of contaminant sorbed to dissolved material *j* per unit mass of dissolved material (i.e. sorbed to DOC) (mg/g),

ϕ is the porosity (dimensionless),

ψ_w is the fraction of the water dissolved contaminant available for sorption (dimensionless),

K_a is the adsorption rate of sediment (S) or dissolved material (D),

K_d is the desorption rate ,

S_C^E Is the external source/sink of the contaminant (mg/s), and

S_C^I Is the internal source/sink of the contaminant (mg/s) due to degradation, volatilization and conversions to/from other contaminants (mg/s).

6.2.1. Contaminant Partitioning

The sorption kinetics are based on the Langmuir isotherm (Chapra, 1997) with $\hat{X}$ denoting the saturation sorbed mass per carrier mass. The sediment and dissolved material concentrations, S and D are defined as mass per unit total volume.

Introducing sorbed concentrations defining sorbed mass per unit total volume

$$C_D^j = D^j \chi_D^j \tag{6.2}$$

$$C_S^i = S^i \chi_S^i \tag{6.3}$$

The EFDC sorbed contaminant transport formulation currently employees equilibrium partitioning with the adsorption and desorption terms

$$\left(K_{sD}^j D^j\right)\left(\psi_w \frac{C_w}{\phi}\right)\left(\hat{\chi}_D^j - \chi_D^j\right) = K_{dD}^j C_D^j \tag{6.4}$$

$$\left(K_{aS}^i S^i\right)\left(\psi_w \frac{C_w}{\phi}\right)\left(\hat{\chi}_S^i - \chi_S^i\right) = K_{dS}^i C_S^i \tag{6.5}$$

Solving (6.4) and (6.5) for the sorbed to water phase concentration ratios gives

$$\begin{aligned} \frac{C_D^j}{C_w} &= \frac{f_D^j}{f_w} = P_D^j \frac{D^j}{\phi} \\ P_D^j &= P_{Do}^j \left(1 + P_{Do}^j \left(\frac{C_w}{\hat{\chi}_D^j \phi}\right)\right)^{-1} \\ P_{Do}^j &= \frac{\psi_w K_{aD}^j \hat{\chi}_D^j}{K_{dD}^j} \end{aligned} \tag{6.6}$$

$$\begin{aligned} \frac{C_S^i}{C_w} &= \frac{f_S^i}{f_w} = P_S^i \frac{S^i}{\phi} \\ P_S^i &= P_{So}^i \left(1 + P_{So}^i \left(\frac{C_w}{\hat{\chi}_S^i \phi}\right)\right)^{-1} \\ P_{So}^i &= \frac{\psi_w K_{aS}^i \hat{\chi}_S^i}{K_{dS}^i} \end{aligned} \tag{6.7}$$

where P denotes the partition coefficient, and P_o is its linear equilibrium value. For linear equilibrium partitioning, P is set to P_o, which in effect approximates $(\ )^{-1}$ terms in (6.6) and (6.7) by unity. Requiring the mass fractions to sum to unity

$$f_w + \sum_i f_S^i + \sum_j f_D^j = 1 \tag{6.8}$$

gives

$$f_w = \frac{C_w}{C} = \frac{\phi}{\phi + \sum_i P_S^i S^i + \sum_j P_D^j D^j}$$
$$f_D^j = \frac{C_D^j}{C} = \frac{P_D^j D^j}{\phi + \sum_i P_S^i S^i + \sum_j P_D^j D^j} \quad (6.9)$$
$$f_S^i = \frac{C_S^i}{C} = \frac{P_S^i S^i}{\phi + \sum_i P_S^i S^i + \sum_j P_D^j D^j}$$

The dissolved concentrations can be alternately expressed by mass per unit volume of the water phase

$$C_{w:w} = \frac{C_w}{\phi}$$
$$C_{D:w}^j = \frac{C_D^j}{\phi} \quad (6.10)$$
$$D_{:w}^J = \frac{D^j}{\phi}$$

with (6.9) becoming

$$\frac{C_{w:w}}{C} = \frac{1}{\phi + \sum_i P_S^i S^i + \sum_j P_D^j \phi D_{:w}^j}$$
$$\frac{C_{D:w}^j}{C} = \frac{P_D^j D_{:w}^j}{\phi + \sum_i P_S^i S^i + \sum_j P_D^j \phi D_{:w}^j} \quad (6.11)$$
$$\frac{C_S^i}{C} = \frac{P_S^i S^i}{\phi + \sum_i P_S^i S^i + \sum_j P_D^j \phi D_{:w}^j}$$

which is a generalization of Chapra's (1997) formulation for sorption to dissolved and particulate organic carbon.

6.2.2. Water Column Transport

The partitioning relationships shown in Eq. (6.2) and Eq. (6.3) allows equation (6.1) to be expanded into the transport equation for each contaminant fractions

$$\frac{\partial}{\partial t}(m_x m_y H C_w) + \frac{\partial}{\partial x}(m_y H u C_w) + \frac{\partial}{\partial y}(m_x H v C_w) + \frac{\partial}{\partial z}(m_x m_y w C_w)$$
$$= \frac{\partial}{\partial z}\left[m_x m_y \frac{A_b}{H}\frac{\partial}{\partial z}(C_w)\right] + m_x m_y H\left(\sum_i (K_{dS}^j C_S^i) + \sum_i (K_{dD}^j C_D^i)\right)$$
$$- m_x m_y H\left(\sum_i (K_{aS}^i S^i)(\Psi_w \frac{C_w}{\phi})(\hat{\chi}_S^i - \chi_S^i) + \sum_i (K_{aD}^j D^j)(\Psi_w \frac{C_w}{\phi})(\hat{\chi}_D^j - \chi_D^j)\right. \quad (6.12)$$
$$\left. + \gamma C_w\right)$$

$$\frac{\partial}{\partial t}\left(m_x m_y H C_D^j\right)+\frac{\partial}{\partial x}\left(m_y H u C_D^j\right)+\frac{\partial}{\partial y}\left(m_x H v C_D^j\right)+\frac{\partial}{\partial z}\left(m_x m_y w C_D^j\right)$$
$$=\frac{\partial}{\partial z}\left[m_x m_y \frac{A_b}{H}\frac{\partial}{\partial z}\left(C_D^j\right)\right]+m_x m_y H\left(K_{sD}^j D^j\right)\left(\psi_w \frac{C_w}{\phi}\right)\left(\hat{\chi}_D^j-\chi_D^j\right) \quad (6.13)$$
$$-m_x m_y H\left(K_{dD}^j+\gamma\right)C_D^j$$

$$\frac{\partial}{\partial t}\left(m_x m_y H C_S^i\right)+\frac{\partial}{\partial x}\left(m_y H u C_S^i\right)+\frac{\partial}{\partial y}\left(m_x H v C_S^i\right)+\frac{\partial}{\partial z}\left(m_x m_y w C_S^i\right)+\frac{\partial}{\partial z}\left(m_x m_y w_S^i C_S^i\right)$$
$$=\frac{\partial}{\partial z}\left(m_x m_y \frac{A_b}{H}\frac{\partial}{\partial z}C_S^i\right)+m_x m_y H\left(K_{aS}^i S^i\right)\left(\psi_w \frac{C_w}{\phi}\right)\left(\hat{\chi}_S^i-\chi_S^i\right) \quad (6.14)$$
$$-m_x m_y H\left(K_{dS}^i+\gamma\right)C_S^i$$

Where Eq. (6.12) is for the dissolved fraction, Eq. (6.13) is for the fraction sorbed to the dissolved material and Eq. (6.14) is for the fraction sorbed to the sediments.

The transport equation for the portion of material *sorbed to a dissolved constituent D* is,

$$\frac{\partial}{\partial t}\left(m_x m_y H D^j \chi_D^j\right)+\frac{\partial}{\partial x}\left(m_y H u D^j \chi_D^j\right)+\frac{\partial}{\partial y}\left(m_x H v D^j \chi_D^j\right)+\frac{\partial}{\partial z}\left(m_x m_y w D^j \chi_D^j\right)$$
$$=\frac{\partial}{\partial z}\left[m_x m_y \frac{A_b}{H}\frac{\partial}{\partial z}\left(D^j \chi_D^j\right)\right]+m_x m_y H\left(K_{sD}^j D^j\right)\left(\psi_w \frac{C_w}{\phi}\right)\left(\hat{\chi}_D^j-\chi_D^j\right) \quad (6.15)$$
$$-m_x m_y H\left(K_{dD}^j+\gamma\right)\left(D^j \chi_D^j\right)$$

The transport equation for the portion of material *sorbed to a suspended constituent S* is,

$$\frac{\partial}{\partial t}\left(m_x m_y H S^i \chi_S^i\right)+\frac{\partial}{\partial z}\left(m_y H u S^i \chi_S^i\right)+\frac{\partial}{\partial y}\left(m_x H v S^i \chi_S^i\right)+\frac{\partial}{\partial z}\left(m_x m_y w S^i \chi_S^i\right)$$
$$+\frac{\partial}{\partial z}\left(m_x m_y w_S^i S^i \chi_S^i\right) \quad (6.16)$$
$$=\frac{\partial}{\partial z}\left[m_x m_y \frac{A_b}{H}\frac{\partial}{\partial z}\left(S^i \chi_S^i\right)\right]+m_x m_y H\left(K_{aS}^i S^i\right)\left(\psi_w \frac{C_w}{\phi}\right)\left(\hat{\chi}_S^i-\chi_S^i\right)$$
$$-m_x m_y H\left(K_{aS}^i+\gamma\right)\left(S^i \chi_S^i\right)$$

Adding equations (6.12), (6.13), and (6.14), using the equilibrium partitioning relationships (6.4) and (6.5) gives

$$\frac{\partial}{\partial t}\left(m_x m_y H C\right)+\frac{1}{m_x m_y}\frac{\partial}{\partial x}\left(m_y H u C\right)+\frac{1}{m_x m_y}\frac{\partial}{\partial y}\left(m_x v C\right)+\frac{\partial}{\partial z}\left(m_x m_y w C\right)$$
$$-\frac{\partial}{\partial z}\left(m_x m_y \sum_i w_S^i f_S^i C\right)=\frac{\partial}{\partial z}\left(m_x m_y \frac{A_b}{H}\frac{\partial C}{\partial z}\right)-m_x m_y H\gamma \quad (6.17)$$

the equation for the total concentration, *C*. The boundary condition at the water column-sediment bed interface, $z = 0$, is

$$
\begin{aligned}
&-\frac{A_b}{H}\frac{\partial C}{\partial z}-\sum_i w_S^i f_S^i C \\
&\quad=\sum_i\left(\left(\max\left(J_{SBS}^i\chi_S^i,0\right)+\varepsilon\max\left(\frac{J_{SBS}^i}{\rho_S^i},0\right)\right)\left(\frac{C_w+\sum_j C_D^j}{\phi}\right)\right)_{SB} \\
&\quad+\sum_i\left(\left(\min\left(J_{SBS}^i\chi_S^i,0\right)+\varepsilon_{dep}\min\left(\frac{J_{SBS}^i}{\rho_S^i},0\right)\right)\left(\frac{C_w+\sum_j C_D^j}{\phi_{dep}}\right)\right)_{WC} \\
&\quad+\sum_i\left(\left(\varepsilon\max\left(\frac{J_{SBB}^i}{\rho_S^i},0\right)\right)\left(\frac{C_w+\sum_j C_D^j}{\phi}\right)\right)_{SB} \\
&\quad+\sum_i\left(\left(\varepsilon_{dep}\min\left(\frac{J_{SBB}^i}{\rho_S^i},0\right)\right)\left(\frac{C_w+\sum_j C_D^j}{\phi_{dep}}\right)\right)_{WC} \\
&\quad+\left(\max(q_w,0)\left(\frac{C_w+\sum_j C_D^j}{\phi}\right)\right)_{SB}+\left(\min(q_w,0)\left(\frac{C_w+\sum_j C_D^j}{\phi_{dep}}\right)\right)_{WC} \\
&\quad-q_{dif}\left(\left(\frac{C_w+\sum_j C_D^j}{\phi_{dep}}\right)_{WC}-\left(\frac{C_w+\sum_j C_D^j}{\phi}\right)_{SB}\right)
\end{aligned}
\tag{6.18}
$$

where J_{SBS} and J_{SBB} are the suspended load and bedload sediment fluxes between the sediment bed and the water column, defined as positive from the bed, ρ_s is the sediment density, q_w is the water specific discharge due to bed consolidation and groundwater interaction, defined as positive from the bed, and q_{dif} is a diffusion velocity incorporating the effects of molecular diffusion, hydrodynamic dispersion, and biological induced mixing. The subscript *SB* denotes conditions in the top layer of the sediment bed, while the subscript *WC* denotes condition in the water column immediately above the bed, with the exception that the specific discharge and diffusion velocity are defined at the water column-bed interface. The subscript, *dep*, is used to denote the void ratio and porosity of newly depositing sediment. Equation (6.11) indicates that contaminant flux between the bed and water column includes, a flux of suspended sediment sorbed material; fluxes of water dissolved and sorbed to water dissolved material due to the specific discharge of water associated with consolidation and ground water interaction and water entrainment and expulsion associated with both suspended and bedload sediment deposition and resuspension; and a flux of water dissolved and sorbed to water dissolved material due to diffusion like processes. Transport of bedload sediment sorbed material is represented by direct transport between horizontally adjacent top bed layers and is included in the contaminant mass conservation equations for the sediment bed. The boundary condition at the water free surface is

$$
-\frac{A_b}{H}\frac{\partial C}{\partial z}-\sum_i w_S^i f_S^i C=0:\ z=1 \tag{6.19}
$$

Using the relationship between the porosity and void ratio

$$
\phi=\frac{\varepsilon}{1+\varepsilon} \tag{6.20}
$$

and (6.3) allows (6.18) to be written as

$$
\begin{aligned}
&-\frac{A_b}{H}\frac{\partial C}{\partial z}-\sum_i w_S^i f_S^i C \\
&\quad=\sum_i\left(\left(\max\left(J_{SBS}^i\frac{C_S^i}{S^i},0\right)+(1+\varepsilon)\max\left(\frac{J_{SB}^i}{\rho_S^i},0\right)\right)\left(C_w+\sum_j C_D^j\right)\right)_{SB} \\
&\quad+\sum_i\left(\left(\min\left(J_{SBS}^i\frac{C_S^i}{S^i},0\right)+(1+\varepsilon_{dep})\min\left(\frac{J_{SBS}^i}{\rho_S^i},0\right)\right)\left(C_w+\sum_j C_D^j\right)\right)_{WC} \\
&\quad+\sum_i\left(\left((1+\varepsilon)\max\left(\frac{J_{SBB}^i}{\rho_S^i},0\right)\right)\left(C_w+\sum_j C_D^j\right)\right)_{SB} \\
&\quad+\sum_i\left(\left((1+\varepsilon_{dep})\min\left(\frac{j_{SBB}^i}{\rho_S^i},0\right)\right)\left(C_w+\sum_j C_D^j\right)\right)_{WC} \\
&\quad+\left(\left(\max(q_w,0)+q_{dif}\right)\frac{1}{\phi}\left(C_w+\sum_j C_D^j\right)\right)_{SB} \\
&\quad+\left(\left(\min(q_w,0)-q_{dif}\right)\frac{1}{\phi}\left(C_w+\sum_j C_D^j\right)\right)_{WC}
\end{aligned}
\tag{6.21}
$$

The sediment concentration can be expressed in terms of the sediment density and void ratio by

$$S^i=\frac{F^i\rho_s^i}{1+\varepsilon} \tag{6.22}$$

where F^i is the fraction of the total sediment volume occupied by each sediment class

$$F^i=\left(\sum_i\left(\frac{S^i}{\rho_s^i}\right)\right)^{-1}\left(\frac{S^i}{\rho_s^i}\right) \tag{6.23}$$

Introducing (6.9) and (6.22) into (6.21) gives the final form of the bottom boundary

$$
\begin{aligned}
&-\frac{A_b}{H}\frac{\partial C}{\partial z}-\sum_i w_S^i f_S^i C \\
&\quad=\sum_i\left(\max\left(\frac{J_{SBS}^i f_S^i}{S^i},0\right)C+\max\left(\frac{F^i J_{SBS}^i}{S^i},0\right)\left(f_w+\sum_j f_D^j\right)C\right)_{SB} \\
&\quad+\sum_i\left(\min\left(\frac{J_{SBS}^i f_S^i}{S^i},0\right)C+\min\left(\frac{F_{dep}^i J_{SBS}^i}{S_{dep}^i},0\right)\left(f_w+\sum_j f_D^j\right)C\right)_{WC} \\
&\quad+\sum_i\left(\left((1+\varepsilon)\max\left(\frac{J_{SBB}^i}{\rho_S^i},0\right)\right)\left(C_w+\sum_j C_D^j\right)\right)_{SB} \\
&\quad+\sum_i\left(\left((1+\varepsilon_{dep})\min\left(\frac{J_{SBB}^i}{\rho_S^i},0\right)\right)\left(C_w+\sum_j C_D^j\right)\right)_{WC} \\
&\quad+\left(\left(\max(q_w,0)+q_{dif}\right)\frac{1}{\phi}\left(f_w+\sum_j f_D^j\right)C\right)_{SB} \\
&\quad+\left(\left(\min(q_w,0)-q_{dif}\right)\frac{1}{\phi}\left(f_w+\sum_j f_D^j\right)C\right)_{WC}
\end{aligned}
\tag{6.24}
$$

Note that the form of the bed flux associated with bedload transport remains unmodified since a sediment concentration in the water column cannot be readily defined for sediment being transported as bedload.

The transport equation (6.17) for the total contaminant concentration in the water column is solved using a fractional step procedure which sequentially treats advection; settling, deposition, and resuspension; pore water advection and diffusion; and reactions. The fractional phase distribution of the contaminant is recalculated between the advection, settling, deposition and resuspension, and pore water advection and diffusion steps using (6.9). The advection step is

$$
(HC)^{n+1/4}-(HC)^n+\frac{\theta}{m_x m_y}\frac{\partial}{\partial x}\left(m_y HuC\right)+\frac{\theta}{m_x m_y}\frac{\partial}{\partial y}\left(m_x HvC\right)+\theta\frac{\partial(wC)}{\partial z}=0 \tag{6.25}
$$

with the vertical boundary conditions

$$
wC=0:\ z=0,1 \tag{6.26}
$$

The fractional time level in (6.25) and subsequent equations is used to denote an intermediate result in the fractional step procedure. The spatially discrete from of (6.25) is solved using one of the standard high order, flux limited, advective transport solvers in the EFDC model.

The settling, deposition, and resuspension step is

$$
(H)^{n+1/2}-(HC)^{n+1/4}=\theta\frac{\partial}{\partial z}\left(\sum_i w_S^i f_S^i C\right) \tag{6.27}
$$

with the boundary conditions

$$
\begin{aligned}
-\sum_i w_S^i f_S^i C = &\sum_i \left(\max\left(\frac{J_{SBS}^i f_S^i}{S^i}, 0\right) C + \max\left(\frac{F^i J_{SB}^i}{S^i}, 0\right) \left(f_w + \sum_j f_D^j \right) C \right)_{SB} \\
&+ \sum_i \left(\min\left(\frac{J_{SBS}^i f_S^i}{S^i}, 0\right) C + \min\left(\frac{F_{dep}^i J_{SBS}^i}{S_{dep}^i}, 0\right) \left(f_w + \sum_j f_D^j \right) C \right)_{WC} \\
&+ \sum_i \left(\left((1+\varepsilon) \max\left(\frac{J_{SBB}^i}{\rho_S^i}, 0\right) \right) \left(C_w + \sum_j C_D^j \right) \right)_{SB} \\
&+ \sum_i \left(\left((1+\varepsilon_{de}) \min\left(\frac{J_{SB}^i}{\rho_S^i}, 0\right) \right) \left(C_w + \sum_j C_D^j \right) \right)_{WC} : z = 0
\end{aligned}
\tag{6.28}
$$

$$
w_S^i f_S^i C = 0 : \; z = 1 \tag{6.29}
$$

Integrating (6.27) over a water column layer and using upwind differencing for the settling gives,

$$
\begin{aligned}
&\Delta_k (HC)_k^{n+\frac{1}{2}} - \Delta_k (HC)_k^{n+\frac{1}{4}} \\
&\qquad = \theta \sum_i \left(\frac{\left(w_S^i S^i\right)_{k+}}{H} \left(\frac{f_S^i}{S^i}\right)_{k+1} \right)^{n+\frac{1}{2}} (HC)_{k+1}^{n+\frac{1}{2}} \\
&\qquad - \theta \sum_i \left(\frac{\left(w_S^i S^i\right)_{k-}}{H} \left(\frac{f_S^i}{S^i}\right)_{k} \right)^{n+\frac{1}{2}} (HC)_{k}^{n+\frac{1}{2}}
\end{aligned}
\tag{6.30}
$$

for a layer not adjacent to the bed, and,

$$\Delta_1(HC)_1^{n+\frac{1}{2}} - \Delta_1(HC)_1^{n+\frac{1}{4}}$$
$$= \theta \sum_i \left(\left(w_S^i S^i\right)_{1+} \left(\frac{f_S^i}{S^i}\right)_2 \right)^{n+\frac{1}{2}} C_2^{n+\frac{1}{2}}$$
$$+ \theta \sum_i \left(\max\left(\frac{J_{SBS}^i f_S^i}{S^i}, 0\right) + \max\left(\frac{J_{SBS}^i F^i}{S^i}, 0\right)\left(f_w + \sum_j f_D^j\right) \right)_{sb}^{n+\frac{1}{2}} C_{sb}^{n+\frac{1}{2}}$$
$$+ \theta \sum_i \left(\min\left(\frac{J_{SBS}^i f_S^i}{S^i}, 0\right) + \min\left(\frac{J_{SBS}^i F^i}{S^i}, 0\right)\left(f_w + \sum_j f_D^j\right) \right)_1^{n+\frac{1}{2}} C_1^{n+\frac{1}{2}} \quad (6.31)$$
$$+ \theta \sum_i \left((1+\varepsilon) \max\left(\frac{J_{SBB}^i}{\rho_S^i}, 0\right)\left(f_w + \sum_j f_D^j\right) \right)_{sb}^{n+\frac{1}{2}} C_{sb}^{n+\frac{1}{2}}$$
$$+ \theta \sum_i \left((1+\varepsilon) \min\left(\frac{J_{SBB}^i}{\rho_S^i}, 0\right)\left(f_w + \sum_j f_D^j\right) \right)_1^{n+\frac{1}{2}} C_1^{n+\frac{1}{2}}$$

for the first layer adjacent to the bed. Note that (6.31) is also the appropriate form for single layer or depth average application. Since the sediment settling flux is zero at the top of the free surface adjacent layer, (6.27) is integrated downward from the top layer to the bottom layer. The bottom layer equation (6.31) is solved simultaneously with a corresponding equation for the top layer of the sediment bed. The settling fluxes, $w_S S$, and water column-sediment bed fluxes, J_{SB}, in (6.30) and (6.31) are known from the preceding solution for sediment settling, deposition and resuspension. Terms containing the sediment sorbed fraction divided by the sediment concentration in (6.30) and (6.31)

$$\frac{f_S^i}{S^i} = \frac{P_S^i}{\theta + \sum_i P_S^i S^i + \sum_j P_D^j D^j} \quad (6.32)$$

The diffusion step is given by

$$(HC)^{n+\frac{3}{4}} - (HC)^{n+\frac{1}{2}} = \theta \frac{\partial}{\partial z}\left(\frac{A_b}{H}\partial_z C\right) \quad (6.33)$$

with boundary conditions

$$-\frac{A_b}{H}\frac{\partial C}{\partial z} = \left(\left(\max(q_w, 0) + q_{dif}\right)\frac{1}{\phi}\left(f_w + \sum_j f_D^j\right)C \right)_{SB}$$
$$+ \left(\left(\min(q_w, 0) - q_{dif}\right)\frac{1}{\phi_{dep}}\left(f_w + \sum_j f_D^j\right)C \right)_{WC} : z = 0 \quad (6.34)$$

$$-\frac{A_b}{H}\frac{\partial C}{\partial z} = 0 : \; z = 1 \tag{6.35}$$

For the first layer adjacent to the bed

$$\begin{aligned}(HC)_1^{n+3/4} - (HC)_1^{n+1/2} \\ = \frac{\theta}{\Delta_1}\left(\frac{A_b}{H}\frac{\partial C}{\partial z}\right)_{1+}^{n+\frac{3}{4}} + \frac{\theta}{\Delta_1}\left(\max(q_w,0) + q_{dif}\right)\left(\left(f_w + \sum_j f_D^j\right)\frac{1}{\phi}\right)_{SB}^{n+\frac{1}{2}} C_{SB}^{n+\frac{3}{4}} \\ + \frac{\theta}{\Delta_1}\left(\min(q_w,0) - q_{dif}\right)\left(\left(f_w + \sum_j f_D^j\right)\frac{1}{\phi_{dep}}\right)_1^{n+\frac{1}{2}} C_1^{n+\frac{1}{2}}\end{aligned} \tag{6.36}$$

It is noted that the bed concentrations are advanced to the *n+3/4* intermediate time level before the advance of the water column concentrations. While for layers not adjacent to the bed,

$$(HC)_k^{n+\frac{3}{4}} - (HC)_k^{n+\frac{1}{2}} = \frac{\theta}{\Delta_1}\left(\frac{A_b}{H}\frac{\partial C}{\partial z}\right)_{k+}^{n+\frac{3}{4}} - \frac{\theta}{\Delta_1}\left(\frac{A_b}{H}\frac{\partial C}{\partial z}\right)_{k-}^{n+\frac{3}{4}} \tag{6.37}$$

The solution is completed by

$$(HC)_k^{n+1} - (H)_k^{n+3/4} = -\theta\gamma(HC)_k^{n+1} \tag{6.38}$$

an implicit reaction step.

6.2.3. Toxic Bed Processes

Contaminant transport in the sediment bed is represented using the discrete layer formulation developed for bed geomechanical processes. The conservation of mass for the total contaminant concentration in a layer of the sediment bed is given by

$$
\begin{aligned}
&\frac{\partial}{\partial t}(BC)_k = -\gamma(BC)_k \\
&-\delta(k,kt)\sum_i\left(\max\left(\frac{J^i_{SBS}f^i_S}{BS^i},0\right)+\max\left(\frac{J^i_{SBS}F^i}{BS^i},0\right)\left(f_w+\sum_j f^j_D\right)\right)_{kt}(BC)_{kt} \\
&-\delta(k,kt)\sum_i\left(\min\left(\frac{J^i_{SBS}f^i_S}{S^i},0\right)C+\min\left(\frac{J^i_{SBS}F^i_{dep}}{S^i_{dep}},0\right)\left(f_w+\sum_j f^j_D\right)C\right)_{WC} \\
&-\delta(k,kt)\sum_i\left(\left(J^i_{SBB}\chi^i_{SBL},0\right)\right) \\
&-\delta(k,kt)\sum_i\left((1+\varepsilon)\max\left(\frac{J^i_{SBB}}{B\rho^i_S},0\right)\left(f_w+\sum_j f^j_D\right)\right)_{kt}(BC)_{kt} \\
&-\delta(k,kt)\sum_i\left((1+\varepsilon_{dep})\min\left(\frac{J^i_{SBB}}{\rho^i_S},0\right)\left(f_w+\sum_j f^j_D\right)C\right)_{WC} \\
&-\left(\left(\max(q_w,0)+q_{dif}\right)_{k+}-\left(\min(q_w,0)-q_{dif}\right)_{k-}\right)\left(\frac{1}{\phi B}\left(f_w+\sum_j f^j_D\right)\right)_k(BC)_k \\
&-\delta(k,kt)\left(\min(q_w,0)-q_{dif}\right)_{kt+}\left(\frac{1}{\phi}\left(f_w+\sum_j f^j_D\right)C\right)_{WC} \\
&-\left(1\delta(k,kt)\right)\left(\min(q_w,0)-q_{dif}\right)_{k+}\left(\frac{1}{\phi B}\left(f_w+\sum_j f^j_D\right)\right)_{k+1}(BC)_{k+1} \\
&+\left(\max(q_w,0)+q_{dif}\right)_{k-}\left(\frac{1}{\phi B}\left(f_w+\sum_j f^j_D\right)\right)_{k-1}(BC)_{k-1}
\end{aligned}
\tag{6.39}
$$

where

$$
\delta(k,kt)=\begin{cases}0: & k=kt\\ 1: & k\neq kt\end{cases} \tag{6.40}
$$

is used to distinguish processes specific to the top, water column adjacent layer of the bed, *kt*. Advective fluxes associated with pore water advection in (6.40) are represented in upwind form. In the sediment bed, the actual computational variables for sediment, contaminant, and dissolved material are their concentrations times the thickness of the bed layer. Consistent with this formulation, the fractional phase components in the bed are defined by

$$
\begin{aligned}
(f_w)_k &= \left(\frac{BC_w}{BC}\right)_k = \left(\frac{B\phi}{B\phi+\sum_i P^i_S BS^i+\sum_j P^j_D BD^j}\right)_k \\
(f^j_D)_k &= \left(\frac{BC^j_D}{BC}\right)_k = \left(\frac{P^j_D BD^j}{B\phi+\sum_i P^i_S BS^i+\sum_j P^j_D BD^j}\right)_k \\
(f^i_S)_k &= \left(\frac{BC^i_S}{BC}\right)_k = \left(\frac{P^i_S BS^i}{B\phi+\sum_i P^i_S BS^i+\sum_j P^j_D BD^j}\right)_k
\end{aligned}
\tag{6.41}
$$

6.2.3.1 Bedload Transport

The contaminant fluxes associated bedload sediment transport are determined as follows.

$$m_x m_y J^i_{SBB} = \frac{\partial}{\partial x}\left(m_y Q^i_{SBLx}\right) + \frac{\partial}{\partial x}\left(m_x Q^i_{SBLy}\right) \tag{6.42}$$

is used to evaluate the flux associated with pore water entrainment and expulsion in (6.25) and (6.40). The transport equation for material sorbed to the bedload is

$$\frac{\partial}{\partial x}\left(m_y Q^i_{SBLx}\chi^i_{SBL}\right) + m_x\left(m_x Q^i_{SBy}\chi^i_{SBL}\right) = m_x m_y J^i_{SB}\chi^i_{SL} \tag{6.43}$$

Since the contaminant mass per sediment mass in the transport divergence corresponds to conditions in the top layer of the sediment bed, (6.43) can be written as

$$\frac{\partial}{\partial x}\left(m_y Q^i_{SBLx}\frac{f^i_S}{S^i}C\right) + \frac{\partial}{\partial x}\left(m_x Q^i_{SBLy}\frac{f^i_S}{S^i}C\right) = m_x m_y J^i_{SBB}\chi^i_{SBL} \tag{6.44}$$

and solved using an upwind approximation

$$\begin{aligned} m_x m_y J^i_{SB}\chi^i_{SBL} = {} & \max\left(m_y Q^i_{SBLx}\right)_E\left(\frac{f^i_S}{S^i}C\right)_C + \min\left(m_y Q^i_{SBLx}\right)_E\left(\frac{f^i_S}{S^i}C\right)_E \\ & - \max\left(m_y Q^i_{SBLx}\right)_W\left(\frac{f^i_S}{S^i}C\right)_W - \min\left(m_y Q^i_{SBLx}\right)_W\left(\frac{f^i_S}{S^i}C\right)_C \\ & + \max\left(m_x Q^i_{SBLy}\right)_N\left(\frac{f^i_S}{S^i}\right)_C + \min\left(m_x Q^i_{SBy}\right)_N\left(\frac{f^i_S}{S^i}C\right)_N \\ & - \max\left(m_x Q^i_{SBLy}\right)_S\left(\frac{f^i_S}{S^i}C\right)_S - \min\left(m_x Q^i_{SBLy}\right)_S\left(\frac{f^i_S}{S^i}C\right)_C \end{aligned} \tag{6.45}$$

to evaluate the transport of bedload sorbed material between horizontally adjacent top layers of the sediment bed.

6.2.3.2 Settling, Depostion and Resuspension

Equation (6.39) is solved using a fractional step procedure consistent with that used for the water column transport. Equation (6.41) is used to update the fractional distribution in the bed between the settling, deposition, and resuspension step and the pore water advection and diffusion step. The settling, deposition and resuspension step applies only to the top layer of the bed and is

$$
\begin{aligned}
&(BC)_{kt}^{n+\frac{1}{2}} - (BC)_{kt}^{n} \\
&\quad = -\theta \sum_i \left(\max\left(\frac{J_{SBS}^i f_S^i}{BS^i}, 0\right) \right. \\
&\quad + \max\left(\frac{J_{SBS}^i F^i}{BS^i}, 0\right)\left(f_w + \sum_j f_D^j\right) \Bigg)_{kt}^{n+\frac{1}{2}} (BC)_{kt}^{n+\frac{1}{2}} \\
&\quad - \theta \sum_i \left(\min\left(\frac{J_{SBS}^i f_S^i}{S^i}, 0\right) C + \min\left(\frac{J_{SBS}^i F_{dep}^i}{S_{dep}^i}, 0\right)\left(f_w + \sum_j f_D^j\right) C \right)_{WC}^{n+\frac{1}{2}} \\
&\quad - \theta \sum_i \left(\left(J_{SBB}^i \chi_{SBL}^i, 0\right) \right) \\
&\quad - \theta \sum_i \left((1+\varepsilon) \max\left(\frac{J_{SBB}^i}{B\rho_S^i}, 0\right)\left(f_w + \sum_j f_D^j\right) \right)_{kt}^{n+\frac{1}{2}} (BC)_{kt}^{n+\frac{1}{2}} \\
&\quad - \theta \sum_i \left((1+\varepsilon_{dep}) \min\left(\frac{J_{SBB}^i}{\rho_S^i}, 0\right)\left(f_w + \sum_j f_D^j\right) C \right)_{WC}^{n+\frac{1}{2}}
\end{aligned}
\tag{6.46}
$$

This equation is solved simultaneously with equation (6.31) for the bottom layer of the water column. The solution is represented by

$$
\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{Bmatrix} (BC)_{kt}^{n+\frac{1}{2}} \\ (HC)_1^{n+\frac{1}{2}} \end{Bmatrix} = \begin{Bmatrix} (BC)_{kt}^{n} - \theta \sum_i \left(J_S^i \chi_{SBL}^i\right)^{n+\frac{1}{2}} \\ i = ib \\ \Delta_1 (C)_1^{n+\frac{1}{4}} + \theta \sum_i \left(\left(w_S^i S^i\right)_{1+} \left(\frac{f_S^i}{S^i}\right)_2 \right)^{n+\frac{1}{2}} C_2^{n+\frac{1}{2}} \end{Bmatrix}
\tag{6.47}
$$

where the coefficients are given by

$$
\begin{aligned}
a_{11} = 1 + \theta \sum_i \left(\max\left(\frac{J_{SBS}^i f_S^i}{BS^i}, 0\right) + \max\left(\frac{J_{SBS}^i F^i}{BS^i}, 0\right)\left(f_w + \sum_j f_D^j\right) \right)_{kt}^{n+\frac{1}{2}} \\
+ \theta \sum_i \left((1+\varepsilon) \max\left(\frac{J_{SBB}^i}{B\rho_S^i}, 0\right)\left(f_w + \sum_j f_D^j\right) \right)_{kt}^{n+\frac{1}{2}}
\end{aligned}
\tag{6.48}
$$

$$a_{12} = \frac{\theta}{H}\sum_i \left(\min\left(\frac{J_{SBS}^i f_S^i}{S^i}, 0\right) C + \min\left(\frac{J_{SBS}^i F_{dep}^i}{S_{dep}^i}, 0\right)\left(f_w + \sum_j f_D^j\right)\right)_1^{n+\frac{1}{2}}$$

$$+ \frac{\theta}{H}\sum_i \left((1+\varepsilon_{dep}) \min\left(\frac{J_{SBB}^i}{\rho_S^i}, 0\right)\left(f_w + \sum_j f_D^j\right)\right)_1^{n+\frac{1}{2}}$$

$$a_{21} = -\theta\sum_i \left(\max\left(\frac{J_{SBS}^i f_S^i}{BS^i}, 0\right) + \max\left(\frac{J_{SBS}^i F^i}{BS^i}, 0\right)\left(f_w + \sum_j f_D^j\right)\right)_{kt}^{n+\frac{1}{2}}$$

$$- \theta\sum_i \left((1+\varepsilon) \max\left(\frac{J_{SBB}^i}{B\rho_S^i}, 0\right)\left(f_w + \sum_j f_D^j\right)\right)_{kt}^{n+\frac{1}{2}}$$

$$a_{22} = \Delta_1 - \frac{\theta}{H}\sum_i \left(\min\left(\frac{J_{SBS}^i f_S^i}{S^i}, 0\right) C + \min\left(\frac{J_{SBS}^i F^i}{S^i}, 0\right)\left(f_w + \sum_j f_D^j\right)\right)_1^{n+\frac{1}{2}}$$

$$- \frac{\theta}{H}\sum_i \left((1+\varepsilon) \min\left(\frac{J_{SBB}^i}{\rho_S^i}, 0\right)\left(f_w + \sum_j f_D^j\right)\right)_1^{n+\frac{1}{2}}$$

Adding the two equations in (6.46) gives

$$(BC)_{kt}^{n+\frac{1}{2}} + \Delta_1 (HC)_1^{+\frac{1}{2}}$$

$$= (BC)_{kt}^n + \Delta_1 (HC)_1^{n+\frac{1}{4}} + \theta\sum_i \left(\left(w_S^i S^i\right)_{1+} \left(\frac{f_S^i}{S^i}\right)_2 \right)^{n+\frac{1}{2}} C_2^{n+\frac{1}{2}}$$

$$- \theta\sum_i \left(J_S^i \chi_{SBL}^i\right)^{n+\frac{1}{2}} \qquad (6.49)$$

This equation verifies the consistency of the water column-sediment bed exchange since the source and sinks on the right side include only settling into the top of the water column layer, and transfer of bedload sediment sorbed contaminant between horizontal sediment bed cells.

6.2.3.3 Porewater Advection and Diffusion

The pore water advection and diffusion step for the top, water column adjacent, layer is

$$
\begin{aligned}
(BC)_{kt}^{n+3/4} = (BC)_{kt}^{n+1/2} &- \theta\left(\max(q_w,0)+q_{dif}\right)_{kt+}\left(\frac{1}{\phi}\left(f_w+\sum_j f_D^j\right)\right)_{kt}^{n+1/2}(BC)_{kt}^{n+3/4} \\
&+ \theta\left(\min(q_w,0)-q_{di}\right)_{kt-}\left(\frac{1}{\phi B}\left(f_w+\sum_j f_D^j\right)\right)_{kt}^{+1/2}(BC)_{kt}^{n+3/4} \\
&- \theta\left(\min(q_w,0)-q_{dif}\right)_{kt+}\left(\frac{1}{\phi H}\left(f_w+\sum_j f_D^j\right)\right)_{1}^{n+1/2}(HC)_1^{n+1/2} \\
&+ \theta\left(\max(q_w,0)+q_{dif}\right)_{kt-}\left(\frac{1}{\phi B}\left(f_w+\sum_j f_D^j\right)\right)_{kt-1}^{n+1/2}(BC)_{kt-1}^{n+1/2}
\end{aligned}
\tag{6.50}
$$

which is an implicit form. Writing (6.36) in the form

$$
\begin{aligned}
\Delta_1(HC)_1^{n+3/4} = \Delta_1(HC)_1^{n+1/2} &+ \theta\left(\frac{A_b}{H}\frac{\partial C}{\partial z}\right)_{1+}^{n+3/4} \\
&+ \theta\left(\max(q_w,0)+q_{dif}\right)_{kt+}\left(\frac{1}{\phi B}\left(f_w+\sum_j f_D^j\right)\right)_{SB}^{n+1/2}(BC)_{kt}^{n+3/4} \\
&+ \theta\left(\min(q_w,0)-q_{dif}\right)_{kt+}\left(\frac{1}{\phi H}\left(f_w+\sum_j f_D^j\right)\right)_{1}^{n+1/2}(HC)_1^{n+1/2}
\end{aligned}
\tag{6.51}
$$

and combining with (6.49) gives

$$
\begin{aligned}
(BC)_{kt}^{n+3/4} + \Delta_1(HC)_1^{n+3/4} &= (BC)_{kt}^{n+1/2} + \Delta_1(HC)_1^{n+1/2} + \theta\left(\frac{A_b}{H}\partial_z C\right)_{1+}^{n+3/4} \\
&+ \theta\left(\min(q_w,0)-q_{dif}\right)_{kt-}\left(\frac{1}{\phi B}\left(f_w+\sum_j f_D^j\right)\right)_{kt}^{n+1/2}(BC)_{kt}^{n+3/4} \\
&+ \theta\left(\max(q_w,0)+q_{dif}\right)_{kt-}\left(\frac{1}{\phi B}\left(f_w+\sum_j f_D^j\right)\right)_{kt-1}^{n+1/2}(BC)_{kt-1}^{n+1/2}
\end{aligned}
\tag{6.52}
$$

This equation verifies the consistency of the representation of pore water advection and diffusion across water column-sediment bed interface since the source and sink terms on the right side of (6.52) represent fluxes at the top to the water column cell and the bottom of the bed cell.

The pore water diffusion and advection step for the remaining bed layers is given by

$$(BC)_k^{n+3/4} = (BC)_k^{n+1/2} - \theta\left(\min(q_w,0) - q_{dif}\right)_{k+}\left(\frac{1}{\phi B}\left(f_w + \sum_j f_D^j\right)\right)_{k+1}^{n+1/2}(BC)_{k+1}^{n+3/4}$$
$$- \theta\left(\max(q_w,0) + q_{dif}\right)_{k+}\left(\frac{1}{\phi B}\left(f_w + \sum_j f_D^j\right)\right)_{k}^{n+1/2}(BC)_{k}^{n+3/4}$$
$$+ \theta\left(\min(q_w,0) - q_{dif}\right)_{k-}\left(\frac{1}{\phi B}\left(f_w + \sum_j f_D^j\right)\right)_{k}^{n+1/2}(BC)_{k}^{n+3/4} \qquad (6.53)$$
$$+ \theta\left(\max(q_w,0) + q_{dif}\right)_{k-}\left(\frac{1}{\phi B}\left(f_w + \sum_j f_D^j\right)\right)_{k-1}^{n+1/2}(BC)_{k-1}^{n+3/4}$$

For the bottom layer of the bed, $k = 1$, the bottom, k-, specific discharge and diffusion velocity must be specified as well as the total contaminant concentration, C_0. The corresponding thickness of the unresolved layer, $k = 0$, is set to unity without loss of generality. The system of equations represented by (6.49) and (6.52) is implicit and is solved using a tri-diagonal linear equation solver. It is noted that the *n+3/4* time level layer thickness is actually the *n+1* time level thickness determined by the solution of (6.23). The specific discharges in (6.49) and (6.52) are given by (6.41) and represent those appearing in (6.23) and guarantee mass conservation for the pore water advection.

The bed transport solution is completed by

$$(BC)_k^{n+1} - (BC)_k^{n+3/4} = -\theta\gamma(BC)_k^{n+1} \qquad (6.54)$$

an implicit reaction step.

6.3 Toxic Contaminant Loss Terms

6.3.1. Bulk Degradation

Bulk degradation can be included in the water column and/or the sediment bed for any toxic constituent. The bulk degradation in the water column follows a first order decay rate

$$\frac{dC_k}{dt} = -KC_k \qquad (6.55)$$

where

C is the toxic concentration in mg/m³ in layer k,

K is the first order decay rate in 1/s, and

t is time is seconds.

For bulk degradation there is no temperature effects on the degradation rate.

In the sediment bed, bulk degradation can be applied for sediment thickness up to a maximum sediment depth

$$\frac{dC_{b,k}}{dt} = -KC_{b,k}, \qquad \text{for} \sum_{k}^{kt} H_{bed,k} \le D_{max} \tag{6.56}$$

where

$C_{b,k}$ is the sediment bed toxic contaminant concentration in layer k (mg/g),

K is the bulk decay rate (1/s),

$H_{bed,k}$ is the sediment bed layer thickness (m), and

D_{max} is the maximum depth to use bulk degradation (m).

6.3.2. Biodegradation

Bacterial degradation, sometimes referred to as microbial transformation, biodegradation or biolysis, is the breakdown of a compound by the enzyme systems in bacteria. Although these transformations can detoxify and mineralize toxins and defuse potential toxins, they can also activate potential toxins.

Biodegradation in EFDC+ follows the bulk degradation approach shown previously

$$\frac{dC_k}{dt} = -K_{w,bio}C_k \tag{6.57}$$

and

$$\frac{dC_{b,k}}{dt} = -K_{b,bio}C_{b,k}, \qquad \text{for} \sum_{k}^{kt} H_{bed,k} \le D_{bio} \tag{6.58}$$

where

C_k is the toxic concentration in the water column in layer k (mg/m^3),

$C_{b,k}$ is the sediment bed toxic contaminant concentration in layer k (mg/g),

$K_{w,bio}$ is the water column biodegradation rate (1/s),

$K_{b,bio}$ is the sediment bed biodegradation rate (1/s),

$H_{bed,k}$ is the sediment bed layer thickness (m), and

D_{bio} is the maximum depth to apply biodegradation (m).

And where the degradation coefficient be temperature dependent according to

$$K_{bio} = K_{bio,ref} Q_{10}^{(T-20)/10} \tag{6.59}$$

where

$K_{bio,ref}$ is respective reference biodegradation rate at 20 °C (1/s),

Q_{10} is temperature correction factor for biodegradation, and

T is the relevant current temperature (°C).

The temperature correction factors represent the increase in the biodegradation rate constants resulting from a 10 °C temperature increase. Values in the range of 1.5 to 2 are common.

6.3.3. Volatilization

Volatilization is the movement of chemical across the air-water interface as the dissolved neutral concentration attempts to equilibrate with the gas phase concentration. Equilibrium occurs when the partial pressure exerted by the chemical in solution equals the partial pressure of the chemical in the overlying atmosphere. The rate of exchange is proportional to the gradient between the dissolved concentration and the concentration in the overlying atmosphere and the conductivity across the interface of the two fluids. The conductivity is influenced by both chemical properties (molecular weight, Henry's Law constant) and environmental conditions at the air-water interface (turbulence-controlled by wind speed, current velocity, and water depth).

In EFDC+ volatilization of a dissolved toxic constituent is computed by

$$\left.\frac{\partial C}{\partial t}\right|_{volat} = \frac{K_v}{H_{KC}}\left(f_d C - \frac{C_a}{\frac{H_L}{RT_K}} \right) \tag{6.60}$$

where

C is the water column concentration in layer KC, mg/m³,

K_v is the transfer rate, m/day

H_{kc} is the water column thickness of the layer KC, m

f_d is the fraction of the total chemical that is dissolved,

C_a is the atmospheric concentration, mg/m³,

R is the universal gas constant, 8.206x10-5 atm-m³/mole °K

T_K is the water temperature, °K

H_L is the Henry's law coefficient for the air-water partitioning of the toxic, atm-m³/mole.

Equilibrium occurs when the dissolved concentration equals the partial pressure divided by Henry's Law Constant.

The dissolved concentration of a chemical in a surface water column segment can volatilize at a rate determined by the two-layer resistance model (Whitman, 1923). The two-resistance method assumes that two "stagnant films" are bounded on either side by well mixed compartments. Concentration differences serve as the driving force for the water layer diffusion. Pressure differences drive the diffusion for the air layer. From mass balance considerations, it is obvious that the same mass must pass through both films, thus the two resistances combine in series, so that the conductivity is the reciprocal of the total resistance:

$$K_v = \left(R_L + R_G\right)^{-1} = \left[K_L^{-1} + \left(K_G \frac{H_L}{RT_K} \right)^{-1} \right]^{-1} \tag{6.61}$$

where

R_L is the liquid phase resistance, day/m

K_L is the liquid phase transfer coefficient, m/day

R_G is the gas phase resistance, day/m

K_G is the gas phase transfer coefficient, m/day.

There is yet another resistance involved, the transport resistance between the two interfaces, but it is assumed to be negligible. This may not be true in two cases: very turbulent conditions and in the presence of surface-active contaminants. Although this two-resistance method, the Whitman model, is rather simplified in its assumption of uniform layers, it has been shown to be as accurate as more complex models.

The value of K_v, the conductivity, depends on the intensity of turbulence in a water body and in the overlying atmosphere. Mackay and Leinonen (1975) have discussed conditions under which the value of K_v is primarily determined by the intensity of turbulence in the water. As the Henry's Law coefficient increases, the conductivity tends to be increasingly influenced by the intensity of turbulence in water. As the Henry's Law coefficient decreases, the value of the conductivity tends to be increasingly influenced by the intensity of atmospheric turbulence.

The computed volatilization rate from Eq. (6.61) is for a temperature of 20 °C. It is adjusted for water temperature using the equation:

$$K_{v,T} = K_v \Theta^{T-20} \tag{6.62}$$

where

Θ is the temperature correction factor,

T is the water temperature, ˚C.

The liquid and gas film transfer coefficients computed under this option vary with the type of waterbody. The type of waterbody is specified as one of the volatilization constants and can either be a flowing stream, river or estuary or a stagnant pond or lake. The primary difference is that in a flowing waterbody the turbulence is primarily a function of the stream velocity, while for stagnant waterbodies wind shear may dominate. The formulations used to compute the transfer coefficients vary with the waterbody type as shown below.

EFDC+ automatically determines which flow regime to apply based on the following criteria

$$\begin{cases} H > H_{max}, & \text{Lake conditions} \\ H \le H_{max}, & \text{River conditions} \end{cases} \tag{6.63}$$

or

$$\begin{cases} U \le U_{max}, & \text{Lake conditions} \\ U > U_{max}, & \text{River conditions} \end{cases} \tag{6.64}$$

where

H is total depth (m),

H_{max} is maximum depth allowed for river conditions (m),

U is the depth averaged velocity magnitude (m/s), and

U_{max} is the maximum lake velocity magnitude (m/s).

6.3.3.1 *Flowing Stream, River or Estuary*

For a flowing system the transfer coefficients are controlled by flow induced turbulence. For flowing conditions, the liquid film transfer coefficient (K_L) is computed using the Covar method (Covar, 1976) in which the equation used varies with the velocity and depth of the cell.

For cells with depths less than 0.61 m the Owens formula is used to calculate the oxygen reaeration rate:

$$K_L = \frac{5.349}{86400}\frac{U^{0.67}}{H^{1.5}} \tag{6.65}$$

where

U is the depth averaged water velocity magnitude (m/s), and

H is cell depth (m).

For segments with a velocity less than 0.518 m/s or a depth (m) greater than 13.584 $U^{2.9135}$ the O'Connor-Dobbins formula is used:

$$K_L = D_w\frac{U^{0.5}}{H^{1.5}} \tag{6.66}$$

where D_w is the diffusivity of the chemical in water (m²/s), computed from

$$D_w = \frac{22\cdot 10^{-9}}{M_w^{0.6667}} \tag{6.67}$$

In all other cases, the Churchill formula is used to calculate reaeration rate:

$$K_L = \frac{5.049}{86400}\frac{U^{0.969}}{H^{1.673}} \tag{6.68}$$

The gas transfer coefficient (K_G) is assumed constant at 100 m/day for flowing systems.

6.3.3.2 *Lake or Pond*

For more quiescent conditions the transfer coefficients are controlled by wind induced turbulence. For these systems, the liquid film transfer coefficient (K_L) is computed using either the O'Connor equations or Mackay and Yeun (1983).

(1) Option 1 O'Connor Approach

$$K_L = u_*\left(\frac{\rho_a}{\rho_w}\right)^{0.5}\frac{\kappa^{0.33}}{\lambda_2}S_{cw}^{-0.67} \tag{6.69}$$

$$K_G = u_*\frac{\kappa^{0.33}}{\lambda_2}S_{ca}^{-0.67} \tag{6.70}$$

where u_* is the shear velocity (m/s) computed from

$$u_* = C_d^{0.5} W_{10} \tag{6.71}$$

where

C_d is drag coefficient (0.0011),

W_{10} is wind velocity at 10 m above water surface (m/s),

ρ_a is density of air, internally calculated from air temperature (kg/m^3),

ρ_w is density of water, internally calculated from water temperature (kg/m^3),

κ is von Karman's constant (0.44),

λ_2 is dimensionless viscous sublayer thickness (4),

S_{ca} and S_{cw} are air and water Schmidt Numbers, computed from

$$S_{ca} = \frac{\mu_a}{\rho_a D_a} \tag{6.72}$$

$$S_{cw} = \frac{\mu_w}{\rho_w D_w} \tag{6.73}$$

where

D_a is diffusivity of chemical in air (m^2/s),

D_w is diffusivity of chemical in water (m^2/s),

μ_a is viscosity of air, internally calculated from air temperature (kg/m-sec),

μ_w is viscosity of water, internally calculated from water temperature (kg/m-sec).

The diffusivity of the chemical in water is computed using Eq. (6.67) while the diffusivity of the chemical in air (D_a, m^2/sec) is computed from

$$D_a = \frac{1.9 \cdot 10^{-4}}{M_w^{2/3}} \tag{6.74}$$

This K_G is proportional to wind and inversely proportional to molecular weight to the 4/9 power.

(2) Option 2 Mackay and Yeun Approach

Under this option, the liquid and gas film transfer coefficients are computed using formulations described by Mackay and Yeun (1983). The Mackay equations are:

$$K_L = \begin{cases} 10^{-6} + 0.00341 u_* S_{cw}^{-0.5}, & u_* > 0.3 \text{ m/s} \\ 10^{-6} + 0.01441 u_*^{2.2} S_{cw}^{-0.5}, & u_* < 0.3 \text{ m/s} \end{cases} \tag{6.75}$$

$$K_G = 10^{-3} + 0.0462 u_* S_{ca}^{-0.67} \tag{6.76}$$

6.3.3.3 *Volatilization Input Data*

Although there are many calculations involved in determining volatilization, most are performed internally using a small set of data. Volatilization data specifications are summarized in Table 6.1. Not all of the constants are required. Volatilization is only active for the surface layer.

Table 6.1: Volatilization Input Data

Description	Notation	Range	Units
Measured or calibrated conductance	K_v	0.6 – 25	m/day
Henry's Law Constant	H	10^{-7} – 10^{-1}	atm-m^3/mole
Concentration of chemical in atmosphere	C_a	0 – 1000	µg/L
Molecular weight	M_w	10 – 10^3	g/mole
Reaeration coefficient (conductance of oxygen)	K_a	0.6 – 25	m/day
Experimentally measured ratio of volatilization to reaeration	k_{vo}	0 – 1	
Current velocity	u_x	0.2	m/s
Water depth	D	0.1 – 10	m
Water temperature	T	4 – 30	°C
Wind speed 10m above surface	W_{10}	0 – 20	m/s

7 EFDC WATER QUALITY MODEL

7.1 Introduction

This section summarizes the basic theory of the water quality-eutrophication component of the EFDC model. The kinetic processes included in the EFDC water quality model are derived from the CE-QUAL-ICM water quality model (Cerco and Cole, 1995) as described in Park *et al.* (1995). This document describes the current model formulation including comparisons with subsequent published documentation of CE-QUAL-ICM model applications. Table 7.1 lists the model's complete set of state variables and their interactions are illustrated in Figure 7-1. As opposed to earlier water quality models such as WASP (Ambrose *et al.*, 1992), which use biochemical oxygen demand to represent oxygen demanding organic material, the EFDC water quality model is carbon based. The four algae species are represented in carbon units. The three organic carbon variables play an equivalent role to BOD. Organic carbon, nitrogen and phosphorous can be represented by up to three reactive sub-classes, refractory particulate, labile particulate and labile dissolve. The use of the sub-classes allows a more realistic distribution of organic material by reactive classes when data are to estimate distribution factors. The following sub-sections discuss the role of each variable and summarize their kinetic interaction processes. The kinetic processes include the exchange of fluxes at the sediment-water interface, including sediment oxygen demand. The description of the EFDC water column water quality model in this section closely follows Park *et al.* (1995).

Table 7.1: EFDC model water quality state variables

#	Water quality state variable	Acronyms	Units	Group
1	Cyanobacteria	B_c	mg/l C	Algae
2	Diatom algae	B_d	mg/l C	
3	Green algae	B_g	mg/l C	
4	Refractory particulate organic carbon	RPOC	mg/l	Organic carbon
5	Labile particulate organic carbon	LPOC	mg/l	
6	Dissolved organic carbon	DOC	mg/l	
7	Refractory particulate organic phosphorus	RPOP	mg/l	Phosphorus
8	Labile particulate organic phosphorus	LPOP	mg/l	
9	Dissolved organic phosphorus	DOP	mg/l	
10	Total phosphate	PO4	mg/l	
11	Refractory particulate organic nitrogen	RPON	mg/l	Nitrogen
12	Labile particulate organic nitrogen	LPON	mg/l	
13	Dissolved organic nitrogen	DON	mg/l	
14	Ammonia nitrogen	NH4	mg/l	
15	Nitrate and nitrite nitrogen	NO3, NO23, NOX	mg/l	
16	Particulate biogenic silica	SU	mg/l	Silica
17	Dissolved available silica	SA	mg/l	
18	Chemical oxygen demand	COD	mg/l	Others
19	Dissolved oxygen	DO	mg/l	
20	Total active metal	TAM	mole/m^3	
21	Fecal coliform bacteria	FCB	MPN/100ml	
22	Macroalgae/Periphyton	B_m	mg/l C	

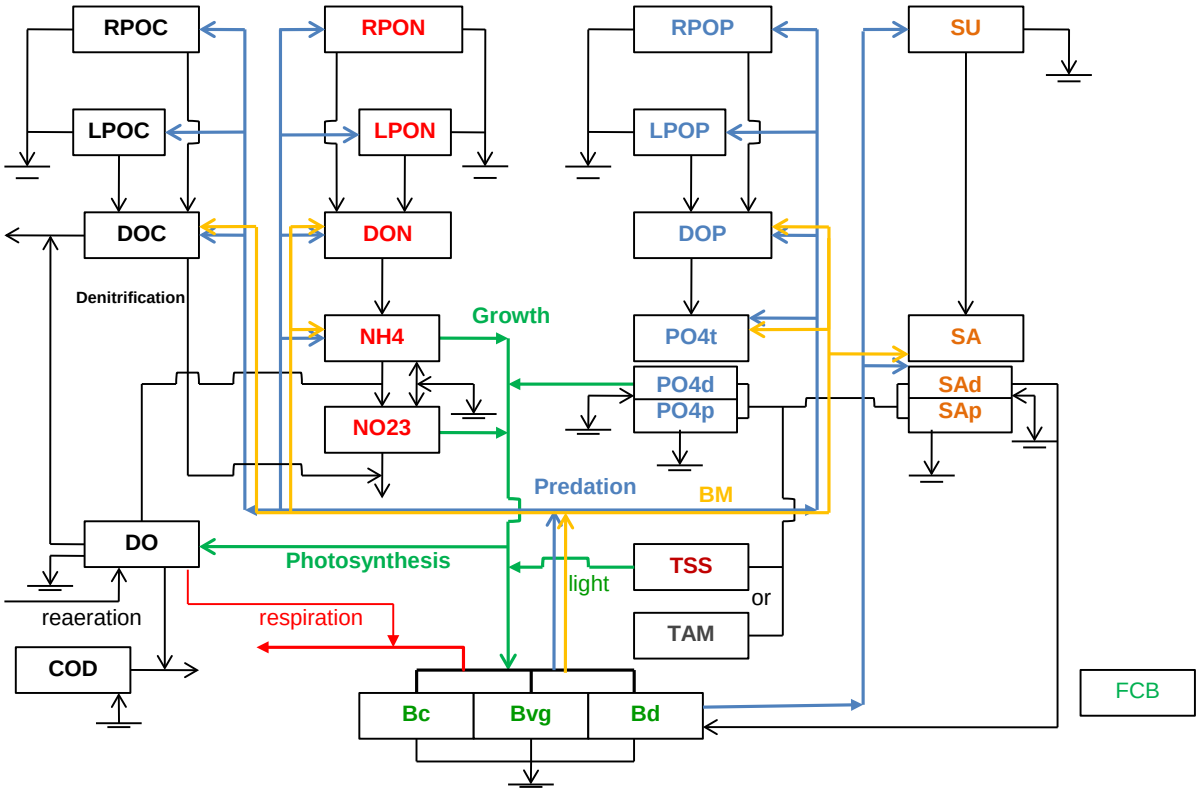


Figure 7-1: Schematic diagram of EFDC Water Quality Model Structure.

7.2 Water Column Eutrophication Formulation

7.2.1. Model State Variables

7.2.1.1 Algae

Algae are grouped into four model classes: cyanobacteria, diatoms, greens, and stationary. The grouping is based upon the distinctive characteristics of each class and upon the significant role the characteristics play in the ecosystem. Cyanobacteria, commonly called blue-green algae, are characterized by their abundance (as picoplankton) in saline water and by their bloom-forming characteristics in fresh water. Cyanobacteria are unique in that some species fix atmospheric nitrogen, although nitrogen fixers are not believed to be predominant in many river systems. Diatoms are distinguished by their requirement of silica as a nutrient to form cell walls. Diatoms are large algae characterized by high settling velocities. Settling of spring diatom blooms to the sediments may be a significant source of carbon for sediment oxygen demand. Algae that do not fall into the preceding two groups are lumped into the heading of green algae. Green algae settle at a rate intermediate between cyanobacteria and diatoms and are subject to greater grazing pressure than cyanobacteria.

A stationary or non-transported algae (constituent 22) variable is included in the model and has been used to simulate macroalgae/peripyton. The stationary algae variable has the same kinetic formulation as the original algae groups, with the exception that it is not transported. The stationary algae group can also be used various types of bottom substrate attached or floating periphyton. It is also noted that in applications requiring simulation of multiple

classes of stationary algae, one or more of the three transported algae groups can be switched to stationary model, under the restriction that the total number of transport and stationary algae classes remains less than or equal to four. Appendix A provides additional specifics with respect to model configuration simulation of macroalgae and periphyton.

7.2.1.2 Organic Carbon

Three organic carbon state variables are considered: dissolved, labile particulate, and refractory particulate. Labile and refractory distinctions are based upon the time scale of decomposition. Labile organic carbon decomposes on a time scale of days to weeks whereas refractory organic carbon requires more time. Labile organic carbon decomposes rapidly in the water column or the sediments. Refractory organic carbon decomposes slowly, primarily in the sediments, and may contribute to sediment oxygen demand years after deposition.

7.2.1.3 Nitrogen

Nitrogen is first divided into organic and mineral fractions. Organic nitrogen state variables are dissolved organic nitrogen, labile particulate organic nitrogen, and refractory particulate organic nitrogen. Two mineral nitrogen forms are considered: ammonium and nitrate. Both are utilized to satisfy algal nutrient requirements, although ammonium is preferred from thermodynamic considerations. The primary reason for distinguishing the two is that ammonium is oxidized by nitrifying bacteria into nitrate. This oxidation can be a significant sink of oxygen in the water column and sediments. An intermediate in the complete oxidation of ammonium, nitrite, also exists. Nitrite concentrations are usually much less than nitrate, and for modeling purposes, nitrite is combined with nitrate. Hence, the nitrate state variable actually represents the sum of nitrate plus nitrite.

7.2.1.4 Phosphorus

As with carbon and nitrogen, organic phosphorus is considered in three states: dissolved, labile particulate, and refractory particulate. Only a single mineral form, total phosphate, is considered. Total phosphate exists as several states within the model ecosystem: dissolved phosphate, phosphate sorbed to inorganic solids, and phosphate incorporated in algal cells. Equilibrium partition coefficients are used to distribute the total among the three states.

7.2.1.5 Silica

Silica is divided into two state variables: available silica and particulate biogenic silica. Available silica is primarily dissolved and can be utilized by diatoms. Particulate biogenic silica cannot be utilized. In the model, particulate biogenic silica is produced through diatom mortality. Particulate biogenic silica undergoes dissolution to available silica or else settles to the bottom sediments.

7.2.1.6 Chemical Oxygen Demand

In the context of this study, chemical oxygen demand is the concentration of reduced substances that are oxidizable by inorganic means. The primary component of chemical oxygen demand is sulfide released from sediments. Oxidation of sulfide to sulfate may remove substantial quantities of dissolved oxygen from the water column.

7.2.1.7 Dissolved Oxygen

Dissolved oxygen is required for the existence of higher life forms. Oxygen availability determines the distribution of organisms and the flows of energy and nutrients in an ecosystem. Dissolved oxygen is a central component of the water quality model.

7.2.1.8 Total Active Metal

Both phosphate and dissolved silica adsorb to inorganic solids, primarily iron and manganese. Sorption and subsequent settling is one pathway for removal of phosphate and silica from the water column. Consequently, the concentration and transport of iron and manganese are represented in the model. However, limited data do not allow a complete treatment of iron and manganese chemistry. Rather, a single-state variable, total active metal, is defined as the total concentration of metals that are active in phosphate and silica transport. Total active metal is partitioned between particulate and dissolved phases by an oxygen-dependent partition coefficient. Inorganic suspended solids can be used, in lieu of total active metal, as a sorption sited for phosphate and silica. Inorganic suspended solids concentration is provided by the sediment transport component of the EFDC modeling system.

7.2.1.9 Salinity

Salinity is a conservative tracer that provides verification of the transport component of the model and facilitates examination of conservation of mass. Salinity also influences the dissolved oxygen saturation concentration and is used in the determination of kinetics constants that differ in saline and fresh water. Salinity is simulated in the hydrodynamic component of the model.

7.2.1.10 Temperature

Temperature is a primary determinant of the rate of biochemical reactions. Reaction rates increase as a function of temperature, although extreme temperatures result in the mortality of organisms. Temperature is simulated in the hydrodynamic component of the model.

7.2.2. Conservation of Mass Equation

The governing mass-balance equation for each of the water quality state variables may be expressed as:

$$\frac{\partial}{\partial t}(m_x m_y HC) + \frac{\partial}{\partial x}(m_y HuC) + \frac{\partial}{\partial y}(m_x HvC) + \frac{\partial}{\partial z}(m_x m_y wC) = \frac{\partial}{\partial x}\left(\frac{m_y HA_x}{m_x}\frac{\partial C}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{m_x HA_y}{m_y}\frac{\partial C}{\partial y}\right) + \frac{\partial}{\partial z}\left(m_x m_y \frac{A_z}{H}\frac{\partial C}{\partial z}\right) + m_x m_y HS_c \quad (7.1)$$

where

C is the concentration of a water quality state variable,

u, v, w are velocity components in the curvilinear, sigma, x-, y-, and z-directions, respectively,

A_x, A_y, A_z are turbulent diffusivities in the x-, y-, and z-directions, respectively,

S_c is the internal and external sources and sinks per unit volume,

H is the water column depth, and

m_x, m_y are horizontal curvilinear coordinate scale factors.

The last three terms on the left-hand side (LHS) of (7.1) account for the advective transport, and the first three terms on the right-hand side (RHS) account for the diffusive transport. These six terms for physical transport are analogous to, and thus the numerical method of solution is the same as, those in the mass-balance equation for salinity in the hydrodynamic model (Hamrick, 1992). The last term in (7.1) represents the kinetic processes and external loads for each of the state variables. The present model solves equation (7.1) using a fractional step procedure which decouples the kinetic terms from the physical transport terms.

$$\frac{\partial}{\partial t_p}(m_x m_y HC) + \frac{\partial}{\partial x}(m_y HuC) + \frac{\partial}{\partial y}(m_x HvC) + \frac{\partial}{\partial z}(m_x m_y wC) = \frac{\partial}{\partial x}\left(\frac{m_y HA_x}{m_x}\frac{\partial C}{\partial x}\right) + \frac{\partial}{\partial y}\left(\frac{m_x HA_y}{m_y}\frac{\partial C}{\partial y}\right) + \frac{\partial}{\partial z}\left(m_x m_y \frac{A_z}{H}\frac{\partial C}{\partial z}\right) + m_x m_y H S_{CP} \quad (7.2)$$

$$\frac{\partial C}{\partial t_K} = S_{CK} \quad (7.3)$$

with

$$\frac{\partial}{\partial t}(m_x m_y HC) = \frac{\partial}{\partial t_p}(m_x m_y HC) + (m_x m_y H)\frac{\partial C}{\partial t_K} \quad (7.4)$$

In equation (7.2) the source sink term has been split into physical sources and sinks which are associated in volumetric inflows and outflows, and kinetic sources and sinks. Since variations in the water column depth are coupled with the divergence of the volume transport field, the kinetic step is made at a constant water column depth corresponding to the depth field at the end for the physical transport step. This allows the depth and scale factors to be eliminated from the kinetic step (7.3) which can be further split into reactive and internal sources and sinks.

$$\frac{\partial C_K}{\partial t} = KC + R \quad (7.5)$$

Where K is kinetic rate (time^{-1}) and R represents internal source/sink term (mass volume^{-1} time^{-1}). Equation (7.5) is obtained by linearizing some terms in the kinetic equations, mostly Monod type expressions. Hence, K and R are known values in equation (7.5). Equation (7.2) is identical to, and thus its numerical method of solution is the same as, the mass-balance equation for salinity (Hamrick, 1992). The solution scheme for both the physical transport (Hamrick, 1992) and the kinetic equations is second-order accurate.

7.2.3. Kinetic Equations for State Variables

The remainder of this chapter details the kinetics portion of the mass-conservation equation for each state variable. Parameters are defined where they first appear. All parameters are listed, in alphabetical order, in an appendix. For consistency with reported rate coefficients, kinetics are detailed using a temporal dimension of days. Within the CE-QUAL-ICM computer code, kinetics sources and sinks are converted to a dimension of seconds before employment in the mass-conservation equation.

7.2.3.1 Algae

Algae, which occupy a central role in the model Figure 7-1: Schematic diagram of EFDC Water Quality Model Structure.), are grouped into three model state variables: cyanobacteria (blue-green algae), diatoms, and green algae. The subscript, x, is used to denote four algal groups: c for cyanobacteria, d for diatoms, g for green algae, and m for macroalgae. Sources and sinks included in the model are:

- Growth (production)
- Basal metabolism

- Predation
- Settling
- External loads

Equations describing these processes are largely the same for the four algal groups with differences in the values of parameters in the equations. The kinetic equation describing these processes is:

$$\frac{\partial B_x}{\partial t} = (P_x - BM_x - PR_x)B_x + \frac{\partial}{\partial Z}(WS_x B_x) + \frac{WB_x}{V} \tag{7.6}$$

where

B_x is the algal biomass of algal group x (g C/m^3),

t is the time (days)

P_x is the production rate of algal group x (1/day),

BM_x is the basal metabolism rate of algal group x (1/day),

PR_x is the predation rate of algal group x (1/day),

WS_x is the positive settling velocity of algal group x (m/day),

WB_x is the external loads of algal group x (g C/day),

V is the cell volume (m^3).

(1) Production (Algal Growth)

Algal growth depends on nutrient availability, ambient light, and temperature. The effects of these processes are considered to be multiplicative:

$$P_x = PM_x f_1(N) f_2(I) f_3(T) f_4(S) \tag{7.7}$$

where

PM_x is the maximum growth rate under optimal conditions for algal group x (1/day),

$f_1(N)$ is the effect of suboptimal nutrient concentration ($0 \le f_1 \le 1$),

$f_2(I)$ is the effect of suboptimal light intensity ($0 \le f_2 \le 1$),

$f_3(T)$ is the effect of suboptimal temperature ($0 \le f_3 \le 1$),

$f_4(S)$ is the effect of salinity on cyanobacteria growth ($0 \le f_4 \le 1$).

The freshwater cyanobacteria may undergo rapid mortality in salt water, e.g., freshwater organisms in the Potomac River (Thomann *et al.*, 1985). For the freshwater organisms, the increased mortality is included in the model by the salinity toxicity term in the growth equation for cyanobacteria. Activation of the salinity toxicity term, f_4 (S), is an option in the source code.

(2) Effect of Nutrients on Algal Growth

Using Liebig's "law of the minimum" (Odum, 1971) that growth is determined by the nutrient in least supply, the nutrient limitation for growth of cyanobacteria and green algae is expressed as:

$$f_1(N) = \left(\frac{NH4 + NO3}{KHN_x + NH4 + NO3}, \frac{PO4d}{KHP_x + PO4d}, \frac{SAd}{KHS + SAd}\right) \quad (7.8)$$

where

NH4 is the ammonium nitrogen concentration (g N/m³),

NO3 is the nitrate nitrogen concentration (g N/m³),

KHN_x is the half-saturation constant for nitrogen uptake for algal group *x* (g N/m³),

PO4d is the dissolved phosphate phosphorus concentration (g P/m³),

KHP_x is the half-saturation constant for phosphorus uptake for algal group *x* (g P/m³),

SAd is the concentration of dissolved available silica (g Si/m³),

KHS is the half-saturation constant for silica uptake for diatoms (g Si/m³).

Some cyanobacteria (e.g., *Anabaena*) can fix nitrogen from atmosphere and thus are not limited by nitrogen. Hence, equation (7.8) is not applicable to the growth of nitrogen fixers. Since diatoms require silica as well as nitrogen and phosphorus for growth, the nutrient limitation for diatoms includes silica limitation.

(3) Effect of Light on Algal Growth

The Light Field

The light field in the water column is governed by

$$\frac{\partial I}{\partial Z_*} = -Kess\ I \quad (7.9)$$

where

I is the light intensity (W/m²),

Kess is the light extinction coefficient (1/m),

*Z** is the depth below the water surface (m),

with the light extinction coefficient being a function of the depth below the water surface. Integration of (7.9) gives

$$I = I_{ws} \exp\left(-\int_0^{Z_*} Kess\ dZ_*\right) \quad (7.10)$$

The light intensity at the water surface, I_{ws}, is given by

$$I_{ws} = I_o \min\left(\exp\left(-Keme(H_{RPS} - H)\right), 1\right) \quad (7.11)$$

where

I_o is the light intensity at the top of the emergent aquatic plant canopy for emergent shoots or the light intensity at the water surface for submerged shoots (W/m²),

Keme is the light extinction coefficient for emergent shoots (1/m),

H_{RPS} is the rooted plant shoot height (m),

H is the water column depth (m).

When submerged aquatic plants are simulated, it is assumed that the light extinction coefficient in the water column above the canopy is given by

$$Kessac = Ke_b + Ke_{ISS} \cdot ISS + Ke_{VSS} \cdot VSS + Ke_{Chl} \sum_{m=1}^{M} \left(\frac{B_m}{CChl_m} \right) \tag{7.12}$$

and that the light extinction coefficient in the water column within the canopy is given by

$$Kessic = Ke_b + Ke_{ISS} \cdot SED + Ke_{VSS} \cdot VSS + Ke_{Chl} \sum_{m=1}^{M} \left(\frac{B_m}{CChl_m} \right) + Ke_{RPS} \cdot RPS \tag{7.13}$$

where

Ke_b is the background light extinction (1/m)

Ke_{ISS} is the light extinction coefficient for inorganic suspended solid (1/m per g/m^3)

SED is the inorganic suspended solid concentration (g/m^3) provided from the hydrodynamic model

Ke_{VSS} is the light extinction coefficient for volatile suspended solid (1/m per g/m^3)

VSS is the volatile suspended solid concentration (g/m^3) provided from the water quality model

$CChl_{RPE}$ is the carbon-to-chlorophyll ratio for epiphytes (g C per mg Chl).

Ke_{Chl} is the light extinction coefficient for algae chlorophyll (1/m per mg Chl/m^3)

B_m is the concentration of algae group m (g C per ml)

$CChl_m$ is the carbon-to-chlorophyll ratio in algal group m (g C per mg Chl).

Ke_{RPS} is the light extinction coefficient for rooted plant shoots (1/m per gm C/m^2)

RPS is the concentration of plant shoots (g C per m^2)

The forms (7.12) and (7.13) are quite general and readily allow inclusion of algae biomass into the volatile suspended solids or visa-versa. The form of (7.13) assumes that the rooted plant shoots are primarily self-shading and that epiphyte effect are manifest only on the shoot surface.

The solutions of (7.10) above and in the rooted plant shoot canopy are

$$I = I_{ws} \exp(-Kessac\ Z_*)\ ;\ 0 \le Z_* \le H - H_{RPS} \tag{7.14}$$

$$I = I_{ct} \exp\left(-Kesic\ (Z_* - H + H_{RPS})\right)$$

$$I_{ct} = I_{ws} \exp\left(-Kessac\ (H - H_{RPS})\right) \tag{7.15}$$

$$H - H_{RPS} \le Z_* \le H$$

Steele's Equation for Light Limitation

The original version of CE-QUAL-ICM (Cerco and Cole, 1995) used Steele's equation

$$f_2(I) = \frac{I}{I_{sx}} \exp\left(1 - \frac{I}{I_{sx}}\right) \tag{7.16}$$

to express light limitation. A daily and vertically integrated form of Steele's equation, in the absences of a plant canopy is:

$$f_2 = \frac{\exp(1)\ FD}{Kess\ (ZB - ZT)} (\exp(-\alpha_b) - \exp(-\alpha_T)) \tag{7.17}$$

$$\alpha_B = \left(\frac{I_{wsavg}}{FDI_{sx}}\right) \exp(-Kess\ ZB) \tag{7.18}$$

$$\alpha_T = \left(\frac{I_{wsavg}}{FDI_{sx}}\right) \exp(-Kess\ ZT) \tag{7.19}$$

where

FD is the fractional day length ($0 \le FD \le 1$),

$Kess$ is the total light extinction coefficient (1/m),

ZT is the distance from water surface to layer top (m),

ZB is the distance from water surface to layer bottom (m),

I_{wsavg} is the daily total light intensity at water surface (langleys/day),

I_{sx} is the optimal light intensity for algal group x (langleys/day).

Optimal light intensity I_{sx} for photosynthesis depends on algal taxonomy, duration of exposure, temperature, nutritional status, and previous acclimation. Variations in I_s are largely due to adaptations by algae intended to maximize production in a variable environment. Steel (1962) noted the result of adaptations is that optimal intensity is a consistent fraction (approximately 50 percent) of daily intensity. Kremer and Nixon (1978) reported an analogous finding that maximum algal growth occurs at a constant depth (approximately 1 m) in the water column. Their approach is adopted so that optimal intensity is expressed as:

$$I_{sx} = \min\left(I_{wsavg} \exp\left(-Kess\ D_{optx}\right), I_{sxmin}\right) \tag{7.20}$$

where

D_{optx} is the depth of maximum algal growth for algal group x (m),

I_{wsavg} is the adjusted surface light intensity (W/m^2),

I_{sxmin} is the minimum optimum light intensity (W/m^2).

A minimum, I_{sxmin}, in equation (7.20) is specified so that algae do not thrive at extremely low light levels. The time required for algae to adapt to changes in light intensity is recognized by estimating I_{sxmin} based on a time-weighted average of daily light intensity:

$$I_{0avg} = CI_a I_0 + CI_b I_1 + CI_c I_2 \tag{7.21}$$

where

I_1 is the daily light intensity 1 day preceding model day (langleys/day),

I_2 is the daily light intensity 2 days preceding model day (langleys/day),

CI_a, CI_b, CI_c are the weighting factors for $I0$, I_1 and I_2, respectively: $CI_a + CI_b + CI_c = 1$.

Equations ((7.17)-(7.19)) can be applied instantaneously by setting the fraction of daylight to unity. It can also be applied within a canopy by replacing I_{WS} with I_{Ct} defined by (7.15).

Alternate Formulations for Light Limitation

Subsequent applications of CE-QUAL-ICM (Bunch *et al.*, 2000) used a Monod type limitation

$$f_2(I) = \frac{I}{KHI + I} \tag{7.22}$$

or a modified Monod limitation (Cerco *et al.*, 2000, 2002, 2004; Tillman *et al.*, 2004)

$$f_2(I) = \frac{I}{\sqrt{KHI^2 + I^2}} \tag{7.23}$$

where

KHI is the half saturation for light limitation (W/m^2).

Equation (7.23) was used in the CE-QUAL-ICM Florida Bay water quality modeling study (Cerco *et al.*, 2000). Equation (7.22) can be directly averaged over a water column layer to give

$$f_{2avg} = \frac{1}{Kess(ZB - ZT)} \ln\left(\frac{KHI + I_{ws}\, exp(-Kess\ ZT)}{KHI + I_{ws}\, exp(-Kess\ ZB)}\right) \tag{7.24}$$

while the average of (7.23) is

$$f_{2avg} = \frac{1}{Kess} \frac{1}{ZB - ZT} \left(\sqrt{1 + \left(\frac{I_{ws}}{KHI} \exp(-Kess\ ZT)\right)^2} - \sqrt{1 + \left(\frac{I_{ws}}{KHI} \exp(-Kess\ ZB)\right)^2} \right) \tag{7.25}$$

with ZT and ZB defined following equation (7.15). Equations (7.24) and (7.25) can be applied within a canopy by replacing I_{WS} with I_{Ct} and use of the appropriate light extinction coefficient.

(4) Effect of Temperature on Algal Growth

A Gaussian probability curve is used to represent temperature dependency of algal growth:

$$f_3(T) = \begin{cases} \exp(-KTG1_x(T - TM1_X)^2) : T \le TM1_X \\ 1 : TM1_X < T < TM2_X \\ \exp(-KTG2_x(T - TM2_X)^2) : T \ge TM2_X \end{cases} \quad (7.26)$$

where

T is the temperature (°C) provided from the hydrodynamic model,

TM_x is the optimal temperature for algal growth for algal group x (°C),

$KTG1_x$ is the effect of temperature below $TM1_x$ on growth for algal group x (1/°C²),

$KTG2_x$ is the effect of temperature above $TM2_x$ on growth for algal group x (1/°C²).

The formulation (7.26) represents a modification to the CE-QUAL-ICM formulation to allow for temperature range specification of optimum growth.

(5) Effect of Salinity on Growth of Freshwater Cyanobacteria

The growth of freshwater cyanobacteria in salt water is limited by:

$$f_4(S) = \frac{STOXS^2}{STOXS^2 + S^2} \quad (7.27)$$

where

$STOX$ is the salinity at which Microcystis growth is halved (psu),

S is the salinity in water column (psu) provided from the hydrodynamic model.

(6) Effect of Temperature on Algal Basal Metabolism

Algal biomass in the present model decreases through basal metabolism (respiration and excretion) and predation. Basal metabolism in the present model is the sum of all internal processes that decrease algal biomass and consists of two parts; respiration and excretion. In basal metabolism, algal matter (carbon, nitrogen, phosphorus, and silica) is returned to organic and inorganic pools in the environment, mainly to dissolved organic and inorganic matter. Respiration, which may be viewed as a reversal of production, consumes dissolved oxygen. Basal metabolism is considered to be an exponentially increasing function of temperature:

$$BM_x = BMR_x \exp(KTB_x[T - TR_x]) \quad (7.28)$$

where

BMR_x is the basal metabolism rate at TR_X for algal group x (1/day),

KTB_x is the effect of temperature on metabolism for algal group x (1/°C),

TR_x is the reference temperature for basal metabolism for algal group x (°C).

(7) Effect of Algal Biomass and Temperature on Algal Predation

The present model does not include zooplankton. Instead, a constant rate can be specified for algal predation, which implicitly assumes zooplankton biomass is a constant fraction of algal biomass. Alternately, the predation rate can be taken as proportional to the algae biomass. Using a temperature effect similar to that for metabolism, the predation rate is given

$$PR_x = PRR_x \left(\frac{B_x}{B_{xP}}\right)^{\alpha_P} \exp(KTP_x[T - TR_x]) \tag{7.29}$$

where

PRR_x is the reference predation rate at BxP and TRx for algal group x (1/day),

B_{xP} is the reference algae concentration for predation (g C/m^3),

a_P is the exponential dependence factor,

KTP_x is the effect of temperature on predation for algal group x (1/°C).

The difference between predation and basal metabolism lies in the distribution of the end products of the two processes. In predation, algal matter (carbon, nitrogen, phosphorus, and silica) is returned to organic and inorganic pools in the environment, mainly to particulate organic matter. It is also noted that predation in the EFDC water quality model follows the original formulation in CE-QUAL-ICM (Cerco and Cole, 1995) which uses a predation rate constant with total predation loss being proportional to algae concentration. Subsequent CE-QUAL-ICM documentation, Cerco *et al.*, (2000), appear to define predation independent of algae concentration.

(8) Algal Settling

Settling velocities for four algal groups, WS_c, WS_d , WS_g, and WS_m, are specified as an input. Seasonal variations in settling velocity of diatoms can be accounted for by specifying time-varying WS_d.

7.2.3.2 Organic Carbon

The present model has three state variables for organic carbon: refractory particulate, labile particulate, and dissolved.

(1) Particulate Organic Carbon

Labile and refractory distinctions are based on the time scale of decomposition. Labile particulate organic carbon with a decomposition time scale of days to weeks decomposes rapidly in the water column or in the sediments. Refractory particulate organic carbon with a longer- than-weeks decomposition time scale decomposes slowly, primarily in the sediments, and may contribute to sediment oxygen demand years after decomposition. For labile and refractory particulate organic carbon, sources and sinks included in the model are (Figure 7-1):

- Algal predation
- Dissolution to dissolved organic carbon
- Settling
- External loads.

The governing equations for refractory and labile particulate organic carbons are:

$$\frac{\partial RPOC}{\partial t} = \sum_{x=c,d,g,m} FCRP_X PR_X B_X - K_{RPOC} RPOC + \frac{\partial}{\partial Z}(WS_{RP} RPOC) + \frac{WRPOC}{V} \tag{7.30}$$

$$\frac{\partial LPOC}{\partial t} = \sum_{x=c,d,g,m} FCLP_X PR_X B_X - K_{LPOC} LPOC + \frac{\partial}{\partial Z}(WS_{LP} LPOC) + \frac{WLPOC}{V} \tag{7.31}$$

where

RPOC is the concentration of refractory particulate organic carbon (g C/m³),

LPOC is the concentration of labile particulate organic carbon (g C/m³),

FCRP is the fraction of predated carbon produced as refractory particulate organic Carbon,

FCLP is the fraction of predated carbon produced as labile particulate organic carbon,

KRPOC is the dissolution rate of refractory particulate organic carbon (1/day),

KLPOC is the dissolution rate of labile particulate organic carbon (1/day),

WS_{RP} is the settling velocity of refractory particulate organic matter (m/day),

WS_{LP} is the settling velocity of labile particulate organic matter (m/day),

WRPOC is the external loads of refractory particulate organic carbon (g C/day),

WLPOC is the external loads of labile particulate organic carbon (g C/day).

(2) Dissolved Organic Carbon

Sources and sinks for dissolved organic carbon included in the model are (Figure 7-1):

- Algal excretion (exudation) and predation
- Dissolution from refractory and labile particulate organic carbon
- Heterotrophic respiration of dissolved organic carbon (decomposition)
- Denitrification
- External loads

The kinetic equation describing these processes is:

$$\frac{\partial DOC}{\partial t} = \sum_{x=c,d,g,m} \left(\left[FCD_X + (1 - FCD_X)\left(\frac{KHR_X}{KHR_X + DO}\right) \right] + FCDP_X PR_X \right) B_X + K_{RPOC} RPOC + K_{LPOC} LPOC - K_{HR} DOC - Denit \cdot DOC + \frac{WDOC}{V} \tag{7.32}$$

where

DOC is the concentration of dissolved organic carbon (g C/m³),

FCD_x is the fraction of basal metabolism exuded as dissolved organic carbon at infinite dissolved oxygen concentration for algal group *x*,

KHR_x is the half-saturation constant of dissolved oxygen for algal dissolved organic carbon excretion for group *x* (g O2/ m³),

DO is the dissolved oxygen concentration (g O2/ m³),

FCDP is the fraction of predated carbon produced as dissolved organic carbon,

K_{HR} is the heterotrophic respiration rate of dissolved organic carbon (1/day),

Denit is the denitrification rate (1/day),

BFDOC is the benthic flux of dissolved organic carbon in bottom layer only (g C/m²/day),

WDOC is the external loads of dissolved organic carbon (g C/day).

The remainder of this section explains each term in equations (2.29-31).

(3) Effect of Algae on Organic Carbon

The terms within summation (Σ) in equations ((7.30)-(7.32)) account for the effects of algae on organic carbon through basal metabolism and predation.

(4) Basal Metabolism.

Basal metabolism, consisting of respiration and excretion, returns algal matter (carbon, nitrogen, phosphorus, and silica) back to the environment. Loss of algal biomass through basal metabolism is

$$\frac{\partial B_x}{\partial t} = -BM_x B_x \tag{7.33}$$

that indicates that the total loss of algal biomass due to basal metabolism is independent of ambient dissolved oxygen concentration. In this model, it is assumed that the distribution of total loss between respiration and excretion is constant as long as there is sufficient dissolved oxygen for algae to respire. Under that condition, the losses by respiration and excretion may be written as:

$$(1 - FCD_x)BM_x B_x : \text{ respiration} \tag{7.34}$$

$$FCD_x BM_x B_x : \text{ excretion} \tag{7.35}$$

where FCD_x is a constant of value between 0 and 1. Algae cannot respire in the absence of oxygen, however. Although the total loss of algal biomass due to basal metabolism is oxygen-independent, equation (7.33), the distribution of total loss between respiration and excretion is oxygen-dependent. When oxygen level is high, respiration is a large fraction of the total. As dissolved oxygen becomes scarce, excretion becomes dominant. Thus, equation (7.34) represents the loss by respiration only at high oxygen levels. In general, equation (7.34) can be decomposed into two fractions as a function of dissolved oxygen availability:

$$(1 - FCD_x)\left(\frac{DO}{KHR_x + DO}\right)BM_x B_x : \text{ respiration} \tag{7.36}$$

$$(1 - FCD_x)\left(\frac{KHR_x}{KHR_x + DO}\right)BM_x B_x : \text{ excretion} \tag{7.37}$$

where

KHR_x is the metabolic DO coefficient (g/m³ O_2).

Equation (7.36) represents the loss of algal biomass by respiration, and equation (7.37) represents additional excretion due to insufficient dissolved oxygen concentration. The parameter KHR_x,

which is defined as the half-saturation constant of dissolved oxygen for algal dissolved organic carbon excretion in equation (7.32), can also be defined as the half-saturation constant of dissolved oxygen for algal respiration in equation (7.37).

Combining equations (7.35) and (7.37) the total loss due to excretion is:

$$\left[FCD_X + (1 + FCD_X)\left(\frac{KHR_X}{KHR_X + DO}\right)\right] BM_X B_X \tag{7.38}$$

Equations (7.36) and (7.38) combine to give the total loss of algal biomass due to basal metabolism, BM_xAB_x, equation (7.33). The definition of FCD_x in equation (7.32) becomes apparent in equation (7.38), (i.e., fraction of basal metabolism exuded as dissolved organic carbon at infinite dissolved oxygen concentration). At zero oxygen level, 100 percent of total loss due to basal metabolism is by excretion regardless of FCD_x. The end carbon product of respiration is primarily carbon dioxide, an inorganic form not considered in the present model, while the end carbon product of excretion is primarily dissolved organic carbon. Therefore, equation (7.38), that appears in equation (7.32), represents the contribution of excretion to dissolved organic carbon, and there is no source term for particulate organic carbon from algal basal metabolism in equations (7.30) and (7.31).

Although this general formulation is incorporate for consistency with the original CE-QUAL-IMC formulation (Cerco and Cole, 1995), most of the subsequent applications of CE-QUAL-ICM have simplified the basal metabolism in the published DOC and DO equations or specified input parameters which effective set KHR_x and FCD_x to zero, see Table 7.2, which result in simplifying the DOC equation to

$$\frac{\partial DOC}{\partial t} = \sum_{x=c,d,g,m} FCDP_X PR_X B_X + K_{RPOC} RPOC + K_{LPOC} LPOC - K_{HR} DOC - Denit \cdot DOC + \frac{WDOC}{V} \tag{7.39}$$

(5) Predation

Algae produce organic carbon through the effects of predation. Zooplankton take up and redistribute algal carbon through grazing, assimilation, respiration, and excretion. Since zooplankton are not included in the model, routing of algal carbon through zooplankton predation is simulated by empirical distribution coefficients in equations (7.30) to (7.32); *FCRP*, *FCLP*, and *FCDP*. The sum of these three predation fractions should be unity.

Table 7.2: Basal Metabolism Formulations and Parameter in CE-QUAL-ICM

Study	FCD_x and KHR_x in DOC Equation	FCDx and KHR_x in from DO Equation
Cerco and Cole, 1995 (Chesapeake Bay)	General	General
Bunch, 2000 (San Juan Bay, PR)	General (used FCD=0, KHRx=0.5)	General (used FCD=0, KHRx=0.5)
Cerco *et al.*, 2000	No BMx source in equation,	Consistent with FCD_x=0,

(Florida Bay)	implies FCD_x=0, KHR_x=0	KHR_x=0
Cerco *et al.*, 2002 (Chesapeake Bay, Trib. Refinements)	No BMx source in equation, implies FCD_x=0, KHR_x=0	Consistent with FCD_x=0, KHR_x=0
Cerco *et al.*, 2004 (Lake Washington)	Equation implies KHR_x =0 (used FCD_x=0)	Consistent with KHR_x =0 (used FCD_x=0)
Tillman *et al.*, 2004 (St. Johns River)	No BM_x source in equation, implies FCD_x=0, KHR_x=0	Consistent with FCD_x=0, KHR_x=0

(6) Heterotrophic Respiration and Dissolution

The refractory and labile particulate organic carbon equations, (7.30) and (7.32) contain decay terms that represent dissolution of particulate material into dissolved material. These terms appear in equation (7.32) as sources. The third sink term in the DOC equation (7.32) represents heterotrophic respiration of dissolved organic carbon. The oxic heterotrophic respiration is a function of dissolved oxygen: the lower the dissolved oxygen, the smaller the respiration term becomes. Heterotrophic respiration rate, therefore, is expressed using a Monod function of dissolved oxygen:

$$K_{HR} = \left(\frac{DO}{KHOR_{DO} + DO}\right) K_{DOC} \tag{7.40}$$

where

$KHOR_{DO}$ is the oxic respiration half-saturation constant for dissolved oxygen ($g\,O_2/m^3$),

K_{DOC} is the heterotrophic respiration rate of dissolved organic carbon at infinite dissolved oxygen concentration ($1/day$).

Dissolution and heterotrophic respiration rates depend on the availability of carbonaceous substrate and on heterotrophic activity. Algae produce labile carbon that fuels heterotrophic activity: dissolution and heterotrophic respiration do not require the presence of algae though, and may be fueled entirely by external carbon inputs. In the model, algal biomass, as a surrogate for heterotrophic activity, is incorporated into formulations of dissolution and heterotrophic respiration rates. Formulations of these rates require specification of algal-dependent and algal-independent rates:

$$K_{RPOC} = \left(K_{RC} + K_{RCalg} \sum_{x=c,d,g} B_x\right) \exp\left(KT_{HDR}(T - TR_{HDR})\right) \tag{7.41}$$

$$K_{LPOC} = \left(K_{LC} + K_{LCalg} \sum_{x=c,d,g} B_x\right) \exp\left(KT_{HDR}(T - TR_{HDR})\right) \tag{7.42}$$

$$K_{DOC} = \left(K_{DC} + K_{DCalg} \sum_{x=c,d,g} B_x\right) \exp\left(KT_{MIN}(T - TR_{MIN})\right) \tag{7.43}$$

where

K_{RC} is the minimum dissolution rate of refractory particulate organic carbon (1/day),

K_{LC} is the minimum dissolution rate of labile particulate organic carbon (1/day),

K_{DC} is the minimum respiration rate of dissolved organic carbon (1/day),

K_{RCalg}, K_{LCalg} are the constants that relate dissolution of refractory and labile particulate organic carbon, respectively, to algal biomass (1/day per g C/m^3),

K_{DCalg} is the constant that relates respiration to algal biomass (1/day per g C/m^3) ,

KT_{HDR} is the effect of temperature on hydrolysis of particulate organic matter (1/°C),

TR_{HDR} is the reference temperature for hydrolysis of particulate organic matter (°C),

KT_{MIN} is the effect of temperature on mineralization of dissolved organic matter (1/°C),

TR_{MIN} is the reference temperature for mineralization of dissolved organic matter (°C).

Equations (7.41) to (7.43) have exponential functions that relate rates to temperature.

In the present model, the term "hydrolysis" is defined as the process by which particulate organic matter is converted to dissolved organic form, and thus includes both dissolution of particulate carbon and hydrolysis of particulate phosphorus and nitrogen. Therefore, the parameters, KT_{HDR} and TR_{HDR}, are also used for the temperature effects on hydrolysis of particulate phosphorus (equations (7.55) and (7.56)) and nitrogen (equations (7.70) and (7.71)). The term "mineralization" is defined as the process by which dissolved organic matter is converted to dissolved inorganic form, and thus includes both heterotrophic respiration of dissolved organic carbon and mineralization of dissolved organic phosphorus and nitrogen. Therefore, the parameters, KT_{MIN} and TR_{MIN}, are also used for the temperature effects on mineralization of dissolved phosphorus (7.57) and nitrogen (7.72).

(7) Effect of Denitrification on Dissolved Organic Carbon

As oxygen is depleted from natural systems, organic matter is oxidized by the reduction of alternate electron acceptors. Thermodynamically, the first alternate acceptor reduced in the absence of oxygen is nitrate. The reduction of nitrate by a large number of heterotrophic anaerobes is referred to as denitrification, and the stoichiometry of this reaction is (Stumm and Morgan, 1981):

$$4NO_3^- + 4H^+ + 5CH_2O \rightarrow 2N_2 + 7H_2O + 5CO_2 \tag{7.44}$$

The last term in equation (7.32) accounts for the effect of denitrification on dissolved organic carbon. The kinetics of denitrification in the model are first-order:

$$Denit = \left(\frac{KROR_{DO}}{KROR_{DO} + DO}\right)\left(\frac{NO3}{KHDN_N + NO3}\right) AANOX \cdot K_{DOC} \tag{7.45}$$

where

$KROR_{DO}$ is the denitrification half-saturation constant for dissolved oxygen (g O/m^3),

$KHDN_N$ is the denitrification half-saturation constant for nitrate (g N/m^3),

$AANOX$ is the ratio of denitrification rate to oxic dissolved organic carbon respiration rate.

In equation (7.45), the dissolved organic carbon respiration rate, K_{DOC}, is modified so that significant decomposition via denitrification occurs only when nitrate is freely available and dissolved oxygen is depleted. The ratio, *AANOX*, makes the anoxic respiration slower than oxic respiration. Note that K_{DOC}, defined in equation (7.43), includes the temperature effect on denitrification.

(8) Labile and Refractory Splitting of Dissolved Organic Carbon

A number of water quality models, including the CE-QUAL-ICM application to the St. Johns River, Florida (Tillman *et al.*, 2004) split dissolved organic carbon into labile and refractory components. The refractory component equation is

$$\frac{\partial RDOC}{\partial t} = \sum_{x=c,d,g,m} FCRDP_X \, PR_X \, B_X - K_{RDOC} RDOC + \frac{WRDOC}{V} \tag{7.46}$$

where

RDOC is the concentration of dissolved organic carbon (g C/m³),

FCRDP is the fraction of predated carbon produced as dissolved organic carbon,

K_{RDOC} is the respiration rate of refractory dissolved organic carbon (1/day),

WRDOC is the external loads of dissolved organic carbon (g C/day).

The decay term in (7.46) can include a photoreaction component. The labile component equation retains the general form of the DOC equation

$$\frac{\partial LDOC}{\partial t} = \sum_{x=c,d,g,m} FCLDP \, PR_X \, B_X + K_{RPOC} RPOC + K_{LPOC} LPOC - K_{LDOC} LDOC - Denit \, LDOC + \frac{WLDOC}{V} \tag{7.47}$$

where

LDOC is the concentration of dissolved organic carbon (g C/m³),

FCLDP is the fraction of predated carbon produced as dissolved organic carbon,

K_{LDOC} is the respiration rate of labile dissolved organic carbon (1/day),

WLDOC is the external loads of dissolved organic carbon (g C/day).

Equations (7.46) and (7.47) follow from Tillman *et al.*, (2004), but are not currently implemented in the EFDC water quality model.

7.2.3.3 Phosphorus

The present model has four state variables for phosphorus: three organic forms (refractory particulate, labile particulate, and dissolved) and one inorganic form representing the sum of dissolved and particulate phosphate in the water phase, but exclude phosphate in algae cells.

(1) Particulate Organic Phosphorus

For refractory and labile particulate organic phosphorus, sources and sinks included in the model are (Figure 7-1):

- Algal basal metabolism and predation
- Dissolution to dissolved organic phosphorus
- Settling
- External loads.

The kinetic equations for refractory and labile particulate organic phosphorus are:

$$\frac{\partial RPOP}{\partial t} = \sum_{x=c,d,g,m} (FPR_X\, BM_X + FPRP_X PR_X) APC_X B_X - K_{RPOP} RPOP + \frac{\partial}{\partial Z}(WS_{RP} RPOP) + \frac{WRPOP}{V} \tag{7.48}$$

$$\frac{\partial LPOP}{\partial t} = \sum_{x=c,d,g,m} (FPL_X\, BM_X + FPLP_X\, PR_X) APC_X B_X - K_{LPOP} LPOP + \frac{\partial}{\partial Z}(WS_{RP} LPOP) + \frac{WLPOP}{V} \tag{7.49}$$

where

RPOP is the concentration of refractory particulate organic phosphorus (g P/m^3),

LPOP is the concentration of labile particulate organic phosphorus (g P/m^3),

FPR_x is the fraction of metabolized phosphorus by algal group *x* produced as refractory particulate organic phosphorus,

FPL_x is the fraction of metabolized phosphorus by algal group *x* produced as labile particulate organic phosphorus,

FPRP is the fraction of predated phosphorus produced as refractory particulate organic phosphorus,

FPLP is the fraction of predated phosphorus produced as labile particulate organic phosphorus,

APC is the mean algal phosphorus-to-carbon ratio for all algal groups (g P per g C),

K_{RPOP} is the hydrolysis rate of refractory particulate organic phosphorus (1/day),

K_{LPOP} is the hydrolysis rate of labile particulate organic phosphorus (1/day),

WRPOP is the external loads of refractory particulate organic phosphorus (g P/day),

WLPOP is the external loads of labile particulate organic phosphorus (g P/day).

(2) Dissolved Organic Phosphorus

Sources and sinks for dissolved organic phosphorus included in the model are (Figure 7-1):

- Algal basal metabolism and predation
- Dissolution from refractory and labile particulate organic phosphorus

- Mineralization to phosphate phosphorus
- External loads.

The kinetic equation describing these processes is:

$$\frac{\partial DOP}{\partial t} = \sum_{x=c,d,g,m} (FPD_X\, BM_X + FPDP_X\, PR_X) APC_X B_X + K_{RPOP} RPOP + K_{LPOP} LPOP - K_{DOP} DOP + \frac{WDOP}{V} \quad (7.50)$$

where

DOP is the concentration of dissolved organic phosphorus (g P/m^3),

FPD_x is the fraction of metabolized phosphorus by algal group x produced as dissolved organic phosphorus,

$FPDP_x$ is the fraction of predated phosphorus produced as dissolved organic phosphorus,

K_{DOP} is the mineralization rate of dissolved organic phosphorus (1/day),

$WDOP$ is the external loads of dissolved organic phosphorus (g P/day).

(3) Total Water Phase Phosphate

For total phosphate that includes both dissolved and sorbed phosphate in the water phase, sources and sinks included in the model are (Figure 2.1):

- Algal basal metabolism, predation, and uptake
- Mineralization from dissolved organic phosphorus
- Settling of sorbed phosphate
- Sediment-water exchange of dissolved phosphate for the bottom layer only
- External loads

The kinetic equation describing these processes is:

$$\frac{\partial}{\partial t}(PO4p + PO4d) = \sum_{x=c,d,g,m} (FPI_X\, BM_X + FPIP_X\, PR_X - P_X) APC_X B_X + K_{DOP} DOP + \frac{\partial}{\partial Z}(WS_{TSS} PO4p) + \frac{BFPO4d}{\Delta Z} + \frac{WPO4p}{V} + \frac{WPO4d}{V} \quad (7.51)$$

where

$PO4t = PO4d + PO4p$ *is the* total phosphate (g P/m^3),

$PO4d$ is the dissolved phosphate (g P/m^3),

$PO4p$ is the particulate (sorbed) phosphate (g P/m^3),

FPI_x is the fraction of metabolized phosphorus by algal group x produced as inorganic phosphorus,

$FPIP$ is the fraction of predated phosphorus produced as inorganic phosphorus,

WS_{TSS} is the settling velocity of suspended solid (m/day), provided by the hydrodynamic model,

$BFPO4d$ is the sediment-water exchange flux of phosphate (g P/m²/day), applied to the bottom layer only,

$WPO4t$ is the external loads of total phosphate (g P/day).

In equation (7.51), if total active metal is chosen as a measure of sorption site, the settling velocity of total suspended solid, WS_{TSS}, is replaced by that of particulate metal, WS_s. The remainder of this section explains each term in equations ((7.48)-(7.51)). The benthic flux term is discussed in Section 7.4. Alternate forms of the total phosphate equation are discussed in Sub-section (8) below.

(4) Total Phosphate System

Suspended and bottom sediment particles (clay, silt, and metal hydroxides) adsorb and desorb phosphate in river and estuarine waters. This adsorption-desorption process buffers phosphate concentration in the water column and enhances the transport of phosphate away from its external sources (Carritt and Goodgal, 1954; Froelich, 1988). To ease the computational complication due to the adsorption-desorption of phosphate, dissolved and sorbed phosphate are treated and transported as a single state variable. Therefore, the model phosphate state variable, total phosphate is defined as the sum of dissolved and sorbed phosphate, equation (7.51), and the concentrations for each fraction are determined by equilibrium partitioning of their sum.

In CE-QUAL-ICM, sorption of phosphate to particulate species of metals including iron and manganese was considered based on a phenomenon observed in the monitoring data from the mainstem of the Chesapeake Bay: phosphate was rapidly depleted from anoxic bottom waters during the autumn reaeration event (Cerco and Cole, 1994). Their hypothesis was that reaeration of bottom waters caused dissolved iron and manganese to precipitate, and phosphate sorbed to newly formed metal particles and rapidly settled to the bottom. One state variable, total active metal, in CE-QUAL-ICM was defined as the sum of all metals that act as sorption sites, and the total active metal was partitioned into particulate and dissolved fractions via an equilibrium partitioning coefficient. Then phosphate was assumed to sorb to only the particulate fraction of the total active metal.

In the treatment of phosphate sorption in CE-QUAL-ICM, the particulate fraction of metal hydroxides was emphasized as a sorption site in bottom waters under anoxic conditions. Phosphorus is a highly particle-reactive element, and phosphate in solution reacts quickly with a wide variety of surfaces, being taken up by and released from particles (Froelich 1988). The present model has two options, total suspended solid and total active metal, as a measure of a sorption site for phosphate, and dissolved and sorbed fractions are determined by equilibrium partitioning of their sum as a function of total suspended solid or total active metal concentration:

$$PO4p = \left(\frac{K_{PO4p}SORPS}{1 + K_{PO4p}SORPS}\right)(PO4p + PO4d)$$

$$PO4p = \left(\frac{1}{1 + K_{PO4p}SORPS}\right)(PO4p + PO4d) \qquad (7.52)$$

$$SORPS = SED \; or \; TAM_P$$

where

KPO4p is the empirical coefficient relating phosphate sorption to total suspended solid (per g/m³) or particulate total active metal (per mol/m³) concentration,

SED is the inorganic sediment concentration (mg/l),

TAMp is the particulate total active metal (mol/m³).

The definition of the partition coefficient alternately follows form (7.52)

$$K_{PO4p} = \frac{PO4p}{PO4d}\frac{1}{TSS} \tag{7.52}$$

or

$$K_{PO4p} = \frac{PO4p}{PO4d}\frac{1}{TAMp} \tag{7.53}$$

where the meaning of K_{PO4p} becomes apparent, i.e., the ratio of sorbed to dissolved phosphate per unit concentration of total suspended solid or particulate total active metal (i.e., per unit sorption site available).

(5) Algal Phosphorus-to-Carbon Ratio (APC)

Algal biomass is quantified in units of carbon per volume of water. In order to express the effects of algal biomass on phosphorus and nitrogen, the ratios of phosphorus-to-carbon and nitrogen-to-carbon in algal biomass must be specified. Although global mean values of these ratios are well known (Redfield *et al.*, 1963), algal composition varies especially as a function of nutrient availability. As phosphorus and nitrogen become scarce, algae adjust their composition so that smaller quantities of these vital nutrients are required to produce carbonaceous biomass (DiToro, 1980; Parsons *et al.*, 1984). Examining the field data from the surface of upper Chesapeake Bay, Cerco and Cole (1993) showed that the variation of nitrogen-to-carbon stoichiometry was small and thus used a constant algal nitrogen-to-carbon ratio, ANC_X. Large variations, however, were observed for algal phosphorus-to-carbon ratio indicating the adaptation of algae to ambient phosphorus concentration (Cerco and Cole, 1993): algal phosphorus content is high when ambient phosphorus is abundant and is low when ambient phosphorus is scarce. Thus, a variable algal phosphorus-to-carbon ratio, *APC*, is used in model formulation. A mean ratio for all algal groups, *APC*, is described by an empirical approximation to the trend observed in field data (Cerco and Cole, 1994):

$$APC = \left(CP1_{prm} + CP2_{prm}\exp\left(-CP3_{prmPO4d}\right)\right)^{-1} \tag{7.54}$$

where

$CP1_{prm}$ is the minimum carbon-to-phosphorus ratio (g C per g P),

$CP2_{prm}$ is the difference between minimum and maximum carbon-to-phosphorus ratio (g C per g P),

$CP3_{prm}$ is the effect of dissolved phosphate concentration on carbon-to-phosphorus ratio (per g P/m³).

(6) Effect of Algae on Phosphorus

The terms within summation in equations ((7.48)-(7.51)) account for the effects of algae on phosphorus. Both basal metabolism (respiration and excretion) and predation are considered, and thus formulated, to contribute to organic and phosphate phosphorus. That is, the total loss by basal metabolism ($BMxABx$) is distributed using distribution coefficients ($FPRx$, $FPLx$, $FPDx$, and $FPIx$). The total loss by predation ($PRxABx$), is also distributed using distribution coefficients ($FPRP$, $FPLP$, $FPDP$, and $FPIP$). The sum of four distribution coefficients for basal metabolism should be unity, and as is the sum for predation. Algae take up dissolved phosphate for growth, and algae uptake of phosphate is represented by ($-3\,PxAAPCABx$) in equation (7.51).

(7) Mineralization and Hydrolysis

The third term on the RHS of equations (7.48) and (7.49) represents hydrolysis of particulate organic phosphorus and the last term in equation (7.50) represents mineralization of dissolved organic phosphorus. Mineralization of organic phosphorus is mediated by the release of nucleotidase and phosphatase enzymes by bacteria (Chróst and Overbek, 1987) and algae (Boni *et al.*, 1989). Since the algae themselves release the enzymes and bacterial abundance is related to algal biomass, the rate of organic phosphorus mineralization is related to algal biomass in model formulation. Another mechanism included in the model formulation is that algae stimulate production of an enzyme that mineralizes organic phosphorus to phosphate when phosphate is scarce (Chróst and Overbek, 1987, Boni *et al.*, 1989). The formulations for hydrolysis and mineralization rates including these processes are:

$$K_{RPOP} = \left(K_{RP} + \left(\frac{KHP}{KHP + PO4d} \right) K_{RPalg} \sum_{x=c,d,g,m} B_x \right) \exp\left(KT_{HDR}(T - TR_{HDR}) \right) \tag{7.55}$$

$$K_{LPOP} = \left(K_{LP} + \left(\frac{KHP}{KHP + PO4d} \right) K_{LPalg} \sum_{x=c,d,g,m} B_x \right) \exp\left(KT_{HDR}(T - TR_{HDR}) \right) \tag{7.56}$$

$$K_{DOP} = \left(K_{DP} + \left(\frac{KHP}{KHP + PO4d} \right) K_{DPalg} \sum_{x=c,d,g,m} B_x \right) \exp\left(KT_{MIN}(T - TR_{MIN}) \right) \tag{7.57}$$

where

K_{RP} is the minimum hydrolysis rate of refractory particulate organic phosphorus (1/day),

K_{LP} is the minimum hydrolysis rate of labile particulate organic phosphorus (1/day),

K_{DP} is the minimum mineralization rate of dissolved organic phosphorus (1/day),

K_{RPalg}, K_{LPalg} are the constants that relate hydrolysis of refractory and labile particulate organic phosphorus, respectively, to algal biomass (1/day per g C/m^3),

K_{DPalg} is the constant that relates mineralization to algal biomass (1/day per g C/m^3),

KHP is the mean half-saturation constant for algal phosphorus uptake (g P/m^3).

$$KHP = \frac{\sum_{x=c,d,g,m} KHP_X}{\sum_{x=c,d,g,m} x} \tag{7.58}$$

When phosphate is abundant relative to *KHP*, the rates are close to the minimum values with little influence from algal biomass. When phosphate becomes scarce relative to *KHP*, the rates increase with the magnitude of increase depending on algal biomass. Equations ((7.55)-(7.57)) have exponential functions that relate rates to temperature.

(8) Alternate Forms of the Total Phosphate Equation

In the CE-QUAL-ICM model (Cerco and Cole, 1995), total phosphate is defined to include dissolved phosphate in algae cells. The phosphate in algae cells is given by

$$\begin{aligned}\frac{\partial PO4a}{\partial t} &= \sum_{x=c,d,g,m}(APC\ B_X)\\ &= \sum_{x=c,d,g,m}(P_X - FPI_X\ BM_X - FPIP\ PR_X)\ APC_X\ B_X\\ &- \sum_{x=c,d,g,m}(FPR_X\ BM_X + FPRP\ PR_X)\ APC_X\ B_X\\ &- \sum_{x=c,d,g,m}(FPL_X\ BM_X + FPLP\ PR_X)\ APC_X\ B_X\\ &- \sum_{x=c,d,g,m}(FPD_X\ BM_X + FPDP\ PR_X)\ APC_X\ B_X\\ &+ \frac{\partial}{\partial Z}\left(\sum_{x=c,d,g,m} WS_{AlgX}\ APC_X B_X\right)\end{aligned} \tag{7.59}$$

Where the first term on the right side represents net uptake of phosphate from the water column, and the subsequent three terms represent loss of organic phosphorous. Noting that the distribution factors for basal metabolism and predation must sum to unity, (7.59) reduces to

$$\frac{\partial PO4a}{\partial t} = \sum_{x=c,d,g,m}(P_X - BM_X - PR_X)APC_X\ B_X + \frac{\partial}{\partial Z}\left(\sum_{x=c,d,g,m} WS_{AlgX}\ APC_X B_X\right) \tag{7.60}$$

which is simply equation (7.6) multiplied by the algae phosphorous to carbon ratio and summed over all algae species. Combining (7.60) with (7.61) gives

$$\begin{aligned}&\frac{\partial}{\partial t}(PO4p + PO4d + PO4a)\\ &= K_{DOP}DOP - \sum_{x=c,d,g,m}((1 - FPI_X)BM_X + (1 - FPIP)PR_X)APC_X\ B_X\\ &+ \frac{\partial}{\partial Z}\left(WS_{TSS}PO4p + \sum_{x=c,d,g,m} WS_{AlgX}\ APC_X B_X\right) + \frac{BFPO4d}{\Delta Z} + \frac{WPO4p}{V}\\ &+ \frac{WPO4d}{V}\end{aligned} \tag{7.61}$$

It is noted that this equation differs from equation (3-51) in Cerco and Cole (1995), which is in error, but is identical to the subsequently corrected equation (35) in Cerco *et al.*, (2000) which documents the Florida Bay CE-QUAL-ICM model application. Thus, either equations (7.51) or (7.61) can be used for total phosphate as long as partitioning between particulate and dissolved phosphate in the water phase is appropriately represented by equations (7.52) and (7.63).

(9) Labile and Refractory Splitting of Dissolve Organic Phosphorous

A number of water quality models, including the CE-QUAL-ICM application to the St. Johns River, Florida (Tillman *et al.*, 2004) split dissolved organic phosphorous into labile and refractory components. The refractory component equation is

$$\frac{\partial RDOP}{\partial t} = \sum_{x=c,d,g,m} (FPRD_X\, BM_X + FPRDP_X\, PR_X)\, APC_X\, B_X - K_{RDOP} RDOP + \frac{WRDOP}{V} \qquad (7.62)$$

where

RDOP is the concentration of refractory dissolved organic phosphorus (g P/m^3),

$FPRD_x$ is the fraction of metabolized phosphorus by algal group *x* produced as refractory dissolved organic phosphorus,

FPRDP is the fraction of predated phosphorus produced as refractory dissolved organic phosphorus,

K_{RDOP} is the mineralization rate of refractory dissolved organic phosphorus (1/day),

WRDOP is the external loads of refractory dissolved organic phosphorus (g P/day).

The labile component equation is:

$$\frac{\partial LDOP}{\partial t} = \sum_{x=c,d,g,m} (FPLD_X\, BM_X + FPLDP_X\, PR_X)\, APC_X\, B_X - K_{RDOP} RDOP + K_{LPOP} LPOP - K_{LDOP} LDOP + \frac{WLDOP}{V} \qquad (7.63)$$

where

LDOP is the concentration of dissolved organic phosphorus (g P/m^3),

$FPLD_x$ is the fraction of metabolized phosphorus by algal group *x* produced as dissolved organic phosphorus,

FPLDP is the fraction of predated phosphorus produced as dissolved organic phosphorus,

K_{LDOP} is the mineralization rate of dissolved organic phosphorus (1/day),

WLDOP is the external loads of dissolved organic phosphorus (g P/day).

Equations (7.62) and (7.63) follow from Tillman *et al*, (2004), but are not currently implemented in the EFDC water quality model.

7.2.3.4 Nitrogen

The present model has five state variables for nitrogen: three organic forms (refractory particulate, labile particulate, and dissolved) and two inorganic forms (ammonium and nitrate). The nitrate state variable in the model represents the sum of nitrate and nitrite.

(1) Particulate Organic Nitrogen

For refractory and labile particulate organic nitrogen, sources and sinks included in the model are (Figure 7-1):

- Algal basal metabolism and predation
- Dissolution to dissolved organic nitrogen
- Settling
- External loads

The kinetic equations for refractory and labile particulate organic nitrogen are:

$$\frac{\partial RPON}{\partial t} = \sum_{x=c,d,g,m} (FNR_X\, BM_X + FNRP_X\, PR_X)\, ANC_X\, B_X - K_{RPON} RPON + \frac{\partial}{\partial Z}(WS_{RP} RPON) + \frac{WRPON}{V} \tag{7.64}$$

$$\frac{\partial LPON}{\partial t} = \sum_{x=c,d,g,m} (FNL_X\, BM_X + FNLP_X\, PR_X)\, ANC_X\, B_X - K_{LPON} LPON + \frac{\partial}{\partial Z}(WS_{LP} LPON) + \frac{WLPON}{V} \tag{7.65}$$

where

RPON is the concentration of refractory particulate organic nitrogen (g N/m^3)

LPON is the concentration of labile particulate organic nitrogen (g N/m^3),

FNR_x is the fraction metabolized nitrogen by algal group x as refractory particulate organic nitrogen,

FNL_x is the fraction of metabolized nitrogen by algal group x produced as labile particulate organic nitrogen,

FNRP is the fraction of predated nitrogen produced as refractory particulate organic nitrogen,

FNLP is the fraction of predated nitrogen produced as labile particulate organic nitrogen,

ANC_x is the nitrogen-to-carbon ratio in algal group *x* (g N per g C),

K_{RPON} is the hydrolysis rate of refractory particulate organic nitrogen (1/day),

K_{LPON} is the hydrolysis rate of labile particulate organic nitrogen (1/day),

WRPON is the external loads of refractory particulate organic nitrogen (g N/day),

WLPON is the external loads of labile particulate organic nitrogen (g N/day).

(2) Dissolved Organic Nitrogen

Sources and sinks for dissolved organic nitrogen included in the model are (Figure 4-1)

- Algal basal metabolism and predation
- Dissolution from refractory and labile particulate organic nitrogen
- Mineralization to ammonium
- External loads.

The kinetic equation describing these processes is:

$$\frac{\partial DON}{\partial t} = \sum_{x=c,d,g,m} (FND_X\, BM_X + FNDP_X\, PR_X)\, ANC_X\, B_X + K_{RPON} RPON + K_{LPON} LPON - K_{DON} DON + \frac{BFDON}{\Delta Z} + \frac{WDON}{V} \tag{7.66}$$

where

DON is the concentration of dissolved organic nitrogen (g N/m^3),

FND_x is the fraction of metabolized nitrogen by algal group x produced as dissolved organic nitrogen,

$FNDP$ is the fraction of predated nitrogen produced as dissolved organic nitrogen,

K_{DON} is the mineralization rate of dissolved organic nitrogen (1/day),

$BFDON$ is the benthic flux of dissolved organic nitrogen in bottom layer only (g C/m^2/day),

$WDON$ is the external loads of dissolved organic nitrogen (g N/day).

(3) Ammonium Nitrogen

Sources and sinks for ammonia nitrogen included in the model are (Figure 7-1):

- Algal basal metabolism, predation, and uptake
- Mineralization from dissolved organic nitrogen
- Nitrification to nitrate
- Sediment-water exchange for the bottom layer only
- External loads

The kinetic equation describing these processes is:

$$\frac{\partial NH4}{\partial t} = \sum_{x=c,d,g,m} (FNI_X\, BM_X + FNIP_X\, PR_X - PN_X\, P_X)\, ANC_X\, B_X + K_{DON} DON - KNit\, NH4 + \frac{BFNH4}{\Delta Z} + \frac{WNH4}{V} \tag{7.67}$$

where

FNI_x is the fraction of metabolized nitrogen by algal group *x* produced as inorganic nitrogen,

$FNIP$ is the fraction of predated nitrogen produced as inorganic nitrogen,

PN_x is the preference for ammonium uptake by algal group x ($0 \le PN_x \le 1$),

KNit is the nitrification rate (1/day) given in equation (2.74),

BFNH4 is the sediment-water exchange flux of ammonium (g N/m²/day), applied to the bottom layer only,

WNH4 is the external loads of ammonium (g N/day).

The form of the nitrification sink in (7.67) and the subsequent source in the nitrate equation (7.68) differ from that in CE-QUAL-ICM.

(4) Nitrate Nitrogen

Sources and sinks for nitrate nitrogen included in the model are (Figure 4-1):

- Algal uptake
- Nitrification from ammonium
- Denitrification to nitrogen gas
- Sediment-water exchange for the bottom layer only
- External loads

The kinetic equation describing these processes is:

$$\frac{\partial NO3}{\partial t} = \sum_{x=c,d,g,m} (PN_X - 1) P_X \, ANC_X \, B_X + KNit \, NH4 - ANDC \, Denit \, DOC + \frac{BFNO3}{\Delta Z} + \frac{WNO3}{V} \tag{7.68}$$

where

ANDC is the mass of nitrate nitrogen reduced per mass of dissolved organic carbon oxidized (0.933 g N per g C),

BFNO3 is the sediment-water exchange flux of nitrate (g N/m²/day), applied to the bottom layer only,

WNO3 is the external loads of nitrate (g N/day).

The remainder of this section explains each term in equations (7.64)-(7.68) with the benthic fluxes, *BFNH4* and *BFNO3* described in Chapter 6.

(5) Effect of Algae on Nitrogen

The terms within summation in equations (7.64)-(7.68) account for the effects of algae on nitrogen. As in phosphorus, both basal metabolism (respiration and excretion) and predation are considered, and thus formulated to contribute to organic and ammonium nitrogen. That is, algal nitrogen released by both basal metabolism and predation are represented by distribution coefficients (FNR_x, FNL_x, FND_x, FNI_x, *FNRP*, *FNLP*, *FNDP*, and *FNIP*). The sum of the four distribution coefficients for basal metabolism should be unity; the sum of the predation distribution coefficients should also be unity.

Algae take up ammonium and nitrate for growth, and ammonium is preferred from thermodynamic considerations. The preference of algae for ammonium is expressed as:

$$PN_X = NH4\left(\frac{NO3}{(KHN_X + NH4)(KHN_X + NO3)}\right) + NH4\left(\frac{KHN_X}{(NH4 + NO3)(KHN_X + NO3)}\right) \tag{7.69}$$

This equation forces the preference for ammonium to be unity when nitrate is absent, and to be zero when ammonium is absent.

(6) Mineralization and Hydrolysis

The third term on the RHS of equations (7.64) and (7.65) represents hydrolysis of particulate organic nitrogen and the last term in equation (7.66) represents mineralization of dissolved organic nitrogen. Including a mechanism for accelerated hydrolysis and mineralization during nutrient-limited conditions, the formulations for these processes are:

$$K_{RPON} = \left(K_{RN} + \left(\frac{KHN}{KHN + NH4 + NO3}\right)K_{RNalg}\sum_{x=c,d,g,m} B_x\right)\exp\left(KT_{HDR}(T - TR_{HDR})\right) \tag{7.70}$$

$$K_{LPON} = \left(K_{LN} + \left(\frac{KHN}{KHN + NH4 + NO3}\right)K_{LNalg}\sum_{x=c,d,g,m} B_x\right)\exp\left(KT_{HDR}(T - TR_{HDR})\right) \tag{7.71}$$

$$K_{DON} = \left(K_{DN} + \left(\frac{KHN}{KHN + NH4 + NO3}\right)K_{DNalg}\sum_{x=c,d,g,m} B_x\right)\exp\left(KT_{MIN}(T - TR_{MIN})\right) \tag{7.72}$$

where

K_{RN} is the minimum hydrolysis rate of refractory particulate organic nitrogen (1/day),

K_{LN} is the minimum hydrolysis rate of labile particulate organic nitrogen (1/day),

K_{DN} is the minimum mineralization rate of dissolved organic nitrogen (1/day),

K_{RNalg}, K_{LNalg} are the constants that relate hydrolysis of refractory and labile particulate organic nitrogen, respectively, to algal biomass (1/day per g C/m³),

K_{DNalg} is the constant that relates mineralization to algal biomass (1/day per g C/m³),

KHN is the mean half-saturation constant for algal nitrogen uptake (g N/m³).

$$KHN = \frac{\sum_{x=c,d,g,m} KHN_X}{\sum_{x=c,d,g,m} x} \tag{7.73}$$

Equations (7.70)-(7.72) have exponential functions that relate rates to temperature.

(7) Nitrification

Nitrification is a process mediated by autotrophic nitrifying bacteria that obtain energy through the oxidation of ammonium to nitrite and of nitrite to nitrate. The stoichiometry of complete reaction is (Bowie *et al.*, 1985):

$$NH_4^+ + 2O_2 \rightarrow NO_3^- + H_2O + 2H^+ \quad (7.74)$$

The first term in the second line of equation (7.65) and its corresponding term in equation (7.68) represent the effect of nitrification on ammonium and nitrate, respectively. The kinetics of the complete nitrification process are formulated as a function of available ammonium, dissolved oxygen and temperature:

$$KNit\ NH4 = fNit\ (T)\left(\frac{DO}{KHNit_{DO} + DO}\right)\left(\frac{NH4}{KHNit_N + NH4}\right)Nit_m \quad (7.75)$$

where

$KHNit_{DO}$ is the nitrification half-saturation constant for dissolved oxygen (g O2/ m^3),

$KHNit_N$ is the nitrification half-saturation constant for ammonium (g N/m^3),

Nit_m is the maximum nitrification rate at *TNit* (g N/m^3/day).

This follows that in the CE-QUAL-ICM model formulation for nitrification. The Monod function of dissolved oxygen in equation (7.73) indicates the inhibition of nitrification at low oxygen level. The Monod function of ammonium indicates that when ammonium is abundant, the nitrification rate is limited by the availability of nitrifying bacteria.

In the EFDC water quality model a reference value of KNit is input into the model instead of Nit_m by writing (7.75) as

$$KNit = fNit\ (T)\left(\frac{DO}{KHNit_{DO} + DO}\right)\left(\frac{KHNit_N}{KHNit_N + NH4}\right)KNit_m \quad (7.76)$$

where

$$KNit_m = \frac{Nit_m}{KHNit_N} \quad (7.77)$$

is interpreted as the linear kinetic rate corresponding to $KHNit_N$ equal to unity, since NH4 is always must less than unit, and DO effects eliminated by setting $KHNit_{DO}$ to zero. In certain applications, particularly those having long-term BOD and Nitrogen series test results, $KNit_m$ is observable.

The temperature function for nitrification in equation (7.75) is given by

$$fNit\ (T) = \begin{cases} \exp(-KNit1(T - TNit1)^2) & : T \le TNit1 \\ 1 & : TNit1 \le T \le TNit2 \\ exp(-KNit2(T - TNit12)^2) & : T \ge TNit2 \end{cases} \quad (7.78$$

where

Tnit1 is the lower optimum temperature for nitrification (°C),

Tnit2 is the upper optimum temperature for nitrification (°C),

KNit1 is the effect of temperature below *TNit* on nitrification rate (1/°C^2),

KNit2 is the effect of temperature above *TNit* on nitrification rate (1/°C²).

The effect of suboptimal temperature is represented using Gaussian form.

(8) Denitrification

The effect of denitrification on dissolved organic carbon was described in Section 2.4.2.7. Denitrification removes nitrate from the system in stoichiometric proportion to carbon removal as determined by equation (7.44). The sink term in (7.68) represents this removal of nitrate.

(9) Labile and Refractory Splitting of Dissolved Organic Nitrogen

A number of water quality models, including the CE-QUAL-ICM application to the St. Johns River, Florida (Tillman *et al.*, 2004) split dissolved organic phosphorous into labile and refractory components. The refractory component equation is

$$\frac{\partial RDON}{\partial t} = \sum_{x=c,d,g,m} (FNRD_X\, BM_X + FNRDP_X\, PR_X)\, ANC_X\, B_X - K_{RDON} RDON + \frac{WRDON}{V} \tag{7.79}$$

where

RDON is the concentration of dissolved organic nitrogen (g N/m³),

$FNRD_x$ is the fraction of metabolized nitrogen by algal group *x* produced as dissolved organic nitrogen,

FNRDP is the fraction of predated nitrogen produced as dissolved organic nitrogen,

K_{RDON} is the mineralization rate of dissolved organic nitrogen (1/day),

WRDON is the external loads of dissolved organic nitrogen (g N/day).

The equation for the labile component is

$$\frac{\partial LDON}{\partial t} = \sum_{x=c,d,g,m} (FNLD_X\, BM_X + FNLDP_X\, PR_X)\, ANC_X\, B_X + K_{RPON} RPON + K_{LPON} LPON - K_{LDON} LDON + \frac{\partial}{\partial Z}(WS_{LP} LPON) + \frac{WDON}{V} \tag{7.80}$$

where

DON is the concentration of dissolved organic nitrogen (g N/m³),

FND_x is the fraction of metabolized nitrogen by algal group *x* produced as dissolved organic nitrogen,

FNDP is the fraction of predated nitrogen produced as dissolved organic nitrogen,

K_{DON} is the mineralization rate of dissolved organic nitrogen (1/day),

WDON is the external loads of dissolved organic nitrogen (g N/day).

Equations (7.79) and (7.80) follow from Tillman *et al.*, (2004), but are not currently implemented in the EFDC water quality model.

7.2.3.5 Silica

The present model has two state variables for silica: particulate biogenic silica and available silica.

(1) Particulate Biogenic Silica

Sources and sinks for particulate biogenic silica included in the model are (Figure 7-1):

- Diatom basal metabolism and predation
- Dissolution to available silica
- Settling
- External loads

The kinetic equation describing these processes is:

$$\frac{\partial SU}{\partial t} = (FSP_d\ BM_d + FSPP\ PR_d)ASC_d\ B_d + K_{SUA}SU + \frac{\partial}{\partial Z}(WS_d SU) + \frac{WSU}{V} \tag{7.81}$$

where

SU is the concentration of particulate biogenic silica (g Si/m^3),

FSP_d is the fraction of metabolized silica by diatoms produced as particulate biogenic silica,

$FSPP$ is the fraction of predated diatom silica produced as particulate biogenic silica,

ASC_d is the silica-to-carbon ratio of diatoms (g Si per g C),

K_{SUA} is the dissolution rate of particulate biogenic silica (1/day),

WSU is the external loads of particulate biogenic silica (g Si/day).

(2) Available Silica

Sources and sinks for available silica included in the model are (Figure 2.1):

- Diatom basal metabolism, predation, and uptake
- Settling of sorbed (particulate) available silica
- Dissolution from particulate biogenic silica
- Sediment-water exchange of dissolved silica for the bottom layer only
- External loads.

The kinetic equation describing these processes is:

$$\frac{\partial SA}{\partial t} = (FSI_d\ BM_d + FSIP\ PR_d - P_d)ASC_d\ B_d + K_{SUA}SU + \frac{\partial}{\partial Z}(WS_{TSS}SAp) + \frac{BFSAd}{\Delta Z} + \frac{WSA}{V} \tag{7.82}$$

where

$SA = SAd + SAp$ is the concentration of available silica (g Si/m^3),

SAd is the dissolved available silica (g Si/m³),

SAp is the particulate (sorbed) available silica (g Si/m³),

FSI_d is the fraction of metabolized silica by diatoms produced as available silica,

FSIP is the fraction of predated diatom silica produced as available silica,

BFSAd is the sediment-water exchange flux of available silica (g Si/m²/ day), applied to bottom layer only,

WSA is the external loads of available silica (g Si/day).

In equation (7.82), if total active metal is chosen as a measure of sorption site, the settling velocity of total suspended solid, WS_{TSS}, is replaced by that of particulate metal, WS_S.

(3) Available Silica System

Analysis of Chesapeake Bay monitoring data indicates that silica shows similar behavior as phosphate in the adsorption-desorption process (Cerco and Cole, 1993). As in phosphate, therefore, available silica is defined to include both dissolved and sorbed fractions. Treatment of available silica is the same as total phosphate, and the same method to partition dissolved and sorbed phosphate is used to partition dissolved and sorbed available silica:

$$SAp = \left(\frac{K_{SAp}SORPS}{1 + K_{SAp}SORPS}\right)SA$$

$$SAd = \left(\frac{1}{1 + K_{SAp}SORPS}\right)SA \tag{7.83}$$

$$SORPS = TSS \; or \; TAM_p$$

$$SA = SAp + SAd \tag{7.84}$$

where

$KSAp$ is the empirical coefficient relating available silica sorption to total suspended solid (per g/m³) or particulate total active metal (per mol/m³) concentration.

(4) Effect of Diatoms on Silica

In equations (7.81) and (7.83), those terms expressed as a function of diatom biomass (B_d) account for the effects of diatoms on silica. As in phosphorus and nitrogen, both basal metabolism (respiration and excretion) and predation are considered, and thus formulated, to contribute to particulate biogenic and available silica. That is, diatom silica released by both basal metabolism and predation are represented by distribution coefficients (FSP_d, FSI_d, *FSPP*, and *FSIP*). The sum of two distribution coefficients for basal metabolism should be unity and so is that for predation. Diatoms require silica as well as phosphorus and nitrogen, and diatom uptake of available silica is represented by ($-P_dAASC_dAB_d$) in equation (7.82).

(5) Dissolution

The term (- $K_{SUA}ASU$) in equation (7.81) and its corresponding term in equation (7.82) represent dissolution of particulate biogenic silica to available silica. The dissolution rate is expressed as an exponential function of temperature:

$$K_{SUA} = K_{SU} \exp\left(KT_{SUA}(T - TR_{SUS})\right) \tag{7.85}$$

where

K_{SU} is the dissolution rate of particulate biogenic silica at TR_{SUA} (1/day),

KT_{SUA} is the effect of temperature on dissolution of particulate biogenic silica (1/°C),

TR_{SUA} is the reference temperature for dissolution of particulate biogenic silica (°C).

7.2.3.6 Chemical Oxygen Demand

In the present model, chemical oxygen demand is the concentration of reduced substances that are oxidizable through inorganic means. The source of chemical oxygen demand in saline water is sulfide released from sediments. A cycle occurs in which sulfate is reduced to sulfide in the sediments and reoxidized to sulfate in the water column. In fresh water, methane is released to the water column by the sediment process model. Both sulfide and methane are quantified in units of oxygen demand and are treated with the same kinetic formulation. The kinetic equation, including external loads, if any, is:

$$\frac{\partial COD}{\partial t} = -\left(\frac{DO}{KH_{COD} + DO}\right) K_{COD} COD + \frac{BFCOD}{\Delta Z} + \frac{WCOD}{V} \tag{7.86}$$

where

COD is the concentration of chemical oxygen demand (g O_2-equivalents/m^3),

KH_{COD} is the half-saturation constant of dissolved oxygen required for oxidation of chemical oxygen demand (g O_2/m^3),

K_{COD} is the oxidation rate of chemical oxygen demand (1/day),

BFCOD is the sediment flux of chemical oxygen demand (g O_2-equivalents/m^2/day), applied to bottom layer only,

WCOD is the external loads of chemical oxygen demand (g O_2-equivalents/day).

An exponential function is used to describe the temperature effect on the oxidation rate of chemical oxygen demand:

$$K_{COD} = K_{CD} \exp\left(KT_{COD}(T - TR_{COD})\right) \tag{7.87}$$

where

K_{CD} is the oxidation rate of chemical oxygen demand at $TRCOD$ (1/day),

KT_{COD} is the effect of temperature on oxidation of chemical oxygen demand (1/°C),

TR_{COD} is the reference temperature for oxidation of chemical oxygen demand (°C).

7.2.3.7 Dissolved Oxygen

Sources and sinks of dissolved oxygen in the water column included in the model are (Figure 7-1):

- Algal photosynthesis and respiration
- Nitrification
- Heterotrophic respiration of dissolved organic carbon
- Oxidation of chemical oxygen demand
- Surface reaeration for the surface layer only
- Sediment oxygen demand for the bottom layer only
- External loads

The kinetic equation describing these processes is:

$$\frac{\partial DO}{\partial t} = \sum_{x=c,d,g,m} \left(\left(1 + 0.3(1 - PN_x)\right) P_x - (1 - FCD_x) \left(\frac{DO}{KHR_x + DO} \right) BM_x \right) AOCR\, B_x$$
$$- AONT\ Nit\ NH4 - AOCR\, K_{HR}\, DOC - \left(\frac{DO}{KH_{COD} + DO} \right) K_{COD} COD \qquad (7.88)$$
$$+ K_R(DO_S - DO) + \frac{SOD}{\Delta Z} + \frac{WDO}{V}$$

where

AONT is the mass of dissolved oxygen consumed per unit mass of ammonium nitrogen nitrified (4.33 g O_2 per g N; see Section Effect of Nitrification on Dissolved Oxygen),

AOCR is the dissolved oxygen-to-carbon ratio in respiration (2.67 g O_2 per g C; see Section Effect of Algae on Dissolved Oxygen),

KR is the reaeration coefficient (1/day): the reaeration term is applied to the surface layer only,

DO_S is the saturated concentration of dissolved oxygen (g O_2/m^3),

SOD is the sediment oxygen demand (g O_2/m^2/day), applied to the bottom layer only; positive is to the water column,

WDO is the external loads of dissolved oxygen (g O2/day),

PN_X is the preference for ammonia uptake by algae group x ($0<PN_X<1$).

The two sink terms in equation (7.88), heterotrophic respiration and chemical oxygen demand, are explained in Section 2.4.2.6 and Section 2.4.6 respectively. The remainder of this section explains the effects of algae, nitrification, and surface reaeration.

(1) Effect of Algae on Dissolved Oxygen

The first line on the RHS of equation (7.88) accounts for the effects of algae on dissolved oxygen. Algae produce oxygen through photosynthesis and consume oxygen through respiration. The quantity produced depends on the form of nitrogen utilized for growth. Equations describing production of dissolved oxygen are (Morel 1983):

$$106CO_2 + 16NH_4^+ + H_2PO_4^- + 106H_2O \rightarrow protoplasm + 106O_2 + 15H^+ \quad (7.89)$$

$$106CO_2 + 16NO_3^- + H_2PO_4^- + 122H_2O \rightarrow protoplasm + 138O_2 \quad (7.90)$$

When ammonium is the nitrogen source, one mole of oxygen is produced per mole of carbon dioxide fixed. When nitrate is the nitrogen source, 1.3 moles of oxygen are produced per mole of carbon dioxide fixed. The quantity, (1.3 - 0.3APN_x), in the first term of equation (7.88) is the photosynthesis ratio and represents the molar quantity of oxygen produced per mole of carbon dioxide fixed. It approaches unity as the algal preference for ammonium approaches unity.

The last term in the first line of equation (7.88) accounts for the oxygen consumption due to algal respiration. A simple representation of respiration process is:

$$CH_2O + O_2 = CO_2 + H_2O \quad (7.91)$$

from which, *AOCR* = 2.67 g O_2 per g C.

(2) Effect of Nitrification on Dissolved Oxygen

The stoichiometry of nitrification reaction, equation (7.74) indicates that two moles of oxygen are required to nitrify one mole of ammonium into nitrate. However, cell synthesis by nitrifying bacteria is accomplished by the fixation of carbon dioxide so that less than two moles of oxygen are consumed per mole ammonium utilized (Wezernak and Gannon, 1968), i.e., *AONT* = 4.33 g O2 per g N.

(3) Effect of Surface Reaeration on Dissolved Oxygen

The reaeration rate of dissolved oxygen at the air-water interface is proportional to the oxygen gradient across the interface, (DO_S - *DO*), when assuming the air is saturated with oxygen. The saturated concentration of dissolved oxygen, which decreases as temperature and salinity increase, is specified using an empirical formula (Genet *et al.*, 1974):

$$DO_s = 14.5532 - 0.38217T + 5.4258 \times 10^{-3}T^2 - CL(1.665 \times 10^{-4} - 5.866 \times 10^{-6}T + 9.796 \times 10^{-8}T^2) \quad (7.92)$$

CL = S/1.80655 is the chloride concentration (mg/l).

The reaeration coefficient includes the effect of turbulence generated by bottom friction (O'Connor and Dobbins, 1958) and that by surface wind stress (Banks and Herrera, 1977):

$$K_r = \frac{1}{\Delta z}\left(K_{ro}\sqrt{\frac{ueq}{heq}} + W_{rea}\right)(KT_r)^{T-20} \quad (7.93)$$

where

K_{RO} = 3.933 is the proportionality constant in SI unit,

$ueq = \sum(ukVk)/3(Vk)$ is the weighted velocity over cross-section (m/s),

$heq = \sum (Vk)/B$ is the weighted depth over cross-section (m),

B is the width at the free surface (m),

$Wrea$ is the wind-induced reaeration (m/day).

$$W_{rea} = 0.728\sqrt{U_w} - 0.317U_w + 0.0372U_w^2 \tag{7.94}$$

Uw is the wind speed (m/s) at the height of 10 m above surface,

KTr is the constant for temperature adjustment of dissolved oxygen reaeration rate.

(4) Simplified Equation for Dissolved Oxygen

The simplified DO equation for KHRx and FCDx equal to zero is

$$\frac{\partial DO}{\partial t} = \sum_{x=c,d,g,m} \left((1.3 - 0.3PN_x)P_x - BM_x\right)AOCR\ B_x - AONT\ Nit\ NH4 - AOCR\ K_{HR}\ DOC - \left(\frac{DO}{KH_{COD} + DO}\right)K_{COD}COD + K_R(DO_S - DO) + \frac{SOD}{\Delta Z} + \frac{WDO}{V} \tag{7.95}$$

which is consistent with equation (7.39).

7.2.3.8 Total Active Metal

The present model requires simulation of total active metal for adsorption of phosphate and silica if that option is chosen (Figure 2.1). The total active metal state variable is the sum of iron and manganese concentrations, both particulate and dissolved. In the model, the origin of total active metal is benthic sediments. Since sediment release of metal is not explicit in the sediment model (see Chapter 6), release is specified in the kinetic portion of the water column model. The only other term included is settling of the particulate fraction. Then the kinetic equation for total active metal, including external loads, if any, may be written as:

$$\frac{\partial TAM}{\partial t} = \left(\frac{KHbmf}{KHbmf + DO}\right)\left(\exp\left(Ktam(T - Ttam)\right)\right)\frac{BFTAM}{\Delta z} + \frac{\partial}{\partial Z}(WS_sTAMp) + \frac{WTAM}{V} \tag{7.96}$$

where

$TAM = TAMd + TAMp$ is the total active metal concentration (mol/m^3),

$TAMd$ is the dissolved total active metal (mol/m^3),

$TAMp$ is the particulate total active metal (mol/m^3),

$KHbmf$ is the dissolved oxygen concentration at which total active metal release is half the anoxic release rate (g O2/ m^3),

$BFTAM$ is the anoxic release rate of total active metal (mol/m^2/day), applied to the bottom layer only,

$Ktam$ is the effect of temperature on sediment release of total active metal (1/°C),

$Ttam$ is the reference temperature for sediment release of total active metal (°C),

WS_S is the settling velocity of particulate metal (m/day),

WTAM is the external loads of total active metal (mol/day).

In estuaries, iron and manganese exist in particulate and dissolved forms depending on dissolved oxygen concentration. In oxygenated water, most of the iron and manganese exist as particulate while under anoxic conditions, large fractions are dissolved, although solid-phase sulfides and carbonates exist and may predominate. The partitioning between particulate and dissolved phases is expressed using a concept that total active metal concentration must achieve a minimum level, which is a function of dissolved oxygen, before precipitation occurs:

$$TAMd = \min(TAMdmx\, exp(-Kdotam\, DO)\, , TAM) \tag{7.97}$$

$$TAMp = TAM - TAMd \tag{7.98}$$

where

TAMdmx is the solubility of total active metal under anoxic conditions (mol/m^3),

Kdotam is the constant that relates total active metal solubility to dissolved oxygen (per g O2/ m^3).

7.2.3.9 Fecal Coliform Bacteria

The fecal coliform variable is completely decoupled from the rest of the water quality model and is included in the model for convenience in TMDL applications which consider both nutrient and pathogen impairments. Fecal coliform bacteria are indicative of organisms from the intestinal tract of humans and other animals and can be used as an indicator bacteria as a measure of public health (Thomann and Mueller, 1987). In the present model, fecal coliform bacteria have no interaction with other state variables, and have only one sink term, die-off. The kinetic equation, including external loads, may be written as:

$$\frac{\partial FCB}{\partial t} = KFCB(TFCB^{T-20})FCB + \frac{WFCB}{V} \tag{7.99}$$

where

FCB is the bacteria concentration (MPN per 100 ml),

KFCB is the first order die-off rate at 20°C (1/day),

TFCB is the effect of temperature on decay of bacteria (1/°C),

WFCB is the external loads of fecal coliform bacteria (MPN per 100 ml m^3/day).

7.2.4. Settling, Deposition and Resuspension of Particulate Matter

The kinetic equations for particulate matter, including particulate organic matter, total phosphate, the two silica state variables and total active metal, contain settling term. A representative generic equation is

$$\frac{\partial PM}{\partial t} = \frac{\partial}{\partial z}(WS_{PM}PM) + PM_{SS} \tag{7.100}$$

where PMSS represents the additional terms in the equation. Integration of equation (7.100) over the bottom layer gives

$$\frac{\partial PM_1}{\partial t} = \frac{WS_{PM}}{\Delta Z_1} PM_2 - \frac{WS_{PM}}{\Delta Z_1} PM_1 + PM_{SS1} \tag{7.101}$$

The original CE-QUAL-ICM and EFDC water quality models were formulated with settling velocities representing long-term average net settling. In the subsequent application of CE-QUAL-ICM to Florida Bay (Cerco *et al.*, 2000), the resuspension or erosion of particulate material form the sediment bed was added and has also been added to the EFDC water quality model.

The EFDC model allows the use of the net settling formulation (7.101) and a formulation allowing resuspension with equation (7.101) modified

$$\frac{\partial PM_1}{\partial t} = \frac{WS_{POM}}{\Delta Z_1} PM_2 - \frac{P_{depPM} WS_{PM}}{\Delta Z_1} PM_1 + \frac{E_{PM}}{\Delta Z_1} + PM_{SS1} \tag{7.102}$$

to include a probability of deposition factor and an erosion term, EPM having units of mass per unit time-unit area. For EFDC model applications with erosion of particulate material in the water quality module, sediment transport must be active in the hydrodynamic model. The erosion term is then defined by

$$E_{PM} = \left(\frac{PM_{bed}}{SED_{bed}}\right) \max(J_{ERO}, 0) \tag{7.103}$$

where

PMbed is the particulate material concentration in bed (g PM/m² or g PM/m³),

SEDbed is the concentration finest sediment class in bed (g PM/m² or g PM/m³),

P_{depPM} is the probability of deposition of the specific particulate matter variable ($0 \le P_{depPM} \le 1$),

JERO is the mass rate of erosion or resuspension of the finest sediment class (g SED/day/m²).

Use the ratio of the water quality model particulate state variable concentration to the finest sediment size class concentration rather than the total solids concentration is based on the reality that finest sediment class (general less than 63 μ m) includes both inorganic and organic material and field observations of settling, deposition and resuspension, when available for model calibration account for this. If simultaneous deposition and erosion are not permitted, the probability of deposition is defined as zero when the sediment erosion flux is greater than zero.

In conclusion, it is noted that in the CE-QUAL-ICM documentation which includes particulate matter resuspension (Cerco *et al.*, 2000), resuspension is explicitly included in various state variable equations, while in this document it is included implicitly as described in this section.

7.2.5. Method of Solution for Kinetics Equations

The kinetic equations for the 20 state variables, excluding fecal coliform, in the EFDC water column water quality model can be expressed in a system of 20 × 20 partial differential equations in each model cell, after linearizing some terms, mostly Monod type expressions:

$$\frac{\partial \mathbf{C}}{\partial t} = \mathbf{KC} + \frac{\partial}{\partial z}\left(\mathbf{WC}\right) + \mathbf{R} \tag{7.104}$$

where **C** is the vector of the state variables in [ML⁻³], **K** is a matrix in [T⁻¹], **W** is in [LT⁻¹], and **R** is a vector in [ML⁻³T⁻¹]. The ordering of variables follows that in Table 7.1 which results in **K** being lower triangular. Integrating (7.104) over layer *k*, gives

$$\frac{\partial \mathbf{C}_k}{\partial t} = \mathbf{K}1_k \mathbf{C}_k + \delta_k \mathbf{K}2_k \mathbf{C}_{k+1} + \mathbf{R}_k$$

$$\mathbf{K}1_k = \mathbf{K}_k - \frac{1}{\Delta_k}\mathbf{W} \tag{7.105}$$

$$\mathbf{K}2_k = \frac{1}{\Delta_k}\mathbf{W}$$

which indicates that the settling of particulate matter from the overlying cell acts as an input for a given cell. For the layer of cells adjacent to the bed, the erosion term in (7.102) is included in the vector **R**. The matrices and vectors in ((7.104) and (7.105)) are defined in Appendix A of Park *et al.* (1995). The layer index *k* increases upward with *KC* vertical layers; $k = 1$ is the bottom layer and $k = KC$ is the surface layer. Then $\delta k = 0$ for k = *KC*; otherwise, $\delta k = 1$. The matrix **K**2 is a diagonal matrix, and the non-zero elements account for the settling of particulate matter from the overlying cell.

Equation (7.105) is solved using a generalized trapezoidal scheme over a time step of θ, which may be expressed as:

$$\begin{aligned}\mathbf{C}_k^{n+1} - \mathbf{C}_k^n = {} & \lambda\theta\left(\mathbf{K}1_k^n \mathbf{C}_k^{n+1} + \delta_k \mathbf{K}2_k^n \mathbf{C}_{k+1}^{n+1} + \mathbf{R}_k^{n+1}\right) \\ & + \left(1-\lambda\right)\theta\left(\mathbf{K}1_k^n \mathbf{C}_k^n + \delta_k \mathbf{K}2_k^n \mathbf{C}_{k+1}^n + \mathbf{R}_k^n\right)\end{aligned} \tag{7.106}$$

or

$$\begin{aligned}&\left(\mathbf{I} - \lambda\theta\mathbf{K}1_k^n\right)\mathbf{C}_k^{n+1} = \left(\mathbf{I} + \left(1-\lambda\right)\theta\mathbf{K}1_k^n\right)\mathbf{C}_k^n \\ &+\theta\delta_k \mathbf{K}2_k^n\left(\lambda\mathbf{C}_{k+1}^{n+1} + \left(1-\lambda\right)\mathbf{C}_{k+1}^n\right) + \theta\left(\lambda\mathbf{R}_k^{n+1} + \left(1-\lambda\right)\mathbf{R}_k^n\right)\end{aligned} \tag{7.107}$$

where λ is an implicitness factor ($0 \le \lambda \le 1$), $\theta = 2 \cdot m \cdot \Delta t$ is the time step for the kinetic equations; **I** is the identity matrix; the superscripts *n* and *n+1* designate the variables before and after being adjusted for the relevant kinetic processes. Since equation (2.105) is solved from the surface layer downward, the term with $\mathbf{C}_{k+1}^{n+1}$ is known for the kth layer and thus placed on the RHS. In equation (2.106), inversion of a matrix can be avoided when the 20 state variables are solved in the order given in Table 7.1.

7.3 Rooted Aquatic Plants Formulation

7.3.1. Introduction

Rooted macrophyte beds are commonly observed along the banks of many rivers and the accuracy of a water quality model may be improved by simulating submerged aquatic vegetation (epiphytic algae and macrophytes) if a waterbody has documented macrophyte occurrences. EFDC's generic Rooted Aquatic Plant and Epiphyte Algae Sub-Model (RPEM) uses kinetic mass balance equations for rooted plant shoots, roots and epiphyte algae growing on the shoots. The user may enable or disable a variety of combinations for RPEM, including enabling simulation of rooted plants or epiphytes; enabling epiphytes growing on rooted plants; enabling the RPEM – Water Column Nutrient Interaction; and enabling RPEM – Sediment Diagenesis Interaction.

The state variables in the sub-model are rooted plant shoots, roots, epiphyte algae biomass and rooted plant shoot detritus biomass (Hamrick, 2006). The kinetic mass balance of these variables depends mainly on production, respiration and non-respiration loss rates. These rates in turn are mainly controlled by nutrients, carbon, oxygen, light field and temperature.

7.3.2. State Variable Equations

The kinetic mass balance equations for rooted plant shoots, roots and epiphyte algae growing on the shoots are

$$\frac{\partial(RPS)}{\partial t} = \left((1 - F_{PRPR}) \cdot P_{RPS} - R_{RPS} - L_{RPS}\right)RPS + JRP_{RS} \tag{7.108}$$

$$\frac{\partial(RPR)}{\partial t} = F_{PRPR} \cdot P_{RPS} \cdot RPS - (R_{RPR} + L_{RPR})RPR + JRP_{RS} \tag{7.109}$$

$$\frac{\partial(RPE)}{\partial t} = (P_{RPE} - R_{RPE} - L_{RPE})RPE \tag{7.110}$$

where

t is the time (day)

RPS (T_a, H_a) is the Rooted Plant Shoot Biomass (g C/m²),

F_{PRPR} (χ_{Ta}, χ_{Ha}) is the fraction of production directly transferred to roots (0< $FPGR$ <1),

P_{RPS} (g_{Ta}, g_{Ha}) is the production rate for plant shoots (1/day),

R_{RPS} (r_{Ta}, r_{Ha}) is the respiration rate for plant shoots (1/day),

L_{RPS} (m_{Ta}, m_{Ha}) is the non-respiration loss rate for plant shoots (1/day),

JRP_{RS} ($\chi_{Tb}T_b$, $\chi_{Hb}H_b$) is the carbon transport positive from roots to shoots (g C/m²/day)

RPR (T_b, H_b) is the Rooted Plant Root Biomass (g C/m²),

R_{RPR} (r_{Tb}, r_{Hb}) is the respiration rate for plant roots (1/day),

L_{RPR} (m_{Tb}, m_{Hb}) is the non-respiration loss rate for plant roots (1/day)

RPE (E) is the Rooted Plant Epiphyte Biomass (g C/m²),

P_{RPE} (g_E) is the production rate for epiphytes (1/day),

R_{PRE} ($r_E E$) is the respiration rate for epiphytes (1/day),

L_{RPE} (r_{Ta} +$m_E E$) is the non-respiration loss rate for epiphytes (1/day).

Equivalent notation used in the Florida Bay sea grass model appears in ( ). For comparison, Table 7.3 and Table 7.4 show generic and Florida Bay sea grass model parameters

Table 7.3: Generic and Florida Bay Sea Grass Model Parameters for Thalassia and Halodule

Parameter	Dimension	Generic	Thalassia	Halodule
F_{PRPR} (χ_{Ta}, χ_{Ha})	none	constant	0.4	0.34
P_{RPS} (g_{Ta}, g_{Ha})	1/day	Function of N, P, Light, Temp, Salt	Function of N, P, Light, Temp, Salt	Function of N, P, Light, Temp, Salt
R_{RPS} (r_{Ta}, r_{Ha})	1/day	Function of Temperature	0.01 (base) Temperature Function	0.029 (base) Temperature Function
L_{RPS} (m_{Ta}, m_{Ha})	1/day	Function of Temperature	0.001 (base) Temperature Function	0.004 (base) Temperature Function
R_{RPR} (r_{Tb}, r_{Hb})	1/day	Function of Temperature	0.0025 (base) Temperature Function	0.011(base) Temperature Function
L_{RPR} (m_{Tb}, m_{Hb})	1/day	Function of Temperature	0.0001 (base) Temperature Function	0.0004 (base) Temperature Function
JRP_{RS} ($\chi_{Tb}T_b$, $\chi_{Hb}H_b$)	g C/m²/day	$KRP_{RS} \cdot RPR$ or Eqs (16&17)	$\chi Tb Tb$ (χTb = 0.0005)	$\chi Hb Hb$ (χHb = 1.E-5)

Table 7.4: Generic and Florida Bay Sea Grass Model Parameters for Epiphytes

Parameter	Dimension	Generic	Epiphytes
P_{RPE} (g_E)	none	Function of N, P, Light, Temp, Salt	Function of N, P, Light, Temp, Salt
R_{RPE} ($r_E E$)	1/day	Function of Temperature	$r_E E$ r_E = (0.01 m²/gm-day)
L_{RPE} (m_{Ta} +$m_E E$)	1/day	constant	m_{Ta} +$m_E E$ r_E = (0.05 m²/gm-day)

An additional state variable is also added to account for shoot detritus at the bottom of the water column

$$\frac{\partial(RPD)}{\partial t} = F_{PRSD} \cdot P_{RPS} \cdot RPS - L_{RPD} RPS \tag{7.111}$$

where

RPD is the Rooted Plant Shoot Detritus Biomass (g C/m²),

F_{RPSD} is the fraction of shoot loss to detritus (0< *FRPSD*<1),

L_{RPD} is the decay rate of detritus (1/day).

The Florida Bay sea grass model does not include this variable.

7.3.2.1 Production Rate for Plant Shoots

The production or growth rate for plant shoots is given by

$$P_{RPS} = PM_{RPS} \cdot f_{1W}(N) \cdot f_{1B}(N) \cdot f_2(I) \cdot f_3(T) \cdot f_4(S) \cdot f_5(RPS) \tag{7.112}$$

where

PM RPS (*VT*, *VH*) is the maximum growth rate under optimal conditions for plant shoots (1/day),

f_1(N) is the effect of suboptimal nutrient concentration ($0 \le f_1 \le 1$),

f_2(I) is the effect of suboptimal light intensity ($0 \le f_2 \le 1$),

f_3(T) is the effect of suboptimal temperature ($0 \le f_3 \le 1$),

f_4(S) is the effect of salinity on fresh water plant shoot growth ($0 \le f_4 \le 1$),

f_5(RPS) is the carrying capacity effect on shoot growth ($0 \le f_5 \le 1$).

Maximum growth rates for the Florida Bay sea grass model are shown in Table 7.5.

Table 7.5: Maximum Growth Rate

Parameter	Units	Generic	Thalassia	Halodule
PM_{RPS} (V_T, V_H)	/day	constant	0.208	0.29

(1) Effect of Nutrients on Production

Nutrient limitation is specified in terms of both water column and bed nutrient levels by

$$f_{1W}(N) = \min\left(\frac{(NH4+NO3)_w}{KHN_{RPS}+(NH4+NO3)_w}, \frac{PO4d_w}{KHP_{RPS}+PO4d_w}\right)$$
$$f_{1b}(N) = \min\left(\frac{(NH4+NO3)_b}{KHN_{RPS}+(NH4+NO3)_b}, \frac{PO4d_b}{KHP_{RPS}+PO4d_b}\right) \tag{7.113}$$

where

NH4 is the ammonium nitrogen concentration (g N/m³)

NO3 is the nitrate + nitrite nitrogen concentration (g N/m³)

KHN_{RPS} is the half-saturation constant for nitrogen uptake from water column (g N/m³)

KHN_{RPR} (K_{TN},K_{HN}) is the half-saturation constant for nitrogen uptake from bed (g N/m³)

PO4d is the dissolved phosphate phosphorus concentration (g P/m³)

KHP_{RPS} is the half-saturation constant for phosphorus uptake from water column (g P/m³)

KHP_{RPR} (K_{TP},K_{HP}) is the half-saturation constant for phosphorus uptake from bed (g P/m³).

Table 7.6: List of nutrient limitation parameters for the Florida Bay sea grass model.

Parameter	Units	Generic	Thalassia	Halodule
KHN_{RPS}	gm/m³ (mg/l)	constant	0.0	0.0
KHN_{RPR}	gm/m³ (mg/l)	constant	0.04 µM 0.00056 mg/l	0.04 µM 0.00056 mg/l
KHP_{RPS}	gm/m³ (mg/l)	constant	0.0	0.0
KHP_{RPR}	gm/m³ (mg/l)	constant	0.1 µM 0.0031 mg/l	0.1 µM 0.0031 mg/l

(2) The Light Field

The light field in the water column is governed by

$$\frac{\partial I}{\partial Z_*} = -Kess \cdot I \tag{7.114}$$

where

I is the light intensity (Langley/day),

Kess is the light extinction coefficient (1/m),

Z_* is the depth below the water surface (m).

With the light extinction coefficient being a function of the depth below the water surface. Integration of (7) gives

$$I = I_{ws} \exp\left(-\int_0^{Z_*} Kess \cdot dZ_*\right) \tag{7.115}$$

The light intensity at the water surface, I_{WS}, is given by

$$I_{ws} = I_o \min\left(exp\left(-Kesh \cdot (H_{RPS} - H)\right), 1\right) \tag{7.116}$$

where

I_o is the light intensity at the top of the emergent shoot canopy for emergent shoots or the light intensity at the water surface for submerged shoots (W/m²),

Keme is the light extinction coefficient for emergent shoots (1/m),

HRPS is the shoot height (m),

H is the water column depth (m).

For submerged shoots, it is assumed that the light extinction coefficient in the water column above the shoot canopy is given by

$$Kessac = Ke_b + Ke_{ISS} \cdot ISS + Ke_{VSS} \cdot VSS + Ke_{Chl} \sum_{m=1}^{M} \left(\frac{B_m}{CChl_m}\right) \tag{7.117}$$

And the light extinction coefficient in the water column within the canopy is given by

$$Kessic = Ke_b + Ke_{ISS} \cdot ISS + Ke_{VSS} \cdot VSS + Ke_{Chl} \sum_{m=1}^{M} \left(\frac{B_m}{CChl_m}\right) + Ke_{RPS} \cdot RPS \tag{7.118}$$

where

Ke_b is the background light extinction (1/m),

Ke_{ISS} is the light extinction coefficient for inorganic suspended solid (1/m per g/m³),

ISS is the inorganic suspended solid concentration (g/m³) provided from the hydrodynamic model,

Ke_{VSS} is the light extinction coefficient for volatile suspended solid (1/m per g/m³),

VSS is the volatil suspended solid concentration (g/m³) provided from the water quality model,

$CChl_{RPE}$ is the carbon-to-chlorophyll ratio for epiphytes (g C per mg Chl),

Ke_{Chl} is the light extinction coefficient for algae chlorophyll (1/m per mg Chl/m³),

B_m is the concentration of algae group m (g C per ml),

$CChl_m$ is the carbon-to-chlorophyll ratio in algal group m (g C per mg Chl),

Ke_{RPS} is the light extinction coefficient for rooted plant shoots (1/m per gm C/m²),

RPS is the concentration of plant shoots (g C per m²).

The forms (7.117) and (7.118) readily allow for inclusion of algae biomass into the volatile suspended solids or visa-versa. The form of (7.118) assumes that the shoots are primarily self shading and that epiphyte effect are manifest on the shoot surface.

The solutions of (7.115) above and in the canopy are

$$I = I_{ws} \exp(-Kessac \cdot Z_*) \;;\; 0 \le Z_* \le H - H_{RPS} \tag{7.119}$$

$$I = I_{ct} \cdot \exp\left(-Kessic \cdot (Z_* - H + H_{RPS})\right)$$
$$I_{ct} = I_{ws} \cdot \exp\left(-Kessac \cdot (H - H_{RPS})\right) \quad (7.120)$$
$$H - H_{RPS} \le Z_* \le H$$

Since rooted plants are represented as carbon mass per unit area, the average light intensity over the shoot canopy is an appropriate light measure. For emergent shoots, the average of (7.120) over the water column depth, noting that $H = HRPS$, is

$$I_{icwa} = \frac{I_{ws}}{Kessic \cdot H}(1 - \exp(-Kessac \cdot H)) \quad (7.121)$$

For submerged shoots, the average of (3.13) over the canopy is

$$I_{icwa} = \frac{I_{ws}}{Kessic \cdot H_{RPS}} \exp\left(-Kessac \cdot (H - H_{RPS})\right) \cdot (1 - \exp(-Kessic \cdot H_{RPS})) \quad (7.122)$$

where I_{icwa} in both (7.121) and (7.122) is the average in canopy water column light intensity.

When epiphytes grow on the shoot surface, the light intensity at the shoot surface is further reduced according to

$$I_{RPS} = I_{icw} \exp(-Ke_{RPE} \cdot RPE) \quad (7.123)$$

where

I_{RPS} is the light intensity on the plant shoots (W/m²),

I_{icw} is the average water column light intensity in the shoot canopy (W/m²),

Ke_{RPE} is the light extinction coefficient for epiphyte (m² per gm C).

For the Florida Bay sea grass model, the epiphyte light extinction coefficient is given by

$$Ke_{RPE} = 0.11 \frac{\delta_{RPE}}{\sum_{Nspecies}\left(\frac{2 \cdot RPS \cdot \delta_{RPS}}{W_{RPS}}\right)} \quad (7.124)$$

where

δ_{RPE} is the Epiphyte dry mass to carbon mass ratio,

δ_{RPS} is the rooted plant shoot dry mass to carbon massratio,

W_{RPS} is the rooted plant shoot mass per unit shoot area.

Values of these parameters for Florida Bay are listed in Table 7.7. It is noted that the expression in (7.124) is not dimensionally homogeneous with the numerical coefficient 0.11 having implied units of (cm² leaf surface area)/(mg dry weight). Equation (7.124) can be made dimensionally consistent by use of the alternative form

$$Ke_{RPE} \cdot RPE = \frac{RPE}{\sum_{Nspecies}(KRPSE \cdot RPS)} \tag{7.125}$$

Where the dimensionless parameter *KRPSE* is also defined in Table 7.7.

Table 7.7: Epiphyte Light Attenuation Parameter for Florida Bay Sea Grass Model

Parameter	Units	Thalassia	Halodule
δ_{RPE}	Dry mass/ Carbon mass	9	9
δ_{RPS}	Dry mass/ Carbon mass	2.94	2.4
W_{RPS}	Mg dry mass/ Cm² leaf area	1.7	2
KRPSE	Dimensionless	3.49	2.42

Using (7.123) and (7.120), the light intensity on the shoot surface can be expressed as

$$I_{RPS} = I_{ws} \cdot \exp\begin{pmatrix} -Kessac \cdot (H - H_{RPS}) \\ -Ke_{RPE} \cdot RPE \end{pmatrix} \cdot \exp\left(-Kessic \cdot (Z_* - H + H_{RPS})\right) \tag{7.126}$$

While (7.123) and (7.122) give the canopy average light intensity on the shoot surface

$$I_{RPSA} = \frac{I_{ws}}{Kessic \cdot H_{RPS}} \exp\begin{pmatrix} -Kessac \cdot (H - H_{RPS}) \\ -Ke_{RPE} \cdot RPE \end{pmatrix} \cdot (1 - \exp(-Kessic \cdot H_{RPS})) \tag{7.127}$$

(3) Effects of Light on Growth

The EFDC generic rooted plant model includes three options to specify the effect of light on rooted plant growth. The first option is based on Steele's equation

$$f_2(I) = \frac{I}{I_{RSPopt}} \exp\left(1 - \frac{I}{I_{RSPopt}}\right) \tag{7.128}$$

Which can be applied in terms of the average light intensity reaching the shoots (20) to give

$$f_2(I) = \frac{I_{RPSA}}{I_{RSPopt}} \exp\left(1 - \frac{I_{RPSA}}{I_{RSPopt}}\right) \tag{7.129}$$

or due to its unique mathematical form directly averaged over the shoot canopy. The average is given by

$$f_{2avg}(I) = \frac{F_2}{H_{RPS}} \int_{H-H_{RPS}}^{H} \exp\begin{pmatrix} 1 - Kessic \cdot (Z_* - H + H_{RPS}) \\ -F_2 \cdot exp\left(-Kessic \cdot (Z_* - H + H_{RPS})\right) \end{pmatrix} dZ_* \tag{7.130}$$

With the results being

$$f_{2avg}(I) = \frac{\exp(1)}{Kessic \cdot H_{RPS}}[\exp(-F_2 \cdot exp(-Kessic \cdot H_{RPS})) - \exp(F_2)]$$

$$F_2 = \frac{I_{ws}}{I_{RSPopt}} \cdot \exp\begin{pmatrix} -Kessac \cdot (H - H_{RPS}) \\ -Ke_{RPE} \cdot RPE \end{pmatrix} \quad (7.131)$$

The second option for the effect of light on growth or production is a Monod type formulation adapted from the Chesapeake Bay SAV model

$$f_2(I) = \left(\frac{I_{RPS}}{I_{RPS} + KHI_{RPS}}\right) \quad (7.132)$$

where

I_{RPS} is the light intensity at the shoot surface (W/m²),

KHI_{RPS} is the shoot surface light intensity half saturation (W/m²).

Equation (7.132) can be applied using the average light intensity over the shoot canopy or averaged over the shoot canopy to give

$$f_{2avg}(I) = \frac{1}{Kessic \cdot H} \ln\left(\frac{KHI + I_{ws}\, exp(Ke_{RPE} \cdot RPE)}{KHI + I_{ws}\, exp(-Kessic \cdot H - Ke_{RPE} \cdot RPE)}\right) \quad (7.133)$$

For an emergent canopy and

$$f_{2avg}(I) = \frac{1}{Kessic \cdot H_{RSP}} \ln\left(\frac{KHI + I_{ct}\, exp(-Kessic \cdot (H - H_{RSP}) - Ke_{RPE} \cdot RPE)}{KHI + I_{ct}\, exp(-Kessic \cdot H - Ke_{RPE} \cdot RPE)}\right) \quad (7.134)$$

for a submerged canopy.

The final option is adapted from the Florida Bay sea grass model and is

$$f_2(I) = \tanh\left(\frac{I_{RPSavg}}{l_{RSP}}\right) \quad (7.135)$$

where the average intensity is used since a closed analytical average of this equation is not possible. Values for the parameters in equations (7.132) and (7.135) for Florida Bay are listed in Table 7.8.

Table 7.8: Light Limitation Parameters for Equations (7.132) and (7.135)

Parameter	Units	Thalassia	Halodule
KHI_{RPS}	W/m²	142(444 µmoles/m²-s)	82(255 µmoles/m²-s)

l_{RPS}	W/m²	130(407 µmoles/m²-s)	102(319 µmoles/m²-s)

Note on conversion: the conversion factor for 1 µmoles/m²-s to 0.3 to 0.34 W/m². Conversions in table are based on 0.32.

(4) Effect of Temperature on Shoot Growth

The effect of temperature on shoot growth is given by a Gaussian function

$$f_3(T) = \begin{cases} \exp(-KTP1_{RPS}[T - TP1_{RPS}]^2) \ if \ T \le TP1_{RPS} \\ 1 \ if \ TP1_{RPS} < T < TP2_{RPS} \\ \exp(-KTP2_{RPS}[T - TP2_{RPS}]^2) \ if \ T \ge TP1_{RPS} \end{cases} \quad (7.136)$$

where

T is the temperature (°C) provided from the hydrodynamic model,

$TP1_{RPS} < T < TP2_{RPS}$ is the optimal temperature range for shoot production (°C),

$KTP1_{RPS}$ is the effect of temperature below $TM1RPS$ on shoot production (1/°C²),

$KTP2_{RPS}$ is the effect of temperature above $TM2RPS$ on shoot production (1/°C²).

or an exponential function.

$$f_3(T) = \exp(KTP_{RPS}[T - TPREF_{RPS}]) \quad (7.137)$$

where

$TPREF_{RPS}$ is the reference temperature for shoot production (°C),

KTP_{RPS} is the effect of temperature on shoot production (1/°C).

With parameters for the Florida Bay sea grass model given in Table 7.9.

Table 7.9: Parameters for Temperature Effect on Growth for Equation (7.137)

Parameter	Units	Thalassia	Halodule
$TPREF_{RPS}$	°C	28	31
KTP_{RPS}	1/°C	0.07	0.07

(5) Effect of Salinity

The effect of salinity on fresh water plant shoot growth is given by

$$f_4(S) = \frac{STOXS^2}{STOXS^2 + S^2} \quad (7.138)$$

where

$STOXS$ is the salinity at which growth is halved (ppt),

S is the salinity in water column (ppt) provided from the hydrodynamic model.

(6) Effect of Rooted Plant Density

The effect of rooted plant density on growth is give by

$$f_5(RPS) = 1 - \left(\sum_{species} \frac{RPS}{RPS_{sat}} \right)^2 \tag{7.139}$$

where

RPS_{sat} is the density saturation parameter (g C/m²).

The summation indicates when multiple species are simulated, the total density of all species effects each individual species

Table 7.10: Parameters for Plant Density Effect on Growth for Equation (7.139)

Parameter	Units	Thalassia	Halodule
RPS_{sat}	(g Cn/m²)	400	667

(7) Respiration Rate for Plant Shoots

The respiration rate for plant shoots is assumed to be temperature dependent

$$R_{RPS} = RREF_{RPS} \cdot \exp(KTR_{RPS}[T - TRREF_{RPS}]) \tag{7.140}$$

where

$RREF_{RPS}$ is the reference respiration rate for shoots (1/day),

T is the temperature (°C) provided from the hydrodynamic model,

$TRREF_{RPS}$ is the reference temperature for shoot respiration (°C),

KTR_{RPS} is the effect of temperature on shoot respiration (1/°C²).

Table 7.11: Parameters for Shoot Respiration in Equation (7.140)

Parameter	Units	Thalassia	Halodule
$RREF_{RPS}$	1/day	0.01	0.029
KTR_{RPS}	dimensionless	0.07	0.07
$TRREF_{RPS}$	°C	28	31

(8) Non-Respiration Loss Rate for Plant Shoots

The non-respiration loss rate for shoots is assumed to be temperature dependent.

$$L_{RPS} = LREF_{RPS} \cdot \exp(KTL_{RPS}[T - TLREF_{RPS}]) \quad (7.141)$$

where

$LREF_{RPS}$ is the reference loss rate for shoots (1/day),

T is the temperature (°C) provided from the hydrodynamic model,

$TLREF_{RPS}$ is the reference temperature for shoot loss (°C),

KTL_{RPS} is the effect of temperature on shoot loss (1/°C²).

Table 7.12: Parameters for Shoot Mortality of non-respiration loss in (7.141)

Parameter	Units	Thalassia	Halodule
$LREF_{RPS}$	1/day	0.001	0.004
KLR_{RPS}	dimensionless	0.07	0.07
$TLREF_{RPS}$	°C	28	28

(9) Carbon Transport from Roots to Shoots

The carbon transport from roots to shoots is defined as positive to the shoots. Two formulations can be utilized, the first is based on observed shoot to root biomass ratios

$$JRP_{RS} = KRPO_{RS} \cdot (RPR - RORS \cdot RPS)$$
$$RORS = \frac{RPR_{obs}}{RPS_{obs}} \quad (7.142)$$

where

$KRPO_{RS}$ is the root to shoot transfer rate to follow observed ratio (1/day),

$RORS$ is the observed ratio of root carbon to shoot carbon (dimensionless).

The second formulation transfers root carbon to shoot carbon under unfavorable light conditions for the shoots

$$JRP_{RS} = KRP_{RS}\left(\frac{I_{SS}}{I_{SS} + I_{SSS}}\right)RPR \quad (7.143)$$

where

KRP_{RS} (χ_{Tb}, χ_{Hb}) is the root to shoot transfer rate (1/day),

I_{SS} is the solar ratio at shoot surface (W/m²),

I_{SSS} is the half-saturation solar ratio at shoot surface (W/m²).

Table 7.13: Root to Shoot Transport Parameters in Equation (7.143)

Parameter	**Dimension**	**Generic**	**Thalassia**	**Halodule**

$KRPO_{RS}$	1/day	constant	0.0005	0.00001
$RORS$	dimensionless	constant	0.0	0.0
KRP_{RS}	1/day	constant	0.0005	0.00001
I_{SSS}	W/m²	constant	0.0	0.0

(10) Respiration Rate for Plant Roots

The respiration rate for plant roots is assumed to be temperature dependent

$$R_{RPR} = RREF_{RPR} \cdot \exp(KTR_{RPR}[T - TRREF_{RPR}]) \tag{7.144}$$

where

$RREF_{RPR}$ is the reference respiration rate for roots (1/day),

T is the temperature (°C) provided from the hydrodynamic model,

$TRREF_{RPR}$ is the reference temperature for root respiration (°C),

KTR_{RPR} is the effect of temperature on shoot respiration (1/°C²).

Table 7.14: Parameters for Root Respiration in Equation (7.144)

Parameter	Units	**Thalassia**	**Halodule**
$RREF_{RPR}$	1/day	0.0025	0.011
KTR_{RPR}	dimensionless	0.07	0.07
$TRREF_{RPR}$	°C	28	31

(11) Non-Respiration Loss Rate for Plant Roots

The non-respiration loss rate for shoots is assumed to be temperature dependent.

$$L_{RPR} = LREF_{RPR} \cdot \exp(KTL_{RPR}[T - TLREF_{RPR}]) \tag{7.145}$$

Table 7.15: Parameters for Root Mortality in Equation (7.145)

Parameter	Units	**Thalassia**	**Halodule**
$LREF_{RPR}$	1/day	0.0001	0.0004
KLR_{RPR}	dimensionless	0.07	0.07
$TLREF_{RPR}$	°C	28	28

7.3.2.2 Production Rate for Epiphytes

The production or growth rate for epiphytes on plant shoots is given by

$$P_{RPE} = PM_{RPE} \cdot f_w(N) \cdot f_2(I) \cdot f_3(T) \cdot f_4(RPE, RPS) \tag{7.146}$$

where

PM_{RPE} is the maximum growth rate under optimal conditions for plant shoots (1/day),

$f_1(N)$ is the effect of suboptimal nutrient concentration ($0 \le f_1 \le 1$),

$f_2(I)$ is the effect of suboptimal light intensity ($0 \le f_2 \le 1$),

$f_3(T)$ is the effect of suboptimal temperature ($0 \le f_3 \le 1$),

$f_4(RPE,RPS)$ is the effect of epiphyte and host rooted density ($0 \le f_4 \le 1$).

(1) Effect of Nutrients on Epiphyte Growth

Nutrient limitation for epiphytes is given by

$$f_1(N) = \min\left(\frac{NH4 + NO3}{KHN_{RPE} + NH4 + NO3}, \frac{PO4d}{KHP_{RPE} + PO4d}\right) \tag{7.147}$$

where

$NH4$ is the ammonium nitrogen concentration (g N/m³),

$NO3$ is the nitrate + nitrite nitrogen concentration (g N/m³),

KHN_{RPE} is the half-saturation constant for nitrogen uptake for epiphytes (g N/m³),

$PO4d$ is the dissolved phosphate phosphorus concentration (g P/m³),

KHP_{RPE} is the half-saturation constant for phosphorus uptake for epiphytes (g P/m³).

(2) Effect of Light on Epiphyte Growth

Light limitation for epiphyte growth is based on a Monod type equation similar to (7.132)

$$f_2(I) = \left(\frac{I_{RPE}}{I_{RPE} + KHI_{RPE}}\right) \tag{7.148}$$

where

KHI_{RPE} is the half-saturation constant for epiphyte light limitation (W/m²).

The average light intensity over the shoot canopy

$$I_{RPEA} = \frac{I_{ws}}{Kessic \cdot H_{RPS}} \exp\left(-Kessac \cdot (H - H_{RPS})\right) \cdot (1 - \exp(-Kessic \cdot H_{RPS})) \tag{7.149}$$

Which follows from (7.130) with $KeRPE = 0$, can be used or (7.148) can be averaged over the shoot canopy to give

$$f_{2avg}(I) = \frac{1}{Kessic \cdot H_{RSP}} \ln\left(\frac{KHI + I_{ct}\, exp\left(-Kessic \cdot (H - H_{RSP})\right)}{KHI + I_{ct}\, exp(-Kessic \cdot H)}\right)$$
$$I_{ct} = I_{ws} \cdot \exp\left(-Kessic \cdot (H - H_{RSP})\right) \tag{7.150}$$

which follows from (7.134) with $Ke_{RPE} = 0$.

(3) Effect of Temperature on Epiphyte Growth

The effect of temperature on epiphyte growth is given by

$$f_3(T) = \begin{cases} \exp(-KTP1_{RPE}[T - TP1_{RPE}]^2) \ if \ T \le TP1_{RPE} \\ 1 \ if \ TP1_{RPE} < T < TP2_{RPE} \\ \exp(-KTP2_{RPE}[T - TP2_{RPE}]^2) \ if \ T \ge TP1_{RPE} \end{cases} \quad (7.151)$$

where

T is the temperature (°C) provided from the hydrodynamic model,

$TP1_{RPE} < T < TP2_{RPE}$ is the optimal temperature range for epiphyte production (°C),

$KTP1_{RPE}$ is the effect of temperature below $TM1RPS$ on epiphyte production (1/°C²),

$KTP2_{RPE}$ is the effect of temperature above $TM2RPS$ on epiphyte production (1/°C²).

or an exponential function.

$$f_3(T) = \exp(KTP_{RPE}[T - TPREF_{RPE}]) \quad (7.152)$$

where

$TPREF_{RPE}$ is the reference temperature for shoot production (°C),

KTP_{RPE} is the effect of temperature on shoot production (1/°C).

(4) Effect of Epiphyte and Rooted Plant Density on Epiphyte Growth

The effect of rooted plant density on growth is give by

$$f_4(RPE, RPS) = 1 - \left(\frac{RPE \cdot \delta_{RPE}}{W_{RPE} \sum_{Nspecies} \left(\frac{2 \cdot RPS \cdot \delta_{RPS}}{W_{RPS}} \right)} \right)^2 \quad (7.153)$$

where

δ_{RPE} is the Epiphyte dry mass to carbon mass ratio,

W_{RPE} (κe) is the maximum epiphyte mass per unit shoot area.

(5) Respiration Rate for Epiphytes

The respiration rate for epiphytes is assumed to be temperature dependent

$$R_{RPE} = RREF_{RPE} \cdot \exp(KTR_{RPE}[T - TRREF_{RPE}]) \quad (7.154)$$

where

$RREF_{RPE}$ is the reference respiration rate for epiphytes (1/day),

T is the temperature (°C) provided from the hydrodynamic model,

$TRREF_{RPE}$ is the reference temperature for epiphytes respiration (°C),

KTR_{RPE} is the effect of temperature on epiphytes respiration (1/°C²).

(6) Non-Respiration Loss Rate for Epiphytes

The non-respiration loss rate for shoots is assumed to have a temperature dependency

$$L_{RPE} = LREF_{RPE} \cdot \exp(KTL_{RPE}[T - TLREF_{RPE}]) \tag{7.155}$$

where

$LREF_{RPE}$ is the reference non-respiration loss rate for epiphytes (1/day),

T is the temperature (°C) provided from the hydrodynamic model,

$TLREF_{RPE}$ is the reference temperature for epiphytes loss (°C),

KTL_{RPE} is the effect of temperature on epiphytes loss (1/°C²).

7.3.2.3 Decay Rate for Shoot Detritus

The decay rate for shoot detritus is assumed to have a temperature dependency

$$L_{RPD} = LREF_{RPD} \cdot \exp(KTL_{RPD}[T - TLREF_{RPD}]) \tag{7.156}$$

where

$LREF_{RPD}$ is the reference non-respiration loss rate for epiphytes (1/day),

T is the temperature (°C) provided from the hydrodynamic model,

$TLREF_{RPD}$ is the reference temperature for epiphytes loss (°C),

KTL_{RPD} is the effect of temperature on epiphytes loss (1/°C²).

7.3.2.4 Coupling with Organic Carbon

The interaction between rooted plants and epiphytes and water column and bed organic carbon species is given by

$$\frac{\partial RPOC_W}{\partial t} = \frac{1}{H}(FCR_{RPS} \cdot R_{RPS} + (1 - F_{RPSD}) \cdot FCRL_{RPS} \cdot L_{RPS})RPS + \frac{1}{H}(FCR_{RPE} \cdot R_{RPE} + FCRL_{RPE} \cdot L_{RPE})RPE + \frac{1}{H}FCRL_{RPD} \cdot L_{RPD} \cdot RPD \tag{7.157}$$

$$\frac{\partial RPOC_B}{\partial t} = \frac{1}{B}(FCR_{RPR} \cdot R_{RPR} + FCRL_{RPR} \cdot L_{RPR})RPR \tag{7.158}$$

$$\frac{\partial LPOC_W}{\partial t} = \frac{1}{H}(FCL_{RPS} \cdot R_{RPS} + (1 - F_{RPSD}) \cdot FCLL_{RPS} \cdot L_{RPS})RPS + \frac{1}{H}(FCL_{RPE} \cdot R_{RPE} + FCLL_{RPE} \cdot L_{RPE})RPE + \frac{1}{H}FCLL_{RPD} \cdot L_{RPD} \cdot RPD \tag{7.159}$$

$$\frac{\partial LPOC_B}{\partial t} = \frac{1}{B}(FCL_{RPR} \cdot R_{RPR} + FCLL_{RPR} \cdot L_{RPR})RPR \tag{7.160}$$

$$\frac{\partial DOC_W}{\partial t} = \frac{1}{H}(FCD_{RPS} \cdot R_{RPS} + (1 - F_{RPSD}) \cdot FCDL_{RPS} \cdot L_{RPS})RPS + \frac{1}{H}(FCD_{RPE} \cdot R_{RPE} + FCDL_{RPE} \cdot L_{RPE})RPE + \frac{1}{H}FCDL_{RPD} \cdot L_{RPD} \cdot RPD \quad (7.161)$$

$$\frac{\partial DOC_B}{\partial t} = \frac{1}{B}(FCD_{RPR} \cdot R_{RPR} + FCDL_{RPR} \cdot L_{RPR})RPR \quad (7.162)$$

where

RPOC is the concentration of refractory particulate organic carbon (g C/m³),

LPOC is the concentration of labile particulate organic carbon (g C/m³),

DOC is the concentration of dissolved organic carbon (g C/m³),

FCR is the fraction of respired carbon produced as refractory particulate organic carbon,

FCL is the fraction of respired carbon produced as labile particulate organic carbon,

FCD is the fraction of respired carbon produced as dissolved organic carbon,

FCRL is the fraction of non-respired carbon produced as refractory particulate organic carbon,

FCLL is the fraction of non-respired carbon produced as labile particulate organic carbon,

FCDL is the fraction of non-respired carbon produced as dissolved organic carbon,

H is the depth of water column,

B is the depth of bed.

7.3.2.5 Coupling with Dissolved Oxygen

The interaction between rooted plants and epiphytes and dissolved oxygen is given by

$$\frac{\partial DO_W}{\partial t} = \frac{1}{H}(P_{RPS} \cdot RPSOC \cdot RPS + P_{RPE} \cdot RPEOC \cdot RPE) \quad (7.163)$$

where

DO is the concentration of dissolved oxygen (g O_2/ m³),

RPSOC is the oxygen to carbon ratio for plant shoots (g O_2 per g C),

RPEOC is the oxygen to carbon ratio for epiphytes (g O_2 per g C).

7.3.2.6 Coupling with Phosphorous

The interaction between rooted plants and epiphytes and water column and bed phosphorous is given by

$$\frac{\partial RPOP_W}{\partial t} = \frac{1}{H}(FPR_{RPS} \cdot R_{RPS} + (1 - F_{RPSD}) \cdot FPRL_{RPS} \cdot L_{RPS}) \cdot RPSPC \cdot RPS + \frac{1}{H}(FPR_{RPE} \cdot R_{RPE} + FPRL_{RPE} \cdot L_{RPE}) \cdot RPEPC \cdot RPE + \frac{1}{H}FPRL_{RPD} \cdot L_{RPD} \cdot RPSPC \cdot RPD \quad (7.164)$$

$$\frac{\partial RPOP_B}{\partial t} = \frac{1}{B}(FPR_{RPR} \cdot R_{RPR} + FPRL_{RPR} \cdot L_{RPR})RPRPC \cdot RPR \quad (7.165)$$

$$\frac{\partial LPOP_W}{\partial t} = \frac{1}{H}(FPL_{RPS} \cdot R_{RPS} + (1 - F_{RPSD}) \cdot FPLL_{RPS} \cdot L_{RPS}) \cdot RPSPC \cdot RPS + \frac{1}{H}(FPL_{RPE} \cdot R_{RPE} + FPLL_{RPE} \cdot L_{RPE}) \cdot RPEPC \cdot RPE + \frac{1}{H} FPLL_{RPD} \cdot L_{RPD} \cdot RPSPC \cdot RPD \quad (7.166)$$

$$\frac{\partial LPOP_B}{\partial t} = \frac{1}{B}(FPL_{RPR} \cdot R_{RPR} + FPLL_{RPR} \cdot L_{RPR}) RPRPC \cdot RPR \quad (7.167)$$

$$\frac{\partial DOP_W}{\partial t} = \frac{1}{H}(FPD_{RPS} \cdot R_{RPS} + (1 - F_{RPSD}) \cdot FPDL_{RPS} \cdot L_{RPS}) \cdot RPSPC \cdot RPS + \frac{1}{H}(FPD_{RPE} \cdot R_{RPE} + FPDL_{RPE} \cdot L_{RPE}) \cdot RPEPC \cdot RPE + \frac{1}{H} FCDL_{RPD} \cdot L_{RPD} \cdot RPSPC \cdot RPD \quad (7.168)$$

$$\frac{\partial DOP_B}{\partial t} = \frac{1}{B}(FPD_{RPR} \cdot R_{RPR} + FPDL_{RPR} \cdot L_{RPR}) RPRPC \cdot RPR \quad (7.169)$$

$$\frac{\partial PO4t_W}{\partial t} = \frac{1}{H}(FPI_{RPS} \cdot R_{RPS} + (1 - F_{RPSD}) \cdot FPIL_{RPS} \cdot L_{RPS}) \cdot RPSPC \cdot RPS + \frac{1}{H}(FPI_{RPE} \cdot R_{RPE} + FPIL_{RPE} \cdot L_{RPE}) \cdot RPEPC \cdot RPE + \frac{1}{H} FCIL_{RPD} \cdot L_{RPD} \cdot RPSPC \cdot RPD - \frac{1}{H} F_{RPSPW} \cdot R_{RPS} \cdot RPSPC \cdot RPS - \frac{1}{H} P_{RPE} \cdot RPEPC \cdot RPE \quad (7.170)$$

$$\frac{\partial PO4t_B}{\partial t} = \frac{1}{B}(FPI_{RPR} \cdot R_{RPR} + FPIL_{RPR} \cdot L_{RPR}) RPRPC \cdot RPR - \frac{1}{H}(1 - F_{RPSPW}) P_{RPS} \cdot RPRPC \cdot RPS \quad (7.171)$$

$$F_{RPSPW} = \frac{KHP_{RPR} P04d_w}{KHP_{RPR} P04d_w + KHP_{RPS} P04d_b} \quad (7.172)$$

where

RPOP is the concentration of refractory particulate organic phosphorous (g C/m³),

LPOP is the concentration of labile particulate organic phosphorous (g C/m³),

DOP is the concentration of dissolved organic phosphorous (g C/m³),

PO4t = *PO4d* + *PO4p* is the total phosphate (g P/m³),

PO4d is the dissolved phosphate (g P/m³),

PO4p is the particulate (sorbed) phosphate (g P/m³),

FPR is the fraction of respired phosphorous produced as refractory particulate organic phosphorous,

FPL is the fraction of respired phosphorous produced as labile particulate organic phosphorous,

FPD is the fraction of respired phosphorous produced as dissolved organic phosphorous,

FPI is the fraction of respired phosphorous produced as total phosphate,

FPRL is the fraction of non-respired phosphorous produced as refractory particulate organic phosphorous,

FPLL is the fraction of non-respired phosphorous produced as labile particulate organic phosphorous,

FPDL is the fraction of non-respired phosphorous produced as dissolved organic phosphorous,

FPIL is the fraction of non-respired phosphorous produced as total phosphate,

RPSPC is the plant shoot phosphorous to carbon ratio (g P per g C),

RPRPC is the plant root phosphorous to carbon ratio (g P per g C),

RPEPC is the epiphyte phosphorous to carbon ratio (g P per g C),

F_{RPSPW} is the fraction of *PO4d* uptake from water column,

KHP_{RPS} is the half-saturation constant for phosphorus uptake from water column (g P/m³),

KHP_{RPR} is the half-saturation constant for phosphorus uptake from bed (g P/m³).

7.3.2.7 Coupling with Nitrogen

The interaction between rooted plants and epiphytes and water column and bed phosphorous is given by

$$\frac{\partial RPON_W}{\partial t} = \frac{1}{H}(FNR_{RPS} \cdot R_{RPS} + (1 - F_{RPSD}) \cdot FNRL_{RPS} \cdot L_{RPS}) \cdot RPSNC \cdot RPS + \frac{1}{H}(FNR_{RPE} \cdot R_{RPE} + FNRL_{RPE} \cdot L_{RPE}) \cdot RPENC \cdot RPE + \frac{1}{H} FNRL_{RPD} \cdot L_{RPD} \cdot RPSNC \cdot RPD \tag{7.173}$$

$$\frac{\partial RPON_B}{\partial t} = \frac{1}{B}(FNR_{RPR} \cdot R_{RPR} + FNRL_{RPR} \cdot L_{RPR}) RPRNC \cdot RPR \tag{7.174}$$

$$\frac{\partial LPON_W}{\partial t} = \frac{1}{H}(FNL_{RPS} \cdot R_{RPS} + (1 - F_{RPSD}) \cdot FNLL_{RPS} \cdot L_{RPS}) \cdot RPSNC \cdot RPS + \frac{1}{H}(FNL_{RPE} \cdot R_{RPE} + FNLL_{RPE} \cdot L_{RPE}) \cdot RPENC \cdot RPE + \frac{1}{H} FNLL_{RPD} \cdot L_{RPD} \cdot RPSNC \cdot RPD \tag{7.175}$$

$$\frac{\partial LPON_B}{\partial t} = \frac{1}{B}(FNL_{RPR} \cdot R_{RPR} + FNLL_{RPR} \cdot L_{RPR}) RPRNC \cdot RPR \tag{7.176}$$

$$\frac{\partial DON_W}{\partial t} = \frac{1}{H}(FND_{RPS} \cdot R_{RPS} + (1 - F_{RPSD}) \cdot FNDL_{RPS} \cdot L_{RPS}) \cdot RPSNC \cdot RPS + \frac{1}{H}(FND_{RPE} \cdot R_{RPE} + FNDL_{RPE} \cdot L_{RPE}) \cdot RPENC \cdot RPE + \frac{1}{H} FNDL_{RPD} \cdot L_{RPD} \cdot RPSNC \cdot RPD \tag{7.177}$$

$$\frac{\partial DON_B}{\partial t} = \frac{1}{B}(FND_{RPR} \cdot R_{RPR} + FNDL_{RPR} \cdot L_{RPR}) RPRNC \cdot RPR \tag{7.178}$$

$$\frac{\partial NH4_W}{\partial t} = \frac{1}{H}(FNI_{RPS} \cdot R_{RPS} + (1 - F_{RPSD}) \cdot FNIL_{RPS} \cdot L_{RPS}) \cdot RPSNC \cdot RPS + \frac{1}{H}(FNI_{RPE} \cdot R_{RPE} + FNIL_{RPE} \cdot L_{RPE}) \cdot RPENC \cdot RPE + \frac{1}{H} FNIL_{RPD} \cdot L_{RPD} \cdot RPSNC \cdot RPD - \frac{1}{H} PN_{RPS} \cdot F_{RPSNW} \cdot R_{RPS} \cdot RPSNC \cdot RPS - \frac{1}{H} PN_{RPE} \cdot P_{RPE} \cdot RPENC \cdot RPE \tag{7.179}$$

$$\frac{\partial NH4_W}{\partial t} = \frac{1}{B}(FNI_{RPR} \cdot R_{RPR} + FNIL_{RPR} \cdot L_{RPR})RPRNC \cdot RPR - \frac{1}{H}PN_{RPE}(1 - F_{RPSPW})P_{RPS} \cdot RPSNC \cdot RPS \quad (7.180)$$

$$\frac{\partial NO3_W}{\partial t} = -\frac{1}{H}(1 - PN_{RPS})F_{RPSNW} \cdot P_{RPS} \cdot RPSNC \cdot RPS - \frac{1}{H}(1 - PN_{RPE})P_{RPE} \cdot RPENC \cdot RPE \quad (7.181)$$

$$\frac{\partial NO3_B}{\partial t} = -\frac{1}{H}(1 - PN_{RPS})(1 - F_{RPSNW})P_{RPS} \cdot RPSNC \cdot RPS \quad (7.182)$$

$$PN_{RPS} = \frac{NH4 \cdot NO3}{(KHNP_{RPS} + NH4)(KHNP_{RPS} + NO3)} + \frac{NH4 \cdot KHNP_{RPS}}{(NH4 + NO3)(KHNP_{RPS} + NO3)} \quad (7.183)$$

$$PN_{RPE} = \frac{NH4 \cdot NO3}{(KHNP_{RPE} + NH4)(KHNP_{RPE} + NO3)} + \frac{NH4 \cdot KHNP_{RPE}}{(NH4 + NO3)(KHNP_{RPE} + NO3)} \quad (7.184)$$

$$F_{RPSNW} = \frac{KHN_{RPR}(NH4 + NO3)_w}{KHN_{RPR}(NH4 + NO3)_w + KHN_{RPS}(NH4 + NO3)_b} \quad (7.185)$$

where

RPON is the concentration of refractory particulate organic nitrogen (g N/m³),

LPON is the concentration of labile particulate organic nitrogen (g N/m³),

DON is the concentration of dissolved organic nitrogen (g N/m³),

NH4 is the ammonia (g N/m³),

NO3 is the nitrate + nitrite nitrogen (g N/m³),

FNR is the fraction of respired nitrogen produced as refractory particulate organic nitrogen,

FNL is the fraction of respired nitrogen produced as labile particulate organic nitrogen,

FND is the fraction of respired nitrogen produced as dissolved organic nitrogen,

FNI is the fraction of respired nitrogen produced as ammonia,

FNRL is the fraction of non-respired nitrogen produced as refractory particulate organic nitrogen,

FNLL is the fraction of non-respired nitrogen produced as labile particulate organic nitrogen,

FNDL is the fraction of non-respired nitrogen produced as dissolved organic nitrogen,

FNIL is the fraction of non-respired nitrogen produced as ammonia,

RPSNC is the plant shoot nitrogen to carbon ratio (g N per g C),

RPRNC is the plant root nitrogen to carbon ratio (g N per g C),

F_{RPSNW} is the plant shoot fraction of NH4 and NOX uptake from water column,

PN_{RPS} is the ammonia nitrogen preference fraction for plant shoots,

$KHNP_{RPS}$ is the saturation coefficient for nitrogen preference for plant shoots (g N per g C),

PN_{RPE} is the ammonia nitrogen preference fraction for epiphytes,

$KHNP_{RPE}$ is the saturation coefficient for nitrogen preference for epiphytes (g N per g C),

KHN_{RPS} is the half-saturation constant for nitrogen uptake from water column (g N/m³),

KHN_{RPR} is the half-saturation constant for nitrogen uptake from bed (g N/m³),

7.4 Macroalgae (Periphyton) State Variable

The EFDC water quality model was augmented to represent benthic attached algae (often referred to as macroalgae in estuarine waters and periphyton in fresh waters) using the existing framework for phytoplankton growth kinetics. Mathematical relationships based on the impacts of temperature, available light, available nutrients, stream velocity, and density-dependent interactions were incorporated into the algae growth kinetics framework within EFDC. The major difference between modeling techniques for attached and free-floating algae are as follows: (1) attached algae are expressed in terms of areal densities rather than volumetric concentrations; (2) attached algae growth can be limited by the availability of bottom substrate; (3) the availability of nutrients to the macroalgae matrix can be influenced by stream velocity; and (4) macroalgae are not subject to hydrodynamic transport. A good description of periphyton kinetics as it relates to water quality modeling can be found in Warwick et al. (1997) and has been used to develop this section of the report.

A mass balance approach was used to model macroalgae growth with carbon serving as the measure of standing crop size or biomass. For each model grid cell the equation for macroalgae growth is slightly different than the one for free-floating algae (Eq. (7.7)):

$$P_m = PM_m \cdot f_1(N) \cdot f_2(I) \cdot f_3(T) \cdot f_4(V) \cdot f_5(D) \tag{7.186}$$

where

PM_m is the maximum growth rate under optimal conditions for macroalgae,

$f_1(N)$ is the effect of suboptimal nutrient concentration ($0 \le f_1 \le 1$),

$f_2(I)$ is the effect of suboptimal light intensity ($0 \le f_2 \le 1$),

$f_3(T)$ is the effect of suboptimal temperature ($0 \le f_3 \le 1$),

$f_4(V)$ is the velocity limitation factor ($0 \le f_4 \le 1$),

$f_5(D)$ is the density dependent growth rate reduction factor ($0 \le f_5 \le 1$).

The basic growth kinetics for macroalgae were developed from those supplied by EFDC and others developed by Runke (1985). The macroalgae population as a whole is characterized by the total biomass present without considering the different species and their associated environmental processes. The optimum growth for the given temperature is adjusted for light, nutrients, velocity, and density- dependent limitations. Each growth limitation factor can vary from 0 to 1. A value of 1 indicates the factor does not limit growth, and a value of 0 means the factor is so severely limiting that growth is stopped entirely (Bowie et al. 1985).

Stream velocity has a twofold effect on periphyton productivity in freshwater streams: velocity increases to a certain level to enhance biomass accrual, but further increases result in substantial scouring (Horner et al. 1990). A benthic algal population is typified as a plant community with an understory and overstory. The entire community is called a matrix. As the matrix develops, the periphyton community is composed of an outer layer of photosynthetically active cells and inner layers of senescent and decomposing cells. Layering can also develop among different species of periphyton. Environmental conditions within the matrix are altered by the physical structure of the periphyton. This influences nutrient uptake and primary production rates of the algae (Sand-Jensen 1983). Above a certain level, current has a simulating effect on periphyton metabolism by mixing the overlying waters with nutrient- poor waters that develop around cells (Whitford and Schumacher 1964). The physical structure of the periphyton community and nutrient uptake by periphyton interfere with nutrient flux through the microbial matrix (Stevenson and Glover 1993).

Current is constantly scouring periphyton from its substrate. At high enough velocities, shear stress can result in substantial biomass reduction. Even at low velocities, sudden increases in velocity raise

instantaneous loss rates substantially, but these high rates persist only briefly (Horner et al. 1990). An increase in velocity above that to which benthic algae are accustomed leads to increased loss rates and temporarily reduced biomass. However, recolonization and growth after biomass reduction are usually rapid. The effects of suboptimal velocity upon growth rate are represented in the model by a velocity limitation function. Two options are available in the model for specifying the velocity limitation: (1) a Michaelis-Menton (or Monod) equation (7.187) and (2) a five-parameter logistic function (7.188). The Monod equation limits macroalgae growth due to low velocities whereas the five-parameter logistic function can be configured to limit growth due to either low or high velocities (see Figure 4-2).

Velocity limitation option 1, the Michaelis-Menton equation is written as follows:

$$f_4(V) = \frac{U}{KMV + U} \tag{7.187}$$

where

U is the stream velocity (m/s),

KMV is the half-saturation velocity (m/s).

Velocity limitation option 2, the five-parameter logistic function is as follows:

$$f_4(V) = d + \frac{a - d}{\left[1 + \left(\frac{U}{c}\right)^b\right]^e} \tag{7.188}$$

where

U is the stream velocity (m/s),

a is the asymptote at minimum x,

b is the slope after asymptote a,

c is the x-translation,

d is the asymptote at maximum x,

e is the slope before asymptote d.

The half-saturation velocity in Eq. (7.187) is the velocity at which half the maximum growth rate occurs. This effect is analogous to the nutrient limitation because the effect of velocity at suboptimal levels on periphyton growth is due to increasing the exchange of nutrients between the algal matrix and the overlying water (Runke 1985). However, this formula can be too limiting at low velocities. This function does not allow periphyton growth in still waters, but periphyton does grow in still waters such as lakes. Therefore, the function is applied only at velocities above a minimum threshold level (KMVmin). When velocities are at or below this lower level, the limitation function is applied at the minimum level. Above this velocity, the current produces a steeper diffusion gradient around the periphyton (Whitford and Schumacher 1964). A minimum formulation is used to combine the limiting factors for nitrogen, phosphorus, and velocity. The most severely limiting factor alone limits periphyton growth. Note that Eq. (7.188) can be configured so that low velocities are limiting by setting parameter d greater than parameter a, and vice versa to limit growth due to high velocities. In waters that are rich in nutrients, low velocities will not limit growth. However, high velocities may cause scouring and detachment of the macroalgae resulting in a reduction in biomass. The five-parameter logistic function can be configured to approximate this reduction by limiting growth at high velocities.

Macroalgae (periphyton) growth can also be limited by the availability of suitable substrate (Ross 1983). Macroalgae communities reach maximum rates of primary productivity at low levels of biomass (McIntire 1973; Pfeifer and McDiffett 1975). The relationship between standing crop and production employs the Michaelis-Menton kinetic equation:

$$f_5(D) = \frac{KBP}{KBP + P_m} \tag{7.189}$$

where

KBP is the half-saturation biomass level (g C/m²),

P_m is the macroalgae biomass level (g C/m²).

The half-saturation biomass level (KBP) is the biomass at which half the maximum growth rate occurs. Caupp et al. (1991) used a KBP value of 5.0 g C/m² (assuming 50% of ash free dry mass is carbon) for a region of the Truckee River system in California. The function in Eq. (7.189) allows maximum rates of primary productivity at low levels of biomass with decreasing rates of primary productivity as the community matrix expands.

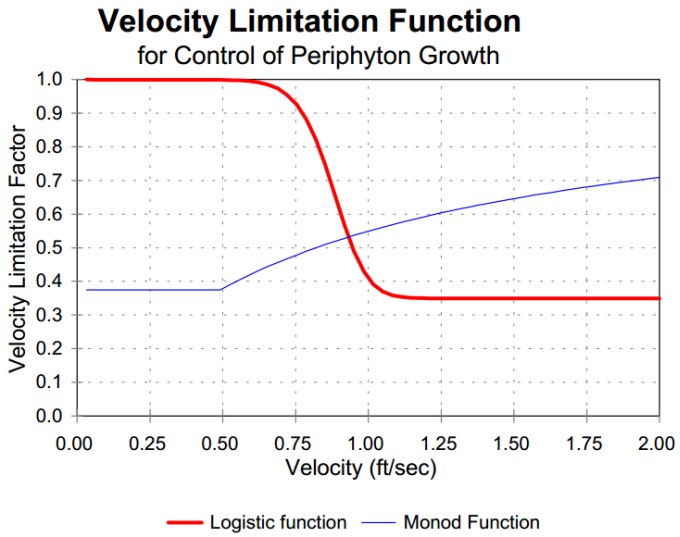


Figure 7-2: Velocity limitation function for (Option 1) the Monod equation where KMV=0.25 m/s and KMVmin=0.15 m/s, and (Option 2) the 5-parameter logistic function where a=1.0, b=12.0, c=0.3, d=0.35, and e=3.0 (high velocities are limiting).

Table 7.16: Parameters related to algae in water column

Parameter	Value[a]	Equation Number[b]
*PM_c (1/day)	2.5 (upper Potomac only)	(7.7)
*PM_d (1/day)	2.25	(7.7)
*PM_g (1/day)	2.5	(7.7)
KHN_x (g N/m^3)	0.01 (all groups)	(7.8)
KHP_x (g P/m^3)	0.001 (all groups)	(7.8)
KHS (g Si/m^3)	0.05	(7.8)
FD	Temporally-varying input	(7.17)
I_o (langleys/day)	Temporally-varying input	(7.11)
*Ke_b (1/m)	spatially-varying input	(7.12)
Ke_{ISS} (1/m per/m^3)	NA[c]	(7.12)
Ke_{Chl} (1/m per mg Chl/m^3)	0.017	(7.12)
$CChl_x$ (g C per mg Chl)	0.06 (all groups)	(7.12)
$(D_{opt})_x$ (m)	1.0 (all groups)	(7.20)
$(I_s)_{min}$ (langleys/day)	40.0	(7.20)
CI_a, CI_b & CI_c	0.7, 0.2 & 0.1	(7.21)
TM_c, TM_d & TM_g (^{o}C)	27.5, 20.0 & 25.0	(7.26)
$KTG1_c$ & $KTG2_c$ (^{o}C^{-2})	0.005 & 0.004	(7.26)
$KTG1_d$ & $KTG2_d$ (^{o}C^{-2})	0.004 & 0.006	(7.26)
$KTG1_g$ & $KTG2_g$ (^{o}C^{-2})	0.008 & 0.01	(7.26)
STOX (ppt)	1.0	(7.27)
*BMR_c (1/day)	0.04	(7.28)
*BMR_d (1/day)	0.01 0.03 (Jan.-May in saltwater only)	(7.28)
*BMR_g (1/day)	0.01	(7.28)
TR_x (^{o}C)	20.0 (all groups)	(7.28)
KTB_x (^{o}C^{-1})	0.069 (all groups)	(7.28)
*PRR_c (1/day)	0.01	(7.29)
*PRR_d (1/day)	0.215	(7.29)

	0.065 (Jan.-May in saltwater only)	
*PRR_g (1/day)	0.215	(7.29)
*WS_c (m/day)	0.0	(7.6)
*WS_d (m/day)	0.35 (January - May) 0.1 (June - December)	(7.6)
*WS_g (m/day)	0.1	(7.6)

[a] The evaluation of these values are detailed in Chapter IX of Cerco & Cole (1994).

[b] The equation number where the corresponding parameter is first shown and defined.

[c] Not available in Cerco & Cole (1994) since their formulations do not include these parameters.

[*] The parameters declared as an array in the source code.

Table 7.17: Parameters related to organic carbon in water column

Parameter	Value[a]	Equation Number[b]
FCRP	0.35	(7.30)
FCLP	0.55	(7.31)
FCDP	0.10	(7.32)
FCD_x	0.0 (all groups)	(7.32)
*WS_{RP} (m/day)	1.0	(7.30)
*WS_{LP} (m/day)	1.0	(7.31)
KHR_x (g O_2/m^3)	0.5 (all groups)	(7.32)
$KHOR_{DO}$ (g O_2/m^3)	0.5	(7.40)
K_{RC} (1/day)	0.005	(7.41)
K_{LC} (1/day)	0.075	(7.42)
K_{DC} (1/day)	0.01	(7.43)
K_{RCalg} (1/day per g C/m^3)	0.0	(7.41)
K_{LCalg} (1/day per g C/m^3)	0.0	(7.42)
K_{DCalg} (1/day per g C/m^3)	0.0	(7.43)
TR_{HDR} (^OC)	20.0	(7.41)
TR_{MIN} (^OC)	20.0	(7.43)
KT_{HDR} ($^OC^{-1}$)	0.069	(7.41)
KT_{MIN} ($^OC^{-1}$)	0.069	(7.43)
$KHDN_N$ (g N/m^3)	0.1	(7.45)
AANOX	0.5	(7.45)

[a] The evaluation of these values are detailed in Chapter IX of Cerco & Cole (1994).

[b] The equation number where the corresponding parameter is first shown and defined

* The parameters declared as an array in the source code.

Table 7.18: Parameters related to phosphorus in water column

Parameter	Value[a]	Equation Number[b]
FPRP	0.1	(7.48)
FPLP	0.2	(7.49)
FPDP	0.5	(7.50)
FPIP	0.2[c]	(7.51)
FPR_x	0.0 (all groups)	(7.48)
FPL_x	0.0 (all groups)	(7.49)
FPD_x	1.0 (all groups)	(7.50)
FPI_x	0.0[c] (all groups)	(7.51)
*WS_s (m/day)	1.0	(7.51)
K_{PO4p} (per g/m³) for TSS	NA[c]	(7.52)
K_{PO4p} (per mol/m³) for TAM	6.0	(7.52)
CP_{prm1} (g C per g P)	42.0	(7.54)
CP_{prm2} (g C per g P)	85.0	(7.54)
CP_{prm3} (per g P/m³)	200.0	(7.54)
K_{RP} (1/day)	0.005	(7.55)
K_{LP} (1/day)	0.075	(7.56)
K_{DP} (1/day)	0.1	(7.57)
K_{RPalg} (1/day per g C/m³)	0.0	(7.55)
K_{LPalg} (1/day per g C/m³)	0.0	(7.56)
K_{DPalg} (1/day per g C/m³)	0.2	(7.57)

[a] The evaluation of these values are detailed in Chapter IX of Cerco & Cole (1994).

[b] The equation number where the corresponding parameter is first shown and defined.

[c] Not available in Cerco & Cole (1994) since their formulations do not include these parameters : FPI_x is estimated from $FPR_x + FPL_x + FPD_x + FPI_x = 1$.

[*] The parameters declared as an array in the source code.

Table 7.19: Parameters related to nitrogen in water column

Parameter	Value[a]	Equation Number[b]
FNRP	0.35	(7.64)
FNLP	0.55	(7.65)
FNDP	0.1	(7.66)
FNIP	0.0[c]	(7.67)
FNR_x	0.0 (all groups)	(7.64)
FNL_x	0.0 (all groups)	(7.65)
FND_x	1.0 (all groups)	(7.66)
FNI_x	0.0 (all groups)	(7.67)
ANC_x (g N per g C)	0.167 (all groups)	(7.64)
ANDC (g N per g C)	0.933	(7.68)
K_{RN} (1/day)	0.005	(7.70)
K_{LN} (1/day)	0.075	(7.71)
K_{DN} (1/day)	0.015	(7.72)
K_{RNalg} (1/day per g C/m³)	0.0	(7.70)
K_{LNalg} (1/day per g C/m³)	0.0	(7.71)
K_{DNalg} (1/day per g C/m³)	0.2	(7.72)
Nit_m (g N/m³/day)	0.07	(7.75)
$KHNit_{DO}$ (g N/m³)	1.0	(7.75)
$KHNit_N$ (g O_2/m³)	1.0	(7.75)
TNit (°C)	27.0	(7.78)
KNit (°C^{-2})	0.0045	(7.78)
KNit (°C^{-2})	0.0045	(7.78)

[a] The evaluation of these values are detailed in Chapter IX of Cerco & Cole (1994).

[b] The equation number where the corresponding parameter is first shown and defined.

Table 7.20: Parameters related to silica in water column

Parameter	Value[a]	Equation Number[b]
FSPP	1.0[c]	(7.81)
FSIP	0.0[c]	(7.82)
FSP_d	1.0[c]	(7.81)
FSI_d	0.0[c]	(7.82)
ASC_d (g Si per g C)	0.5	(7.81)
K_{SAp} (per g/m³) for TSS	NA[c]	(7.83)
K_{SAp} (per mol/m³) for TAM	6.0	(7.83)
K_{SU} (1/day)	0.03	(7.85)
TR_{SUA} (°C)	20.0	(7.85)
KT_{SUA} ($°C^{-1}$)	0.092	(7.85)

[a] The evaluation of these values are detailed in Chapter IX of Cerco & Cole (1994).

[b] The equation number where the corresponding parameter is first shown and defined.

[c] Not available in Cerco & Cole (1994) since their formulations do not include these parameters

: FSPP and FSIP are estimated from FSPP+FSIP = 1

: FSP_d and FSI_d are estimated from FSP_d+FSI_d = 1.

Table 7.21: Parameters related to chemical oxygen demand and dissolved oxygen in water column

Parameter	Value[a]	Equation Number[b]
KH_{COD} (g O_2/m^3)	1.5	(7.86)
K_{CD} (1/day)	20.0	(7.87)
TR_{COD} (^oC)	20.0	(7.87)
KT_{COD} ($^oC^{-1}$)	0.041	(7.87)
AOCR (g O_2 per g C)	2.67	(7.88)
AONT (g O2 per g N)	4.33	(7.87)
K_R (in MKS unit)	3.933[c]	(7.88)
KT_r	1.024[c] (1.005-1.030)	(7.93)

[a] The evaluation of these values are detailed in Chapter IX of Cerco & Cole (1994).

[b] The equation number where the corresponding parameter is first shown and defined.

[c] Not available in Cerco & Cole (1994) since their formulations do not include these parameters

: K_{ro} is from O'Connor & Dobbins (1958)

: KT_r is from Thomann & Mueller (1987).

Table 7.22: Parameters related to total active metal and fecal coliform bacteria in water column

Parameter	Value[a]	Equation Number[b]
KHbmf (g O_2/m^3)	0.5	(7.96)
BFTAM (mol/m^2/day)	0.01	(7.96)
Ttam (^oC)	20.0	(7.96)
Ktam ($^oC^{-1}$)	0.2	(7.96)
TAMdmx (mol/m^3)	0.015	(7.97)
Kdotam (per g O_2/m^3)	1.0	(7.97)
KFCB (1/day)	0.0-6.1[c] (seawater)	(7.99)
TFCB ($^oC^{-1}$)	1.07[c]	(7.99)

[a] The evaluation of these values are detailed in Chapter IX of Cerco & Cole (1994).

[b] The equation number where the corresponding parameter is first shown and defined.

[c] Not available in Cerco & Cole (1994) since their formulations do not include these parameters : KFCB and TFCB are from Thomann & Mueller (1987)

7.5 Sediment Diagenesis and Flux Formulation

7.5.1. Introduction

The EFDC water quality model provides three options for defining the sediment-water interface fluxes for nutrients and dissolved oxygen. The options are: (1) externally forced spatially and temporally constant fluxes; (2) externally forced spatially and temporally variable fluxes; and (3) internally coupled fluxes simulated with the sediment diagenesis model. The water quality state variables that are controlled by diffusive exchange across the sediment-water interface include phosphate, ammonia, nitrate, silica, chemical oxygen demand and dissolved oxygen. The first two options require that the sediment fluxes be assigned as spatial/temporal forcing functions based on either observed site-specific data from field surveys or best estimates based on the literature and sediment bed characteristics. The first two options, although acceptable for model calibration against historical data sets, do not provide the cause-effect predictive capability that is needed to evaluate future water quality conditions that might result from implementation of pollutant load reductions from watershed runoff. The third option, activation of the sediment diagenesis model developed by Di Toro (2001), does provide the cause-effect predictive capability to evaluate how water quality conditions might change with implementation of alternative load reduction or management scenarios. For the Lake Thunderbird model, the third option was selected to implement the sediment diagenesis model so that load allocation scenarios could be evaluated to determine an appropriate load allocation for Lake Thunderbird.

Living and non-living particulate organic carbon deposition, simulated in the EFDC water quality model, is internally coupled with the EFDC sediment diagenesis model. The sediment diagenesis model, based on the sediment flux model of Di Toro (2001), describes the decomposition of particulate organic matter in the sediment bed, the consumption of dissolved oxygen at the

sediment-water interface (SOD) and the exchange of dissolved constituents (ammonia, nitrate, phosphate, silica, COD) across the sediment-water interface, State variables of the EFDC sediment flux model are sediment bed temperature, sediment bed particulate organic carbon (POC), particulate organic nitrogen (PON), particulate organic phosphorus (POP), porewater concentrations of phosphate, ammonia, nitrate, silica and sulfide/methane. The sediment diagenesis model computes sediment-water fluxes of chemical oxygen demand (COD), sediment oxygen demand (SOD), phosphate, ammonium, nitrate, and silica. The state variables modeled for the Lake Thunderbird sediment flux model listed in Table 7.23. An overview of the source and sink terms is presented with a description of each state variable group in this section. The details of the state variable equations, kinetic terms and numerical solution methods for the sediment diagenesis model are presented in Di Toro (2001), Park et al. (1995) and Ji (2008).

Table 7.23: EFDC sediment diagenesis model state variables

No.	Name	Bed Layer	Units
1	POC-G1	Layer-2	g/m³
2	POC-G2	Layer-2	g/m³
3	POC-G3	Layer-2	g/m³
4	PON-G1	Layer-2	g/m³
5	PON-G2	Layer-2	g/m³
6	PON-G3	Layer-2	g/m³
7	POP-G1	Layer-2	g/m³
8	POP-G2	Layer-2	g/m³
9	POP-G3	Layer-2	g/m³
10	Partic-Biogenic-Silica	Layer-2	g/m³
11	Sulfide/Methane	Layer-1	g/m³
12	Sulfide/Methane	Layer-2	g/m³
13	Ammonia-N	Layer-1	g/m³
14	Ammonia-N	Layer-2	g/m³
15	Nitrate-N	Layer-1	g/m³
16	Nitrate-N	Layer-2	g/m³
17	Phosphate-P	Layer-1	g/m³
18	Phosphate-P	Layer-2	g/m³
19	Available-Silica	Layer-1	g/m³
20	Available-Silica	Layer-2	g/m³
21	Ammonia-N-Flux		g/m²-day
22	Nitrate-N-Flux		g/m²-day
23	Phosphate-P-Flux		g/m²-day
24	Silica Flux		g/m²-day
25	SOD		g/m²-day
26	COD Flux		g/m²-day
27	Bed Temperature		°C

A sediment process model developed by DiToro & Fitzpatrick (1993; hereinafter referred to as D&F) and was coupled with CE-QUAL-ICM for Chesapeake Bay water quality modeling (Cerco & Cole 1994). The sediment process model was slightly modified and incorporated into the EFDC water quality model to simulate the processes in the sediment and at the sediment-water interface.

The description of the EFDC sediment process model in this section is from Park et al. (1995). The sediment process model has 27 water quality related state variables and fluxes (Table 7.24).

Table 7.24: EFDC sediment process model state variables and flux terms

1) particulate organic carbon G1 class in layer 2	15) nitrate nitrogen in layer 1
2) particulate organic carbon G2 class in layer 2	16) nitrate nitrogen in layer 2
3) particulate organic carbon G3 class in layer 2	17) phosphate phosphorus in layer 1
4) particulate organic nitrogen G1 class in layer 2	18) phosphate phosphorus in layer 2
5) particulate organic nitrogen G2 class in layer 2	19) available silica in layer 1
6) particulate organic nitrogen G3 class in layer 2	20) available silica in layer 2
7) particulate organic phosphorus G1 class in layer 2	21) ammonia nitrogen flux
8) particulate organic phosphorus G2 class in layer 2	22) nitrate nitrogen flux
9) particulate organic phosphorus G3 class in layer 2	23) phosphate phosphorus flux
10) particulate biogenic silica in layer 2	24) silica flux
11) sulfide/methane in layer 1	25) sediment oxygen demand
12) sulfide/methane in layer 2	26) release of chemical oxygen demand
13) ammonia nitrogen in layer 1	27) sediment temperature
14) ammonia nitrogen in layer 2	

The nitrate state variables, (15), (16) and (22), in the model represent the sum of nitrate and nitrite nitrogen. The three G classes for particulate organic matter (POM) in Layer 2, and the two layers for inorganic substances are described below.

In the sediment model, benthic sediments are represented as two layers (Figure 7-3). The upper layer (Layer 1) is in contact with the water column and may be oxic or anoxic depending on dissolved oxygen concentration in the overlying water. The lower layer (Layer 2) is permanently anoxic. The upper layer depth, which is determined by the penetration of oxygen into the sediments, is at its maximum only a small fraction of the total depth. Because H_1 (~ 0.1 cm) << H_2,

$$H = H_1 + H_2 \approx H_2 \tag{7.190}$$

where H is the total depth (approximately 10 cm), H_1 is the upper layer depth and H_2 is the lower layer depth.

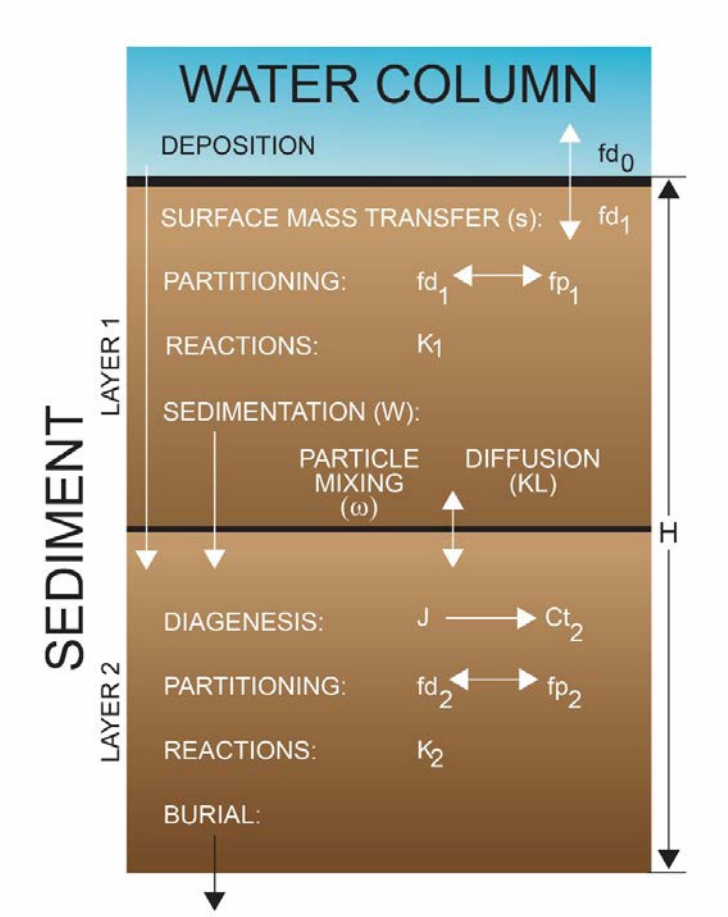


Figure 7-3: Sediment layers and processes included in sediment process model

The model incorporates three basic processes (Figure 7-4): (1) depositional flux of POM, (2) the diagenesis of POM, and (3) the resulting sediment flux. The sediment model is driven by net settling of particulate organic carbon, nitrogen, phosphorus and silica from the overlying water to the sediments (depositional flux). Because of the negligible thickness of the upper layer (Eq. (7.190)), deposition proceeds from the water column directly to the lower layer. Within the lower layer, the model simulates the diagenesis (mineralization or decay) of deposited POM, which produces oxygen demand and inorganic nutrients (diagenesis flux). The third basic process is the flux of substances produced by diagenesis (sediment flux). Oxygen demand, as sulfide (in saltwater) or methane (in freshwater), takes three paths out of the sediments: (1) oxidation at the sediment-water interface as sediment oxygen demand, (2) export to the water column as chemical oxygen demand, or (3) burial to deep, inactive sediments.

Inorganic nutrients produced by diagenesis take two paths out of the sediments: (1) release to the water column or (2) burial to deep, inactive sediments (Figure 7-4).

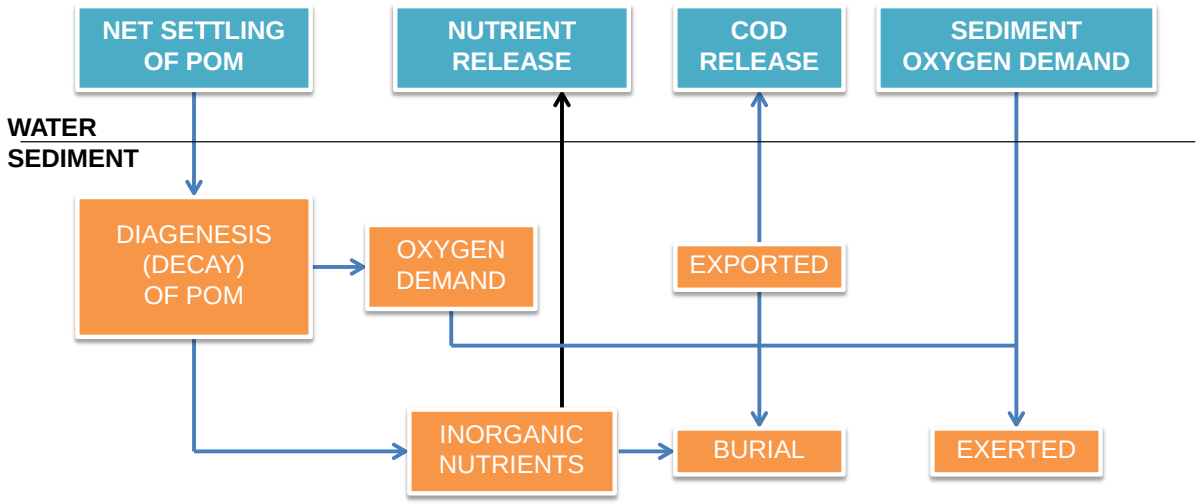


Figure 7-4: Schematic diagram for sediment process model

This section describes the three basic processes with reactions and sources/sinks for each state variable. The method of solution includes finite difference equations, solution scheme, boundary and initial conditions. Complete model documentation can be found in D&F (1993).

7.5.2. Depositional Flux

Deposition is one process that couples the water column model with the sediment model. Consequently, deposition is represented in both the water column and sediment models. In the water column model, the governing mass-balance equations for the following state variables contain settling terms, which represent the depositional fluxes:

- three algal groups, cyanobacteria, diatoms and green algae (Eq. (7.6))
- refractory and labile particulate organic carbon (Equations (7.30)and (7.31))
- refractory and labile particulate organic phosphorus (Equations (7.48) and (7.49)) and total phosphate (Eq. (7.51))
- refractory and labile particulate organic nitrogen (Equations (7.64) and (7.65))
- particulate biogenic silica (Eq. (7.81)) and available silica (Eq. (7.82))

The sediment model receives these depositional fluxes of particulate organic carbon (POC), particulate organic nitrogen (PON), particulate organic phosphorus (POP) and particulate biogenic silica (PSi). Because of the negligible thickness of the upper layer (Eq. (7.190)), deposition is considered to proceed from the water column directly to the lower layer. Since the sediment model has three G classes of POM, G_i (i = 1, 2 or 3), depending on the time scales of reactivity (Section 5.2), the POM fluxes from the water column should be mapped into three G classes based on their reactivity. Then, the depositional fluxes for the ith G class (i = 1, 2 or 3) may be expressed as:

$$J_{POC,i} = FCLP_i \cdot WS_{LP} \cdot LPOC^N + FCRP_i \cdot WS_{RP} \cdot RPOC^N + \sum_{x=c,d,g} FCB_{x,i} \cdot WS_x \cdot B_x^N \qquad (7.191)$$

$$J_{PON,i} = FNLP_i \cdot WS_{LP} \cdot LPON^N + FNRP_i \cdot WS_{RP} \cdot RPON^N + \sum_{x=c,d,g} FNB_{x,i} \cdot ANC_x \cdot WS_x \cdot B_x^N \quad (7.192)$$

$$J_{POP,i} = FPLP_i \cdot WS_{LP} \cdot LPOP^N + FPRP_i \cdot WS_{RP} \cdot RPOP^N + \sum_{x=c,d,g} FPB_{x,i} \cdot APC \cdot WS_x \cdot B_x^N + \gamma_i \cdot WS_{TSS} \cdot PO4p^N \quad (7.193)$$

$$J_{PSi} = WS_d \cdot SU^N + ASC_d \cdot WS_d \cdot B_d^N + WS_{TSS} \cdot SAp^N \quad (7.194)$$

where

$J_{POM,i}$ is the depositional flux of POM (M = C, N or P) routed into the i[th] G class (g/m²/day),

J_{PSi} is the depositional flux of PSi (g Si/m²/day),

$FCLP_i$, $FNLP_i$ & $FPLP_i$ are the fraction of water column labile POC, PON and POP, respectively, routed into the i[th] G class in sediment,

$FCRP_i$, $FNRP_i$ & $FPRP_i$ are the fraction of water column refractory POC, PON and POP, respectively, routed into the i[th] G class in sediment,

$FCB_{x,i}$, $FNB_{x,i}$ & $FPB_{x,i}$ are the fraction of POC, PON and POP, respectively, in the algal group x routed into the i[th] G class in sediment

γ_i= 1 for i = 1, γ_i= 0 for i = 2 or 3.

In the source code, the sediment process model is solved after the water column water quality model, and the calculated fluxes using the water column conditions at $t = t_n$ are used for the computation of the water quality variables at $t = t_n + \theta$. The superscript N indicates the variables after being updated for the kinetic processes, as defined in Eq. (7.105).

The settling of sorbed phosphate is considered to contribute to the labile G_1 pool in Eq. (7.193), and settling of sorbed silica contributes to J_{PSi} in Eq. (7.194) to avoid creation of additional depositional fluxes for inorganic particulates. The sum of distribution coefficients should be unity: $\sum_i FCLP_i = \sum_i FNLP_i = \sum_i FPLP_i = \sum_i FCRP_i = \sum_i FNRP_i = \sum_i FPRP_i = \sum_i FCB_{x,i} = \sum_i FNB_{x,i} = \sum_i FPB_{x,i} = 1$. The settling velocities, WS_{LP}, WS_{RP}, WS_x and WS_{TSS}, as defined in the EFDC water column model (Section 4), are net settling velocities. If total active metal is selected as a measure of sorption site, WS_{TSS} is replaced by WS_s in Equations (7.193) and (7.194) (see Sections 4.5 and 4.7).

7.5.3. Diagenesis Flux

Another coupling point of the sediment model to the water column model is the sediment flux, which is described in Section 7.5.4. The computation of sediment flux requires that the magnitude of the diagenesis flux be known. The diagenesis flux is explicitly computed using mass-balance equations for deposited POC, PON and POP. (Dissolved silica is produced in the sediments as the result of the dissolution of PSi. Since the dissolution process is different from the bacterial-mediated diagenesis process, it is presented separately in Section 7.5.5) In the mass-balance equations, the depositional fluxes of POM are the source terms and the decay of POM in the sediments produces the diagenesis fluxes. The integration of the mass-balance equations for POM provides the diagenesis fluxes that are the inputs for the mass-balance equations for ammonium, nitrate, phosphate and sulfide/methane in the sediments (Section 7.2.3).

The difference in decay rates of POM is accounted for by assigning a fraction of POM to various decay classes (Westrisch & Berner 1984). POM in the sediments is divided into three G classes, or fractions, representing three scales of reactivity. The G_1 (labile) fraction has a half life of 20 days, and

the G_2 (refractory) fraction has a half life of one year. The G_3 (inert) fraction is non-reactive, i.e., it undergoes no significant decay before burial into deep, inactive sediments. The varying reactivity of the G classes controls the time scale over which changes in depositional flux will be reflected in changes in diagenesis flux. If the G_1 class would dominate the POM input into the sediments, then there would be no significant time lag introduced by POM diagenesis and any changes in depositional flux would be readily reflected in diagenesis flux.

Because the upper layer thickness is negligible (Eq. (7.190)) and thus depositional flux is considered to proceed directly to the lower layer (Equations (7.191) to (7.194)), diagenesis is considered to occur in the lower layer only. The mass-balance equations are similar for POC, PON and POP, and for different G classes. The mass-balance equation in the anoxic lower layer for the ith G class (i = 1, 2 or 3) may be expressed as:

$$H_2 \frac{\partial G_{POM,i}}{\partial t} = -K_{POM,i} \cdot \theta_{POM,i}^{T-20} \cdot G_{POM,i} \cdot H_2 - W \cdot G_{POM,i} + J_{POM,i} \tag{7.195}$$

where

$G_{POM,i}$ is the concentration of POM (M=C, N or P) in the ith G class in Layer 2 (g/m^3),

$K_{PMO,i}$ is the decay rate of the ith G class POM at 20°C in Layer 2 (1/day),

$\theta_{POM,i}$ is the constant for temperature adjustment for $K_{POM,i}$

T is the sediment temperature (°C),

W is the burial rate (m/day),

Since the G_3 class is inert, $K_{POM,3} = 0$.

Once the mass-balance equations for $G_{POM,1}$ and $G_{POM,2}$ are solved, the diagenesis fluxes are computed from the rate of mineralization of the two reactive G classes:

$$J_M = \sum_{i=1}^{2} K_{POM,i} \cdot \theta_{POM,i}^{T-20} \cdot G_{POM,i} \cdot H_2 \tag{7.196}$$

J_M is the diagenesis flux ($g/m^2/day$) of carbon (M=C), nitrogen (M=N) or phosphorus (M=P)

7.5.4. Sediment Flux

The mineralization of POM produces soluble intermediates, which are quantified as diagenesis fluxes in the previous section. The intermediates react in the oxic and anoxic layers, and portions are returned to the overlying water as sediment fluxes. Computation of sediment fluxes requires mass-balance equations for ammonium, nitrate, phosphate, sulfide/methane and available silica. This section describes the flux portion for ammonium, nitrate, phosphate and sulfide/methane of the model. Available silica is described in Section 7.2.3.5.

In the upper layer, the processes included in the flux portion are (Fig. 5-1)

- exchange of dissolved fraction between Layer 1 and the overlying water
- exchange of dissolved fraction between Layer 1 and 2 via diffusive transport
- exchange of particulate fraction between Layer 1 and 2 via particle mixing
- loss by burial to the lower layer (Layer 2)

- removal (sink) by reaction
- internal sources.

Since the upper layer is quite thin, H_1 ~ 0.1 cm (Eq. (7.190)) and the surface mass transfer coefficient (s) is on the order of 0.1 m/day, then the residence time in the upper layer is: $H/s \approx 10^{-2}$ days. Hence, a steady-state approximation is made in the upper layer. Then the mass-balance equation for ammonium, nitrate, phosphate or sulfide/methane in the upper layer is:

$$H_1 \frac{\partial Ct_1}{\partial t} = 0 = s(fd_0 \cdot Ct_0 - fd_1 \cdot Ct_1) + KL(fd_2 \cdot Ct_2 - fd_1 \cdot Ct_1) + \omega(fp_2 \cdot Ct_2 - fp_1 \cdot Ct_1) - W \cdot Ct_1 - \frac{K_1^2}{s} Ct_1 + J_1 \quad (7.197)$$

where

Ct_1 & Ct_2 are the total concentrations in Layer 1 and 2, respectively (g/m^3),

Ct_0 is the total concentrations in the overlying water (g/m^3),

s is the surface mass transfer coefficient (m/day),

KL is the diffusion velocity for dissolved fraction between Layer 1 and 2 (m/day),

ω is the particle mixing velocity between Layer 1 and 2 (m/day),

fd_0 is the dissolved fraction of total substance in the overlying water ($0 \le fd_0 \le 1$),

fd_1 is the dissolved fraction of total substance in Layer 1 ($0 \le fd_1 \le 1$),

fp_1 is the Particulate fraction of total substance in Layer 1 ($= 1 - fd_1$),

fd_2 is the dissolved fraction of total substance in Layer 2 ($0 \le fd_2 \le 1$),

fp_2 is the Particulate fraction of total substance in Layer 2 ($= 1 - fd_2$),

K_1 is the reaction velocity in Layer 1 (m/day),

J_1 is the sum of all internal sources in Layer 1 (g/m^2/day).

The first term on the RHS of Eq. (7.197) represents the exchange across sediment-water interface. Then the sediment flux from Layer 1 to the overlying water, which couples the sediment model to the water column model, may be expressed as:

$$J_{aq} = s(fd_1 \cdot Ct_1 - fd_0 \cdot Ct_0) \quad (7.198)$$

where

J_{aq} is the sediment flux of ammonium, nitrate, phosphate or sulfide/methane to the overlying water (g/m^2/day).

The convention used in Eq. (7.198) is that positive flux is from the sediment to the overlying water.

In the lower layer, the processes included in the flux portion are (Figure 7-1)

- exchange of dissolved fraction between Layer 1 and 2 via diffusive transport
- exchange of particulate fraction between Layer 1 and 2 via particle mixing
- deposition from Layer 1 and burial to the deep inactive sediments
- removal (sink) by reaction
- internal sources including diagenetic source.

The mass-balance equation for ammonium, nitrate, phosphate or sulfide/methane in the lower layer is:

$$H_2\frac{\partial Ct_2}{\partial t} = -KL(fd_2\cdot Ct_2 - fd_1\cdot Ct_1) - \omega(fp_2\cdot Ct_2 - fp_1\cdot Ct_1) + W(Ct_1 - Ct_2) - K_2\cdot Ct_2 + J_2 \quad (7.199)$$

where

K_2 is the reaction velocity in Layer 2 (m/day)

J_2 is the sum of all internal sources including diagenesis in Layer 2 (g/m²/day).

The substances produced by mineralization of POM in sediments may be present in both dissolved and particulate phases. This distribution directly affects the magnitude of the substance that is returned to the overlying water. In Equations (7.197) to (7.199), the distribution of a substance between the dissolved and particulate phases in a sediment is parameterized using a linear partitioning coefficient.

The dissolved and particulate fractions are computed from the partitioning equations:

$$fd_1 = \frac{1}{1+m_1\cdot\pi_1} \qquad fp_1 = 1 - fd_1 \quad (7.200)$$

$$fd_2 = \frac{1}{1+m_2\cdot\pi_2} \qquad fp_2 = 1 - fd_2 \quad (7.201)$$

where

m_1 & m_2 are the solid concentrations in Layer 1 and 2, respectively (kg/l),

π_1 & π_2 are the partition coefficient in Layer 1 and 2, respectively (per kg/l).

The partition coefficient is the ratio of particulate to dissolved fraction per unit solid concentration (i.e., per unit sorption site available).

All terms, except the last two terms, in Equations (7.197) and (7.199) are common to all state variables and are described in Section 5.3.1. The last two terms represent the reaction and source/sink terms, respectively. These terms, which take different mathematical formulations for different state variables, are described in Sections 5.3.2 to 5.3.5 for ammonium, nitrate, phosphate and sulfide/methane, respectively.

a) Common Parameters for Sediment Flux

Parameters that are needed for the sediment fluxes are s, ω, KL, W, H_2, m_1, m_2, π_1, π_2, κ_1, κ_2, J_1 and J_2 in Equations (7.197) to (7.201). Of these, κ_1, κ_2 , J_1 and J_2 are variable-specific. Among the other common parameters, W, H_2, m_1 and m_2, are specified as input. The modeling of the remaining three parameters, s, ω, KL, are described in this section.

Surface mass transfer coefficient.

Owing to the observation that the surface mass transfer coefficient, s, can be related to the sediment oxygen demand, SOD (DiToro et al. 1990), s can be estimated from the ratio of SOD and overlying water oxygen concentration:

$$s = \frac{D_1}{H_1} = \frac{SOD}{DO_0} \quad (7.202)$$

where

D is the diffusion coefficient in Layer 1 (m²/day).

Knowing *s*, it is possible to estimate the other model parameters.

Particulate phase mixing coefficient:

The particle mixing velocity between Layer 1 and 2 is parameterized as:

$$\omega = \frac{D_p \cdot \theta_{D_p}^{T-20}}{H_2} \frac{G_{POC,1}}{G_{POC,R}} \frac{DO_0}{KM_{D_p} + DO_0} \tag{7.203}$$

where

D_p is the apparent diffusion coefficient for particle mixing (m^2/day),

θ_{Dp} is the constant for temperature adjustment for D_p,

$G_{POC,R}$ is the reference concentration for $G_{POC,1}$ ($g\ C/m^3$),

KM_{Dp} is the particle mixing half-saturation constant for oxygen ($g\ O_2/m^3$).

The enhanced mixing of sediment particles by macrobenthos (bioturbation) is quantified by estimating D_p. The particle mixing appears to be proportional to the benthic biomass (Matisoff 1982), which is correlated to the carbon input to the sediment (Robbins et al. 1989). This is parameterized by assuming that benthic biomass is proportional to the available labile carbon, $G_{POC,1}$, and $G_{POC,R}$ is the reference concentration at which the particle mixing velocity is at its nominal value. The Monod-type oxygen dependency accounts for the oxygen dependency of benthic biomass.

It has been observed that a hysteresis exists in the relationship between the bottom water oxygen and benthic biomass. Benthic biomass increases as the summer progresses. However, the occurrence of anoxia/hypoxia reduces the biomass drastically and also imposes stress on benthic activities. After full overturn, the bottom water oxygen increases but the population does not recover immediately. Hence, the particle mixing velocity, which is proportional to the benthic biomass, does not increase in response to the increased bottom water oxygen. Recovery of benthic biomass following hypoxic events depends on many factors including severity and longevity of hypoxia, constituent species and salinity (Diaz & Rosenberg 1995).

This phenomenon of reduced benthic activities and hysteresis is parameterized based on the idea of stress that low oxygen imposes on the benthic population. It is analogous to the modeling of the toxic effect of chemicals on organisms (Mancini 1983). A first order differential equation is employed, in which the benthic stress 1) accumulates only when overlying oxygen is below KM_{Dp} and 2) is dissipated at a first order rate (Figure 7-5a):

$$\begin{aligned} \frac{\partial ST}{\partial t} &= -K_{ST} \cdot ST + \left(1 - \frac{DO_0}{KM_{Dp}}\right) && \text{if } DO_0 < KM_{Dp} \\ \frac{\partial ST}{\partial t} &= -K_{ST} \cdot ST && \text{if } DO_0 > KM_{Dp} \end{aligned} \tag{7.204}$$

where

ST is the accumulated benthic stress (day),

K_{ST} is the first order decay rate for ST ($1/day$).

The behavior of this formulation can be understood by evaluating the steady-state stresses at two extreme conditions of overlying water oxygen, DO_0:

as $DO_0 = 0$ $K_{ST} \cdot ST = 1$ $f(ST) = (1 - K_{ST} \cdot ST) = 0$

as $DO_0 \geq KM_{Dp}$ $K_{ST} \cdot ST = 0$ $f(ST) = (1 - K_{ST} \cdot ST) = 1$

The dimensionless expression, $f(ST) = 1 - K_{ST} \cdot ST$, appears to be the proper variable to quantify the effect of benthic stress on benthic biomass and thus particle mixing (Figure 7-5b).

The final formulation for the particle mixing velocity including the benthic stress is:

$$\omega = \frac{D_p \cdot \theta_{Dp}^{T-20}}{H_2} \frac{G_{POC,1}}{G_{POC,R}} \frac{DO_0}{KM_{Dp} + DO_0} f(ST) + \frac{D_{p_{min}}}{H_2} \tag{7.205}$$

where

Dp_{min} is the minimum diffusion coefficient for particle mixing (m²/day).

The reduction in particle mixing due to the benthic stress, $f(ST)$, is estimated by employing the following procedure. The stress, ST, is normally calculated with Eq. (7.204). Once DO_0 drops below a critical concentration, $DO_{ST,c}$, for $NC_{hypoxia}$ consecutive days or more, the calculated stress is not allowed to decrease until t_{MBS} days of $DO_0 > DO_{ST,c}$. That is, only when hypoxic days are longer than critical hypoxia days ($NC_{hypoxia}$), the maximum stress, or minimum $(1 - K_{ST} \cdot ST)$, is retained for a specified period (t_{MBS} days) after DO_0 recovery (Figure 7-5). No hysteresis occurs if DO_0 does not drop below $DO_{ST,c}$ or if hypoxia lasts less than $NC_{hypoxia}$ days. When applying maximum stress for t_{MBS} days, the subsequent hypoxic days are not included in t_{MBS}. This parameterization of hysteresis essentially assumes seasonal hypoxia, i.e., one or two major hypoxic events during summer, and might be unsuitable for systems with multiple hypoxic events throughout a year.

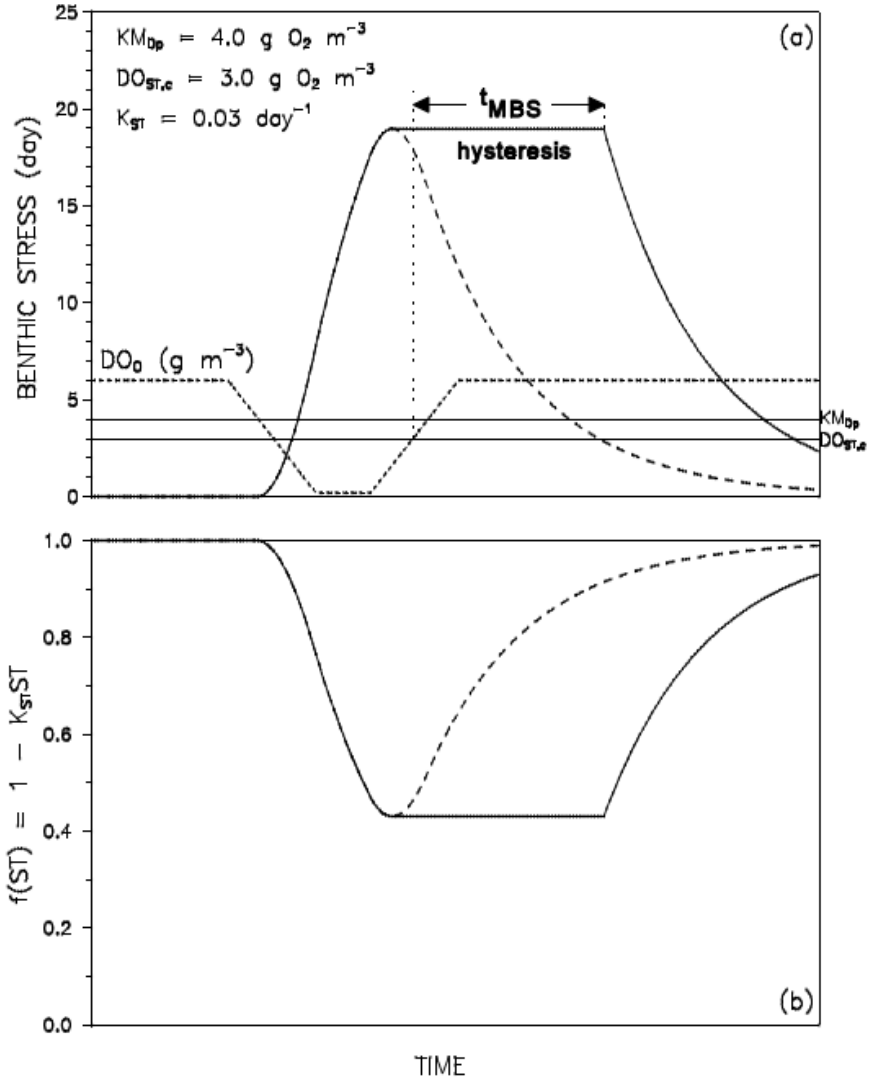


Figure 7-5: Benthic stress (a) and its effect on particle mixing (b) as a function of overlying water column dissolved oxygen concentration.

Three parameters relating to hysteresis, $DO_{ST,c}$, $NC_{hypoxia}$ and t_{MBS}, are functions of many factors including severity and longevity of hypoxia, constituent species and salinity, and thus have site-specific variabilities (Diaz & Rosenberg 1995). The critical overlying oxygen concentration, $DO_{ST,c}$, also depends on the distance from the bottom of the location of $DO_{0.}$ The critical hypoxia days, $NC_{hypoxia}$, depends on tolerance of benthic organisms to hypoxia and thus on benthic community structure (Diaz & Rosenberg, 1995). The time lag for the recovery of benthic biomass following hypoxic events, t_{MBS}, tends to be longer for higher salinity. The above three parameters are considered to be spatially constant input parameters.

Dissolved phase mixing coefficient.

Dissolved phase mixing between Layer 1 and 2 is via passive molecular diffusion, which is enhanced by the mixing activities of the benthic organisms (bio-irrigation). This is modeled by increasing the diffusion coefficient relative to the molecular diffusion coefficient:

$$KL = \frac{D_d \cdot \theta_{Dd}^{T-20}}{H_2} + R_{BI,BT} \cdot \omega \tag{7.206}$$

where

D_d is the diffusion coefficient in pore water (m²/day),

θ_{Dd} is the constant for temperature adjustment for D_d,

$R_{BI,BT}$ is the ratio of bio-irrigation to bioturbation.

The last term in Eq. (7.206) accounts for the enhanced mixing by organism activities.

b) Ammonia Nitrogen

Diagenesis is assumed not to occur in the upper layer because of its shallow depth, and ammonium is produced by diagenesis in the lower layer:

$$J_{1,NH4} = 0 \qquad J_{2,NH4} = J_N \text{ (from Eq. (7.196))} \tag{7.207}$$

Ammonium is nitrified to nitrate in the presence of oxygen. A Monod-type expression is used for the ammonium and oxygen dependency of the nitrification rate. Then, the oxic layer reaction velocity in Eq. (7.197) for ammonium may be expressed as:

$$K_{1,NH4}^2 = \frac{DO_0}{2 \cdot KM_{NH4,O2} + DO_0} \frac{KM_{NH4}}{KM_{NH4} + NH4_1} K_{NH4}^2 \cdot \theta_{NH4}^{T-20} \tag{7.208}$$

and then the nitrification flux becomes:

$$J_{Nit} = \frac{K_{1,NH4}^2}{s} \cdot NH4_1 \tag{7.209}$$

where

$KM_{NH4,O2}$ is the nitrification half-saturation constant for dissolved oxygen (g O_2/m³),

$NH4_1$ is the total ammonium nitrogen concentration in Layer 1 (g N/m³),

KM_{NH4} is the nitrification half-saturation constant for ammonium (g N/m³),

K_{NH4} is the optimal reaction velocity for nitrification at 20°C (m/day),

θ_{NH4} is the constant for temperature adjustment for K_{NH4},

J_{Nit} is the nitrification flux (g N/m²/day).

Nitrification does not occur in the anoxic lower layer:

$$K_{2,NH4} = 0 \tag{7.210}$$

Once Equations (7.197) and (7.199) are solved for $NH4_1$ and $NH4_2$, the sediment flux of ammonium to the overlying water, $J_{aq,NH4}$, can be calculated using Eq. (7.198). Note that it is not $NH4_1$ and $NH4_2$ that determine the magnitude of $J_{aq,NH4}$ (Section X-B-2 in D&F 1993). The magnitude is determined by (1) the diagenesis flux, (2) the fraction that is nitrified, and (3) the surface mass transfer coefficient (s) that mixes the remaining portion.

c) Nitrate Nitrogen

Nitrification flux is the only source of nitrate in the upper layer, and there is no diagenetic source for nitrate in both layers:

$$J_{1,NO3} = J_{Nit} \quad \text{(from Eq. (7.208))} \qquad J_{2,NO3} = 0 \tag{7.211}$$

Nitrate is present in sediments as dissolved substance, i.e., $\pi_{1,NO3} = \pi_{2,NO3} = 0$, making $fd_{1,NO3} = fd_{2,NO3} = 1$ (Equations (7.200) and (7.201)): it also makes R meaningless, hence $R = 0$. Nitrate is removed by denitrification in both oxic and anoxic layers with the carbon required for denitrification supplied by

carbon diagenesis. The reaction velocities in Equations (7.197) and (7.199) for nitrate may be expressed as:

$$K^2_{1,NO3} = K^2_{NO3,1} \cdot \theta^{T-20}_{NO3} \tag{7.212}$$

$$K_{2,NO3} = K^2_{NO3,2} \cdot \theta^{T-20}_{NO3} \tag{7.213}$$

and the denitrification flux out of sediments as a nitrogen gas becomes:

$$J_{N2(g)} = \frac{K^2_{1,NO3}}{s} NO3_1 + K_{2,NO3} \cdot NO3_2 \tag{7.214}$$

where

$K_{NO3,1}$ is the reaction velocity for denitrification in Layer 1 at 20°C (m/day),

$K_{NO3,2}$ is the reaction velocity for denitrification in Layer 2 at 20°C (m/day),

θ_{NO3} is the constant for temperature adjustment for $K_{NO3,1}$ and $K_{NO3,2}$,

$J_{N2(g)}$ is the denitrification flux (g N/m²/day),

$NO3_1$ is the total nitrate nitrogen concentration in Layer 1 (g N/m³),

$NO3_2$ is the total nitrate nitrogen concentration in Layer 2 (g N/m³).

Once Equations (7.197) and (7.199) are solved for $NO3_1$ and $NO3_2$, the sediment flux of nitrate to the overlying water, $J_{aq,NO3}$, can be calculated using Eq. (7.198). The steady-state solution for nitrate showed that the nitrate flux is a linear function of $NO3_0$ (Eq. III-15 in D&F 1993): the intercept quantifies the amount of ammonium in the sediment that is nitrified but not denitrified (thus releases as $J_{aq,NO3}$), and the slope quantifies the extent to which overlying water nitrate is denitrified in the sediment. It also revealed that if the internal production of nitrate is small relative to the flux of nitrate from the overlying water, the normalized nitrate flux to the sediment, $- J_{aq,NO3}/NO3_0$, is linear in s for small s and constant for large s (Section III-C in D&F 1993). For small s (~ 0.01 m/day), H is large (Eq. (7.202)) so that oxic layer denitrification predominates and $J_{aq,NO3}$ is essentially zero independent of $NO3_0$ (Fig. III-4 in D&F 1993).

d) Phosphate Phosphorus

Phosphate is produced by the diagenetic breakdown of POP in the lower layer:

$$J_{1,PO4} = 0 \qquad J_{2,PO4} = J_P \quad \text{(from Eq. (7.196))} \tag{7.215}$$

A portion of the liberated phosphate remains in the dissolved form and a portion becomes particulate phosphate, either via precipitation of phosphate-containing minerals (Troup 1974), e.g., vivianite, $Fe_3(PO_4)_2$(s), or by partitioning to phosphate sorption sites (Lijklema 1980; Barrow 1983; Giordani &

Astorri 1986). The extent of particulate formation is determined by the magnitude of the partition coefficients, $\pi_{1,PO4}$ and $\pi_{2,PO4}$, in Equations (7.200) and (7.201). Phosphate flux is strongly affected by DO_0, the overlying water oxygen concentration. As DO_0 approaches zero, the phosphate flux from the sediments increases. This mechanism is incorporated by making $\pi_{1,PO4}$ larger, under oxic conditions, than $\pi_{2,PO4}$. In the model, when DO_0 exceeds a critical concentration, $(DO_0)_{crit,PO4}$, sorption in the upper layer is enhanced by an amount $\pi_{PO4,1}$:

$$\pi_{1,PO4} = \pi_{2,PO4} \cdot (\Delta\pi_{PO4,1}) \qquad DO_0 > (DO_0)_{crit,PO4} \tag{7.216}$$

When oxygen falls below $(DO_0)_{crit,PO4}$, then:

$$\pi_{1,PO4} = \pi_{2,PO4} \cdot \left(\Delta\pi_{PO4,1}\right)^{DO_0/(DO_0)_{crit,PO4}} \qquad DO_0 \le (DO_0)_{crit,PO4} \tag{7.217}$$

which smoothly reduces $\pi_{1,PO4}$ to $\pi_{2,PO4}$ as DO_0 goes to zero. There is no removal reaction for phosphate in both layers:

$$\kappa_{1,PO4} = \kappa_{2,PO4} = 0 \tag{7.218}$$

Once Equations (7.197) and (7.199) are solved for $PO4_1$ and $PO4_2$, the sediment flux of phosphate to the overlying water, $J_{aq,PO4}$, can be calculated using Eq. (7.198).

e) Sulfide/methane and Oxygen Demand

Sulfide

No diagenetic production of sulfide occurs in the upper layer. In the lower layer, sulfide is produced by carbon diagenesis (Eq. (7.196)) decremented by the organic carbon consumed by denitrification (Eq. (7.214)). Then:

$$J_{1,H2S} = 0 \qquad J_{2,H2S} = a_{O2,C} \cdot J_C - a_{O2,NO3} \cdot J_{N2(g)} \tag{7.219}$$

where

$a_{O2,C}$ is the stoichiometric coefficient for carbon diagenesis consumed by sulfide oxidation (2.6667 g O_2-equivalents per g C),

$a_{O2,NO3}$ is the stoichiometric coefficient for carbon diagenesis consumed by denitrification (2.8571 g O_2-equivalents per g N).

A portion of the dissolved sulfide that is produced in the anoxic layer reacts with the iron to form particulate iron monosulfide, FeS(s) (Morse et al. 1987). The particulate fraction is mixed into the oxic layer where it can be oxidized to ferric oxyhydroxide, Fe_2O_3(s). The remaining dissolved fraction also diffuses into the oxic layer where it is oxidized to sulfate. Partitioning between dissolved and particulate sulfide in the model represents the formation of FeS(s), which is parameterized using partition coefficients, $\%_{1,H2S}$ and $\%_{2,H2S}$, in Equations (7.200) and (7.201).

The present sediment model has three pathways for sulfide, the reduced end product of carbon diagenesis: (1) sulfide oxidation, (2) aqueous sulfide flux, and (3) burial. The distribution of sulfide among the three pathways is controlled by the partitioning coefficients and the oxidation reaction velocities (Section V-E in D&F 1993). Both dissolved and particulate sulfide are oxidized in the oxic layer, consuming oxygen in the process. In the oxic upper layer, the oxidation rate that is linear in oxygen concentration is used (Cline & Richards 1969; Millero 1986; Boudreau 1991). In

the anoxic lower layer, no oxidation can occur. Then, the reaction velocities in Equations (7.197) and (7.199) may be expressed as:

$$K_{1,H2S}^2 = \left(K_{H2S,d1}^2 \cdot fd_{1,H2S} + K_{H2S,p1}^2 \cdot fp_{1,H2S}\right)\theta_{H2S}^{T-20} \frac{DO_0}{2 \cdot KM_{H2S,O2}} \tag{7.220}$$

$$K_{2,H2S}^2 = 0 \tag{7.221}$$

where

$K_{H2S,d1}$ is the reaction velocity for dissolved sulfide oxidation in Layer 1 at 20°C (m/day),

$K_{H2S,p1}$ is the reaction velocity for particulate sulfide oxidation in Layer 1 at 20°C (m/day),

θ_{H2S} is the constant for temperature adjustment for $K_{H2S,d1}$ and $K_{H2S,p1}$,

$KM_{H2S,O2}$ is the constant to normalize the sulfide oxidation rate for oxygen (g O_2/m³).

The constant, $KM_{H2S,O2}$, which is included for convenience only, is used to scale the oxygen concentration in the overlying water. At $DO_0 = KM_{H2S,O2}$, the reaction velocity for sulfide oxidation rate is at its nominal value.

The oxidation reactions in the oxic upper layer cause oxygen flux to the sediment, which exerts SOD. By convention, SOD is positive: SOD = $-J_{aq,O2}$. The SOD in the model consists of two components, carbonaceous sediment oxygen demand (CSOD) due to sulfide oxidation and nitrogenous sediment oxygen demand (NSOD) due to nitrification:

$$SOD = CSOD + NSOD = \frac{K_{1,H2S}^2}{s} H2S_1 + a_{O2,NH4} \cdot J_{Nit} \tag{7.222}$$

where

$H2S_1$ is the total sulfide concentration in Layer 1 (g O_2-equivalents/m³),

$a_{O2,NH4}$ is the stoichiometric coefficient for oxygen consumed by nitrification (4.33 g O_2 per g N).

Equation (7.219) is nonlinear for SOD because the RHS contains s (= SOD/DO_0) so that SOD appears on both sides of the equation: note that J_{Nit} (Eq. (7.209)) is also a function of s. A simple back substitution method is used, as explained in Section 5.6.1.

If the overlying water oxygen is low, then the sulfide that is not completely oxidized in the upper layer can diffuse into the overlying water. This aqueous sulfide flux out of the sediments, which contributes to the chemical oxygen demand in the water column model, is modeled using

$$J_{aq,H2S} = s(fd_{1,H2S} \cdot H2S_1 - COD) \tag{7.223}$$

The sulfide released from the sediment reacts very quickly in the water column when oxygen is available, but can accumulate in the water column under anoxic conditions. The COD, quantified as oxygen equivalents, is entirely supplied by benthic release in the water column model (Eq. (7.86)). Since sulfide also is quantified as oxygen equivalents, COD is used as a measure of sulfide in the water column in Eq. (7.223).

Methane:

When sulfate is used up, methane can be produced by carbon diagenesis and methane oxidation

consumes oxygen (DiToro et al. 1990). Owing to the abundant sulfate in the saltwater, only the aforementioned sulfide production and oxidation are considered to occur in the saltwater. Since the sulfate concentration in the freshwater is generally insignificant, methane production is considered to replace sulfide production in the freshwater. In the freshwater, methane is produced by carbon diagenesis in the lower layer decremented by the organic carbon consumed by denitrification, and no diagenetic production of methane occurs in the upper layer (Eq. (7.219)):

$$J_{1,CH4} = 0 \qquad J_{2,CH4} = a_{O2,C} \cdot J_C - a_{O2,NO3} \cdot J_{N2(g)} \tag{7.224}$$

The dissolved methane produced takes two pathways: (1) oxidation in the oxic upper layer causing CSOD or (2) escape from the sediment as aqueous flux or as gas flux:

$$J_{2,CH4} = CSOD + J_{aq,CH4} + J_{CH4(g)} \tag{7.225}$$

where

$J_{aq,CH4}$ is the aqueous methane flux (g O_2-equivalents/m^2/day),

$J_{CH4(g)}$ is the gaseous methane flux (g O_2-equivalents/m^2/day).

A portion of dissolved methane that is produced in the anoxic layer diffuses into the oxic layer where it is oxidized. This methane oxidation causes CSOD in the freshwater sediment (DiToro et al. 1990):

$$CSOD = CSOD_{max} \cdot \left(1 - \text{sech}\left[\frac{K_{CH4} \cdot \theta_{CH4}^{T-20}}{s}\right]\right) \tag{7.226}$$

$$CSOD_{max} = minimum\left\{\sqrt{2 \cdot KL \cdot CH4_{sat} \cdot J_{2,CH4}},\ J_{2,CH4}\right\} \tag{7.227}$$

$$CH4_{sat} = 100\left(1 + \frac{h + H_2}{10}\right)1.024^{20-T} \tag{7.228}$$

where

$CSOD_{max}$ is the maximum CSOD occurring when all the dissolved methane transported to the oxic layer is oxidized,

K_{CH4} is the reaction velocity for dissolved methane oxidation in Layer 1 at 20°C (m/day) ,

θ_{H2S} is the constant for temperature adjustment for K_{CH4} ,

$CH4_{sat}$ is the saturation concentration of methane in the pore water (g O_2-equivalents/m^3).

The term, (h + H_2)/10 where h and H_2 are in meters, in Eq. (7.228) is the depth from the water surface that corrects for the in situ pressure. Equation (7.228) is accurate to within 3% of the reported methane solubility between 5 and 20(C (Yamamoto et al. 1976).

If the overlying water oxygen is low, the methane that is not completely oxidized can escape the sediment into the overlying water either as aqueous flux or as gas flux. The aqueous methane flux, which contributes to the chemical oxygen demand in the water column model, is modeled using (DiToro et al., 1990):

$$J_{aq,CH4} = CSOD_{max} \cdot \text{sech}\left[\frac{K_{CH4} \cdot \theta_{CH4}^{T-20}}{s}\right] = CSOD_{max} - CSOD \tag{7.229}$$

Methane is only slightly soluble in water. If its solubility, $CH4_{sat}$ given by Eq. (7.228), is exceeded in the pore water, it forms a gas phase that escapes as bubbles. The loss of methane as bubbles, i.e., the gaseous methane flux, is modeled using Eq. (7.225) with $J_{2,CH4}$ from Eq. (7.224), CSOD from Eq. (7.226) and $J_{aq,CH4}$ from Eq. (7.229) (DiToro et al. 1990).

7.5.5. Silica

The production of ammonium, nitrate and phosphate in sediments is the result of the mineralization of POM by bacteria. The production of dissolved silica in sediments is the result of the dissolution of particulate biogenic or opaline silica, which is thought to be independent of bacterial processes.

The depositional flux of particulate biogenic silica from the overlying water to the sediments is modeled using Eq. (7.194). With this source, the mass-balance equation for particulate biogenic silica may be written as:

$$H_2 \frac{\partial PSi}{\partial t} = -S_{Si} \cdot H_2 - W \cdot PSi + J_{PSi} + J_{DSi} \tag{7.230}$$

where

Psi is the concentration of particulate biogenic silica in the sediment (g Si/m³),

S_{Si} is the dissolution rate of PSi in Layer 2 (g Si/m³/day),

J_{Psi} is the depositional flux of PSi (g Si/m³/day) given by Eq. (7.194),

J_{DSi} is the detrital flux of PSi (g Si/m³/day) to account for PSi settling to the sediment that is not associated with the algal flux of biogenic silica.

The processes included in Eq. (7.230) are dissolution (i.e., production of dissolved silica), burial, and depositional and detrital fluxes from the overlying water. Equation (7.230) can be viewed as the analog of the diagenesis equations for POM (Eq. (7.195)). The dissolution rate is formulated using a reversible reaction that is first order in silica solubility deficit and follows a Monod-type relationship in particulate silica:

$$S_{Si} = K_{Si} \cdot \theta_{Si}^{T-20} \frac{PSi}{PSi + KH_{PSi}} (Si_{sat} - fd_{2,Si} \cdot Si_2) \tag{7.231}$$

where

K_{Si} is the first order dissolution rate for PSi at 20°C in Layer 2 (1/day),

θ_{Si} is the constant for temperature adjustment for K_{Si} ,

KM_{PSi} is the silica dissolution half-saturation constant for PSi (g Si/m³),

Si_{sat} is the saturation concentration of silica in the pore water (g Si/m³).

The mass-balance equations for mineralized silica can be formulated using the general forms, Equations (7.197) and (7.199). There is no source/sink term and no reaction in the upper layer:

$$J_{1,Si} = \kappa_{1,Si} = 0 \tag{7.232}$$

In the lower layer, silica is produced by the dissolution of particulate biogenic silica, which is modeled using Eq. (7.231). The two terms in Eq. (7.231) correspond to the source term and reaction term in Eq. (7.199):

$$J_{2,Si} = K_{Si} \cdot \theta_{Si}^{T-20} \frac{PSi}{PSi + KM_{PSi}} Si_{sat} \cdot H_2 \tag{7.233}$$

$$\kappa_{2,Si} = K_{Si} \cdot \theta_{Si}^{T-20} \frac{PSi}{PSi + KM_{PSi}} f_{d2,Si} \cdot H_2 \tag{7.234}$$

A portion of silica dissolved from particulate silica sorbs to solids and a portion remains in the dissolved form. Partitioning using the partition coefficients, $\pi_{1,Si}$ and $\pi_{2,Si}$, in Equations (7.200) and (7.201) controls the extent to which dissolved silica sorbs to solids. Since silica shows similar behavior as phosphate in the adsorption-desorption process, the same partitioning method as applied to phosphate (Section 5.3.4) is used for silica. That is, when DO_0 exceeds a critical concentration, $(DO_0)_{crit,Si}$, sorption in the upper layer is enhanced by an amount $\Delta\pi_{Si,1}$:

$$\pi_{1,Si} = \pi_{2,Si} \cdot (\Delta\pi_{Si,1}) \qquad DO_0 > (DO_0)_{crit,Si} \tag{7.235}$$

When oxygen falls below $(DO_0)_{crit,Si}$, then:

$$\pi_{1,Si} = \pi_{2,Si} \cdot \left(\Delta\pi_{Si,1}\right)^{DO_0/(DO_0)_{crit,Si}} \qquad DO_0 \le (DO_0)_{crit,Si} \tag{7.236}$$

which smoothly reduces $\pi_{1,Si}$ to $\pi_{2,Si}$ as DO_0 goes to zero.

Once Equations (7.197) and (7.199) are solved for Si_1 and Si_2, the sediment flux of silica to the overlying water, $J_{aq,Si}$, can be calculated using Eq. (7.198).

7.5.6. Sediment Temperature

All rate coefficients in the aforementioned mass-balance equations are expressed as a function of sediment temperature, *T*. The sediment temperature is modeled based on the diffusion of heat between the water column and sediment:

$$\frac{\partial T}{\partial t} = \frac{D_T}{H^2}(T_W - T) \tag{7.237}$$

where

D_T is the heat diffusion coefficient between the water column and sediment (m²/s),

T_W is the temperature in the overlying water column (°C) calculated by Eq. (7.105).

The model application in D&F and Cerco & Cole (1994) used D = 1.8 × 10^{-7} m²/s.

7.5.7. Method of Solution

a) Finite-Difference Equations and Solution Scheme

An implicit integration scheme is used to solve the governing mass-balance equations. The finite difference form of Eq. (7.197) may be expressed as:

$$0 = s(fd_0 \cdot Ct'_o - fd_1 \cdot Ct'_1) + KL(fd_2 \cdot Ct'_2 - fd_1 \cdot Ct'_1) + \omega(fp_2 \cdot Ct'_2 - fp_1 \cdot Ct'_1) - W \cdot Ct'_1 - \frac{K_1^2}{s} Ct'_1 + J'_1 \tag{7.238}$$

where the primed variables designate the values evaluated at $t+$ and the unprimed variables are those at t, where θ is defined in Eq. (7.105). The finite difference form of Eq. (7.199) may be expressed as:

$$0 = -KL(fd_2 \cdot Ct_2' - fd_1 \cdot Ct_1') - \omega(fp_2 \cdot Ct_2' - fp_1 \cdot Ct_1') + W(Ct_1' - Ct_2') - \left(K_2 + \frac{H_2}{\theta}\right) Ct_2' + \left(J_2' + \frac{H_2}{\theta} Ct_2\right) \tag{7.239}$$

The two terms, - (H_2/θ)$Ct_2$1 and (H_2/θ)Ct_2, are from the derivative term, $H_2(\partial Ct_2/\partial t)$ in Eq. (7.199), each of which simply adds to the Layer 2 removal rate and the forcing function, respectively. Setting these two terms equal to zero results in the steady-state model. The two unknowns, Ct_1' and Ct_2', can be calculated at every time step using:

$$\begin{bmatrix} s \cdot fd_1 + a_1 + \frac{K_1^2}{s} & -a_2 \\ -a_1 & a_2 + W + K_2 + \frac{H_2}{\theta} \end{bmatrix} \begin{bmatrix} Ct_1' \\ Ct_2' \end{bmatrix} = \begin{bmatrix} J_1' + s \cdot fd_0 \cdot Ct_0' \\ J_2' + \frac{H_2}{\theta} Ct_2 \end{bmatrix} \tag{7.240}$$

$$a_1 = KL \cdot fd_1 + \omega \cdot fp_1 + W \qquad a_2 = KL \cdot fd_2 + \omega \cdot fp_2 \tag{7.241}$$

The solution of Eq. (7.240) requires an iterative method since the surface mass transfer coefficient, s, is a function of the SOD (Eq. (7.202)), which is also a function of s (Eq. (7.222)). A simple back substitution method is used:

(1) Start with an initial estimate of SOD: for example, SOD = $a_{O2,C} J_C$ or the previous time step SOD.

(2) Solve Eq. (7.240) for ammonium, nitrate, and sulfide/methane.

(3) Compute the SOD using Eq. (7.222).

(4) Refine the estimate of SOD: a root finding method (Brent's method in Press et al. 1986) is used to make the new estimate.

(5) Go to (2) if no convergence.

(6) Solve Eq. 5-51 for phosphate and silica.

For the sake of symmetry, the equations for diagenesis, particulate biogenic silica and sediment temperature are also solved in implicit form. The finite difference form of the diagenesis equation (Eq. (7.195)) may be expressed as:

$$G_{POM,i}' = \left(G_{POM,i} + \frac{\theta}{H_2} J_{POM,i}\right)\left(1 + \theta \cdot K_{POM,i} \cdot \theta_{POM,i}^{T-20} + \frac{\theta}{H_2} W\right)^{-1} \tag{7.242}$$

The finite difference form of the PSi equation (Eq. (7.230)) may be expressed as:

$$PSi' = \left(PSi + \frac{\theta}{H_2}(J_{PSi} + J_{DSi})\right)\left(1 + \theta \cdot K_{Si} \cdot \theta_{Si}^{T-20} \frac{Si_{sat} - f_{d2,Si} \cdot Si_2}{PSi + KM_{PSi}} + \frac{\theta}{H_2} W\right)^{-1} \tag{7.243}$$

using Eq. (7.225) for the dissolution term, in which PSi in the Monod-type term has been kept at time level t to simplify the solution. The finite difference form of the sediment temperature equation (Eq. (7.237)) may be expressed as:

$$T' = \left(T + \frac{\theta}{H^2} D_T \cdot T_W\right)\left(1 + \frac{\theta}{H^2} D_T\right)^{-1} \quad (7.244)$$

b) Boundary and Initial Conditions

The above finite difference equations constitute an initial boundary-value problem. The boundary conditions are the depositional fluxes ($J_{POM,i}$ and J_{PSi}) and the overlying water conditions (Ct_0 and T_W) as a function of time, which are provided from the water column water quality model. The initial conditions are the concentrations at t = 0, $G_{POM,i}(0)$, PSi(0), $Ct_1(0)$, $Ct_2(0)$ and T(0), to start the computations. Strictly speaking, these initial conditions should reflect the past history of the overlying water conditions and depositional fluxes, which is often impractical because of lack of field data for these earlier years.

Table 7.25: Assignment of water column particulate organic matter (POM) to sediment G classes used in Cerco & Cole (1994)

WCM Variable	Carbon & Phosphorus			Nitrogen		
	G1	G2	G3	G1	G2	G3
A. "stand alone" model	0.65	0.20	0.15	0.65	0.25	0.10
B. coupled model						
Labile Particulate	1.0	0.0	0.0	1.0	0.0	0.0
Refractory Particulate[a]						
: Bay and Tributary Zones 1	0.0	0.11	0.89	0.0	0.26	0.74
: Bay Zones 2 and 10	0.0	0.43	0.57	0.0	0.54	0.46
: All Other Zones	0.0	0.73	0.27	0.0	0.82	0.18
Algae	0.65	0.255	0.095	0.65	0.28	0.07

[a] See Figure 10-6 in Cerco & Cole (1994) for the definition of Zones.

Table 7.26: Sediment burial rates (W) used in Cerco & Cole (1994)

Bay Zones[a]	Rate (cm/yr)	Tributary Zones[a]	Rate (cm/yr)
1, 2, 10	0.50	1	0.50
3, 6, 9	0.25	2, 3	0.25
7, 8	0.37		

[a] See Figure 10-6 in Cerco & Cole (1994) for the definition of Zones.

8 LAGRANGIAN PARTICLE TRACKING

8.1 Introduction

The section presents a sub-model coded with FORTRAN 90/95 for Lagrangian particle tracking coupled to EFDC (Environmental Fluid Dynamics Code). The module of Lagrangian particle tracking is developed as an effective tool for solving numerous problems in fluid dynamics related to the simulation and prediction of the trajectory of objects traveling in rivers, lakes and marine systems. The model has been calibrated by using a simple analytical calculation for quasi-steady state and uniform flow in an open channel. In addition, several tests with different hydrodynamic regimes and geometries were performed and demonstrated the soundness of the model as well as a variety of applications. In particular, the test involving a hypothetical situation in the movement of insects at Caloosahatchee River Basin was simulated. From the result of computation the drift tendency of insects and their locations corresponding to time moments are quickly determined qualitatively and quantitatively. Through the simulations it is demonstrated that not only the velocity field, but also the randomness and diffusion due to turbulence also considerably impact the dispersion of the cluster and behavior of drifter trajectories.

8.2 Basic Equations

Study of the trajectories of movement of solid particles in a fluid environment appeared very early in mechanics and was considered as movement in a Lagrangian approach. The advantage of this method is that it is possible to track the process of movement for each specific particle in more detail and more accurately in comparison with the method of determining average concentration for grid cells. However, the solution was then too difficult to implement in practice when the number of particles is very large, due to the strong dependence on the speed of computational processing as well as the large amount of memory to be distributed for the variables. Fortunately, nowadays the development of information technology both on hardware and software is considerable and such a problem is completely possible to solve quickly, especially applying parallel processors. The movement of solid particles is decided by a field of fluid velocity, therefore it is necessary to couple it to a fluid flow model.

The Environmental Fluid Dynamics Code (EFDC) has been chosen for this purpose. It is a general-purpose modeling package for simulating three dimensional (3D) flow, transport, and biogeochemical processes in surface water systems including: rivers, lakes, estuaries, reservoirs, wetlands, and near-shore to shelf-scale coastal regions (Hamrick, 1992a, 1992b and 1996). The EFDC model was originally developed at the Virginia Institute of Marine Science for estuarine and coastal applications and is public domain software. The US EPA has continued to support its development and now EFDC is part of a family of models recommended by EPA for TMDL development.

In this paper the module of Lagrangian particle tracking is developed as a tool to solve many problems in fluid dynamics stemming from practical necessity, such as the simulation and prediction of trajectory of floating objects in rivers, lakes and the sea in general. At the same time, the extension of the EFDC model is also to solve a specific problem for the Middle St. Johns River (MSJR) Basin that has water resource issues of importance to the local authorities. The MSJR is located in an area of rapid urbanization and has associated water quality concerns of eutrophication. The possible withdrawal of surface water from the MSJR has raised concerns about changes to flushing that could lead to increased frequency of algal blooms and changes in flow velocity. As a result the siting and design of water withdrawal infrastructure must consider entrainment and impingement of larvae that are considered as neutral solid particles.

8.3 Random Walk

The governing equations used in EFDC are Navier-Stokes for fluid flow, the advection-diffusion equations for salinity, temperature, dye, toxic substances and suspended sediment transport (Hamrick, 1992, 1996; Hamrick and Wu, 1997). The equations are presented in curvi-linear coordinate system for 2DH and Sigma coordinates for the vertical direction. They are discretized with the finite difference method with explicit scheme. A detailed description on EFDC algorithm exceeds the scope of this paper. It should be noted that the hypothesis of hydrostatic pressure is used in EFDC. However, the effect of non-hydrostatic pressure is not important when the vertical velocity of flow is not very large in comparison with the horizontal components as mentioned in Chung and Eppel (2008).

The advection-diffusion equation for mass transport in a three dimensional curvilinear orthogonal coordinate system is:

$$\frac{\partial C}{\partial t}+\frac{\partial (uC)}{\partial x}+\frac{\partial (vC)}{\partial y}+\frac{\partial (wC)}{\partial z}=\frac{\partial }{\partial x}\left(A_H \frac{\partial C}{\partial x}\right)+\frac{\partial }{\partial y}\left(A_H \frac{\partial C}{\partial y}\right)+\frac{\partial }{\partial z}\left(A_b \frac{\partial C}{\partial z}\right) \tag{8.1}$$

where t is time, (x,y,z) are Lagrangian coordinates of a particle, C is concentration,(u,v,w) are velocity components of fluid flow, and A_H, A_b are the horizontal and vertical diffusion coefficients, respectively.

The differential equations for the Lagrangian movement of particles is consistent with Eq.(8.1) and are as follows:

$$dx=\left(u+\frac{\partial A_H}{\partial x}\right) dt+\left(2p-1\right)\sqrt{2A_H dt} \tag{8.2}$$

$$dy=\left(v+\frac{\partial A_H}{\partial y}\right) dt+\left(2p-1\right)\sqrt{2A_H dt} \tag{8.3}$$

$$dz=\left(w+\frac{\partial A_b}{\partial z}\right) dt+\left(2p-1\right)\sqrt{2A_b dt} \tag{8.4}$$

In which dt is the time step and p is a random number from a uniformly distributed random variable generator having mean of 0.5. When transformed using the $2p-1$ the random component has a mean of zero and a range from -1 to 1. The transformed random value allows the diffusion term to move particles +/- about the advected position. Eq's (8.2)-(8.4) follow the 3D random walk approach used by Dunsbergen et al. (1993).

In order to determine the Lagrangian trajectory of the particle, the equations (8.2)-(8.4) were incorporated into EFDC model. The numerical solution was separately divided into the advective transport and random components as described above. This approach allows the user to enable (i.e. turn on random walk) or disable (advective transport only) the random components for either the horizontal and/or the vertical directions.

The Fortran90 module, DRIFTER.F90, has been developed and merged into EFDC to solve the Eqs. (8.2)-(8.4). The module contains the following subroutines and functions:

DRIFTERC:	Solves the Eqs. (8.2)-(8.4) using one of three numerical options
DRIFTER.INP:	Reading the input parameters
READSTR:	Reading the comment lines
CONTAINER:	Determining the cell containing the drifter
AREACAL:	Calculating the area of polygons
DRIFVELCAL:	Interpolating the velocity components at the previous location of the drifter
RANDCAL:	Calculating the random movement
EDGEMOVE:	Dealing with the drifter hitting the land boundary or internal walls
INSIDECELL:	Determining if the drifter is inside a cell
DRIFTERWDEP:	Interpolating the bathymetry and total water depth at the location of drifter
DRIFTERLAYER:	Determining the layer containing the drifter

Three options are available for the solution of the differential equations (8.2)-(8.4). They are explicit Euler, predictor-corrector Euler and forth order Runge-Kutta. Their discretization for the equations are as follows:

- Explicit Euler method: This method is very simple with the approximation of O(Δt):

$$x_{n+1} = x_n + \Delta t u(t_n, x_n, y_n, z_n) \tag{8.5}$$

$$y_{n+1} = y_n + \Delta t v(t_n, x_n, y_n, z_n) \tag{8.6}$$

$$z_{n+1} = z_n + \Delta t w(t_n, x_n, y_n, z_n) \tag{8.7}$$

- Predictor-corrector Euler method: This method has the advantage of explicit and implicit features with the approximation of $O(\Delta t^2)$.

$$x_{n+1} = x_n + 0.5\Delta t\left(u(t_n, x_n, y_n, z_n) + u\left(t_{n+1}, x^p_{n+1}, y^p_{n+1}, z^p_{n+1}\right)\right) \tag{8.8}$$

$$y_{n+1} = y_n + 0.5\Delta t\left(v(t_n, x_n, y_n, z_n) + v\left(t_{n+1}, x^p_{n+1}, y^p_{n+1}, z^p_{n+1}\right)\right) \tag{8.9}$$

$$z_{n+1} = z_n + 0.5\Delta t\left(w(t_n, x_n, y_n, z_n) + w\left(t_{n+1}, x^p_{n+1}, y^p_{n+1}, z^p_{n+1}\right)\right) \tag{8.10}$$

  where $\left(x^p_{n+1}, y^p_{n+1}, z^p_{n+1}\right)$ are calculated by Eqs. (8.4)- (8.6)

- Runge-Kutta 4 method: This method has the approximation of $O(\Delta t^4)$ and has been shown in the testing for this project that it is best option of the three solution techniques provided.

$$x_{n+1} = x_n + \frac{1}{6}\left(\Delta x_1 + 2\Delta x_2 + 2\Delta x_3 + \Delta x_4\right) \tag{8.11}$$

$$y_{n+1} = y_n + \frac{1}{6}\left(\Delta y_1 + 2\Delta y_2 + 2\Delta y_3 + \Delta y_4\right) \tag{8.12}$$

$$z_{n+1} = z_n + \frac{1}{6}\left(\Delta z_1 + 2\Delta z_2 + 2\Delta z_3 + \Delta z_4\right) \tag{8.13}$$

in which

$$\Delta x_1 = \Delta tu\left(t_n, x_n, y_n, z_n\right)$$

$$\Delta x_2 = \Delta tu(t_n + 0.5\Delta t, x_n + 0.5\Delta x_1, y_n + 0.5\Delta y_1, z_n + 0.5\Delta z_1$$
$$\Delta x_3 = \Delta tu(t_n + 0.5\Delta t, x_n + 0.5\Delta x_2, y_n + 0.5\Delta y_2, z_n + 0.5\Delta z_2$$
$$\Delta x_4 = \Delta tu(t_n + \Delta t, x_n + \Delta x_3, y_n + \Delta y_3, z_n + \Delta z_3$$

$$\Delta y_1 = \Delta tv\left(t_n, x_n, y_n, z_n\right)$$
$$\Delta y_2 = \Delta tv\left(t_n + 0.5\Delta t, x_n + 0.5\Delta x_1, y_n + 0.5\Delta y_1, z_n + 0.5\Delta z_1\right)$$
$$\Delta y_3 = \Delta tv\left(t_n + 0.5\Delta t, x_n + 0.5\Delta x_2, y_n + 0.5\Delta y_2, z_n + 0.5\Delta z_2\right)$$
$$\Delta y_4 = \Delta tv\left(t_n + \Delta t, x_n + \Delta x_3, y_n + \Delta y_3, z_n + \Delta z_3\right)$$
$$\Delta z_1 = \Delta tw\left(t_n, x_n, y_n, z_n\right)$$
$$\Delta z_2 = \Delta tw\left(t_n + 0.5\Delta t, x_n + 0.5\Delta x_1, y_n + 0.5\Delta y_1, z_n + 0.5\Delta z_1\right)$$
$$\Delta z_3 = \Delta tw\left(t_n + 0.5\Delta t, x_n + 0.5\Delta x_2, y_n + 0.5\Delta y_2, z_n + 0.5\Delta z_2\right)$$

$$\Delta z_4 = \Delta tw(t_n + \Delta t, x_n + \Delta x_3, y_n + \Delta y_3, z_n + \Delta z_3$$

8.4 Oil spill model

EFDC+ version allows to simulate oil spill based on drifters. Oil is maintained at the surface layer and is moved along with the hydrodynamic impacts similar to Lagrangian particles and broken down by evaporation and biodegradation.

EFDC will cause the drifter to disappear when the oil per drifter is less than 1.0 mm^3. When configuring an oil spill model it should be noted that use of the "Vertical Movement Option" is ignored for groups that are designated as simulating oil. If the density of oil is less than that of water then the particles are always in the surface layer. If the oil is heavier than water, then the fully 3D option is enabled.

For simulation on the oil evaporation process, the theory of surface evaporation presented in the paper of Warren Stiver and Donald Mackay (1984) is used.

EFDC+ allows for effects of biodegradation of the oil based on user defined biodegradation rate with a simple first order decay approach based on Stewart (1993). A rate of 0.011 per day leads to approximately two month for half- life for oil. If temperature of an oil drifter is provided by the user then this is used as the reference rate for the optimal biodegradation.

In order to simulate a conservative oil, the user would set the degradation rate and the vapor pressure as zero.

9 MARINE HYDROKINETICS

Content to be updated a later date.

10 SUMMARY

Content to be updated at a later date.

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Appendix A: Symbol Definitions

Symbol	Units	Definition
A_1, A_2	–	empirical constants in turbulent model
A_b	m²/s	vertical turbulent eddy diffusivity
A_H	m²/s	horizontal turbulent eddy viscosity/diffusivity
A_q	m²/s	vertical turbulent diffusivity for turbulent intensity equation
A_{ql}	m²/s	vertical turbulent diffusivity for turbulent length scale equation
A_v	m²/s	vertical turbulent eddy viscosity
B_1, B_2	–	empirical constants in turbulent model
B_c	–	empirical constant
b	–	buoyancy
C	g/m³	concentration of transport constituent
C		cloud fraction
C_r	–	Courant number
C_1	–	empirical constant in turbulent model
C_b	–	bottom stress coefficient
C_D	–	wind drag coefficient
C_s		wind stress coefficient
c_e		turbulent exchange coefficients
c_h		turbulent exchange coefficients
c_p	J/kg/°C	specific heat of water
c_{pa}	J/kg/°K	specific heat of air
D	m	water depth
E_1, E_2, E_3, E_4, E_5	–	empirical constants
e_a	mb	actual vapor pressure
e_s	mb	saturation vapor pressure at surface water temperature
f	–	Coriolis parameter
g	m/s²	acceleration due to gravitation
h	m	bathymetry
H	m	water depth
H_T	J/s	rate of heat energy exchange
i		
I	W/m²	solar radiation
$I(D)$	W/m²	solar radiation at depth D below the surface
I_s	W/m²	solar radiation at the surface (D = 0)
KC	–	number of horizontal grid cells
K_0	–	constants calculated by Galperin, Kantha and Clayson
K_e	1/m	light extinction coefficient
K_o	–	dimensionless coefficient
k	–	layer index in vertical direction
k_x, k_y		wave numbers in x- and y- directions
L	m	defined at equation (2-15)
l	m	turbulent length scale
L_E	J/kg	latent heat of evaporation
M		
M	m²	Jacobian or square root of the metric tensor determinant or cell area, $m = m_x m_y$
M_+, M_-		
m_x, m_y	m	square roots of the diagonal components of the metric tensor. Cell dimensions in the x and y directions

n		
N		
N_+, N_-		
p	m^2/s^2	physical pressure in excess of the reference density hydrostatic pressure
P_a	mb	atmospheric pressure
p_s	m^2/s^2	physical pressure at the free surface or under the rigid lid divided by the reference density
q^2	m^2/s^2	turbulent intensity
q^2		turbulent kinetic energy
S_c	–	source/sink term for the constituent
Q_c	m^3/s	discharge rate
Q_l	m^4/s^3	source-sink term for turbulent length scale equation
Q_q	m^3/s^3	source-sink term for turbulent intensity equation
Q_T		horizontal turbulent diffusion and external sources / sinks
S_u	m^2/s^2	source/sink terms for the horizontal momentum in the x direction
S_v	m^2/s^2	source/sink terms for the horizontal momentum in the y direction
R_1, R_2, R_3	–	constants calculated by Galperin, Kantha and Clayson
R_q	–	Richardson number
S	ppt	salinity
t	s	time
T	°C	water temperature
T_a	°C	air temperature
T_s	°C	water surface temperature
u, v	m/s	horizontal velocity components in the curvilinear
U, V	m/s	depth averaged horizontal velocity components
U_o, V_o, W_o		
U_w, V_w		x and y components of the wind velocity
U_w, V_w	m/s	x and y components of the wind velocity at 10 meters above the water surface
w	m/s	vertical velocity component in sigma coordinate
W	m/s	wind speed
w_*	m/s	physical vertical velocity
w_{sc}	m/s	positive settling velocity when C represents a suspended material
x, y	m	orthogonal curvilinear coordinates in horizontal direction
Z	m	vertical coordinate
z	–	sigma coordinate
Z^*	m	original physical vertical coordinate
Z^*_o	m	dimensional bottom roughness height
z_o	m	roughness height
β	–	coefficient
Δ_1	m	thickness of the bottom layer
Δ_k	m	thickness of the k layer
ΔT	°C	temperature difference between the discharged water and the receiving water
Δx, Δy	m	grid sizes in x and y directions
ε		emissivity of the waterbody
ζ	m	physical vertical coordinate of the free surface
η_p	–	production efficiency factor
κ	–	von Karman constant
λ		eigenvalue
ρ	kg/m^3	water density
ρ_a	kg/m^3	air density
ρ_o	kg/m^3	water density in standard condition
ρ_s	kg/m^3	sediment density

ρ_w	kg/m³	water density
σ	W/m²/K⁴	Stefan–Boltzmann constant
τ	N/m²	turbulent shear stress
ϕ_A	–	stability viscosity coefficient
ϕ_b	–	coefficient
ϕ_K	–	Stability diffusivity coefficient
ϕ_v	–	coefficient
ω	rad/s	angular frequency

Appendix B: Stability Analysis for 3-Time Level Solution

Some insight into the stability and accuracy of the semi-implicit three time level scheme for solving the external mode equations with a dynamically active free surface can be gained by a Fourier analysis of the linearized discrete equations:

$$\bar{P}^{n+1} = \bar{P}^{n-1} - \Delta t\left(\frac{m_y}{m_x}h_0\right)^u g\delta_x^u\left(\zeta^{n+1}+\zeta^{n-1}\right) + 2\Delta t\left(\frac{m_y}{m_x}\right)^u f\bar{Q}^u \tag{B.1}$$

$$\left(\tau_{yy}\right)_k = 2A_H\frac{1}{m_y}\frac{\partial v_k}{\partial y} \tag{B.2}$$

$$\left(\tau_{yy}\right)_k = 2A_H\frac{1}{m_y}\frac{\partial v_k}{\partial y} \tag{B.3}$$

$$\overline{U}^{n+1} = \overline{U}^{n-1} - \theta\left(\frac{m_y}{m_x}h_0\right)^u g\delta_x^u(\zeta^{n+1}+\zeta^{n-1}) + 2\theta\left(\frac{m_y}{m_x}\right)^u f\overline{V}^u \tag{B.4}$$

$$\overline{V}^{n+1} = \overline{V}^{n-1} - \theta\left(\frac{m_x}{m_y}h_0\right)^v g\delta_y^v(\zeta^{n+1}+\zeta^{n-1}) - 2\theta\left(\frac{m_x}{m_y}\right)^u f\overline{U^v} \tag{B.5}$$

$$\zeta^{n+1} - \zeta^{n-1} + \theta\left(\frac{1}{m}\right)^\zeta\left[\delta_x^\zeta\left(\overline{U}^{n+1}+\overline{U}^{n-1}\right) + \delta_y^\zeta\left(\overline{V}^{n+1}+\overline{V}^{n-1}\right)\right] = 0 \tag{B.6}$$

Introduction of the Fourier representations:

$$\begin{Bmatrix}\overline{U}\\ \overline{V}\\ \overline{W}\end{Bmatrix} = \begin{Bmatrix}U_o\\ V_0\\ W_0\end{Bmatrix} exp(i\omega n\theta + ik_x m_x x + ik_y m_y y) \tag{B.7}$$

$$\overline{W} = \sqrt{gh_0 m}\zeta \tag{B.8}$$

on a rectangular Cartesian grid gives the eigenvalue problem:

$$\lambda^2\mathbf{M}_+ + \lambda\mathbf{M} + \mathbf{M}_- = 0 \tag{B.9}$$

$$M_+ = \begin{bmatrix} 1 & 0 & 2i\frac{m_y}{m_x}C_r sin(0.5k_x m_x) \\ 0 & 1 & 2i\frac{m_x}{m_y}C_r sin(0.5k_y m_y) \\ 2iC_r sin(0.5k_x m_x) & 2iC_r sin(0.5k_y m_y) & 1 \end{bmatrix} \quad \text{(B.10)}$$

$$M = 2f\theta\, cos(0.5k_x m_x)\, cos(0.5k_y m_y) \begin{bmatrix} 0 & -\frac{m_y}{m_x} & 0 \\ \frac{m_x}{m_y} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{(B.11)}$$

$$M_- = \begin{bmatrix} -1 & 0 & 2i\frac{m_y}{m_x}C_r sin(0.5k_x m_x) \\ 0 & -1 & 2i\frac{m_x}{m_y}C_r sin(0.5k_y m_y) \\ 2iC_r sin(0.5k_x m_x) & 2iC_r sin(0.5k_y m_y) & -1 \end{bmatrix} \quad \text{(B.12)}$$

$$C_r = \frac{\sqrt{gh_0}}{\sqrt{m}}\theta \quad \text{(B.13)}$$

where:

ω is the frequency,

k_x and k_y are the wave numbers, and

C_r is the Courant number associated with the shallow water wave speed.

The eigenvalue, λ, is related to the frequency and time step by:

$$\lambda = |\lambda| exp(i\omega\theta) \quad \text{(B.14)}$$

The characteristic polynomial of equation (B.9) is:

$$(\lambda^2 - 1)(\lambda^4 - 2\psi\lambda^2 + 1) = 0 \quad \text{(B.15)}$$

$$\psi = \frac{1 - 4C^2\left[\frac{m_y}{m_x} sin^2\left(\frac{k_x m_x}{2}\right) + \frac{m_x}{m_y} sin^2\left(\frac{k_y m_y}{2}\right)\right] - 2f^2\theta^2\, cos^2\left(\frac{k_x m_x}{2}\right) cos^2\left(\frac{k_y m_y}{2}\right)}{1 + 4C^2\left[\frac{m_y}{m_x} sin^2\left(\frac{k_x m_x}{2}\right) + \frac{m_x}{m_y} sin^2\left(\frac{k_y m_y}{2}\right)\right]} \quad \text{(B.16)}$$

and the roots or eigenvalues are:

$$\lambda = 1, \sqrt{\psi + i\sqrt{1-\psi^2}}, \sqrt{\psi - i\sqrt{1-\psi^2}}, -1, -\sqrt{\psi + i\sqrt{1-\psi^2}}, -\sqrt{\psi - i\sqrt{1-\psi^2}} \quad \text{(B.17)}$$

For linear stability, the absolute values of the eigenvalues must be less than or equal to one. The absolute values of the four complex eigenvalues are identically one, provided the absolute value of ψ is less than one, which requires:

$$\theta \leq |f|^{-1} \quad \text{(B.18)}$$

Thus the linearized three time level semi-implicit scheme is neutrally stable when equation (B.18) is satisfied. Since at mid-latitude the inverse of the Coriolis parameter is on the order of 10,000 sec, the time step is not overly constrained. The overall stability of the scheme will most likely be controlled by the stability of the explicit advective and curvature accelerations. The form of the curvature terms in the momentum equations suggest that they may increase the effective magnitude of the Coriolis parameter and reduce the stable time step. The stability of the explicit scheme for the advective accelerations will be discussed in subsequent sections.

The major computational problem with three time level schemes for systems of first order equations is the doubling of the number of eigenvalues over those physically characterizing the system. The eigenvalues in equation (B.17) are grouped such that the first three correspond to the true physical solution or physical mode of the system while the last three are spurious and give rise to what is referred to as the computational mode (Haltiner and Williams, 1980). Expressing the physical mode eigenvalues of the numerical scheme in terms of $\omega\theta$ using equation (B.14), gives the dispersion relation:

$$\omega\theta = 0, \pm 0.5\, arccos(|\psi|) \quad \text{(B.19)}$$

which can be compared with the dispersion relation

$$\omega\theta = 0, \pm \sqrt{f^2\theta^2 + \frac{m_y}{m_x}(Ck_x m_x)^2 + \frac{m_x}{m_y}\left(Ck_y m_y\right)^2} \quad \text{(B.20)}$$

for the continuous in time and space form of equations (B.4) to (B.6). Both the numerical and continuous solutions are characterized by a steady mode and pairs of waves propagating in opposite directions. The major feature of the computational mode of the numerical solution scheme, represented by the last three eigenvalues in equation (B.17), is an alternating change in sign of the solution at every time step. Since all eigenvalues of the scheme are neutrally stable, the computational mode solution can persist and become a source of error. Two alternatives to eliminating the computational mode are the application of weak time filter or the periodic insertion of a single step using a two time level scheme (Haltiner and Williams, 1980). For the present work, the insertion of a two time level step or possibly more appropriately termed a correction step was selected.

The correction step used to eliminate the computational mode is a trapezoidal scheme. The scheme computes a corrected time level n+1 solution using the initial condition at time level n and the solution at time level n+1 previously computed using the three time level scheme. The momentum and continuity equations, equivalent to equations (2.101)-(2.103) for the trapezoidal correction step are:

$$\begin{aligned}\overline{U}^{n+1} = \overline{U}^{n} &- 0.5\theta\left(\frac{m_y}{m_x}H^{n+\frac{1}{2}}\right)^{u} g\delta_x^u(\zeta^{n+1}+\zeta^{n}) - \theta\left[\left(\frac{m_y}{m_x}H\right)^{u}\delta_x^u p_s\right]^{n+\frac{1}{2}} \\ &+ \theta\left[\left(\frac{m_y}{m_x}H\right)^{u} g\left(\overline{b}^{u}\,\delta_x^u h - \overline{B}^{u}\delta_x^u H - 0.5H^u\delta_x^u\overline{\beta}\right)\right]^{n+\frac{1}{2}} \\ &- \theta\left(\frac{1}{m_x}\right)^{u}\sum_{k=1}^{k}\Delta_k\left[\delta_x^u(U_k u_k) + \delta_y^u(V_k u_k)\right] \\ &+ \theta\left(\frac{1}{m_x}\right)^{u}\sum_{k=1}^{K}\Delta_k\left\{\left[\left(mf + v_k\frac{\partial m_y}{\partial x} - u_k\frac{\partial m_x}{\partial y}\right)Hv_k\right]^{u}\right\}^{n+\frac{1}{2}} \\ &+ \theta m_y^u[(\tau_{xz}^n)_K - (\tau_{xz}^n)_0]^u \\ &+ \theta\left(\frac{1}{m_x}\right)^{u}\sum_{k=1}^{K}\Delta_k\left[\frac{\partial}{\partial x}\left(m_y H\tau_{xx}^n\right) + \frac{\partial}{\partial y}\left(m_x H\tau_{xy}^n\right) + \frac{\partial}{\partial y}m_x H\tau_{xy}^n\right. \\ &\left. - \frac{\partial}{\partial x}m_y H\tau_{yy}^n\right]_k^u\end{aligned} \tag{B.21}$$

$$\begin{aligned}\overline{V}^{n+1} = \overline{V}^{n} &- 0.5\theta\left(\frac{m_x}{m_y}H^{n+\frac{1}{2}}\right)^{v} g\delta_y^v(\zeta^{n+1}+\zeta^{n}) - \theta\left[\left(\frac{m_x}{m_y}H\right)^{v}\delta_y^v p_s\right]^{n+\frac{1}{2}} \\ &+ \theta\left[\left(\frac{m_x}{m_y}H\right)^{v} g\left(\overline{b}^{v}\,\delta_y^v h - \overline{B}^{v}\delta_y^v H - 0.5H^v\delta_y^v\overline{\beta}\right)\right]^{n+\frac{1}{2}} \\ &- \theta\left(\frac{1}{m_y}\right)^{v}\sum_{k=1}^{k}\Delta_k\left[\delta_x^v(U_k v_k) + \delta_y^v(V_k v_k)\right] \\ &- \theta\left(\frac{1}{m_y}\right)^{v}\sum_{k=1}^{K}\Delta_k\left\{\left[\left(mf + v_k\frac{\partial m_y}{\partial x} - u_k\frac{\partial m_x}{\partial y}\right)Hu_k\right]^{v}\right\}^{n+\frac{1}{2}} \\ &+ \theta m_x^v\left((\tau_{yz}^n)_K - (\tau_{yz}^n)_0\right)^v \\ &+ \theta\left(\frac{1}{m_y}\right)^{v}\sum_{k=1}^{K}\Delta_k\left[\frac{\partial}{\partial x}\left(m_y H\tau_{yx}^n\right) + \frac{\partial}{\partial y}\left(m_x H\tau_{yy}^n\right) - \frac{\partial}{\partial y}m_x H\tau_{xx}^n\right. \\ &\left. + \frac{\partial}{\partial x}m_y H\tau_{yx}^n\right]_k^v\end{aligned} \tag{B.22}$$

$$\zeta^{n+1} - \zeta^{n} + 0.5\theta\left(\frac{1}{m}\right)^{\zeta}\left[\delta_x^\zeta\left(\overline{U}^{n+1} + \overline{U}^{n}\right) + \delta_y^\zeta\left(\overline{V}^{n+1} + \overline{V}^{n}\right)\right] = 0 \tag{B.23}$$

where the notation n+1/2 implies:

$$(\quad)^{n+\frac{1}{2}} = 0.5((\quad)^{*} + (\quad)^{n})$$

with * denoting evaluation using time level n+1 results from the previous three time level step. Because of stability restrictions, the advective accelerations must be of the upwind form:

$$\begin{aligned}\delta_x^u[U_k(x)u_k(x)] = {} & 0.5Max\{[U_k(x+1)+U_k(x)],0\}^{n+\frac{1}{2}}u_k^n(x,y) \\ & + 0.5Min\{[U_k(x+1)+U_k(x)],0\}^{n+\frac{1}{2}}u_k^n(x+1,y) \\ & - 0.5Max\{[U_k(x)+U_k(x-1)],0\}^{n+\frac{1}{2}}u_k^n(x-1,y) \\ & - 0.5Min\{[U_k(x)+U_k(x-1)],0\}^{n+\frac{1}{2}}u_k^n(x,y)\end{aligned} \tag{B.24}$$

The solution of semi-implicit equations (B.21) –(B.23) follows that outlined for the three time level scheme.

To analyze the stability of the trapezoidal correction step with a dynamically active free surface, two points of view will be considered. Since a single application of the trapezoidal scheme essentially corrects a step of the three time level scheme, it is actually three time level. The Fourier analysis of the combined linearized three time level, trapezoidal correction scheme on a rectangular Cartesian grid gives the eigenvalue problem:

$$\lambda^2 N_+ + \lambda(N + N_- — NM_+^{-1}M) - NM_+^{-1}M_- = 0 \tag{B.25}$$

$$N_+ = \begin{bmatrix} 1 & 0 & i\frac{m_y}{m_x}Csin(0.5k_xm_x) \\ 0 & 1 & i\frac{m_x}{m_y}Csin(0.5k_ym_y) \\ iCsin(0.5k_xm_x) & iCsin(0.5k_ym_y) & 1 \end{bmatrix} \tag{B.26}$$

$$N = 0.5f\theta\, cos(0.5k_xm_x)\, cos(0.5k_ym_y) \begin{bmatrix} 0 & -\frac{m_y}{m_x} & 0 \\ \frac{m_x}{m_y} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{B.27}$$

$$N_- = \begin{bmatrix} -1 & 0 & i\frac{m_y}{m_x}Csin(0.5k_xm_x) \\ 0 & -1 & i\frac{m_x}{m_y}Csin(0.5k_ym_y) \\ iCsin(0.5k_xm_x) & iCsin(0.5k_ym_y) & -1 \end{bmatrix} \tag{B.28}$$

The sixth order characteristic polynomial of equation (B.25) is algebraically rather complex and only the two roots or eigenvalues, 1 and 0, can be determined in closed form. The eigenvalue of one is associated with the steady physical mode, while the eigenvalue of zero is spurious but serves a useful purpose in eliminating the computational mode. A second point of view is that the successive application of the trapezoidal correction is equivalent to an iterative two time level scheme. The Fourier analysis of such a scheme gives the eigenvalue problem:

$$\lambda(N_+ + N) + (N + N_-) = 0 \tag{B.29}$$

whose characteristic polynomial is:

$$(\lambda - 1)(\lambda^2 - 2\chi\lambda + 1) = 0 \tag{B.30}$$

where

$$\chi = \frac{1 - C^2\left[\frac{m_y}{m_x}sin^2\left(\frac{k_x m_x}{2}\right) + \frac{m_x}{m_y}sin^2\left(\frac{k_y m_y}{2}\right)\right] - \frac{1}{4}f^2\theta^2\, cos^2\left(\frac{k_x m_x}{2}\right)cos^2\left(\frac{k_y m_y}{2}\right)}{1 + C^2\left[\frac{m_y}{m_x}sin^2\left(\frac{k_x m_x}{2}\right) + \frac{m_x}{m_y}sin^2\left(\frac{k_y m_y}{2}\right)\right] + \frac{1}{4}f^2\theta^2\, cos^2\left(\frac{k_x m_x}{2}\right)cos^2\left(\frac{k_y m_y}{2}\right)} \tag{B.31}$$

The roots or eigenvalues of equation (B.25) are:

$$\lambda = 1, \chi + i\sqrt{1 + \chi^2}, \chi - i\sqrt{1 - \chi^2} \tag{B.32}$$

Since the absolute value of χ is always less than one, the absolute value of the complex eigenvalues is identically one and the scheme is neutrally and unconditionally stable.

The results of the Fourier analysis is also useful in accessing the accuracy of the external mode solution scheme with respect to its ability to represent the dispersion relation and the phase and group velocities of shallow water waves (Foreman, 1983). The dispersion relations for the continuous in space and time shallow water equations, and the three and two time level schemes, respectively, are:

$$\omega\theta = \sqrt{f^2\theta^2 + \frac{m_y}{m_x}(Ck_x m_x)^2 + \frac{m_x}{m_y}\left(Ck_y m_y\right)^2} \tag{B.33}$$

$$\omega\theta = 0.5 arccos(\psi) \tag{B.34}$$

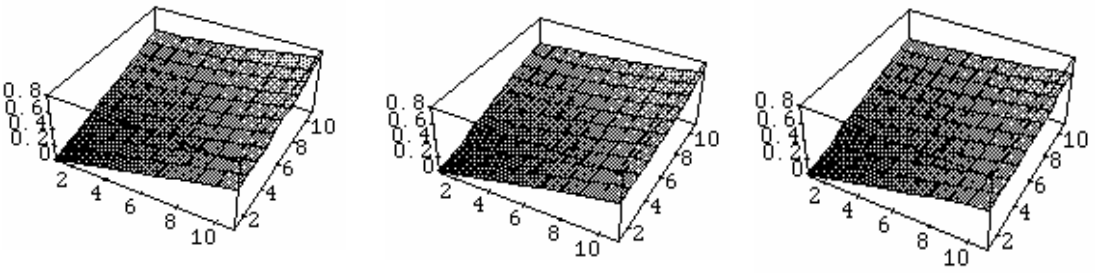
$$\omega\theta = arccos(\chi) \tag{B.35}$$

where ψ and χ are given by equations (B.16), (B.31). The phase and group velocities are given respectively by:

$$\frac{\left(c_x, c_y\right)}{\sqrt{gh_0}} = \frac{\left(k_x, k_y\right)}{\sqrt{m}\left(k_x^2 + k_y^2\right)}\frac{\omega\theta}{C} \tag{B.36}$$

$$\frac{\left(G_x, G_y\right)}{\sqrt{gh_0}} = \frac{1}{\sqrt{m}}\left(\frac{\partial}{\partial k_x}, \frac{\partial}{\partial k_y}\right)\frac{\omega\theta}{C} \tag{B.37}$$

A comparison of the dispersion relations for a square Cartesian grid with a Courant number of 5 and $f\theta$ of 0.01 is shown in Figure B-1.



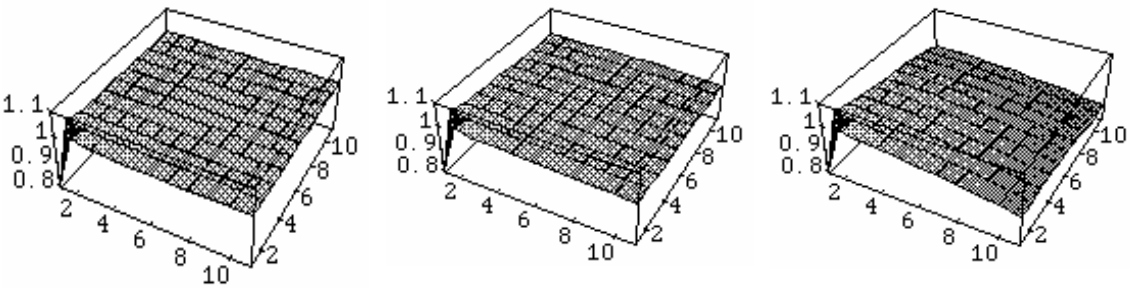
(a) Continuous equations

(b) Two time level trapezoidal difference equations

(c) Three time level leap-frog trapezoidal difference equations

Figure B-1: Dispersion relations: $\omega\theta$, vertical axis, versus $k_x m_x$ and $k_y m_y$, horizontal axes, for wave propagation schemes.

The dispersion relations for the three time level and the two time level, included for comparison, difference schemes show excellent agreement with the continuous equations for dimensionless wave numbers less than approximately 0.05. As the dimensionless wave number magnitude increases toward 0.1, both numerical schemes under predict the dimensionless frequency with the three time level scheme being less accurate than the two time level scheme. The magnitudes of the phase velocity as a function of the dimensionless horizontal wave numbers are shown in Figure B-2. Both numerical schemes increasingly under predict the phase velocity magnitude as the wave number magnitude increases, but provide relatively accurate predictions for dimensionless magnitudes less than 0.05, with the two time level scheme being more accurate. The magnitudes of the group velocity as a function of the dimensionless horizontal wave numbers are shown in Figure B-3. Again the numerical schemes increasingly under predict the group velocity magnitude as the wave number magnitude increases, but provide relatively accurate predictions for dimensionless magnitudes less than 0.05. Although the two time level scheme is shown to be more accurate than the three time level scheme at higher wave number magnitudes, it would be computationally much more costly to implement due to the iterative evaluation of the Coriolis, curvature and advective accelerations.

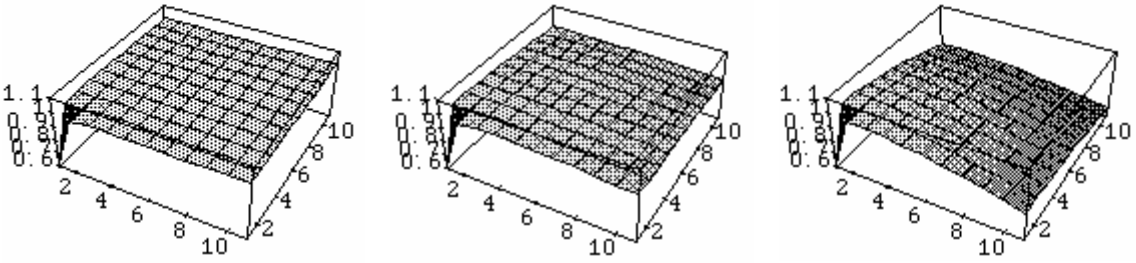


(a) Continuous equations

(b) Two time level trapezoidal difference equations

(c) Three time level leap-frog trapezoidal difference equations

Figure B-2: Magnitude of phase velocity, $|C|/\sqrt{(gh)}$, vertical axis, as a function of horizontal wave number, $k_x m_x$ and $k_y m_y$, horizontal axes.



(a) Continuous equations

(b) Two time level trapezoidal difference equations

(c) Three time level leap-frog trapezoidal difference equations

Figure B-3: Magnitude of group velocity, $|C|/\sqrt{(gh)}$, vertical axis, as a function of horizontal wave number, $k_x m_x$ and $k_y m_y$, horizontal axes.

Appendix C: Stability Analysis for Transport Equations

In this section, stability and accuracy aspects of the advection schemes common to the transport equations and the external and internal horizontal momentum equations are also discussed. The salinity transport equation (3.2) is used as a generic example and the location of variables is shown in Figure 3-1.

The salinity transport equation (3.2) is integrated over a cell layer to give:

$$\frac{\partial}{\partial t}(mHS_k) + \frac{\partial}{\partial x}(P_k S_k) + \frac{\partial}{\partial y}(Q_k S_k) + \Delta_k^{-1}[(WS)_k + (WS)_{k-1}] - \Delta_k^{-1} m(\left(H^{-1}A_b \frac{\partial S}{\partial z}\right)_k - \left(H^{-1}A_b \frac{\partial S}{\partial z}\right)_{k-1} - (Q_S)_k = 0 \quad \text{(C.1)}$$

where P_k, Q_k, and W are defined by equations (2.78) and (2.119). The source, sink, advection, and vertical diffusion portions of equation (C.1) are treated in separate fractional steps, as was done for the internal mode momentum equations in Section 5. The three time level fractional step sequence is given by:

$$S_k^* = S_k^{n-1} + 2\theta(mH^{n-1})^{-1}\,(Q_S)_k^{n-1} \quad \text{(C.2)}$$

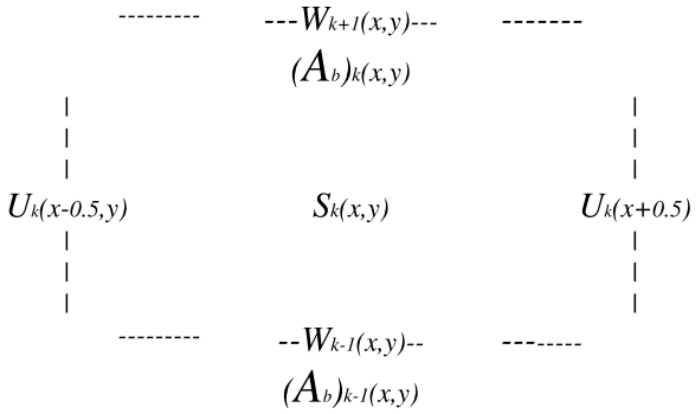


Figure C-4: S centered grid in the vertical (x,z) plane.

$$(mH)^{n+1}S_k^{**} = (mH)^{n-1}S_k^* - 2\theta\left[\delta_x^{\zeta}(U_k S_k) + \delta_y^{\zeta}(V_k S_k) + \Delta_k^{-1}((WS)_k - (WS)_{k-1})\right] \quad \text{(C.3)}$$

$$(HS_k)^{n+1} - 2\theta\left\{\left[\frac{(H^{-1}A_b)_k^n(S_{k+1} - S_k)^{n+1}}{\Delta_k\Delta_{k+1,k}}\right] - \left[\frac{(H^{-1}A_b)_{k-1}^n(S_k - S_{k-1})^{n+1}}{\Delta_k\Delta_{k,k+1}}\right]\right\} = H^{n+1}S_k^{**} \quad \text{(C.4)}$$

The source, sink step, equation (C.2) is explicit and involves no changes in cell volumes. When the source, sink term represents horizontal turbulent diffusion, it is evaluated at time level n-1, for stability (Fletcher, 1988). The advection step, equation (C.3), is explicit and involves changes

in cell volumes. The vertical diffusion step, equation (C.4), which involves no changes in cell volumes, is fully implicit and unconditionally stable (Fletcher, 1988).

Rearranging equation (C.4), the vertical diffusion step, gives:

$$-\frac{2\theta}{\Delta_k\Delta_{k+1,k}}\left(\frac{A_b}{H}\right)^n_{k-1} S^{n+1}_{k-1} + \left[\frac{2\theta}{\Delta_k\Delta_{k,k-1}}\left(\frac{A_b}{H}\right)^n_{k-1} + H^{n+1} + \frac{2\theta}{\Delta_k\Delta_{k+1,k}}\left(\frac{A_b}{H}\right)^n_{k}\right] S^{n+1}_{k} - \frac{2\theta}{\Delta_k\Delta_{k+1,k}}\left(\frac{A_b}{H}\right)^n_{k} S^{n+1}_{k+1} = H^{n+1}S^{**}_k \tag{C.5}$$

For salinity, temperature, and suspended sediment concentration, the generic variable S is defined vertically at cell layer centers, and the diffusivity is defined at cell layer interfaces. Equation (C.5) then represents a system of K equations and the boundary conditions are generally of the specified flux type. Specified surface and bottom flux boundary conditions are most conveniently incorporated in the surface and bottom cell layer source and sink terms allowing A_b at the bottom boundary, k = 0, and the surface boundary, k = K+1, to be set to zero making equation (C.5) tri-diagonal. For turbulence intensity and turbulence length scale, equations (2.23) - (2.24), the generic variable S is defined vertically at cell layer interfaces and the diffusivity is defined at cell layer centers. Equation (C.5) then represents a system of K-1 equations for the variables at internal interfaces with the variable values at the free surface and bottom being provided as boundary conditions. For the turbulence intensity and length scale, the boundary conditions are:

$$\begin{aligned} q^2_0 &= B_1^{2/3}|\tau_0| \\ q^2_K &= B_1^{2/3}|\tau_K| \\ I_0 &= 0 \\ I_K &= 0 \end{aligned} \tag{C.6}$$

where τ_0 and τ_K are the bottom and surface stress vectors respectively. Insertion of these boundary conditions results in equation (C.5) representing tri-diagonal systems of K-1 equations for the turbulence intensity and length scale.

Without loss of generality, the notation used in analyzing the three time level advection step, equation (C.3), is simplified by replacing the double and single asterisk intermediate time level indicators by n+1 and n-1, respectively to give:

$$\begin{aligned} (mHS_k)^{n+1} = (mHS_k)^{n-1} &- 2\theta\{U_k(x+0.5)S_k(x+0.5) - U_k(x-0.5)S_k(x-0.5) \\ &+ V_k(y+0.5)S_k(y+0.5) - V_k(y-0.5)S_k(y-0.5) \\ &+ \Delta_k^{-1}[(WS)_k - (WS)_{k-1}]\} \end{aligned} \tag{C.7}$$

where the horizontal central difference operators have been expanded about the cell volume centroid (x,y), according to equations (2.104) - (2.105). The cell face fluxes can be represented consistent with centered in time and space differencing as was illustrated by equations (2.106), (2.107),(2.120) or forward in time and backward or upwind in space as was illustrated by equations (2.108), (2.121) for the x momentum fluxes. For the centered in time and space form, equation (C.7) becomes:

$$(mHS_k)^{n+1} = (mHS_k)^{n-1} - \theta\{\tilde{U}_k(x+0.5)[S_k(x+1)+S_k(x)] - \tilde{U}_k(x-0.5)[S_k(x)+S_k(x-1)] + \tilde{V}_k(y+0.5)[S_k(y+1)+S_k(y)] - \tilde{V}_k(y-0.5)[S_k(y)+S_k(y-1)] + \Delta_k^{-1}\widetilde{W}_k(S_{k+1}+S_k) - \Delta_k^{-1}\widetilde{W}_{k-1}(S_k + S_{k-1})\} \quad \text{(C.8)}$$

The transports in equation (C.8) are evaluated at the centered time level when used in the external and internal momentum equations, and are averaged to the centered time level using

$$\tilde{U}_k = 0.5(U_k^{n+1} + U_k^{n-1}) \quad \text{(C.9)}$$

when used in the transport equations for scalar variables.

To investigate the stability and accuracy of the centered in time and space scheme, the Fourier representation:

$$S_k = S_0 exp(i\omega n\theta + ik_x m_x x + ik_y m_y x + ik_z H\Delta z) \quad \text{(C.10)}$$

is introduced into equation (C.8) giving the characteristic polynomial

$$\lambda^2 + 2i\psi\lambda - 1 = 0 \quad \text{(C.11)}$$

$$\psi = \frac{u\theta}{m_x} sin(k_x m_x) + \frac{v\theta}{m_x} sin(k_y m_y) + \frac{w\theta}{H\Delta} sin(k_z H\Delta) \quad \text{(C.12)}$$

for a steady and spatially uniform velocity field. The roots of equation (C.11) are:

$$\lambda = \left(\sqrt{1-\psi^2} - i\psi\right), -\left(\sqrt{1-\psi^2} + i\psi\right) \quad \text{(C.13)}$$

and the scheme is neutrally stable if the absolute value of ψ is less than or equal to one. The most restrictive stability condition is then

$$\theta \le \left[\left(\frac{u}{m_x}\right)^2 + \left(\frac{v}{m_y}\right)^2 + \left(\frac{w}{H\Delta}\right)^2\right]^{-\frac{1}{2}} \quad \text{(C.14)}$$

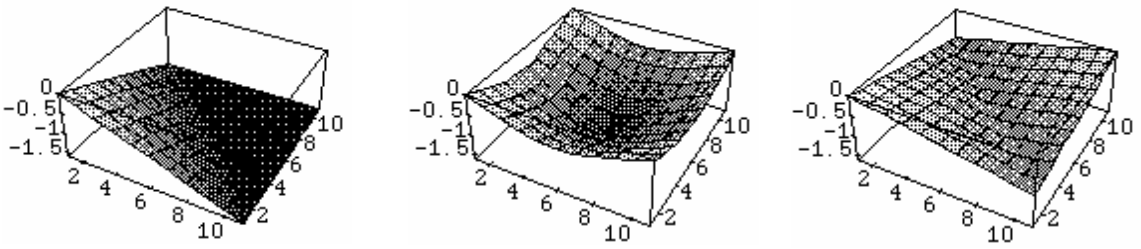
which requires the vector magnitude of the directional Courant Numbers to be less than or equal to unity. Since the centered in time and space scheme is neutrally stable when equation (C.14) is satisfied, the numerical scheme, like the continuous equations, has no dissipation. Since the scheme involves three time levels, a spurious solution mode corresponding the second eigenvalue in equation (C.13) is introduced. Using equations (B.14), (C.12) and (C.13), the dispersion relation for the physical mode of the numerical scheme is:

$$sin(\omega\theta) = -\frac{u\theta}{m_x} sin(k_x m_x) - \frac{v\theta}{m_x} sin(k_y m_y) - \frac{w\theta}{H\Delta} sin(k_z H\Delta) \quad \text{(C.15)}$$

The dispersion relation for the equivalent continuous equation is:

$$\omega\theta = -\frac{u\theta}{m_x}(k_x m_x) - \frac{v\theta}{m_x}(k_y m_y) - \frac{w\theta}{H\Delta}(k_z H\Delta) \qquad \text{(C.16)}$$

Comparison of the dispersion relations shows that errors in the phase and propagation speed of the centered in time and space numerical scheme are smallest for directional Courant numbers near unity in magnitude and for small values of the wave number component, grid spacing products (Fletcher, 1988). Figure C-5 shows equations (C.15), (C.16) for a two-dimensional flow with directional Courant Numbers of 0.5. Although the centered in time and space scheme is desirable because it has no dissipation, its phase errors at high wave numbers are undesirable. For the transport of the horizontal momentum components in regions having large velocity gradients due to topographic variations, the centered in time and space scheme generates high wave number spatial oscillations which can corrupt the solution for the velocity field (Smith and Cheng, 1987). The addition of horizontal diffusion to smooth the local oscillations can result in unrealistic damping of the surface wave propagation in other regions of the solution domain. When used for the transport of positive scalar fields, particularly in regions having high gradients or frontal discontinuities, the dispersive character of the centered in time and space scheme at high wave numbers is undesirable since it can lead to high wave number oscillations and unrealistic negative values of strictly positive scalar field variables.



(a) Continuous equation (values less than -1.5 shown as -1.5)

(b) Three time level centered in time and space

(c) Three time level forward in time and upwind in space

Figure C-5: Dispersion relations $\omega\theta$, vertical axis, versus $k_x m_x$ and $k_y m_y$, horizontal axes, for advection scheme. Courant Numbers = 0.5

Forward in time and backward or upwind in space representation of advective transport provides an alternative to the centered in time and space representation. The forward in time and backward or upwind in space form of equation (C.7) is:

$$
\begin{aligned}
(mHS_k)^{n+1} = (mHS_k)^{n-1} \\
&- \theta\{[\tilde{U}_k(x+0.5) + |\tilde{U}_k(x+0.5)|]S_k^{n-1}(x) \\
&+ +[\tilde{U}_k(x+0.5) - |\tilde{U}_k(x+0.5)|]S_k^{n-1}(x+1) \\
&- [\tilde{U}_k(x-0.5) + |\tilde{U}_k(x-0.5)|]S_k^{n-1}(x-1) \\
&- [\tilde{U}_k(x-0.5) - |\tilde{U}_k(x-0.5)|]S_k^{n-1}(x) \\
&+ [\tilde{V}_k(y+0.5) + |\tilde{V}_k(y+0.5)|]S_k^{n-1}(y) \\
&+ [\tilde{V}_k(y+0.5) - |\tilde{V}_k(y+0.5)|]S_k^{n-1}(y+1) \\
&- [\tilde{V}_k(y-0.5) + |\tilde{V}_k(y-0.5)|]S_k^{n-1}(y-1) \\
&+ [\tilde{V}_k(y-0.5) - |\tilde{V}_k(y-0.5)|]S_k^{n-1}(y) \\
&+ \Delta_k^{-1}\left[\left(\widetilde{W}_k + \left|\widetilde{W}_k\right|\right)S_k^{n-1} + \left(\widetilde{W}_k - \left|\widetilde{W}_k\right|\right)S_{k+1}^{n-1} - \left(\widetilde{W}_{k-1} + \left|\widetilde{W}_{k-1}\right|\right)S_{k-1}^{n-1}\right. \\
&\left. - \left(\widetilde{W}_{k-1} - \left|\widetilde{W}_{k-1}\right|\right)S_k^{n-1}\right]\}
\end{aligned}
\tag{C.17}
$$

The transports in equation (C.17) are evaluated at the centered time level when used in the external and internal momentum equations, and are averaged to the centered time level as illustrated by equation (C.9), when used in the transport equations for scalar variables. A Fourier analysis of equation (C.17) for a steady and spatially uniform velocity field gives the amplification factors or eigenvalues:

$$\lambda = \pm\sqrt{1-\alpha-i\beta} \tag{C.18}$$

$$\alpha = \frac{|u|(2\theta)}{m_x}[1-cos(k_x m_x)] + \frac{|v|(2\theta)}{m_y}\left[1-cos\left(k_y m_y\right)\right] + \frac{|w|(2\theta)}{H\Delta}[1-cos(k_z H\Delta)] \tag{C.19}$$

$$\beta = \frac{u(2\theta)}{m_x}sin(k_x m_x) + \frac{v(2\theta)}{m_y}sin\left(k_y m_y\right) + \frac{w(2\theta)}{H\Delta}sin(k_z H\Delta) \tag{C.20}$$

The stability of the scheme is determined by noting that the absolute value of λ,

$$|\lambda| = [(1-\alpha)^2 + \beta^2]^{\frac{1}{4}} \tag{C.21}$$

is maximum with respect to the three wave numbers when,

$$1-\alpha = \frac{|u|(2\theta)}{m_x}cos(k_x m_x) + \frac{|v|(2\theta)}{m_y}cos\left(k_y m_y\right) + \frac{|w|(2\theta)}{H\Delta}cos(k_z H\Delta) \tag{C.22}$$

Requiring consistency with the one-dimensional results allows it to be shown that the maximum absolute value of λ is equal to the fourth root of the maximum value with respect to the wave numbers of 1-*a*. Thus the most restrictive stability requirement for the absolute value of λ being less than or equal to one is:

$$\theta \le \frac{1}{2}\left[\left(\frac{u}{m_x}\right)^2 + \left(\frac{v}{m_y}\right)^2 + \left(\frac{w}{H\Delta}\right)^2\right]^{-\frac{1}{2}} \tag{C.23}$$

which was previously given by Smolarkiewicz (1984). Smolarkiewicz also showed that when the stability condition is satisfied, the upwind scheme is positive definite and the sign of strictly positive scalar variables is preserved.

It is noted that when the stability condition, equation (C.23), is satisfied, the amplification factor or absolute value of λ will in general be less than one and the scheme is dissipative. Figure C-5 shows the absolute value of the amplification factor, equation (C.22), for two two- dimensional flows with directional Courant Numbers of 0.5 and 0.25. For the case of the directional Courant Numbers equal to 0.5, there is no dissipation of disturbances propagating diagonal to the grid, while dissipation otherwise increases as either wave number increases or the direction of propagation changes from the diagonal. For the lower Courant Number case, dissipation increased with increasing wave number magnitude. The dissipation of high wave number or short wave length disturbances is desirable for controlling noise in the solution, but is undesirable when high wave number features such as strong vertical stratification or horizontal frontal discontinuities are important dynamical features sought in the solution. The dispersion relation for the forward in time and upwind in space scheme is:

$$\sqrt{(1-\alpha)^2+\beta^2}\,sin(2\omega\theta) = -\frac{u(2\theta)}{m_x}sin(k_x m_x) - \frac{v(2\theta)}{m_y}sin\left(k_y m_y\right) - \frac{w(2\theta)}{H\Delta}sin(k_z H\Delta) \qquad \text{(C.24)}$$

Figure C-5 shows the dispersion relations for the continuous advection equation, equation (C.17), the centered in time and space scheme, equation (C.16) and the forward in time and upwind in space scheme, equation (C.24), for a two-dimensional flow with directional Courant Numbers of 0.5. For high wave number disturbances propagating along either grid direction the upwind scheme, although inaccurate relative to the continuous equations, is more accurate than the centered in time and space scheme.

The ideal advective transport scheme for scalar variables in environmental flows would retain the positive definite character of the forward in time and upwind in space scheme but control the dissipation of the scheme. The search for an ideal advective transport scheme has resulted in the development of numerous high order upwind schemes, modified centered in space schemes, and combined schemes as evidenced in the review by Rood (1986). Many of these schemes, although successful, are difficult to apply near boundaries in multi-dimensional flow fields. A high order upwind scheme developed by Smolarkiewicz (Smolarkiewicz, 1984; Smolarkiewicz and Clark, 1986; and Smolarkiewicz and Grabowski, 1990) and referred to as the multi-dimensional positive definite advective transport algorithm is particularly attractive because it is simple to apply near boundaries and has a sound and transparent theoretical basis. Since this scheme is used for scalar advective transport in the environmental fluid dynamics computer code, an outline of the scheme is presented here for completeness.

The central concept in the development of the Smolarkiewicz or MPDATA scheme is the determination of the sources of dissipation or damping in the forward in time and upwind in space scheme and the modification of the scheme to compensate or eliminate a significant portion of the dissipation. The sources of dissipation are identified by a consistency analysis of equation (C.18) which involves the use of Taylor series to determine the actual continuous equation represented by the finite difference equation. The resulting continuous equation, to second order in time and space, is:

$$\begin{aligned}
&\frac{\partial}{\partial t}(m_x m_y hS)+\frac{\partial}{\partial x}(m_y HuS)+\frac{\partial}{\partial y}(m_x HvS)+\frac{\partial}{\partial x}(m_x m_y wS)\\
&\quad=\frac{\partial}{\partial x}\left(m_y H\left(\frac{m_x|u|}{2}\left(1-\frac{|u|(2\theta)}{m_x}\right)\frac{1}{m_x}\frac{\partial S}{\partial x}-\frac{u(2\theta)v}{2}\frac{1}{m_y}\frac{\partial S}{\partial y}-\frac{u(2\theta)w}{2}\frac{1}{H}\frac{\partial S}{\partial z}\right)\right)\\
&\quad+\partial_y\left(m_x H\left(-\frac{v(2\theta)u}{2}\frac{1}{m_y}\frac{\partial S}{\partial x}+\frac{m_y|v|}{2}\left(1-\frac{|v|(2\theta)}{m_y}\right)\frac{1}{m_y}\frac{\partial S}{\partial y}\right.\right.\\
&\quad\left.\left.-\frac{v(2\theta)w}{2}\frac{1}{H}\frac{\partial S}{\partial z}\right)\right)\\
&\quad+\partial_z\left(m_x m_y\left(-\frac{w(2\theta)u}{2}\frac{1}{m_x}\frac{\partial S}{\partial x}-\frac{w(2\theta)v}{2}\frac{1}{m_y}\frac{\partial S}{\partial y}\right.\right.\\
&\quad\left.\left.+\frac{H\Delta|w|}{2}\left(1-\frac{|w|(2\theta)}{H\Delta}\right)\frac{1}{H}\frac{\partial S}{\partial z}\right)\right)
\end{aligned}\tag{C.25}$$

The appearance of the second order diffusion terms in equation (C.25), which are the source of dissipation or damping, indicates that the scheme is only first order accurate in space. The centering of the advective field at time level *n* or as an average between time levels *n*-1 and *n*+1 in equation (C.18) eliminates a second order in time truncation term that would otherwise appear in equation (C.25), (Smolarkiewicz and Clark, 1986). Thus, the scheme is formally second order in time. The diffusion terms in equation (C.25) are generally referred to as numerical diffusion and are represented by a symmetric diffusion coefficient tensor,

$$D=\begin{bmatrix}
\frac{m_x|u|}{2}\left(1-\frac{|u|(2\theta)}{m_x}\right) & -\frac{u(2\theta)v}{2} & -\frac{u(2\theta)w}{2}\\
-\frac{v(2\theta)u}{2} & \frac{m_y|v|}{2}\left(1-\frac{|v|(2\theta)}{m_y}\right) & -\frac{v(2\theta)w}{2}\\
-\frac{w(2\theta)u}{2} & -\frac{w(2\theta)v}{2} & \frac{H\Delta|w|}{2}\left(1-\frac{|w|(2\theta)}{H\Delta}\right)
\end{bmatrix}\tag{C.26}$$

The magnitude of the diagonal components of the tensor increase as the directional Courant Numbers decrease, while off diagonal components or cross wind diffusivities tend to increase in magnitude as the velocity vector becomes diagonal to the grid.

To compensate for the diffusion terms in equation (C.25), the velocity field is modified by the vector addition of an anti-diffusive velocity field,

$$\begin{aligned}
\hat{u}&=\frac{1}{S}\left(\frac{m_x|u|}{2}\left(1-\frac{|u|(2\theta)}{m_x}\right)\frac{1}{m_x}\frac{\partial S}{\partial x}-\frac{u(2\theta)v}{2}\frac{1}{m_y}\frac{\partial S}{\partial y}-\frac{u(2\theta)w}{2}\frac{1}{H}\frac{\partial S}{\partial z}\right)\\
\tilde{v}&=\frac{1}{S}\left(-\frac{v(2\theta)u}{2}\frac{1}{m_x}\frac{\partial S}{\partial x}+\frac{m_y|v|}{2}\left(1-\frac{|v|(2\theta)}{m_y}\right)\frac{1}{m_y}\frac{\partial S}{\partial y}-\frac{v(2\theta)w}{2}\frac{1}{H}\frac{\partial S}{\partial z}\right)
\end{aligned}\tag{C.27}$$

$$\hat{w} = \frac{1}{S}\left(-\frac{w(2\theta)u}{2}\frac{1}{m_x}\frac{\partial S}{\partial x} - \frac{w(2\theta)v}{2}\frac{1}{m_y}\frac{\partial S}{\partial y} + \frac{H\Delta|w|}{2}\left(1 - \frac{|w|(2\theta)}{H\Delta}\right)\frac{1}{H}\frac{\partial S}{\partial z}\right)$$

which in principle exactly cancels the diffusion terms in equation (C.25). As implemented in the MPDATA scheme, the anti-diffusive velocity field is introduced in a fractional step process. The first step involves calculating a low order solution for S at time level n+1 using equation (C.18) and the actual velocity field. Using the actual velocity field and the low order solution at time level n+1, the anti-diffusive velocity field is calculated using equation (C.27). A corrected or high order solution for S at time level n+1 is then calculated using equation (C.18) in the form:

$$\begin{aligned}(mHS_k)^{n+1,H} = (mHS_k)^{n+1,L} \\ - \theta\Big(&\big(\hat{U}_k(x+0.5) + |\hat{U}_k(x+0.5)|\big)S_k^{n+1,L}(x) \\ &+ \big(\hat{U}_k(x+0.5) - |\hat{U}_k(x+0.5)|\big)S_k^{n+1,L}(x+1) \\ &- \big(\hat{U}_k(x-0.5) + |\hat{U}_k(x-0.5)|\big)S_k^{n+1,L}(x-1) \\ &- \big(\hat{U}_k(x-0.5) - |\hat{U}_k(x-0.5)|\big)S_k^{n+1,L}(x) \\ &+ (\tilde{V}_k(y+0.5) + |\tilde{V}_k(y+0.5)|)S_k^{n+1,L}(y) \\ &+ (\tilde{V}_k(y+0.5) - |\tilde{V}_k(y+0.5)|)S_k^{n+1,L}(y+1) \\ &- (\tilde{V}_k(y-0.5) + |\tilde{V}_k(y-0.5)|)S_k^{n+1,L}(y-1) \\ &+ (\tilde{V}_k(y-0.5) - |\tilde{V}_k(y-0.5)|)S_k^{n+1,L}(y) \\ &+ \Delta_k^{-1}\Big(\big(\hat{W}_k + |\hat{W}_k|\big)S_k^{n+1,L} + \big(\hat{W}_k - |\hat{W}_k|\big)S_{k+1}^{n+1,L} \\ &- \big(\hat{W}_{k-1} + |\hat{W}_{k-1}|\big)S_{k-1}^{n+1,L} - \big(\hat{W}_{k-1} - |\hat{W}_{k-1}|\big)S_k^{n+1,L}\Big)\Big)\end{aligned} \tag{C.28}$$

Since the anti-diffusive velocity field, equation (C.27), does not satisfy the continuity equations (2.103), (2.134), the anti-diffusive velocity should be set to zero on all open boundaries such that global continuity of the transported variable is maintained. The application of equation (C.28) will also introduce numerical diffusion, and another anti- diffusive velocity field can be calculated and an additional application of equation (C.28) can be made. In principle this can continue until the numerical diffusion is insignificantly small, however computationally more than two anti-diffusive steps tends to be inefficient.

The MPDATA scheme, althought strictly sign preserving, can suffer from dispersive ripples similar to other higher order advection schemes, (Smolarkiewicz and Grabowski, 1990). The dispersive ripples can be controlled by applying the scheme in conjunction with flux corrected transport methodology (Zalesak, 1979) as described by Smolarkiewicz and Grabowski. When applied in the flux corrected transport form, the solution of equation (C.18) is the low order positive definite solution, while the solution of equation (C.28) is the high order solution. The advective fluxes in the high order solution are however multiplied by a flux limiter, which is less than unity, such that the high order solution is free of dispersive ripples. The calculation of the flux limiter is described by Smolarkiewicz and Grabowski (1990) and Zalesak (1979).

Since the three time level fractional step scheme, equations (3.4) to (3.6), for the transport equations introduces a spurious computational solution mode, periodic insertion of a two time level correction step as discussed in Section 4 is necessary. The two time level fractional step scheme is given by:

$$S_k^* = S_k^n + \theta(nH^n)^{-1}(Q_S)_k^n \tag{C.29}$$

$$(mH)^{n+1}S_k^{**} = (mH)^{n-1}S_k^* - \theta\left(\delta_x^\zeta(U_kS_k) + \delta_y^\zeta(V_kS_k) + \Delta_k^{-1}((WS)_k - (WS)_{k-1})\right)^{n+\frac{1}{2}} \tag{C.30}$$

$$(HS_k)^{n+1} - \theta\left(\left(\frac{(H^{-1}A_b)_k^{n+\frac{1}{2}}(S_{k+1} - S_k)^{n+1}}{\Delta_k\Delta_{k+1,k}}\right) - \left(\frac{(H^{-1}A_b)_{k-1}^{n+\frac{1}{2}}(S_k - S_{k-1})^{n+1}}{\Delta_k\Delta_{k,k+1}}\right)\right) = H^nS_k^{**} \tag{C.31}$$

The vertical diffusion fractional step, equation (C.31) is rearranged to give a tri-diagonal system similar to equation (3.7). The vertical diffusivity divided by depth in equation (C.31) can be an arithmetic or geometric average between the value at time level n and the value at time level n+1 from the three time level step.

The advective fractional step, equation (C.30), is forward in time and upwind in space and is given by:

$$\begin{aligned}(mHS_k)^{n+1} = {} & (mHS_k)^n \\ & -\frac{\theta}{2}\Big((\tilde{U}_k(x+0.5) + |\tilde{U}_k(x+0.5)|)^{n+\frac{1}{2}}S_k^n(x) \\ & + (\tilde{U}_k(x+0.5) - |\tilde{U}_k(x+0.5)|)^{n+\frac{1}{2}}S_k^n(x+1) \\ & - (\tilde{U}_k(x-0.5) + |\tilde{U}_k(x-0.5)|)^{n+\frac{1}{2}}S_k^n(x-1) \\ & - (\tilde{U}_k(x-0.5) - |\tilde{U}_k(x-0.5)|)^{n+\frac{1}{2}}S_k^n(x) \\ & + (\tilde{V}_k(y+0.5) + |\tilde{V}_k(y+0.5)|)^{n+\frac{1}{2}}S_k^n(y) \\ & + (\tilde{V}_k(y+0.5) - |\tilde{V}_k(y+0.5)|)^{n+\frac{1}{2}}S_k^n(y+1) \\ & - (\tilde{V}_k(y-0.5) + |\tilde{V}_k(y-0.5)|)^{n+\frac{1}{2}}S_k^n(y-1) \\ & + (\tilde{V}_k(y-0.5) - |\tilde{V}_k(y-0.5)|)^{n+\frac{1}{2}}S_k^n(y) \\ & + \Delta_k^{-1}\Big((\widetilde{W}_k + |\widetilde{W}_k|)^{n+\frac{1}{2}}S_k^n + (\widetilde{W}_k - |\widetilde{W}_k|)^{n+\frac{1}{2}}S_{k+1}^n \\ & - (\widetilde{W}_{k-1} + |\widetilde{W}_{k-1}|)^{n+\frac{1}{2}}S_{k-1}^n - (\widetilde{W}_{k-1} - |\widetilde{W}_{k-1}|)^{n+\frac{1}{2}}S_k^n\Big)\Big)\end{aligned} \tag{C.32}$$

where the single and double asterisks have been replaced by n and n+1. The flows in equation (C.32) are averages of the values at n and the values at n+1 computed from the three time level scheme. The stability condition and dispersion relation for the two time level advection scheme are given by replacing 2 θ in equations (C.23) (C.24) with θ. The anti-diffusive correction to equation (C.32) follows directly from equations (C.27) (C.28) with 2θ being replaced by θ.

Appendix D: Derived Constituents for Water Quality

Water Quality

1	CHC	cyanobacteria
2	CHG	diatom algae
3	CHD	green algae
4	ROC	refractory particulate organic carbon
5	LOC	labile particulate organic carbon
6	DOC	dissolved organic carbon
7	ROP	refractory particulate organic phosphorus
8	LOP	labile particulate organic phosphorus
9	DOP	dissolved organic phosphorus
10	P4D	total phosphate
11	RON	refractory particulate organic nitrogen
12	LON	labile particulate organic nitrogen
13	DON	dissolved organic nitrogen
14	NHX	ammonia nitrogen
15	NOX	nitrate nitrogen
16	SUU	particulate biogenic silica
17	SAA	dissolved available silica
18	COD	chemical oxygen demand
19	DOX	dissolved oxygen
20	TAM	total active metal
21	FCB	fecal coliform bacteria
22	NWQV+1) MAC	macroalgae and periphyton

Water Quality Derived Parameters

23	Total Organic Carbon	TOC = CHC + CHG + CHD + ROC + LOC + DOC
24	Total N	TN = AlgaeRatio(N:C)*(CHC + CHG + CHD) + RON + LON + DON + NHX + NOX
25	Total P	TP = AlgaeRatio(P:C)*(CHC + CHG + CHD) + ROP + LOP + DOP + P4D
26	Chlorophyll a	Chl-a = AlgaeRatio(Chla:C)*CHC + AlgaeRatio(Chla:C)*CHG + AlgaeRatio(Chla:C)*CHD
27	Total Organic N	TON = AlgaeRatio(N:C)*(CHC + CHG + CHD) + RON + LON + DON
28	Total Inorganic N	TIN = NHX + NOX
29	Total Organic P	TOP = AlgaeRatio(P:C)*(CHC + CHG + CHD) + ROP + LOP + DOP
30	Carlson's Trophic State Index	
31	Particulate Organic Carbon	POC = CHC + CHG + CHD + ROC + LOC
32	Particulate Organic Phosphorus	POP = AlgaeRatio(P:C)*(CHC + CHG + CHD) + ROP + LOP
33	Particulate Organic Nitrogen	PON = AlgaeRatio(N:C)*(CHC + CHG + CHD) + RON + LON
34	Total Suspended Solids (Inorganic + Organic) TSS(io)	TSSio = TSED + TSND + CHC + CHG + CHD + ROC + LOC
35	Total Kjedahl Nitrogen (TKN)	TKN = PON + NHX TKN = AlgaeRatio(N:C)*(CHC + CHG + CHD) + RON + LON + NHX
36	Total Dissolved Phosphorous (DOP+PO4)	TDP = DOP + PO4
37	Macroalgae per unit area ug Chl-a/m²	MAC/m² = AlgaeRatio(Chla:C)*MAC/Thickness[Layer 1]

Appendix E: Application Specific Parameter Tables for Water Quality

This section includes application specific parameter tables for algae (Table 7.2), organic carbon (Table 0.3), phosphorous (Table 0.4), nitrogen (Table 0.5), silica (Table 0.6), chemical oxygen demand and dissolved oxygen (Table 0.7). The Table contents may vary between various application specific distributions of this document. The reader is referred to the references listed in Table 0.1, for a list of various CE-QUAL-ICM and EFDC water quality model application specific parameter tables.

Table 0.1: Water Quality Model Applications

Application Site	Description	Study
Chesapeake Bay	Middle Atlantic Estuary System	Cerco and Cole, 1995; Park *et al.*, 1995
San Juan Bay, Puerto Rico	Sub-Tropical Estuary	Bunch *et al.*, 2000
Florida Bay	Sub-Tropical Estuary/Ocean	Cerco *et al.*, 2000
Chesapeake Bay Refinement	Middle Atlantic Estuary System	Cerco *et al.*, 2002
Lake Washington, WA	Pacific Northwest Freshwater Lake	Cerco *et al.*, 2004
St. Johns River	Southern Atlantic Estuary	Tillman *et al.*, 2004
Peconic Bays, NY	New England Estuary/Ocean	Tetra Tech, 1999
Christina River, DE/PA	Middle Atlantic Freshwater River	Tetra Tech, 2000a
Tenkiller Lake, OK	Mid-West Freshwater Lake	Tetra Tech, 2000b
Cape Fear River, NC	Middle Atlantic Estuary	Tetra Tech, 2001
Mashpouge Pond, MA	New England Freshwater Lake	Tetra Tech, 2002c
Charles River, MA	New England Estuary/River	Tetra Tech, 2006b
Charleston Harbor, SC	Mid-Southern Atlantic Estuary	Tetra Tech, 2006c

Table 0.2: Water Quality Parameters Related to Algae in the Water Column

Parameter	Florida Bay (Cerco *et al.*, 2000)	Equation Ref
PM_c (1/day)	4	
PM_d (1/day)	N/A	
PM_g (1/day)	N/A	
KHN_x (g N/m³)	0.03	
KHP_x (g P/m³)	0.005	
FD	1	
Ke_b (1/m)	0.13	
$KeISS$ (1/m per g/m³)	0.085	
$KeVSS$ (1/m per g/m³)	0.085	
$KeChl$ (1/m/mg Chl/m³)	N/A	
$CChl_x$ (g C/mg Chl)	75	
$(D_{opt})_x$ (m)	N/A	
I_{sx} (W/m²)	N/A	
I_{sxmin} (W/m²)	N/A	
KHI (W/m²)	60 (E/m²-day)	
CI_a, CI_b & CI_c	1.0, 0.0, 0.0	
$TMlow_c$, $TMupp_c$ (°C)	25, 25	
$TMlow_d$, $TMupp_d$ (°C)	N/A	
$TMlow_g$, $TMupp_g$ (°C)	N/A	

$KTG1_c$, $KTG2_c$ (1/°C²)	0.004, 0.012	
$KTG1_d$, $KTG2_d$ (1/°C²)	N/A	
$KTG1_g$, $KTG2_g$ (1/°C²)	N/A	
$STOX$ (psu)	1.0	
BMR_c (1/day)	0.1	
BMR_d (1/day)	N/A	
BMR_g (1/day)	N/A	
TR_x (°C)	20	
KTB_x (1/°C)	0.0322	
PRR_c (1/day)	0.02	(7.29)
PRR_d (1/day)	N/A	
PRR_g (1/day)	N/A	
BxP (g C/m³)	1	(7.29)
aP	1	(7.29)
WS_c (m/day)	0.01	
WS_d (m/day)	N/A	
WS_g (m/day)	N/A	

Table 0.3: Parameters Related to Organic Carbon in the Water Column

Parameter	Florida Bay (Cerco *et al.*, 2000)	Equation Ref
$FCRP$	0.25	
$FCLP$	0.50	
$FCDP$	0.25	
FCD	0.0[a]	
KHR_x (g O2/ m³)	0.0	
$WSRP$ (m/day)	0.03	
$WSLP$ (m/day)	0.03	
$KHORDO$ (g O2/ m³)	0.5	
K_{RC} (1/day)	0.005	
K_{LC} (1/day)	0.02	
KDC (1/day)	0.01	
K_{RCalg} (1/day/ g C/m³)	0.0[a]	
K_{LCalg} (1/day/ g C/m³)	0.0[a]	
$KDCalg$ (1/day/ g C/m³)	0.0[a]	
TR_{HDR} (°C)	20.0	
TR_{MNL} (°C)	20.0	
$KTHDR$ (1/°C)	0.069	
$KTMNL$ (1/°C)	0.069	
$AANOX$	0.0[b]	
$KROR_{DO}$ (g O/m³)	0.0[b]	
$KHDN_N$ (g N/m³)	0.0[b]	

[a] Not reported. Assumed value of zero

[b] Not reported. Assumed no simulation of denitrification

Table 0.4: Parameters Related to Phosphorus in the Water Column

Parameter	Florida Bay (Cerco *et al.*, 2000)	Equation Ref
FPRP	0.03	
FPLP	0.07	
FPDP	0.4	
FPIP	0.5[a]	
FPR_x	0.0	
FPL_x	0.0	
FPD_x	0.5	
FPI_x	0.5[a]	
APC_x (g P per g C)	0.0167	
WS_S (m/day)	0.03	
K_{PO4p} (m^3/g) for TSS	0.2	
CP_{prm1} (g C per g P)	60	
CP_{prm2} (g C per g P)	0[b]	
CP_{prm3} (per g P/m^3)	0[b]	
K_{RP} (1/day)	0.005	
K_{LP} (1/day)	0.12	
K_{DP} (1/day)	0.2	
K_{RPalg}(1/day per g C/m^3)	0.2	
K_{LPalg}(1/day per g C/m^3)	0.0[b]	
K_{DPalg}(1/day per g C/m^3)	0.0[b]	

[a] Not reported. Value inferred from constraint.

[b] Not reported. Value assumed to be zero.

Table 0.5: Parameters Related to Nitrogen in the Water Column

Parameter	Florida Bay (Cerco *et al.*, 2000)	Equation Ref
FNRP	0.15	
FNLP	0.25	
FNDP	0.5	
FNIP	0.1	
FNR_x	0.15	
FNL_x	0.25	
FND_x	0.5	
FNI_x	0.1	
ANC_x (g N per g C)	0.175	
ANDC (g N per g C)	0.933	
K_{RN} (1/day)	0.005	
K_{LN} (1/day)	0.03	
K_{DN} (1/day)	0.01	
K_{RNalg} (1/day/g C/m^3)	0.0[a]	
K_{LNalg} (1/day/g C/m^3)	0.0[a]	
K_{DNalg} (1/day/g C/m^3)	0.0[a]	
Nit_m (g N/m^3/day)	0.01	
K_{Nit} (1/day)	0.01	

KHNitDO (g O2/ m³)	3.0	
KHNitN (g N/m³)	1.0	
TNit1 (°C)	30	
TNit2 (°C)	30	
KNit1 (1/°C²)	0.003	
KNit2 (1/°C²)	0.003	

[a] Not reported. Assumed value of zero.

Table 0.6: Parameters Related to Silica in the Water Column

Parameter	St. Johns River (Tillman *et al.*, 2004)	Equation Ref
FSId	0.5	
FSIP	0.5[a]	
FSPd	0.5	
FSPP	0.5[a]	
ASCd (g Si/g C)	0.8	
KSUA (1/day)	0.03	
KSap (m³/g)	0.2[a]	
KHS (g Si/m³)	0.03	

[a] Not Reported. Values assumed

Table 0.7: Parameters Related to Chemical Oxygen Demand and Dissolved Oxygen in the Water Column

Parameter	Florida Bay (Cerco *et al.*, 2000)	Equation Ref
KHCOD (g O2/ m³)	0.5	
KCD (1/day)	20	
TRCOD (°C)	20a	
KTCOD (1/°C)	0.069a	
AOCR (g O2 per g C)	2.67	
AONT (g O2 per g N)	4.33	
Kro (in SI unit)	3.933	
KTr	20	